Power System Simulation with LTspice
A practical approach to power system simulation

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For everyone who wants to gain a deeper understanding of power systems.

Preface

This book presents practical methods for modelling and analysing electrical power systems with LTspice. The emphasis is placed on reproducible examples that connect physical interpretation, manual calculation and numerical simulation.

The material is intended for students, engineers and technically interested readers. Each chapter combines a concise theoretical introduction with worked calculations, circuit models, simulation results and practical remarks.

The book presents nine simulation examples, ranging from simple to more complex cases. The formatting of the book and, in some cases, the translation from German to English were done with the help of ChatGPT. All examples and technical details are, of course, provided by the author. I wish readers every success with their simulations and hope that they, too—like me—will find great enjoyment in them. I would also like to thank the developers of LTspice for this excellent software, which made it possible for me to write this book. Although it is designed for simulating electronic circuits, it is so flexible that it allowed me to perform high-voltage analyses with it as well. Using a mathematical trick, I was also able to adapt LTspice for steady-state load-flow calculations, which led to the development of my second toolbox. This toolbox is also freely available and will be presented in more detail in a future book.

Friedrich Schmidt
Austria, July 2026

Acknowledgements

I would like to express my sincere thanks for the excellent examples in the book ”Dynamisches Verhalten elektrischer Maschinen” by Prof. Otto Justus, which provided me with a great deal of inspiration for my own book.

How to Use This Book

The chapters are written so that the reader can reproduce the examples directly. The following elements are used consistently:

Introduction	Introduces the examples
Remark	Adds interpretation, limitations or practical advice.
Example	Presents a worked application.
Calculation	Aligns formulas, substituted values and results.
Simulation	Describes model settings and simulation steps.
Chapter Summary	Collects the main conclusions at the end of a chapter.

It is recommended that you work through the book from beginning to end, while reproducing the examples in LTspice. A prerequisite for successful simulations is a good understanding of how to use the LTspice simulator. A solid understanding of SPICE syntax is absolutely essential for a thorough understanding of how the models are constructed. The complex models of the transformer and the synchronous generator are explained in the appendices of this book. To better understand the detailed modeling, it is recommended to review the LTspice-library files.

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1 Transformer and Shunt Reactor

1.1 General

In the first example, a three-phase circuit is examined. The case of an inductive load, namely the switching-on of a shunt reactor, and its impact on the voltage are investigated. The load is energized first by one transformer and then by two transformers. Finally, a special case with two transformers in series is presented. Such a configuration can occur when a de-energized part of a network is restored from a network section that is already energized (for instance after a blackout).

1.2 Circuit Design

The example file `Transformer_and_shunt_reactor.asc` is for the first transient (.TRAN) analysis. Additional measurements defined with `.meas` in the LTspice schematic are written to the `spice-output-logfile` file at the end of the simulation. In LTspice, this file can be opened with *View* → *SPICE Output Log*. To set simulation accuracy for voltages and currents the LTspice netlist settings are:

```
.options vntol=1m
.options abstol=1u
```

These values are much higher as at normal LTspice simulations with electronic circuits.

Remark 1.1 – (Convergence)

If you encounter convergence problems you can try to increase `vntol` and `abstol`. You can also try to set the relative tolerance `reltol` to 0.01 Changing the default integration mode can also be helpful.

The following line for frequency and angular velocity should be included in every schematic:

```
.param tdv=1 tdt=1 fn1=50Hz(60Hz) omega_n={2*pi*fn1}
```

tdv=1 and tdt=1 are LTspice parameters used to avoid convergence problems with electrical power networks. This first example contains buses, an impedance, transformers, and shunt reactors. The bus elements provide external measurement pins and the setting for the line-to-line voltage. Elements with three pins, such as `bus_v`, measure all three line-to-line voltages as instantaneous values (bus_v) or RMS values (bus_Veff). During simulation the RMS values require some time to settle.

For modeling the net-impedance the model impedance_sk is used. The model use nominal voltage and short-circuit power as inputs, assuming equal values for all phases. The values for SN and UN should be:

```
SN=40e9
UN=400k
```

The transformer model used in this circuit is a simple linear model without a magnetic core, because a nonlinear core is not required for this example. After placing the symbol `trafo_yy`, the model `RUYy0_400k_120k_300MVA_simple` is selected in the SpiceModel field (double-click to open the drop-down list) in the attribute editor. Other default parameter lines are removed.

The three-phase voltage source is started with exponential damping using the parameter `Texp=50ms`. This accelerates the settling of the current of the shunt reactor. The shunt reactor has a rating of 50 Mvar and a nominal voltage of 110 kV. Its nominal current is approximately 250 A RMS. The total simulation time is set to 4 s.
This is set by the .tran command of LTspice

```
.tran 0 4s 0m 0.1m
```

A value of 0.1m specifies a maximum time step of 0.1 ms, which should be sufficiently small to resolve 50 Hz or 60 Hz sinusoidal waveforms accurately. Use a smaller time step for higher frequencies. For the following four examples the ltspice file is `Transformer_and_shunt_reactor.asc`

1.2.1 Transformer Without Shunt Reactor

The voltage at the bus is approximately 119.7 kV. Thus, the nominal voltage of 120 kV is nearly reached. The small difference is caused by the series impedance of the network, which slightly reduces the voltage.

1.2.2 One Transformer and Shunt Reactor

In this case, the shunt reactor reduces the voltage to approximately 115.1 kV. The RMS current is approximate 275 A. This is higher than the rough estimate of 250 A, because the operating voltage is higher than the nominal voltage of the shunt reactor.

1.2.3 Two Transformers in Parallel and Shunt Reactor

With two transformers connected in parallel, the equivalent internal impedance is reduced by a factor of two. Consequently, the voltage drop is smaller, and the voltage reaches approximately 117.3 kV. The current through the shunt reactor increases to approximately 280 A because of the higher operating voltage.

1.2.4 Two Transformers in Series and Shunt Reactor

The shunt reactor is now supplied from a 120 kV source through two transformers connected in series. The transformer impedance is therefore doubled, and the voltage is reduced from 119.2 kV to approximately 110.5 kV corresponding to a voltage drop of 8.7 kV. The shunt-reactor current is approximately 264 A. This configuration can arise when a de-energized part of a network is restored from a network section that remains energized (for instance after a blackout). Because of the large combined transformer impedance, the voltage is particularly sensitive to load changes in this configuration.

1.2.5 Transformer and Shunt Reactor for AC-Analysis

For an AC analysis the filename of the example is `Transformer_and_shunt_reactor_AC.asc`. For the AC analysis, the voltage-source model is changed to `V_3phase_AC`. The transient-analysis directive `.tran` is replaced by an `.ac` directive. For measuring the bus voltages the transient bus models must be replaced with the appropriate ac bus models `bus_V` with `bus_V_AC` to measure line to line voltages and `bus_Veff` with `bus_V0_AC` to measure line to earth voltages.

Important 1.1 – (AC-Analysis)

Linear AC analysis linearizes nonlinear elements at the operating point and is therefore not suitable for large-signal analysis.

1.3 Manual Calculation of the Line-to-Line Voltage Drop with Two Transformers in Series

The internal impedance of a 400 kV/120 kV transformer is approximately 9.6 Ω inductive when referred to the secondary side. (See equation 1.9). For two transformers connected in series, the total impedance is therefore approximately

Calculation 1.1 – (Voltage reduction)

$$X_{\mathrm{T,tot}} = 2 \cdot 9.6\,\Omega = 19.2\,\Omega. \tag{1.1}$$

With a current of approximately 264 A, the phase-voltage drop is

$$\begin{aligned} \Delta U_{\mathrm{ph}} &= I X_{\mathrm{T,tot}} && (1.2)\\ &= 264\,\mathrm{A} \cdot 19.2\,\Omega && (1.3)\\ &= 5\,\mathrm{kV}. && (1.4) \end{aligned}$$

The corresponding line-to-line voltage drop is approximately

$$\begin{aligned} \Delta U_{\mathrm{LL}} &= \sqrt{3}\,\Delta U_{\mathrm{ph}} && (1.5)\\ &\approx 1.73 \cdot 5\,\mathrm{kV} && (1.6)\\ &\approx \boxed{8.65\,\mathrm{kV}} && (1.7) \end{aligned}$$

This result agrees well with the simulated voltage reduction in Section 1.2.4.

To get the value of the leakage impedance of transformer we can see the wxmaxima file `transformer_400kV_120kV_yy.wxmx` and search for xkus.

Calculation 1.2 – (Leakage impedance)

$$X_{1s} = L_{1s}\,\omega$$
$$= 0.1697 \cdot 314.159$$
$$= 53.33\,\Omega \tag{1.8}$$

$$X_{2s} = X_{1s}\frac{(120\,\mathrm{kV})^2}{(400\,\mathrm{kV})^2}$$
$$= \boxed{9.60\,\Omega} \tag{1.9}$$

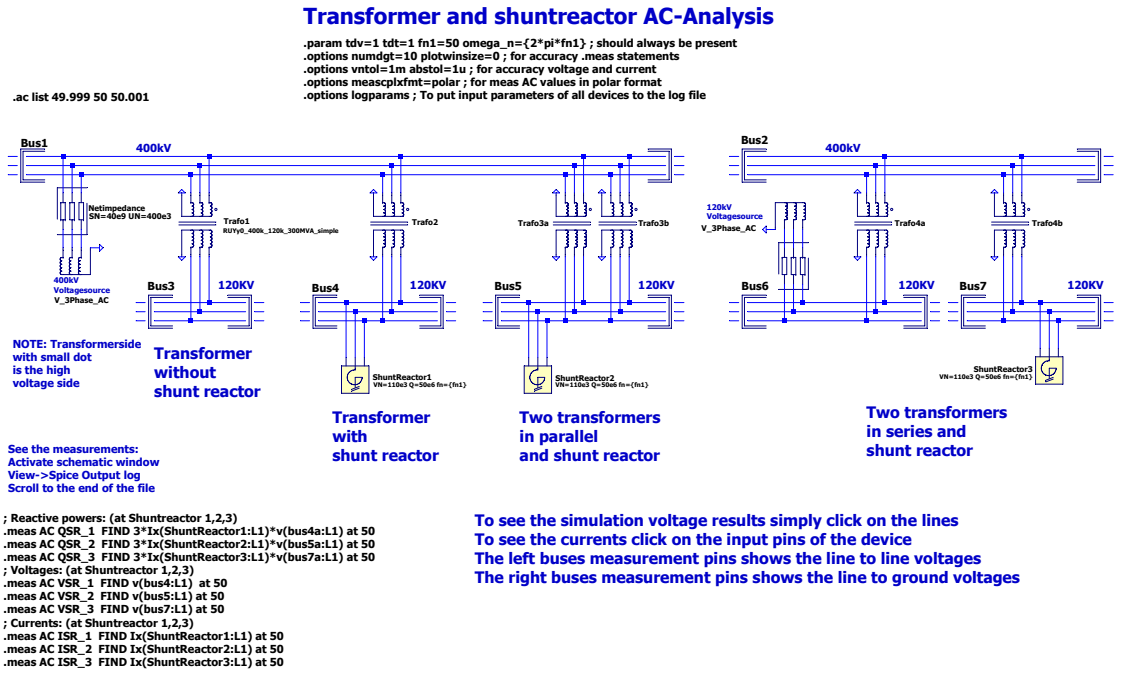


Figure 1.1: AC voltages and currents at the shunt reactors.

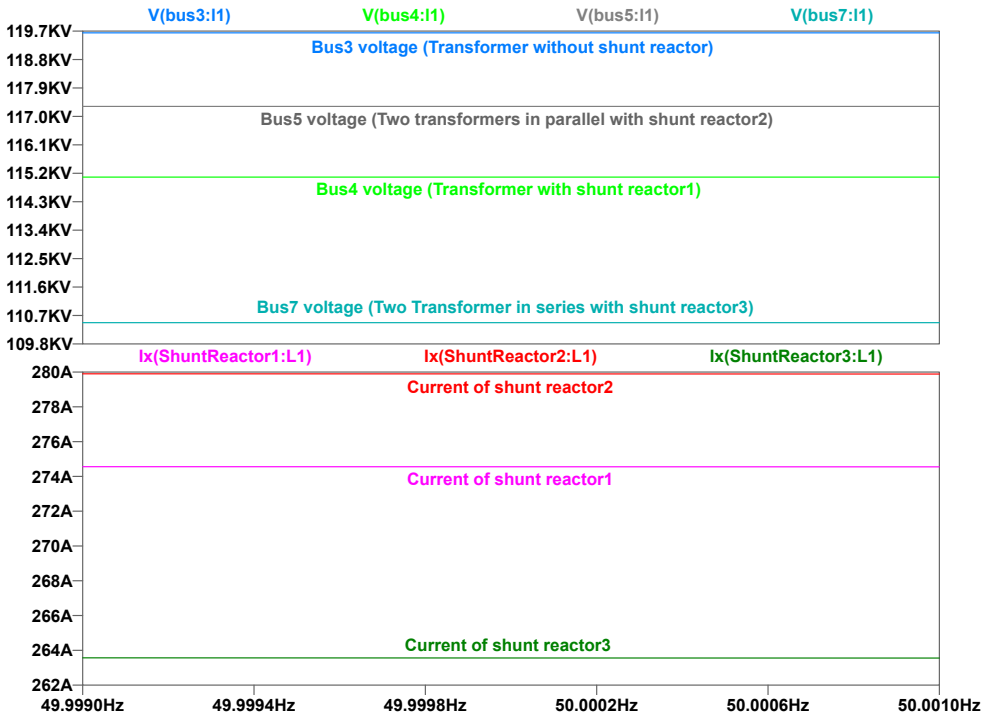


Figure 1.2: AC voltages and currents at the shunt reactors.

Chapter Summary

Shunt reactors are used to regulate the voltage in a power grid. It operates as standalone device or are connected to the tertiary winding of a high voltage transformer. In this chapter, we have seen that loading a circuit with a shunt reactor reduces the voltage towards the shunt reactor. This voltage reduction occurs whenever a current flow through transformers or the line impedance, which is also highly inductive. As a rule of thumb, inductive reactive power generally flows from the higher voltage level to the lower voltage level. When the transformer is loaded with pure active power, active power always flows from the higher voltage phase angle to the lower voltage angle. Real-world example: After energizing a network section via two transformers in series, the voltage is highly volatile. It is best to energize the power district with balanced reactive power. Voltage regulation in the power district should be performed using the regulating transformers and not the shunt reactor.

2 Short-Circuit Current in the Medium-Voltage Grid

2.1 General

This example investigates the short-circuit current close to a substation when two medium-voltage transformers, each rated at 40 MV A, are operated in parallel. The transient simulation is compared with simple manual estimates. The influence of an initial load and an additional short-circuit reactor is then examined, followed by a steady-state AC analysis.

2.2 Circuit Design

We open the circuit-file `Shortcircuit_Current.asc`. This example introduces the line device. Depending on the selected model, this device represents either an overhead line or an underground cable. Here, an oil-insulated underground cable with a conductor cross-section of 500 mm^2 is assumed.

After placing the line element, we opened its attribute window by right-clicking the symbol. We Double-clicked the `SpiceModel` field and selected the required model. We deleted the attributes `Value2`, `SpiceLine`, and `SpiceLine2`. In the `Value` attribute, we changed the length parameter from 1 to 3, corresponding to a line length of 3 km.

The high-voltage transformer uses the model `RUYYo_400k_120k_300MVA`. Its 400 kV side is permanently grounded, whereas the 120 kV side is isolated. The tap changer is set to `L+3` using

```
l_=3 (transformer tap value, pos. sign lower secondary voltage, neg. sign
    higher secondary voltage)
stproz=1.5 (percentage change in nominal voltage per tap step)
```

so that the secondary voltage is approximately 116 kV.

For the medium-voltage transformers, select the model `RUYd5_110k_10k_40MVA`. Their relative short-circuit voltage (ohmic part neglected) is set to 18 % with

```
ux=0.18 (uk=sqrt(ux*ux+ur*ur))
```

The neutral points of the medium-voltage transformers on the 110 kV side are grounded through a Petersen coil, represented by the additional circuit element `pet`. The two 110 kV lines are assumed to be compensated by this coil. One cable produces an earth-fault current of approximately 76 A; This can be verified by deleting all the circuit elements after the line elements. Now simulate the circuit `Shortcircuit_Current.asc` and measure the peak current into the line by dividing the peak current by 1.414 and multiplying the result by three. The earth-fault current can now be seen in the waveform viewer. The Petersen coil is rated at 150 A (~2x76A) and is connected to both transformer neutral points. When measuring the Petersen-coil voltage, you can observe a slowly varying amplitude during the settling process. This is because of the slight undercompensation or overcompensation of the Petersen coil

To obtain a realistic voltage across the Petersen coil, the phase-1 capacitance of line 1 is changed from 400 nF to 401 nF. This introduces a small asymmetry. A Petersen coil is not modelled in this example on the 10 kV side. Because the transformer secondary is delta-connected, a neutral point would be created with the grounding-transformer element `Trafo_gnd` element, connected to the transformer secondary terminals.

The earth-fault element `fault_earth` is also introduced. It is switched on at 500 ms by setting

```
t_start=500m
```

Remark 2.1 – (Model asymmetry)

The deliberate 1 nF capacitance difference prevents a perfectly symmetrical zero-voltage condition across the Petersen coil.

2.3 Transient Simulation

2.3.1 Before the Fault

The Petersen coil does not play a significant role in this short-circuit example. We now start the simulation. Before the fault, the line-to-line voltage is approximately 116 kVand the line currents are predominantly capacitive charging currents with a peak value of approximately 35 A.

2.3.2 Fault Inception

At fault inception, the short-circuit current at the fault position reaches a maximum instantaneous peak value of approximately 53 kA. The 116 kV voltage decreases to approximately 99 kV. The RMS voltage can be checked at the measurement pins of bus4, the bus element on the right side

Remark 2.2 – (Output of measurement values)

The values calculated with .meas statements can be viewed in the SPICE output log. Activate the schematic window, select `View → SPICE Output Log`, and scroll to the end.

2.3.3 Settled Fault Current

After the transient components have largely decayed—approximately 200 ms after fault inception—the steady state short-circuit current is approximately 30 kA peak, or 22 kA RMS. The voltage on the 400 kV side is approximately 396 kV.

Result 2.1 – (Transient simulation results)

The principal simulated values are a first peak of approximately 53 kA and a settled RMS short-circuit current of approximately 22 kA.

2.4 Estimating the Steady-State Short-Circuit Current

The RMS value of the steady-state short-circuit current can be estimated from the transformer rating and relative short-circuit voltage. The rated current on the medium-voltage side of one transformer is

Calculation 2.1 – (Short-circuit current estimate)

$$I_N = \frac{S_N}{\sqrt{3}\,U_N} \tag{2.1}$$

$$= \frac{40\,\mathrm{MV\,A}}{\sqrt{3}\cdot 10.6\,\mathrm{kV}} \tag{2.2}$$

$$\approx 2.18\,\mathrm{kA}. \tag{2.3}$$

For a relative short-circuit voltage of $u_k = 0.18$, the approximate short-circuit-current factor is

$$\frac{1}{u_k} = \frac{1}{0.18} \approx 5.5. \tag{2.4}$$

Thus, one transformer contributes approximately

$$I_{k,1} \approx 5.5 \cdot 2.18\,\mathrm{kA} = 12\,\mathrm{kA}. \tag{2.5}$$

For two identical transformers in parallel,

$$I_{k,2} \approx 2I_{k,1} = 24\,\mathrm{kA}. \tag{2.6}$$

This value applies at the nominal primary voltage of 110 kV. Because the 110kV operating voltage decreases to approximately 100 kV, the corrected estimate is

$$I_k \approx \frac{100}{110}\cdot 24\,\mathrm{kA} = \boxed{21.8\,\mathrm{kA}}. \tag{2.7}$$

Considering the simplifying assumptions, the manual estimate of approximately 21.8 kA agrees well with the simulated RMS value of approximately 22 kA in section 2.3.3.

2.5 Short-Circuit Current with Initial Load

Open the file `Shortcircuit_Current_with_initial_load.asc`. A shunt reactor is used as the initial load. It reduces the operating voltage and therefore also reduces the short-circuit current. The first current peak is slightly lower, and the settled RMS current (Peak-value/1.414) is approximately 20.8 kA.

2.6 Short-Circuit Reactor

Load file `Shortcircuit_Current_with_scr.asc`. An additional short-circuit reactor, denoted by `scr` in the model, is now introduced. For strongly inductive currents, its reactance causes a larger voltage drop and limits the short-circuit current. During normal operation with predominantly resistive current, the voltage drop across the reactor is much smaller. The parameter `delta_un=5%` specifies the percentage voltage drop at rated, purely inductive current.

With this reactor, the first transient current peak is reduced to approximately 11 kA, and the settled RMS short-circuit current is approximately 4.4 kA. The voltages are approximately 113.5 kV on the 110 kV level, 399.8 kV on the 400 kV level and 9 kV on the 10 kV level.

2.6.1 Manual Calculation

For a reactor rated at $U_N = 10\,\text{kV}$ and $I_N = 250\,\text{A}$ and with `delta_un=5%` , the nominal three-phase apparent power is

Calculation 2.2 – (Fault current behind the reactor)

$$S_D = \sqrt{3}\,U_N I_N \tag{2.8}$$
$$= \sqrt{3}\cdot 10\,\text{kV}\cdot 250\,\text{A} \tag{2.9}$$
$$\approx 4.33\,\text{MV A}. \tag{2.10}$$

With the specified relative inductive voltage drop, the reactor impedance is

$$X_D = \frac{\Delta u_N}{100}\frac{U_N^2}{S_D} \approx 1.15\,\Omega. \tag{2.11}$$

The maximum RMS fault current immediately behind the reactor is therefore

$$I_{kD} = \frac{U_N}{\sqrt{3}\,X_D} \tag{2.12}$$
$$= \frac{10\,\text{kV}}{\sqrt{3}\cdot 1.15\,\Omega} \tag{2.13}$$
$$\approx 5\,\text{kA}. \tag{2.14}$$

This assumes a perfectly stiff operating voltage. In the example, the voltage drops from 10.2 kV to 8.8 kV; consequently, the short-circuit current falls to approximately

$$I_{kD} \approx \boxed{4.4\,\text{kA}}. \tag{2.15}$$

2.7 AC Analysis

Load the file `Shortcircuit_Current_AC.asc`. For the AC analysis, we first replaced the transient voltage-source model with the corresponding AC source model. Then we adapted the `.meas` directives, deleted the fault switch, and we placed the new ground element `ground` on the schematic. We added the following option so that complex measurement results are reported in polar form instead of decibels:

```
.options meascplxfmt=polar
```

The magnitudes of the voltages and currents agree well with the settled values from the transient analysis. Directly at the phaseA fault location, however, the current angle is approximately 120° rather than the initially expected −90°. Closer to the voltage source—for example, at the transformer primary side—the expected current angle of approximately −90° and voltage angle of approximately 0° are obtained.

Important 2.1 – (Effect of transformer vector group)

The transformer vector group is Yd5, corresponding to a phase shift of $5 \cdot -30^\circ = -150^\circ$. For an almost purely inductive fault path, the current lags the voltage by approximately -90°. Hence,

$$\varphi_I \approx -150^\circ + (-90^\circ) = -240^\circ = 120^\circ. \tag{2.16}$$

Because the fault resistance is assumed to be purely resistive, the current and the voltage at the fault have approximately the same angle. The theoretical voltage angle can be checked by removing the ground elements and repeating the simulation.

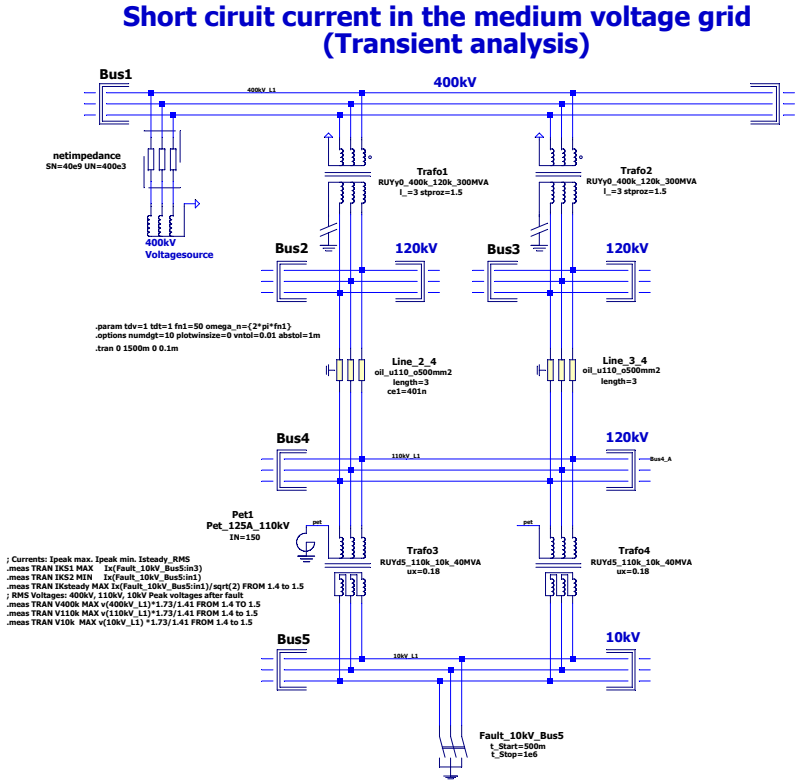


Figure 2.1: Shortcircuit in the medium voltage grid - schematic

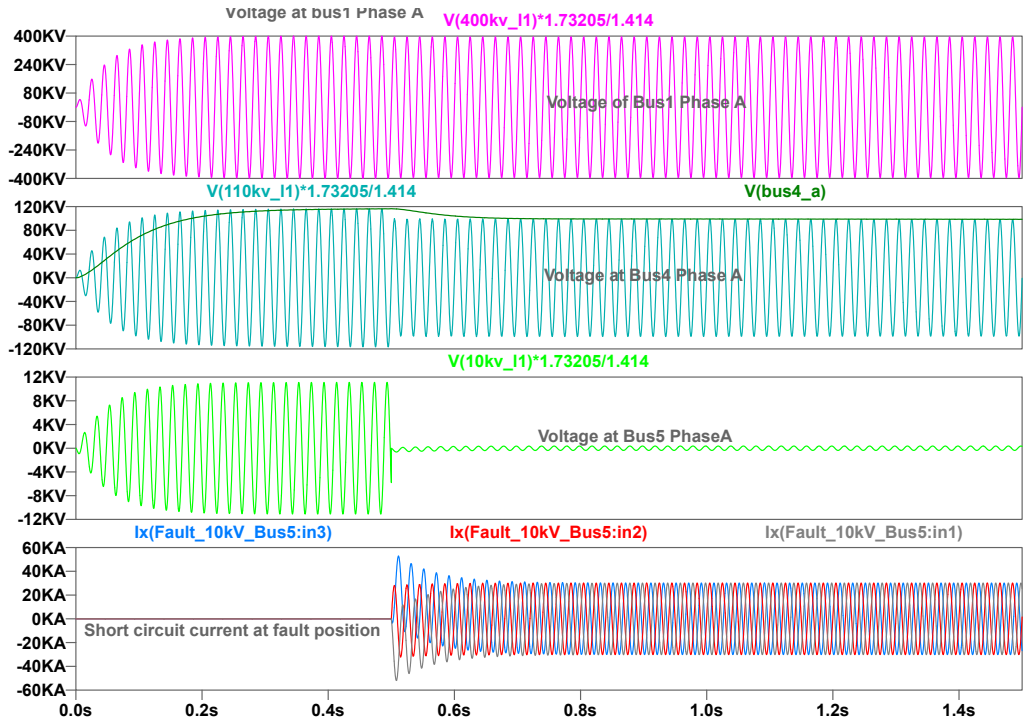


Figure 2.2: Shortcircuit in the medium voltage grid - waveform

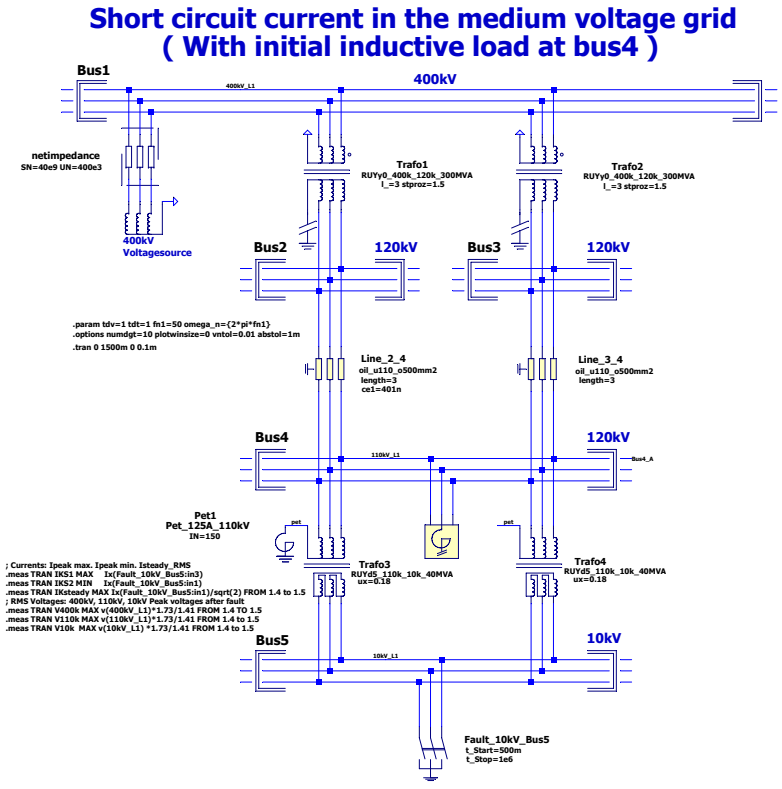


Figure 2.3: Shortcircuit in the medium voltage grid with initial load - schematic

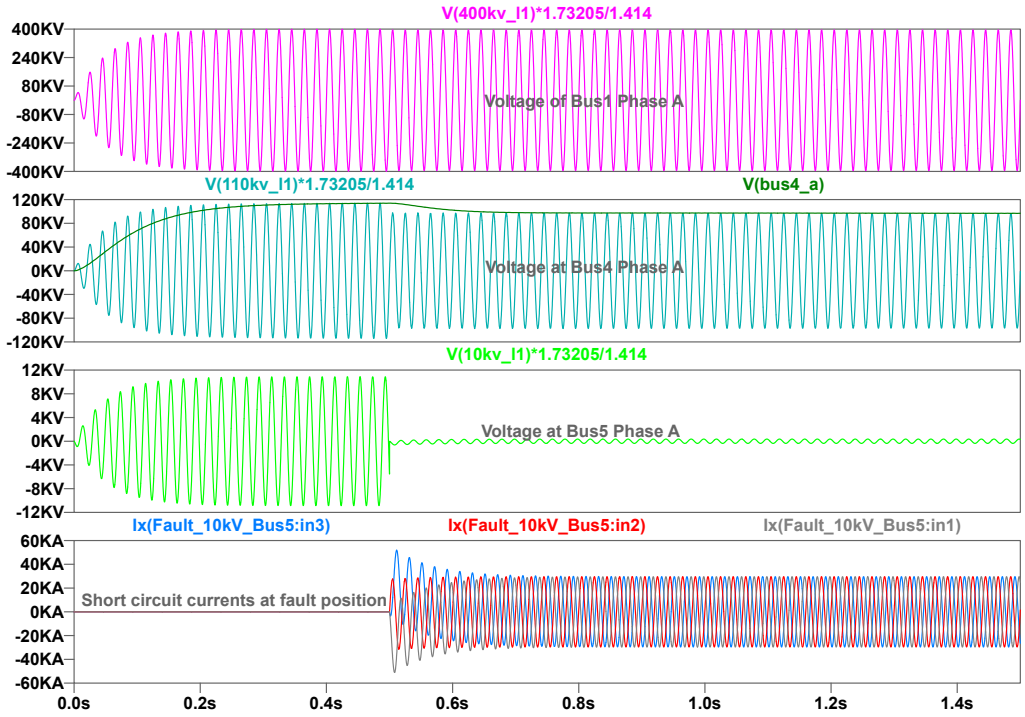


Figure 2.4: Shortcircuit in the medium voltage grid with initial load - waveform

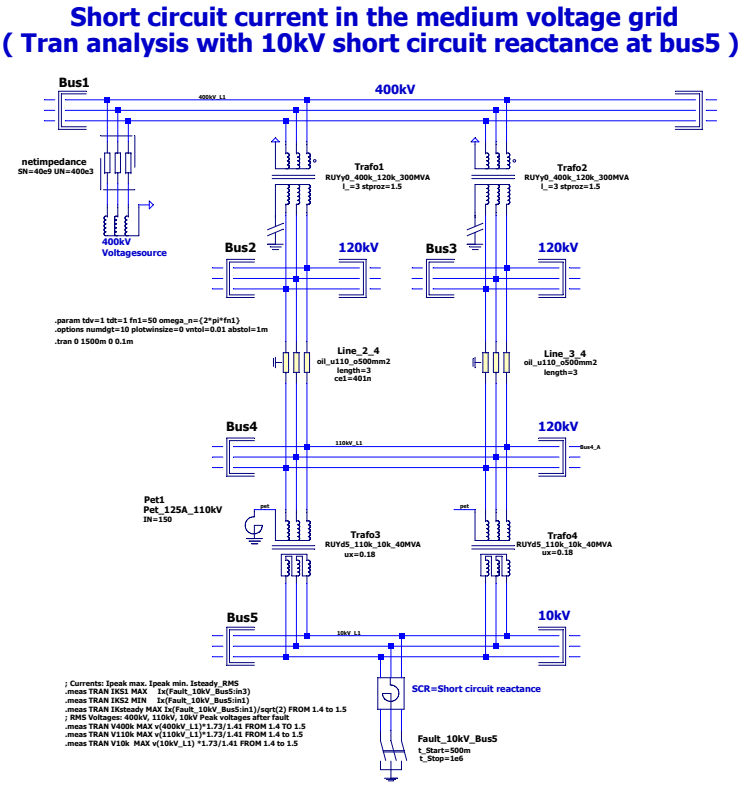


Figure 2.5: Shortcircuit in the medium voltage grid with scr - schematic

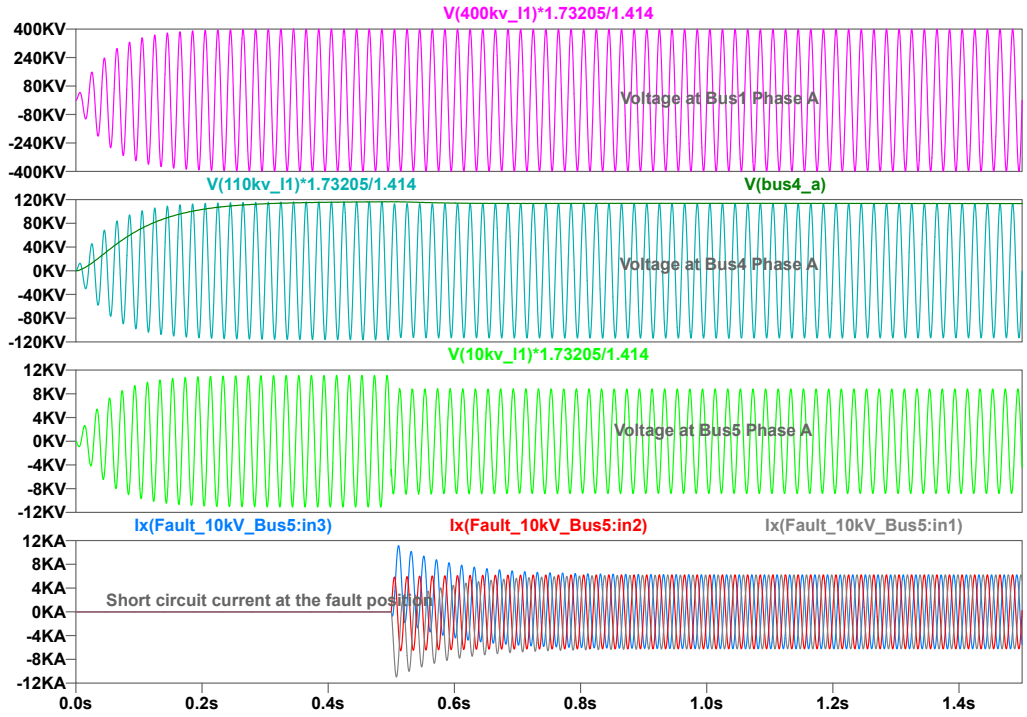


Figure 2.6: Shortcircuit in the medium voltage grid with scr - waveform

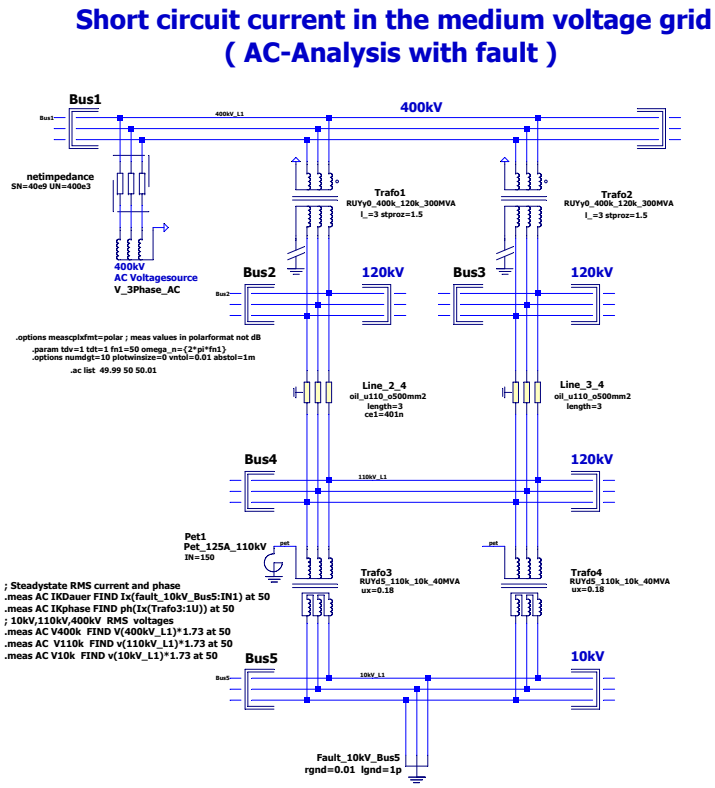


Figure 2.7: Shortcircuit in the medium voltage grid AC analysis - schematic

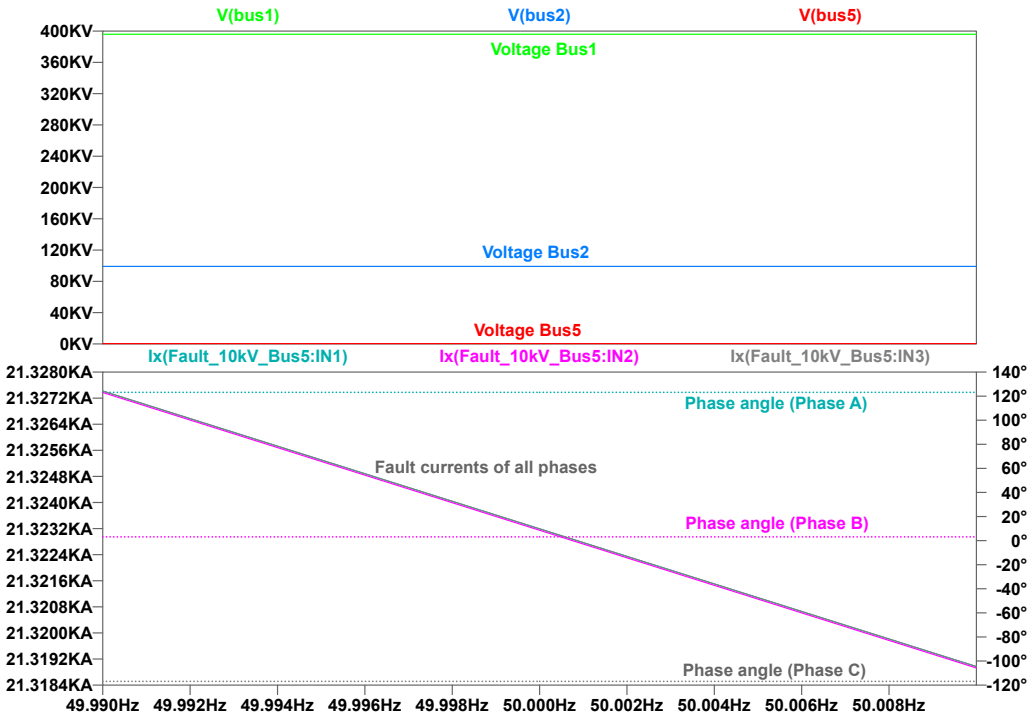


Figure 2.8: Shortcircuit in the medium voltage grid AC analysis - waveform

Chapter Summary

Two 40 MV A medium-voltage transformers operated in parallel produce a settled short-circuit current of approximately 22 kA RMS in the investigated network. A simple estimate based on rated current, relative short-circuit voltage, parallel operation, and the actual operating voltage closely reproduces this result. An initial inductive load reduces the current slightly, whereas an additional short-circuit reactor limits it substantially to approximately 4.4 kA. The AC analysis also demonstrates that transformer vector groups must be considered when interpreting phase angles.

3 Initializing Transformers with Residual Flux

3.1 General

This chapter demonstrates how a transformer model can be initialized with residual magnetic flux. The examples show how the switching instant and the polarity of the residual flux determine the maximum core flux and the resulting magnetizing inrush current.

For an ideal winding, neglecting the winding resistance, the applied voltage and core flux are related by

$$u(t) = N\frac{\mathrm{d}\Phi(t)}{\mathrm{d}t}, \qquad \Phi(t) = \Phi(0) + \frac{1}{N}\int_0^t u(\tau)\,\mathrm{d}\tau. \tag{3.1}$$

Consequently, the flux is determined by the time integral of the imposed voltage and by the initial flux $\Phi(0)$. If a transformer is energized at an unfavourable point on the voltage waveform, the transient flux can reach twice its normal steady-state peak. With full residual flux of the same polarity, it can approach three times the normal peak. Because the core characteristic is nonlinear, this overfluxing can produce a magnetizing current many times larger than the normal no-load current.

Important 3.1 – (Switching instant and residual flux)

The initial flux, the voltage phase angle at switching, and the parameter that applies the initial condition must always be coordinated. A physically reasonable residual-flux vector in a three-leg core also satisfies approximately $\Phi_{r0} + \Phi_{s0} + \Phi_{t0} = 0$.

3.2 Circuit Design

For all cases the ltspice file `Initializing_Transformers_with_residual_flux.asc` is loaded. Four transformer circuits are arranged on the right side of the schematic. They are supplied at 115 kV; each transformer is rated 110 kV/10.6 kV. A switch is placed between each source and transformer.

The first three circuits are energized at a phase-1 voltage zero crossing and use different residual-flux settings:

- no residual flux: `psi0=0`,
- unfavourable residual flux: `psi0=1`,
- favourable residual flux: `psi0=-1`.

The fourth circuit is energized at a different point on the voltage waveform and uses explicitly specified phase residual fluxes. See section 3.6

Remark 3.1 – (Coordination of initialization time)

The transformer parameter `tpsi0` must match the switch closing time `t_start`. For the first three circuits, use

```
t_start=40m  tpsi0=40m
```

For the final circuit, use

```
t_start=45m  tpsi0=45m
```

The `tpsi0` event rapidly initializes the voltage integrator capacitor that represents the residual flux.

3.3 Energization at 40 ms Without Residual Flux

For Trafo1 we set the transformer parameter `psi0` to zero. The parameter `tpsi0` may either be set to 40 ms or placed well outside the simulation interval, for example at 1 000 s, because no initial flux has to be applied to Transformer1.

At 40 ms, the trace v(phaseA1) voltage starts at zero and initially becomes positive. Its integral therefore acquires a unidirectional offset. At 10 ms after energization, the phase-1 flux reaches approximately twice the normal steady-state peak, and the magnetizing current rises accordingly.

The three phase fluxes can be displayed in the LTspice waveform viewer with

```
V(Trafo1:u:_w1:phi)
V(Trafo1:v:_w1:phi)
V(Trafo1:w:_w1:phi)
```

Both flux and current decay slowly because a large power transformer has a high magnetizing inductance and a long electrical time constant. Increasing the resistance of the source impedance accelerates the decay. This can be examined with $R = 10\,\Omega$, $100\,\Omega$, and $1\,000\,\Omega$.

The rated primary current is approximately 210 A RMS. In this case, the peak magnetizing current is approximately 70 A, or about 50 A RMS, and therefore remains below the rated current.

The steady-state peak flux V(trafo1:u:_w1:phi) is 0.3 Wb and the initial flux is 2*0.3 Wb.

Calculation 3.1 – (Rated steady-state flux)

$$\hat{\Phi}_N = \frac{\sqrt{2}\,U_{\mathrm{ph}}}{\omega N_1} = \frac{\sqrt{2}\,U_{\mathrm{LL}}}{\sqrt{3}\,\omega N_1}. \tag{3.2}$$

With $U_{\mathrm{LL}} = 115\,\mathrm{kV}$, $\omega = 2\pi \cdot 50\,\mathrm{Hz}$, and the toolbox default $N_1 = 1000$, this gives

$$\hat{\Phi}_N = \frac{\sqrt{2} \cdot 115\,\mathrm{kV}}{\sqrt{3} \cdot 2\pi \cdot 50\,\mathrm{Hz} \cdot 1000} \approx \boxed{0.299\,\mathrm{Wb}}. \tag{3.3}$$

Remark 3.2 – (Line-to-line and phase voltage)

For a star-connected three-phase winding, use the phase voltage $U_{ph} = U_{LL}/\sqrt{3}$ and include the RMS-to-peak factor $\sqrt{2}$ when calculating the peak flux. It is particularly interesting that the magnetic flux depends only on voltage !!!

3.4 Energization at 40 ms with Unfavourable Residual Flux

We set `psi0=1` for Trafo1 . Unless the individual phase parameters are overridden, the model initializes the phase residual-flux values to

$$(\psi_{r0}, \psi_{s0}, \psi_{t0}) = (1, -0.5, -0.5) \text{ p.u.} \tag{3.4}$$

Phase 1 therefore starts with full positive residual flux, while phases 2 and 3 start with half the negative value. Since the phase-1 voltage becomes positive immediately after the switch closes, its integral adds to the positive residual flux. Approximately 10 ms later, the flux reaches approximately three times the steady-state peak.

The phase-1 current reaches approximately 900 A peak. The corresponding RMS value can be calculated in LTspice by zooming in from 40 ms to 60 ms and then left-clicking on the trace name Ix(Trafo2:1U) using the Control button. A pop-up window will then appear, and you can read the RMS value as 380A. This is the procedure to follow if the waveform is not sinusoidal. The inrush current decays comparatively quickly because the high current increases the effective damping of the transient.

The corresponding flux traces are

```
V(Trafo2:u:_w1:phi)
V(Trafo2:v:_w1:phi)
V(Trafo2:w:_w1:phi)
```

The residual-flux values are visible as the initial values at the switching instant.

3.5 Energization at 40 ms with Favourable Residual Flux

Set `psi0=-1`. The internal phase residual-flux vector becomes

$$(\psi_{r0}, \psi_{s0}, \psi_{t0}) = (-1, 0.5, 0.5) \text{ p.u.} \tag{3.5}$$

The maximum negative phaseA residual flux compensates the positive flux increment that follows switching at 40 ms. Apart from a short numerical switching spike, the flux moves directly into its steady-state trajectory and the magnetizing current remains very small.

A similarly smooth startup can be obtained by deleting the switcher and set the Texp parameter of the voltage source to 50ms.

To inspect the small steady-state current in the waveform viewer, manually limit the current-axis range after the switching spike. The nonlinear magnetization curve produces a small but visibly distorted current. The phase fluxes are displayed with ..

```
V(Trafo3:u:_w1:phi)
V(Trafo3:v:_w1:phi)
V(Trafo3:w:_w1:phi)
```

They are nearly symmetrical with respect to the Y-axis and already settled after energization.

Remark 3.3 – (Default residual-flux vector)

The default vectors $(1, -0.5, -0.5)$ and $(-1, 0.5, 0.5)$ produce the behaviour described above only at equivalent points of the 50 Hz waveform, such as 0, 20, and 40 ms. At other switching instants, specify the individual values with `psir`, `psis`, and `psit`.

3.6 Energization at 45 ms with Unfavourable Residual Flux

For switching at 45 ms, the unfavourable residual-flux vector differs from the default vector used at the voltage zero crossing. The required values are as follows:

```
psi0=1
psir=0
psis=0.866
psit=-0.866
t_start=45m
tpsi0=45m
```

With these settings, the phaseA current is the smallest, whereas the currents in phases 2 and 3 are the largest. The currents remain high, but their maxima are lower than in the unfavourable 40 ms case seen in section 3.4.

The phase fluxes can be displayed with ..

```
V(Trafo4:u:_w1:phi)
V(Trafo4:v:_w1:phi)
V(Trafo4:w:_w1:phi)
```

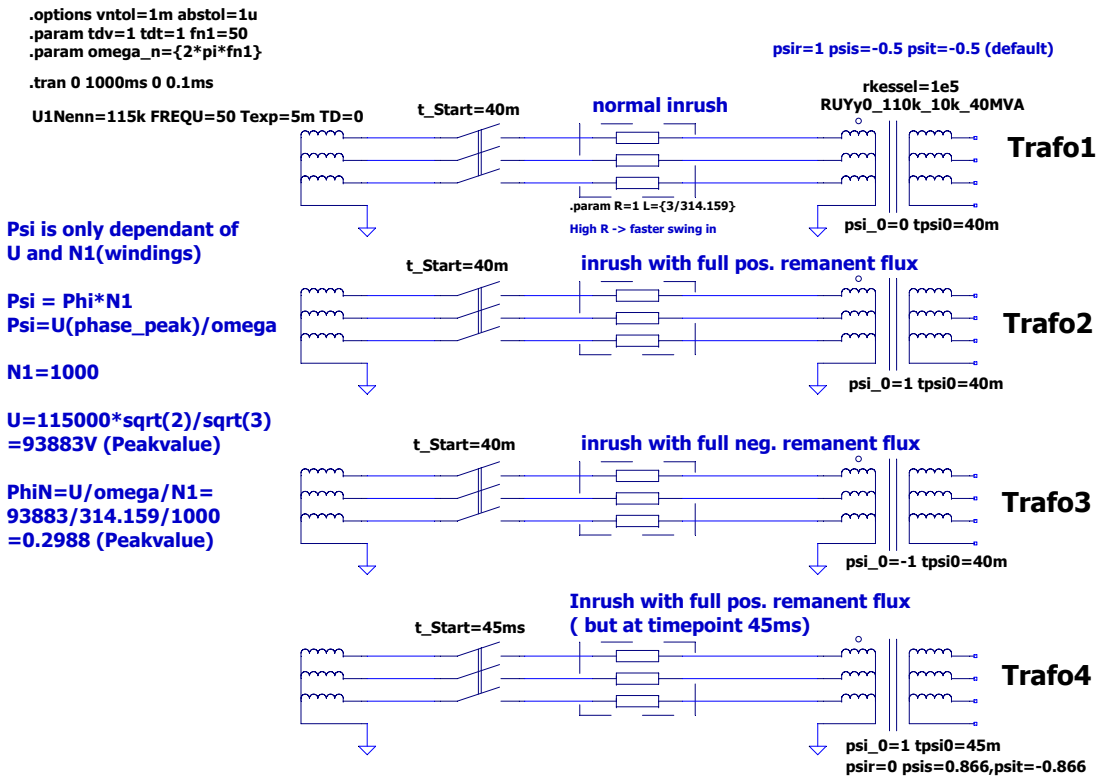


Figure 3.1: Initializing a Transformer with residual flux - schematic

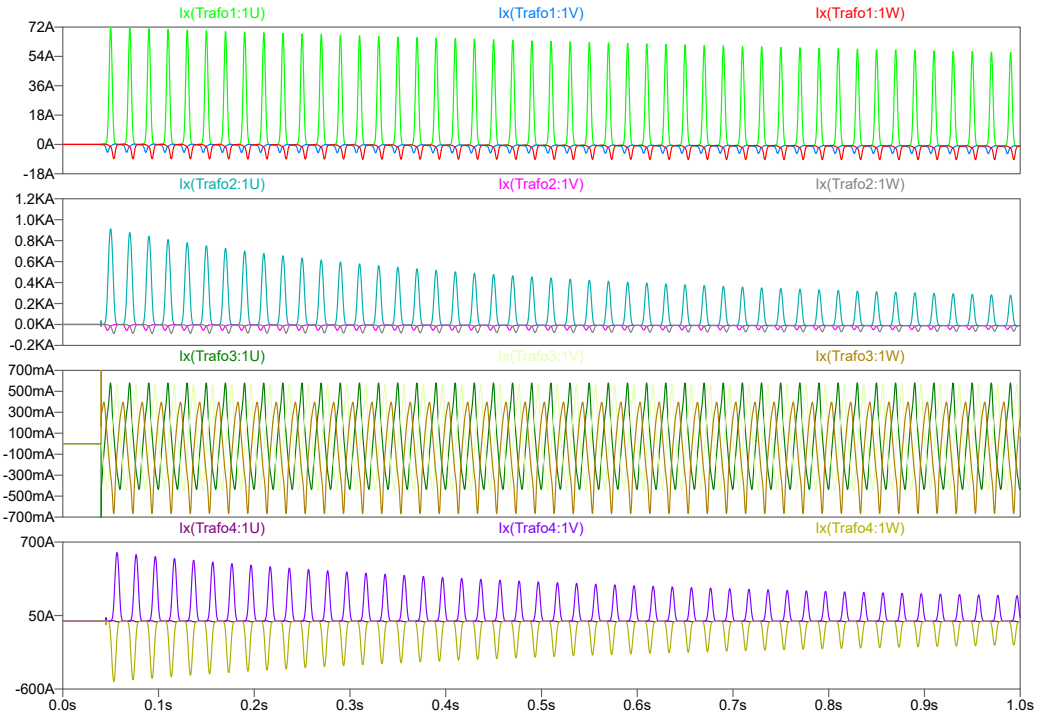


Figure 3.2: Initializing a Transformer with residual flux - Currents

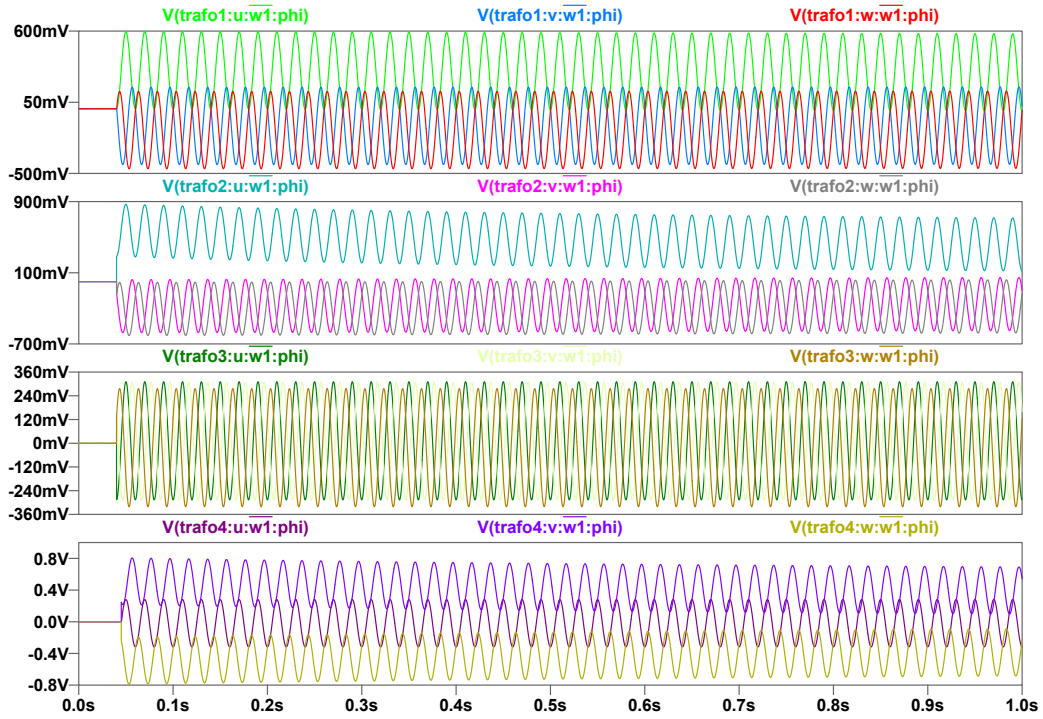


Figure 3.3: Initializing Transformer with residual flux - Fluxes

Chapter Summary

Transformer energization is governed by the integral of the applied voltage and by the residual core flux at the switching instant. Closing at a voltage zero crossing without residual flux can produce approximately twice the normal steady-state peak flux. Unfavourable full residual flux can increase this to about three times the normal peak and cause a large inrush current. Favourable residual flux, controlled switching, or a gradual source-voltage ramp can greatly reduce the transient. The initialization time and all three phase residual-flux values must be coordinated with the switch closing instant. Hint: It is often mentioned in the literature that the inrush current is attenuated by connecting the transformer to a load. However, this is not the case unless the load is truly very high—which, in turn, causes the total current to be very high, thereby masking the inrush current. See LTspice example `Hidden_inrush.asc` in this example folder

4 Switching Lines

4.1 General

This chapter examines the energization of unloaded and loaded cable lines. It also considers the energization of a second line connected to an already energized busbar and the energization of a cable terminated by a transformer. The examples illustrate the influence of switching instant, line capacitance, network inductance, load damping, and transformer residual flux on the resulting transients.

The switching transients are oscillations at the natural frequencies of the electrical network. Such oscillations occur whenever the system state changes abruptly, for example during line energization or at the inception of an earth fault or short circuit.

4.2 Circuit Design

Open the file: `switching_lines.asc`. The LTspice schematic contains four circuits. Each line is a 5 km, 110 kV oil-filled cable with a conductor cross-section of 500 mm^2. The circuits represent:

1. energization of an unloaded cable,
2. energization of a cable supplying a 100 MW load,
3. energization of a second unloaded cable connected to an energized busbar,
4. energization of a cable terminated by a transformer with unfavourable full residual flux.

Each single line is earth-fault compensated by a 125 A Petersen coil. The double-line arrangement initially uses a compensation setting of 125 A, which is increased to250 A at 400 ms.

4.2.1 Adjustment of the Petersen Coil

To determine the required compensation, temporarily reduce the load in the second circuit from `100e6` to `100e-6` and run the simulation. The phase-1 charging-current peak is approximately 60 A. For a shielded single-core cable whose dominant capacitances are phase-to-earth, the earth-fault charging current is approximately

$$I_{\mathrm{E,peak}} \approx 3I_{0,\mathrm{peak}} = 3 \cdot 60\,\mathrm{A} = 180\,\mathrm{A}, \tag{4.1}$$

or approximately 127 A RMS. A 125 A Petersen coil is therefore appropriate. After this adjustment, restore the load to `100e6`.

Remark 4.1 – (Applicability of the threefold charging current)

The relation $I_{\mathrm{E}} \approx 3I_0$ applies to the zero-sequence charging current of the cable. Overhead lines have appreciable phase-to-phase capacitances, so the same simplified interpretation must not be transferred without checking the complete capacitance matrix.

4.3 Energization of an Unloaded 110 kV Cable

NOTE: Simulate `Switching_lines.asc` and choose after simulation `line1.plt`. This is done by activating the waveform-viewer window and click in the PlotSettings the menu item ”open plot settings file”.

For the selected switching instant, the Phase A switch closes near a voltage zero crossing. Its peak current is therefore the smallest, approximately 130 A. Immediately after closing, the cable capacitances behave like short circuits and draw large, rapidly changing charging currents. The phase-2 and phase-3 current peaks reach approximately 1.5 kA. The oscillations have substantially decayed after approximately 200 ms.

The Phase A voltage transient is likewise the smallest. The Phase B and Phase C voltages reach approximately −160 kV peak, about 30 % above the nominal phase-voltage peak. The precise current and voltage maxima depend strongly on the switching instant. This can be investigated by changing the switch parameter `t_start` from 40 ms to 45 ms.

Important 4.1 – (Origin of the voltage transient)

The capacitive charging current flows through the inductive source and line impedances. The resulting voltage drop is superimposed on the source voltage at the cable terminal. Consequently, the phases with the largest charging current also show the largest voltage transient.

4.3.1 Natural Frequency

The natural frequency can be identified in LTspice with View/FFT. For example, an FFT of the Phase B voltage shows peaks at 50 Hz and approximately 1.44 kHz.

The short-circuit power of the network is 7.5 GV A. Its nominal reactance is

Calculation 4.1 – (Natural frequency of the 5 km cable)

$$X_{\mathrm{N}} = \frac{U_{\mathrm{N}}^2}{S_{\mathrm{k}}} = \frac{(110\,\mathrm{kV})^2}{7.5\,\mathrm{GV\,A}} = 1.613\,\Omega, \tag{4.2}$$

$$X_{\mathrm{line}} = 0.11\,\Omega/\mathrm{km} \cdot 5\,\mathrm{km} = 0.55\,\Omega. \tag{4.3}$$

With $\omega = 2\pi \cdot 50\,\mathrm{Hz}$ and $C' = 400\,\mathrm{nF/km}$, a lumped approximation gives

$$L = \frac{X_{\mathrm{N}} + X_{\mathrm{line}}}{\omega} \approx 6.9\,\mathrm{mH}, \tag{4.4}$$

$$C = C'l = 2\,\mu\mathrm{F}, \tag{4.5}$$

$$f_{\mathrm{r}} = \frac{1}{2\pi\sqrt{LC}} \approx 1.35\,\mathrm{kHz}. \tag{4.6}$$

The cable is implemented as a lumped T-section. During the initial charging process, only the source-side half of its series inductance is effective in the relevant resonant loop. Therefore,

$$L_{\text{eff}} = \frac{X_{\text{N}} + X_{\text{line}}/2}{\omega} \approx 6.0\,\text{mH}, \tag{4.7}$$

$$f_{\text{r},T} = \frac{1}{2\pi\sqrt{L_{\text{eff}}C}} \approx \boxed{1.45\,\text{kHz}}, \tag{4.8}$$

which agrees closely with the simulated 1.44 kHz.

The approximate oscillation frequency can also be read directly in the waveform viewer: zoom into two adjacent transient peaks and use the time difference and frequency displayed in the lower-left status line.

4.3.2 Petersen-Coil Oscillation

The Petersen-coil voltage and current do not immediately reach steady state. A slow beat of approximately 1 Hz is visible because the compensated zero-sequence resonant frequency differs slightly from the network frequency. After the transient has decayed, the residual coil current and beat are nearly zero.

4.4 Energization of a Loaded Line

NOTE: Simulate `Switching_lines.asc` and choose after simulation `line2.plt`. This is done by activating the waveform-viewer window and click in the PlotSettings the menu item ”open plot settings file”.

When the cable supplies the load, its voltage and current transients are more strongly damped. The active part of the load absorbs oscillatory energy and reduces the duration of the switching transient.

4.5 Energization of a Second Line

NOTE: Simulate `Switching_lines.asc` and choose after simulation `line3.plt`. This is done by activating the waveform-viewer window and click in the PlotSettings the menu item ”open plot settings file”.

The first, upper line is energized in the same way as the unloaded line in Section 4.3. At 400 ms, the second line is connected to the same busbar. The traces `Ix(Line3a:L1_1)` and `Ix(Line3b:L1_1)` initially show transients of similar magnitude in both lines.

Closer inspection shows that the high-frequency components have opposite signs. They are capacitive currents circulating through the busbar and the phase-to-earth capacitances of the two cables. For two identical 5 km cables, the approximate circulating-current frequency is

Calculation 4.2 – (Circulating-current and network-current frequencies)

$$L_{\text{ring}} = \frac{2 \cdot 0.11\,\Omega/\text{km} \cdot 2.5\,\text{km}}{2\pi \cdot 50\,\text{Hz}} \approx 1.75\,\text{mH}, \tag{4.9}$$

$$C_{\text{ring}} = \frac{400\,\text{nF}}{2} = 200\,\text{nF}, \tag{4.10}$$

$$f_{\text{ring}} = \frac{1}{2\pi\sqrt{L_{\text{ring}} C_{\text{ring}}}} \approx \boxed{3.8\,\text{kHz}}. \tag{4.11}$$

The lower-frequency component drawn from the network source sees the combined capacitance of both lines:

$$L_{\text{net}} = \frac{1.613\,\Omega + 0.11\,\Omega/\text{km} \cdot 2.5\,\text{km}}{2\pi \cdot 50\,\text{Hz}} \approx 6.0\,\text{mH}, \tag{4.12}$$

$$C_{\text{net}} = 2 \cdot 400\,\text{nF} = 800\,\text{nF}, \tag{4.13}$$

$$f_{\text{net}} = \frac{1}{2\pi\sqrt{L_{\text{net}} C_{\text{net}}}} \approx \boxed{1.03\,\text{kHz}}. \tag{4.14}$$

Because this network-current component has the lower frequency, its associated inductive voltage drop and busbar-voltage transient are also smaller. The reactive-power exchange of each cable can be viewed by holding Alt and right-clicking the respective line element. LTspice then displays the instantaneous product of element voltage and current. The lower frequency can be observed by measuring the current at the voltage source. The higher frequency is measured by measuring the current flowing through the respective cable.

4.6 Energization of a Transformer-Terminated Line

NOTE: Simulate `Switching_lines.asc` and choose after simulation `line4.plt`. This is done by activating the waveform-viewer window and click in the PlotSettings the menu item ”open plot settings file”.

In the fourth circuit, the cable supplies a transformer initialized with unfavourable residual flux. The transient current is therefore a combination of capacitive cable-charging current and inductive transformer-inrush current.

In the Phase A closed at a voltage zero crossing, the capacitive contribution is small while the inductive inrush contribution is large. In the other phases, the capacitive component is initially dominant but decays rapidly; the inductive component subsequently prevails. The two components decay at different rates. Now extend the simulation to 15 s (`Switching_lines_15s.asc`) and run it again. During the first few seconds, the voltages on the primary side of the transformer exceed their nominal values. The Phase C voltage is particularly high. The peak value is 240kV compared to the nominal value of 90kV nominal peak voltage of line to ground voltage. You can get this under control by operating the Petersen coil with slight undercompensation or overcompensation (out of resonance). The residual-flux initialization and its coordination with the switching time are described in Chapter 3.

4.7 Extended-Line Study: Charging Current Near Rated Current

NOTE: Simulate `Switching_lines_70km.asc` and choose after simulation for plotting `lines_70km_1.plt` for waveform1 or `lines_70km_2.plt` for waveform2

The cable length is increased until its steady-state charging current is close to the nominal current of approximately 900 A peak, or 630 A RMS. The simulation study uses a line length of 70 km.

For a phase-to-earth capacitance C' and line-to-line RMS voltage U_{LL}, the approximate charging current is

$$I_{\mathrm{c}} = \omega C' l \frac{U_{\mathrm{LL}}}{\sqrt{3}}. \tag{4.15}$$

With $C' = 400\,\mathrm{nF/km}$, the ideal length required for exactly 600 A RMS is about 75 km. The 70 km simulation case is therefore a practical approximate setting, with the exact result determined by the implemented line model.

At 70 km, the current transient reaches approximately 3.5 kA in phase B and C, while its dominant oscillation frequency falls to approximately 230 Hz. Despite the larger current, the voltage transient is smaller and the oscillation decays rapidly. The greater cable resistance provides stronger damping, and the lower oscillation frequency reduces the network impedance.

Calculation 4.3 – (Comparison of inductive voltage drops across the network impedance)

For the 5 km case,

$$L_{\mathrm{N}} = \frac{1.613\,\Omega}{2\pi \cdot 50\,\mathrm{Hz}} \approx 5.1\,\mathrm{mH}, \tag{4.16}$$

$$X_L(1.44\,\mathrm{kHz}) = 2\pi \cdot 1.44\,\mathrm{kHz} \cdot L_{\mathrm{N}} \approx 46.4\,\Omega, \tag{4.17}$$

$$\widehat{U}_L \approx 46.4\,\Omega \cdot 1.5\,\mathrm{kA} \approx 69.6\,\mathrm{kV}. \tag{4.18}$$

For the 70 km case,

$$X_L(230\,\mathrm{Hz}) = 2\pi \cdot 230\,\mathrm{Hz} \cdot L_{\mathrm{N}} \approx 7.4\,\Omega, \tag{4.19}$$

$$\widehat{U}_L \approx 7.4\,\Omega \cdot 3.5\,\mathrm{kA} \approx 25.9\,\mathrm{kV}. \tag{4.20}$$

This explains why the 70 km case produces a smaller voltage transient even though its charging-current transient is larger. In LTspice, the voltage drop can be measured by right-clicking the source-side terminal of the phase B impedance, choosing Mark Reference, and selecting the switch-side terminal of the impedance as the second measurement point. The observed drop is approximately 25 kV, compared with approximately 70 kV in the 5 km case. This is because the dominant frequency is much lower and therefore the impedance is much smaller. HINT: Observe the Petersen-coil current in the 70 km cable network. Its peak value is approximately 90 A.

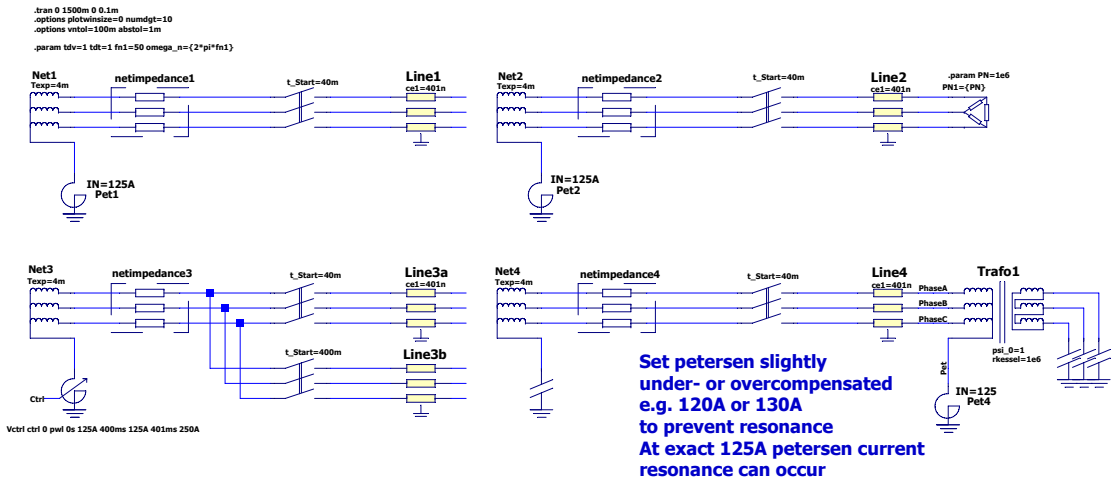


Figure 4.1: Switching lines - schematic

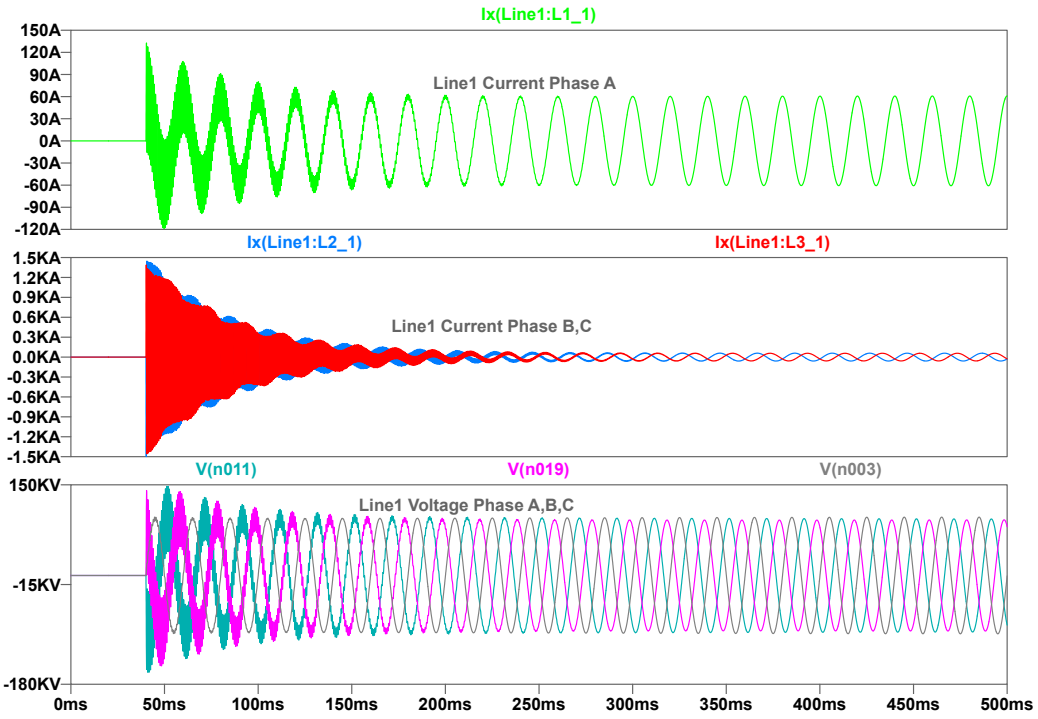


Figure 4.2: Switching lines1 - waveform

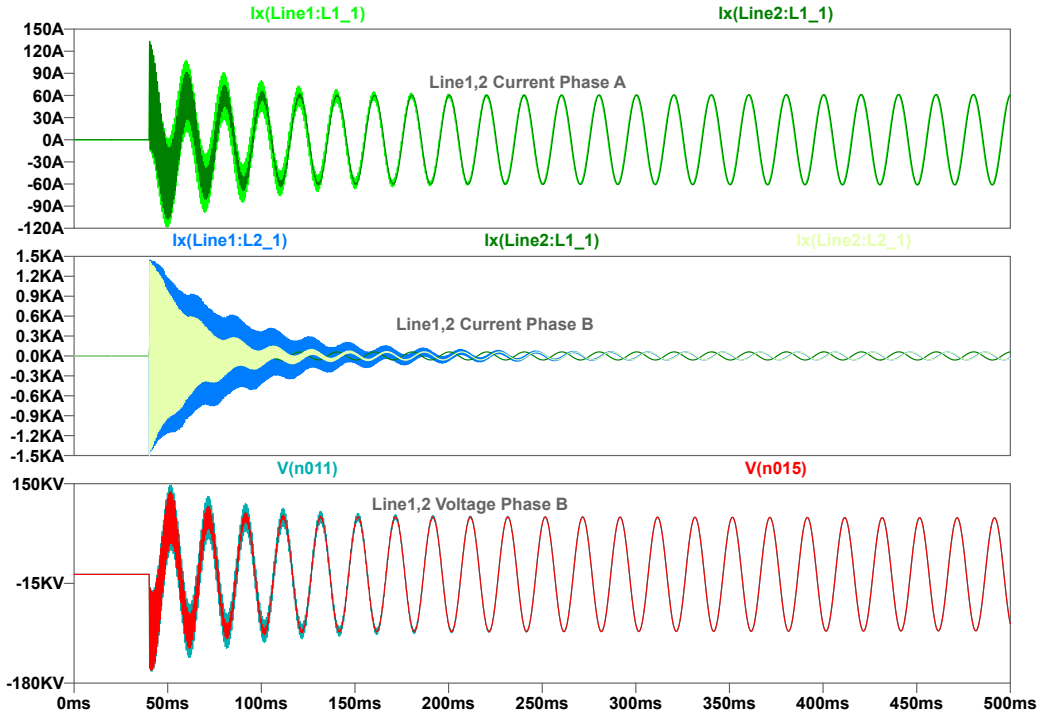


Figure 4.3: Switching lines2 - waveform

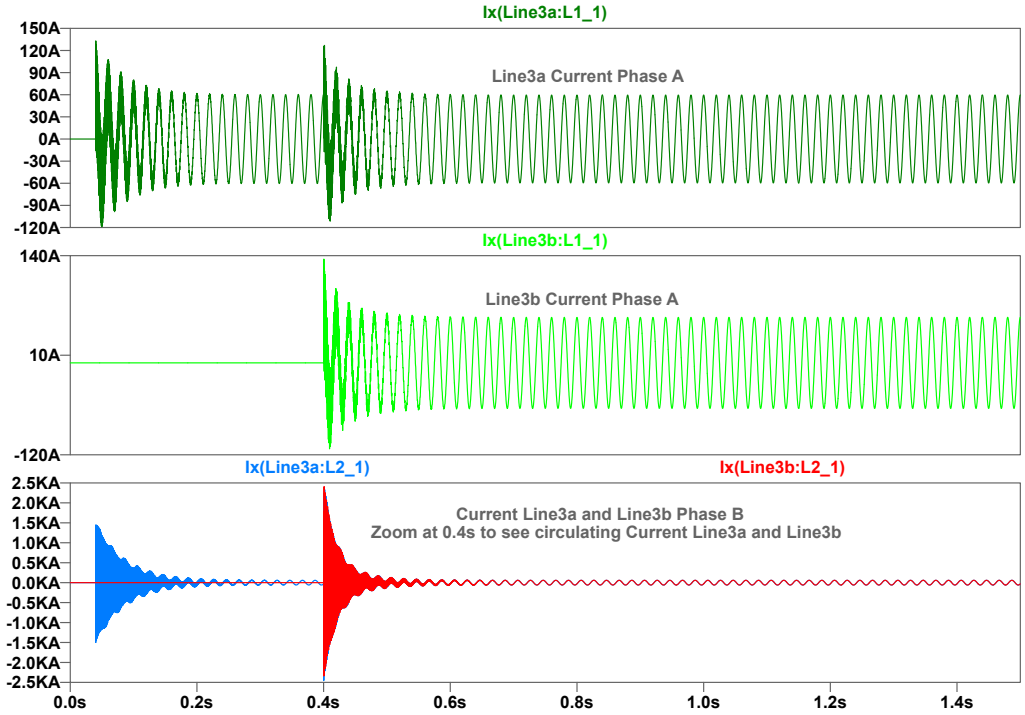


Figure 4.4: Switching lines3 - waveform

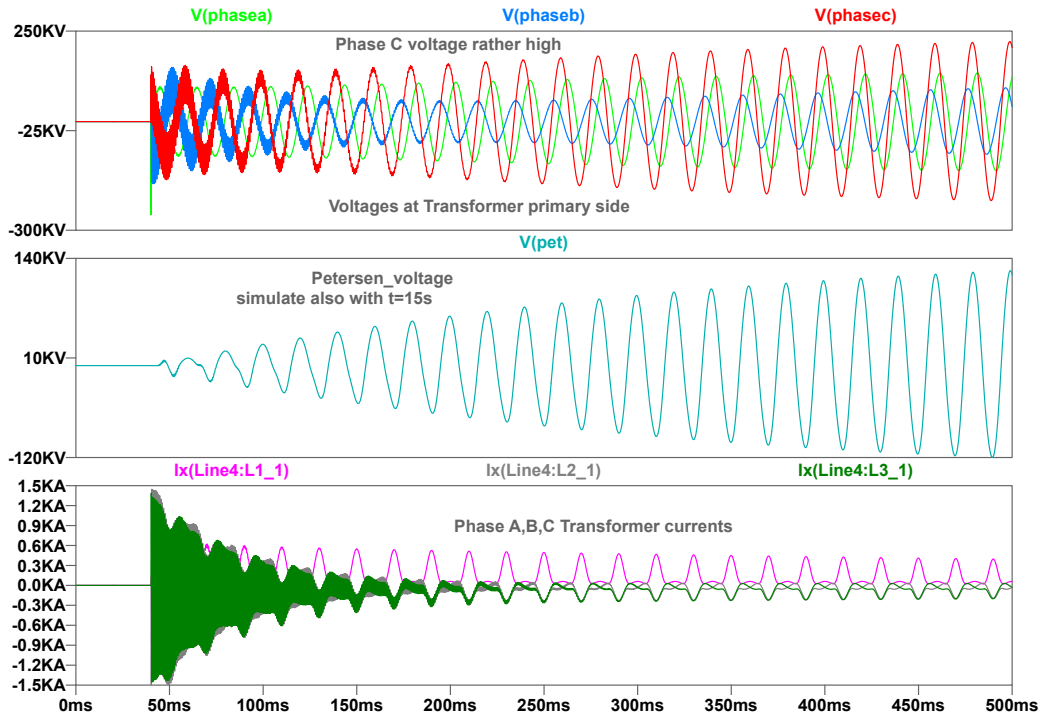


Figure 4.5: Switching lines4 - waveform

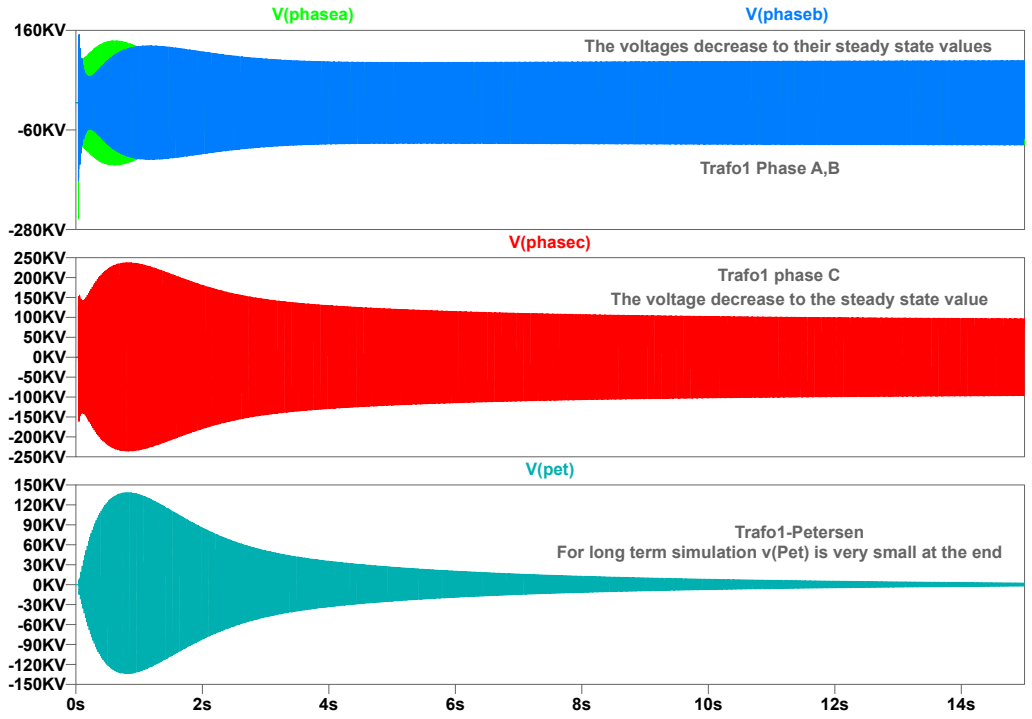


Figure 4.6: Switching lines 15s - waveform

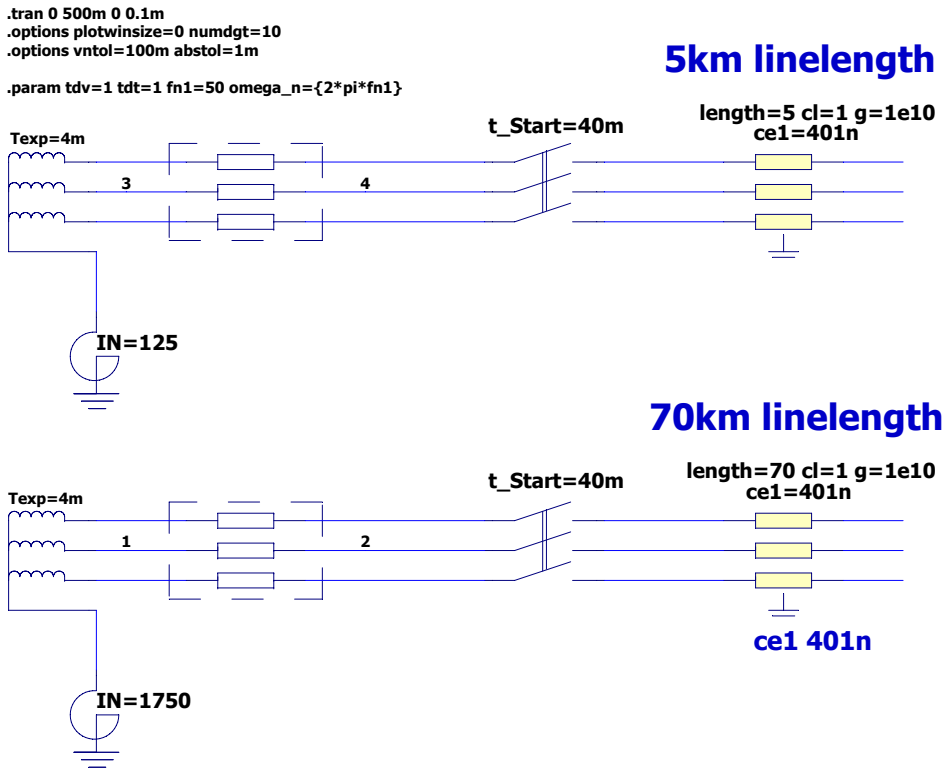


Figure 4.7: Switching lines 70km - schematic

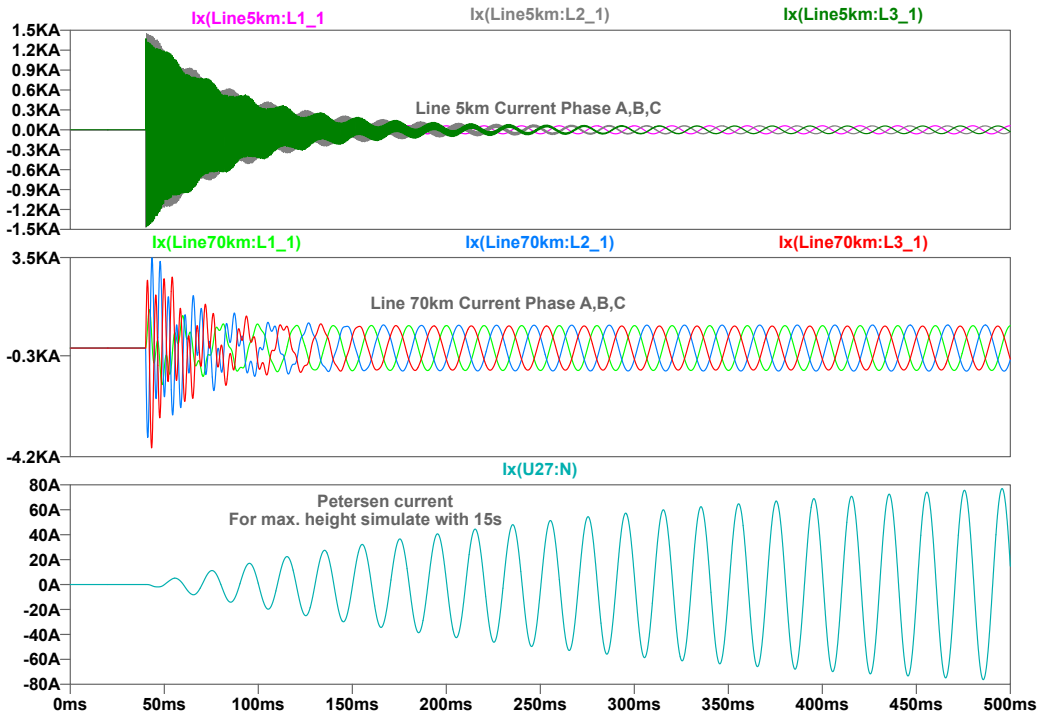


Figure 4.8: Switching lines 70km - waveform1

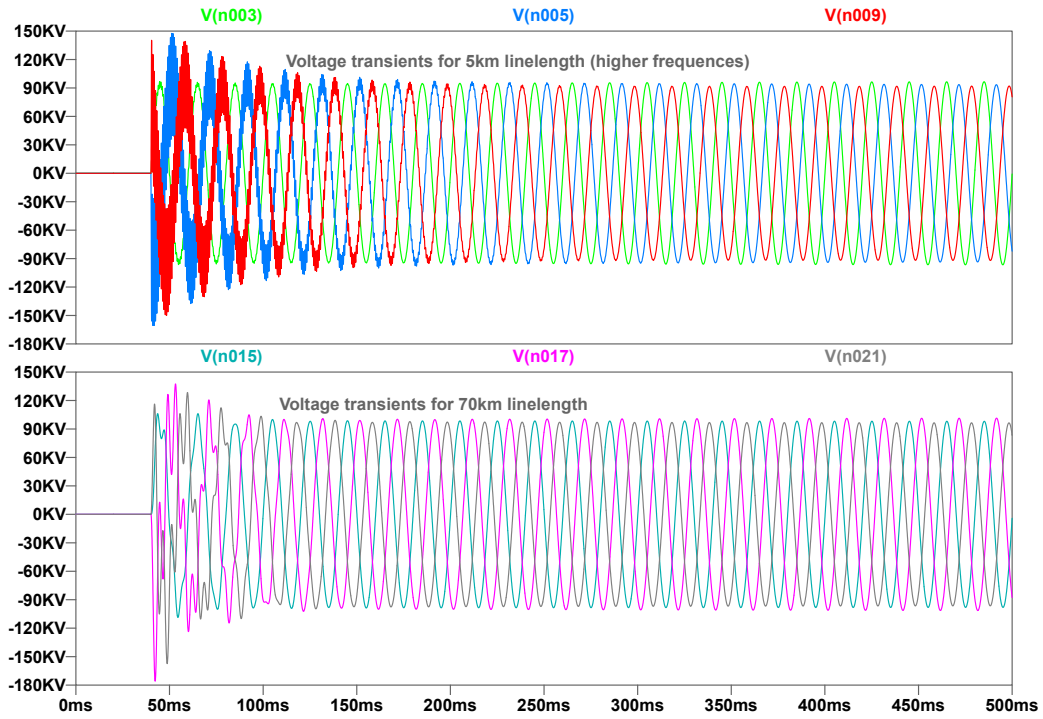


Figure 4.9: Switching lines 70km - waveform2

Chapter Summary

Cable energization produces high-frequency oscillations determined mainly by network inductance and cable capacitance. The switching instant controls the initial phase currents and overvoltages, while a connected load increases damping. Energizing a second cable creates both high-frequency circulating currents and a lower-frequency network-current component. When a cable supplies a transformer, capacitive charging and inductive inrush currents are superimposed. Increasing the cable length raises the charging current but also lowers the resonant frequency and increases resistive damping; the resulting voltage transient can therefore become smaller even when the current transient is larger. As can be seen in the example with the transformer, operating the Petersen coil exactly at the resonance point can cause very high voltages. For this reason, in practice, the Petersen coil is operated with either slight overcompensation or undercompensation. As can be seen in the network with the 70 km cable, the residual current in very large cable networks becomes quite high. The network is therefore usually divided into smaller subnetworks.

5 Earth Faults in 110 kV and 10 kV Power Grids

5.1 General

A Petersen coil is an intelligent protection and stabilization element in medium-voltage networks and, in some cases, also in high-voltage networks. It ensures that single-phase earth faults can be controlled safely and without immediate shutdown. This makes it an indispensable component of modern power supply systems. The Petersen coil is an inductive coil that is connected to the neutral point of the transformer. In a three-phase AC network, capacitive currents flow between conductors and earth. If a ground fault occurs, a capacitive fault current flows through the fault location – this can be dangerously high. In the event of a fault, the Petersen coil generates an inductive current that is opposes the capacitive fault current and, in the ideal case, compensates for it at the fault location.

5.2 Circuit design

This chapter investigates earth faults in small cable networks with resonant neutral grounding. The first example is the simulation file `Petersen_without_Fault_TRAN.asc`. The principal 110 kV example is designed for a nominal earth-fault compensation current of 300 A. In normal operation, network asymmetry produces only a small neutral displacement voltage and therefore a small Petersen-coil current. During a solid earth fault, approximately the full phase-to-earth voltage appears between the transformer neutral and earth, and the coil supplies approximately its rated current.

Unless stated otherwise, the transformer is unloaded. The earth fault is applied at 205 ms. The voltage-source parameter `texp` is used to establish steady state rapidly before the fault. The model contains a Yd5 transformer whose parameters are defined in the LTspice file `Transformer_110k_10k.asc`

5.3 Compensated 110 kV Network Without a Fault

5.3.1 Adjustment of the Petersen Coil

The network has a phase-to-earth capacitance of 5 012 nF per phase. At 50 Hz, its capacitive reactance is ..

Calculation 5.1 – (Petersen-coil rating from the network capacitance)

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi \cdot 50\,\text{Hz} \cdot 5\,012\,\text{nF}} \approx 635.1\,\Omega, \tag{5.1}$$

$$I_C = \frac{U_{\text{LL}}/\sqrt{3}}{X_C} = \frac{110\,\text{kV}/\sqrt{3}}{635.1\,\Omega} \approx 100\,\text{A RMS}. \tag{5.2}$$

For a single-phase earth fault, the total capacitive earth-fault current is approximately three times the phase charging current. Resonant compensation therefore requires

$$I_{\mathrm{P}} \approx 3I_C = 300\,\mathrm{A}, \tag{5.3}$$

$$X_{L,\mathrm{P}} = \frac{X_C}{3} \approx \boxed{211.7\,\Omega}. \tag{5.4}$$

The Petersen-coil model accepts the nominal line-to-line voltage and compensation current directly. We set these to 110 kV and 300 A.

5.3.2 Neutral Displacement Voltage in the Transient Analysis

Even without a fault, unequal phase-to-earth capacitances produce a neutral displacement voltage across the Petersen coil. In this example, its peak value is approximately 1.2 kV peak, much smaller than the rated phase voltage. The greater the difference between the phase capacitances, represented in the model by CE+CE1 and CE, the higher this voltage becomes.

The displacement voltage rises slowly and shows a beat frequency of approximately 1.2 Hz. This indicates that the resonant frequency is not exactly 50 Hz. In the LTspice waveform viewer, the beat frequency can be read by drawing a zoom window between two adjacent maxima and observing the frequency in the status line. (See figure 5.2)

5.3.3 Corrected Petersen-Coil Setting

The capacitance-only calculation in calculation 5.1 is correct for an ideal transformer, but the complete resonant circuit is not optimally compensated with a 300 A coil. The transformer short-circuit reactance on the primary side, $X_{1\mathrm{s}} = 30.25\,\Omega$ (see `Transformer_110k_10k.asc`) , reduces the resonant frequency to 48.8 Hz. The coil nominal current must therefore be increased to approximately 315 A to get again 50 Hz. This can be verified by simulating with `Petersen_without_Fault_AC.asc`. (See figure 5.3)

Now we set the initially choosen value of 1E20 for the magnetic reluctance of the transformer tank to a reasonable value of 1e6. After running the simulation, we find that the resonant frequency is again below 50 Hz.

We now choose 328 A for the Petersen coil. We restart the simulation. Now we see that the resonant point is again at approximately 50 Hz. We have seen now that the compensation current calculated from the total cable capacitance alone differs from the required Petersen-coil setting because of the transformer's internal impedance. We have to add about 10% for the Petersen current value to compensate for the voltage drop at the starpoint.

Remark 5.1 – (Phase information in AC analysis)

In practice, the automatic Petersen-coil controller normally maintains slight overcompensation or undercompensation to avoid resonance. Additionally the controller is disabled when an earth fault occurs to prevent further drift.

5.3.4 Reason about the reduction of the starpoint voltage

Explanation of the Reduced Neutral-Point Voltage when adding the magnetic reluctance of the transformer tank lies in the additional secondary leakage flux and the corresponding voltage

drop across the leakage reactance. This further decreases the effective neutral-point voltage. You can clearly see this in the file `Petersen_with_fault_hierarchical_model_AC.asc`. After simulation click on the transformer symbol then open schematic. Then click on the symbol for the primary winding Phase A and open schematic again. Then measure the current through RPhis. This gives approximately 9 mA (This current models the flux value in Wb). Perform the same measurement for the secondary winding above the primary winding. The current there is 7 mA increasing now the sum of all leakage fluxes. The Resonant frequency is approximately 47.9Hz. (See figure 5.4)

5.3.5 Resonance Frequency at about 50Hz

This is the resonance of all three phase capacitances with the Petersen-coil inductance and the effective transformer leakage inductance in the zero-sequence path. The coil reactance and inductance are

$$X_\mathrm{P} = \frac{U_\mathrm{LL}}{\sqrt{3} I_\mathrm{P}} = \frac{110\,\mathrm{kV}}{\sqrt{3}\cdot 300\,\mathrm{A}} \approx 211.7\,\Omega, \tag{5.5}$$

$$L_\mathrm{P} = \frac{X_\mathrm{P}}{2\pi\cdot 50\,\mathrm{Hz}} \approx 0.674\,\mathrm{H}. \tag{5.6}$$

The transformer leakage reactance gives

$$L1s = \frac{30.25\,\Omega}{2\pi\cdot 50\,\mathrm{Hz}} \approx 0.096\,\mathrm{H}. \tag{5.7}$$

Since the three phase capacitances act in parallel and the three transformer phase inductances contribute as $L1s/3$

Calculation 5.2 – (Resonanz at 50Hz without leakage inductance)

$$C_0 = 3\cdot 5\,012\,\mathrm{nF}, \tag{5.8}$$

$$L_0 = L_\mathrm{P} + \frac{L1s}{3} = 0.674\,\mathrm{H}, \tag{5.9}$$

$$f_{\mathrm{r},0} = \frac{1}{2\pi\sqrt{L_0 C_0}} \approx \boxed{50\,\mathrm{Hz}}. \tag{5.10}$$

Calculation 5.3 – (Resonanz at 48.8Hz with primary leakage inductance)

$$C_0 = 3\cdot 5\,012\,\mathrm{nF}, \tag{5.11}$$

$$L_0 = L_\mathrm{P} + \frac{L1s}{3} = 0.674\,\mathrm{H} + \frac{0.096\,\mathrm{H}}{3}, \tag{5.12}$$

$$f_{\mathrm{r},0} = \frac{1}{2\pi\sqrt{L_0 C_0}} \approx \boxed{48.8\,\mathrm{Hz}}. \tag{5.13}$$

Calculation 5.4 – (Resonanz at 47.9Hz also with secondary leakage inductance)

$$C_0 = 3 \cdot 5\,012\,\text{nF}, \tag{5.14}$$

$$L_0 = L_\text{P} + \frac{L1s}{3} + \frac{L1s * 7}{9} = 0.674\,\text{H} + \frac{0.096\,\text{H}}{3} + \frac{0.096\,\text{H} * 7}{9}, \tag{5.15}$$

$$f_{\text{r},0} = \frac{1}{2\pi\sqrt{L_0 C_0}} \approx \boxed{47.95\,\text{Hz}}. \tag{5.16}$$

7/9 means the relation of the leakage fluxes measured above (9 Wb and 7 Wb). The resonant peaks for 48.8 Hz and 47.95 Hz can also be observed in the file `Petersen_without_fault_AC.asc`. by setting rkessel (this is the parameter for tank reluctance) to 1e6 or 1e20. For the calculation of the resonant frequencies see also the file `Petersen.asc`

5.4 Low-Resistance Earth Fault in the 110 kV Network

5.4.1 Network Compensated with 328 A

The transient simulation is contained in `Petersen_with_fault_TRAN.asc` (see figure 5.6). At fault inception, the phase-R voltage rapidly collapses to zero. The voltages of the two healthy phases rise toward the line-to-line voltage. This transition is accompanied by high-frequency overvoltages and currents. An FFT shows a mixture of dominant components near 260 Hz and 1.2 kHz. (see figure 5.19) .

At this higher frequency, the capacitive reactance decreases and the inductive reactance of the line impedance increases. The capacitive current through the inductive impedance therefore produces an additional voltage rise. The currents flowing toward the transformer are predominantly inductive and approximately in phase with one another. Their inductive character can be recognized because the current maximum lags the corresponding voltage maximum. You can also see that the voltages of the healthy phases have a phase difference going from 120 degrees to 60 degrees (see figure 5.5)

The current at the fault location initially reaches approximately 4.2 kA peak (see figure 5.7). It decays rapidly and approaches approximately a steady-state peak value of approximately 4 A peak. Its minimum also depends on the ohmic resistance of the Petersen coil, set by `r_x` parameter. The prefault capacitive reactive power is approximately −6.3 Mvar per phase, or about −19 Mvar in total. Immediately after the phase A fault, the high-frequency recharging process briefly drives the phase B and phase C values to approximately −40 Mvar. However, immediately after the fault. The capacitive and inductive reactive-power components are superimposed. The reason for the doubling of the capacitive reactive power is that the two capacitances of the healthy phases are suddenly subjected to a voltage √3 times the normal phase-to-earth voltage. At the end of the transient processes, the additional capacitive reactive power is again compensated by the Petersen reactor. In addition, the inductive reactive power decays to zero. In the end, the reactive power is exactly the same as it was at the beginning (see figure 5.7 and 5.8). In `Petersen_with_Fault_AC.asc`, the minimum fault current is approximately 3 A RMS near 50 Hz (see figure 5.8). Set the vertical axis to linear scaling to read this minimum accurately. You can also see that the voltage magnitude of phase A is zero, and that Phase B and Phase C are at

111 kV, which is the phase-to-phase voltage.The phase angle of phase B is −150°, and that of phase C is +150°. The difference is −150 − 150 = −300 = +60 degrees, just as we observed earlier. (see figure 5.5)

Remark 5.2 – (Phase information in AC analysis)

The relative phase angles of voltages and currents are most accurately examined in the AC-analysis files, where the fundamental-frequency phasors are not obscured by the switching transient.

5.4.2 Reactive Power detailed

The capacitive reactive power of the network is approximately

$$Q_C = \sqrt{3}\, U_{\mathrm{LL}} I_C = \sqrt{3} \cdot 110\,\mathrm{kV} \cdot 100\,\mathrm{A} \approx 19\,\mathrm{Mvar}. \tag{5.17}$$

It can be measured with the `qmeter_uvw` element. Each phase contributes approximately −6.3 Mvar, with small differences caused by the unequal capacitances and the resulting unequal phase voltages (This explains the neutral displacement voltage). The waveform viewer shows approximately 110 kV, 111.7 kV, and 110.7 kV after the model's display scaling by $\sqrt{3}/\sqrt{2}$. Their mean is slightly above 110 kV because the capacitive current produces a voltage rise across the inductive line impedance.

Remark 5.3 – (Waveform-viewer units and scaling)

Multiplying a power trace by `1W/1V` can be used to change the displayed unit prefix in the waveform viewer. If `sfactor=3` is selected in `qmeter_uvw`, the measured values are multiplied internally by three, which may simplify comparison with the total three-phase power.

5.4.3 Reactive-Power Increase at Fault Inception

During the earth fault, the two healthy phase-to-earth capacitances are charged to approximately the line-to-line voltage, while the capacitance of the faulted phase is discharged. Since capacitive reactive power is proportional to voltage squared, the total capacitive reactive power of the healthy phases is ideally twice (because of the voltage transients even higher) the prefault three-phase value. The source current therefore also rises correspondingly.

After the transient, one part of this increased capacitive earth-fault current is compensated by the Petersen coil. The ordinary positive-sequence charging current remains supplied by the source.

5.4.4 Additional Transformer Loading Due to Petersen-Coil Current

The Petersen-coil current flows through the transformer windings and therefore adds to the transformer loading. In the example, its equivalent contribution is approximately 100 A per phase, while the rated primary current is approximately 200 A. Because this current is predominantly inductive, it must be added vectorially to the load current. For the assumed load, whose reactive component is about one fifth of its active component, approximately 90 % of the original load capability remains. NOTE: The additional load due to the Petersen is only present when a earth-fault occur. In addition,

we must take into account the high secondary-side circulating current resulting from the inclusion of the transformer tank in the simulation which also puts a certain load on the transformer. This is seen best in the file `Petersen_without_fault_hierarchical_model_TRAN/AC.asc`. Right-click the transformer symbol and select `Open Schematic`. After running the simulation, measure the current in the secondary winding of Phase A.

5.4.5 Loaded and Unloaded Transformer

Changing the load parameter from `20e-6` to `40e6` reduces both the phase voltages and the transformer neutral voltage. Since the voltage across the cable capacitances decreases at the same time, the compensation tuning remains almost unchanged. The comparison is implemented in `Petersen_with_fault_with_load_vs_without_load_AC.asc`. (see figure 5.9 and 5.10)
NOTE: The Petersen current in the unloaded case is higher (304A) because of higher voltage level of 111.4kV compared to the loaded case where the voltage level is 110kV and the current is (300A).

5.5 High-Resistance Earth Fault

The AC-analysis file `Petersen_with_fault_highohmic_AC.asc` compares the earth-fault resistances 50 Ω, 100 Ω, 500 Ω, and 1 000 Ω. At the resonance point, all current curves approach approximately 3 A, representing the residual ohmic component of the compensated earth-fault current. (see figure 5.11 and 5.12)

5.6 Comparison of coupled and uncoupeled line impedance

Open `Petersen_fault_coupled_uncoupled.asc` and run the simulation. After simulation we see that the peak fault current is lower in the coupled-line case than in the uncoupled-line case. The difference is about 1 kA. The steady state current at the end of the simulation is the same for the two cases. This shows that simulating a asymmetric fault the coupling of the lines has to be considered. For the calculation of the coupled impedance values see also the wxmaxima file `Petersen.asc`
(see figure 5.13 and 5.14)

Important 5.1 – (Asymmetric faults)

Coupling must be included when simulating asymmetrical faults in coupled-line systems.

5.7 Compensated 10 kV Network with a Grounding Transformer

A Yd transformer does not provide an accessible neutral on its delta-connected side. An external neutral-forming transformer, represented by `Trafo_gnd` symbol, creates a neutral point in the medium-voltage network without presence of a transformer with starpoint so that a Petersen coil can be connected. The corresponding simulation files are

- `Petersen_Trafo_gnd_TRAN.asc`,
- `Petersen_Trafo_gnd_AC_fault.asc`,
- `Petersen_Trafo_gnd_AC_no_fault.asc`.

We now see a 10 kV compensated network with external grounding transformer. You see a Petersen in a medium voltage network. Due to the star point builder, a YD transformer can establish a star point at the secondary side for connecting a Petersen coil.

5.8 Petersen-coil and Yy-Transformers with three legs

Now consider a Petersen coil connected to a Yy transformer with a three-leg core. Since the YY transformer has no low-reluctance return path for zero-sequence flux in the core of the transformer, the flux must return through the tank and oil. This is not the goal. Therefore, the star point of a wye/wye transformer should only be loaded to no more than 10% of the transformer's rated current. See `ThreeLegCore-Yy-Transformer.asc`

5.9 Solid Neutral Earthing and a Yy-Transformer with a Five-Leg Core

Open the File `FiveLegCore_without_third_winding.asc`. If the Yy-transformer has a five leg core, the star point cannot deliver any current if the second star point is isolated. This is because the outer legs provide a low-reluctance return path for in-phase fluxes. If the second star point is also connected, current can flow from one into the other star points, but current should not be exchanged between the two star points. In short, the two star points of a five leg Yy-transformer should be isolated. Now open file `FiveLegCore_with_third_winding.asc`. Now consider neutral-point loading in a Yy transformer with a five-leg core and a compensating tertiary winding. The transformer can deliver full current at the star points. There is also no current transfer from one to the other star point.

5.10 Characteristic Resonance Frequencies

Several natural frequencies occur after fault inception. The highest is associated with the discharging capacitance of the faulted phase. A lower frequency appears in the healthy-phase voltages and capacitance currents.

5.10.1 Resonance at 1.2 kHz

This resonant path contains the cable series inductance and the faulted phase capacitance. It is driven mainly through capacitor C12 and the fault position. With $L = 3.5\,\text{mH}$ and $C = 5\,022\,\text{nF}$ follows ..

$$f_\text{r} = \frac{1}{2\pi\sqrt{LC}} \approx \boxed{1.2\,\text{kHz}}. \tag{5.18}$$

5.10.2 Resonance at 260 Hz

This is the frequency of the current through the netimpedance and the capacities C10 and C11. With $L = 25\,\text{mH}$ and $C = 5\,012\,\text{nF}$*3

$$f_\text{r} = \frac{1}{2\pi\sqrt{LC}} \approx \boxed{260\,\text{Hz}}. \tag{5.19}$$

For the frequencies see figure 5.19

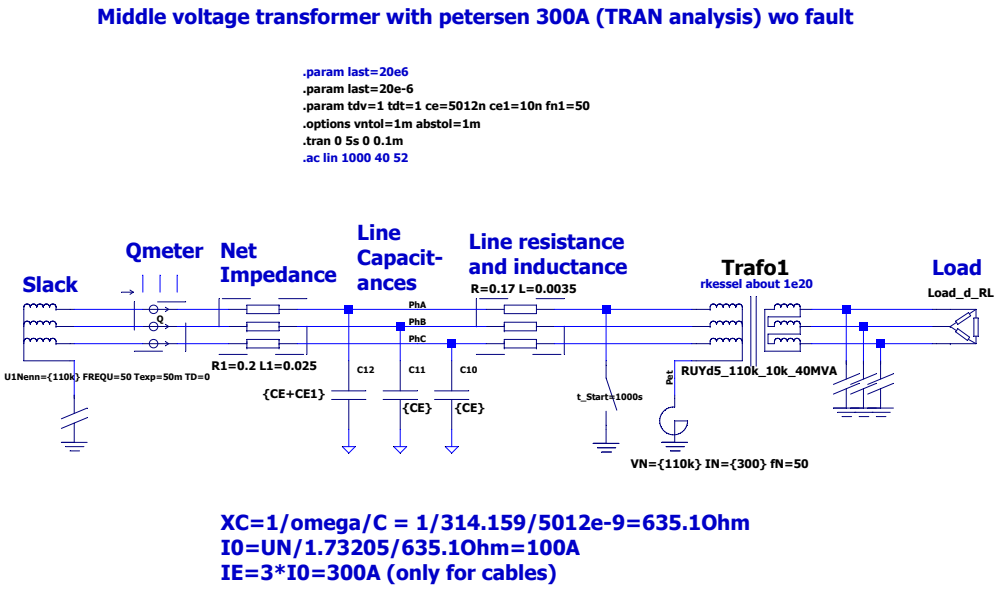


Figure 5.1: Petersen without fault - schematic

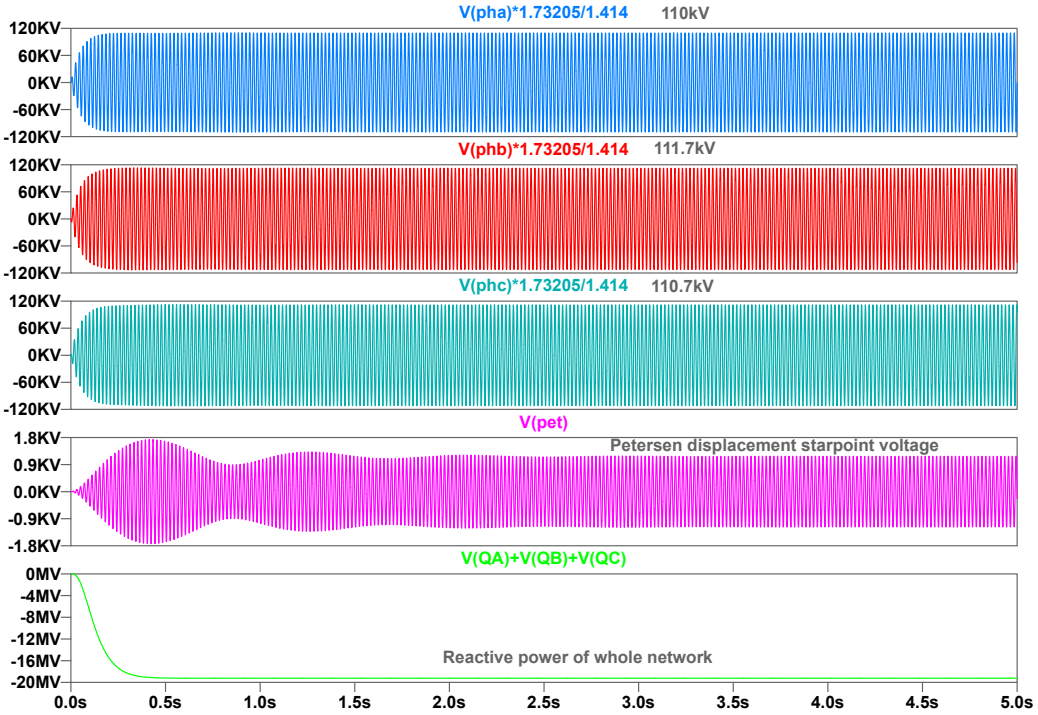


Figure 5.2: Petersen without fault TRAN - waveform

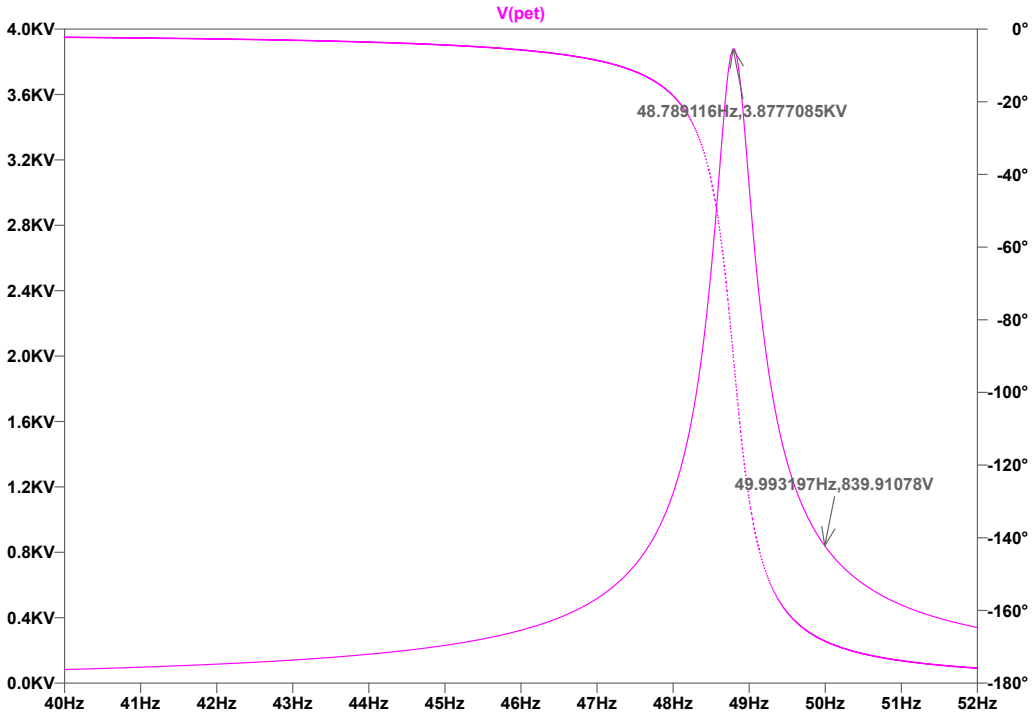


Figure 5.3: Petersen without fault AC starpoint (Petersen) voltage - waveform

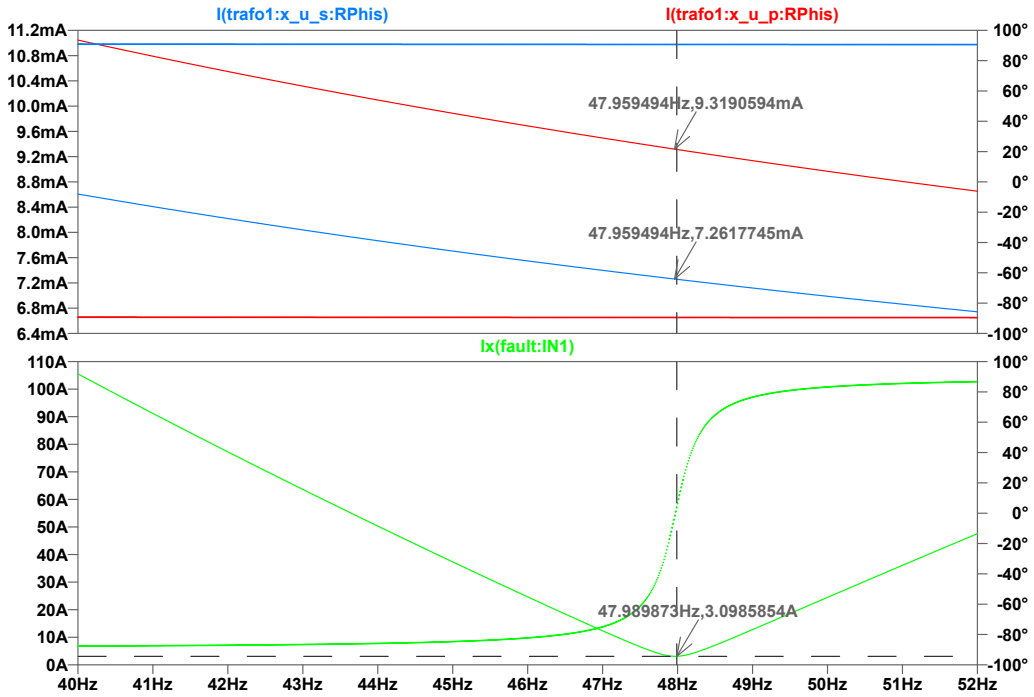


Figure 5.4: Petersen with fault AC hierarchical model - waveform

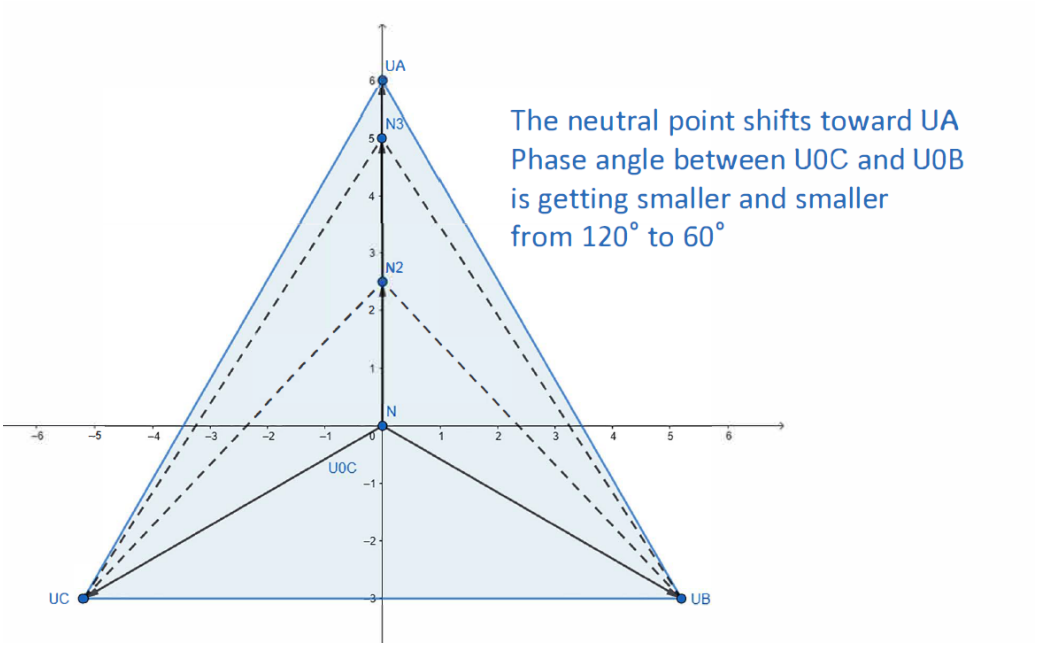


Figure 5.5: Principal phase difference between the voltages U0B and U0C in fault case, plotted with GeoGebra - see file Starpoint_vektor.ggb

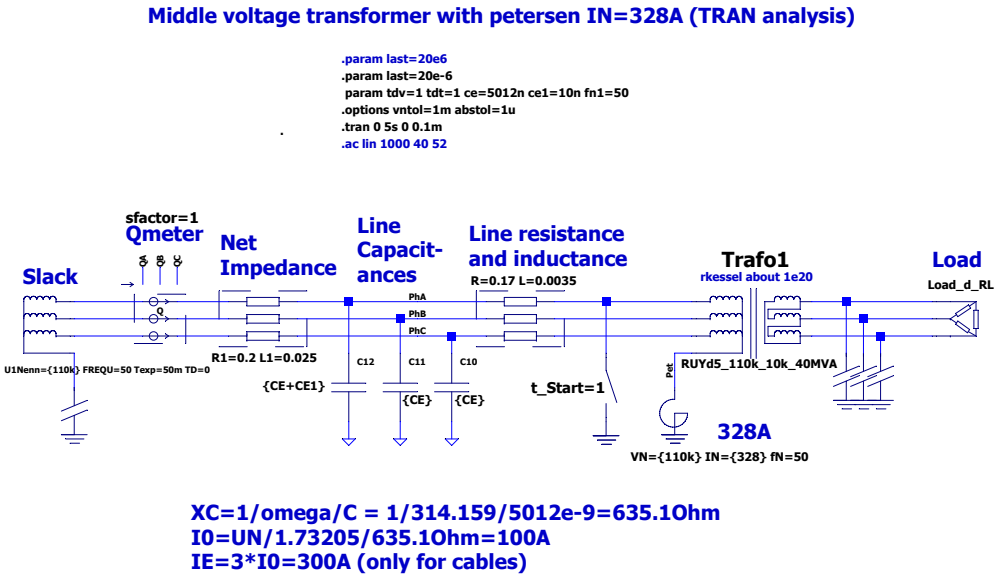


Figure 5.6: Petersen with fault TRAN - schematic

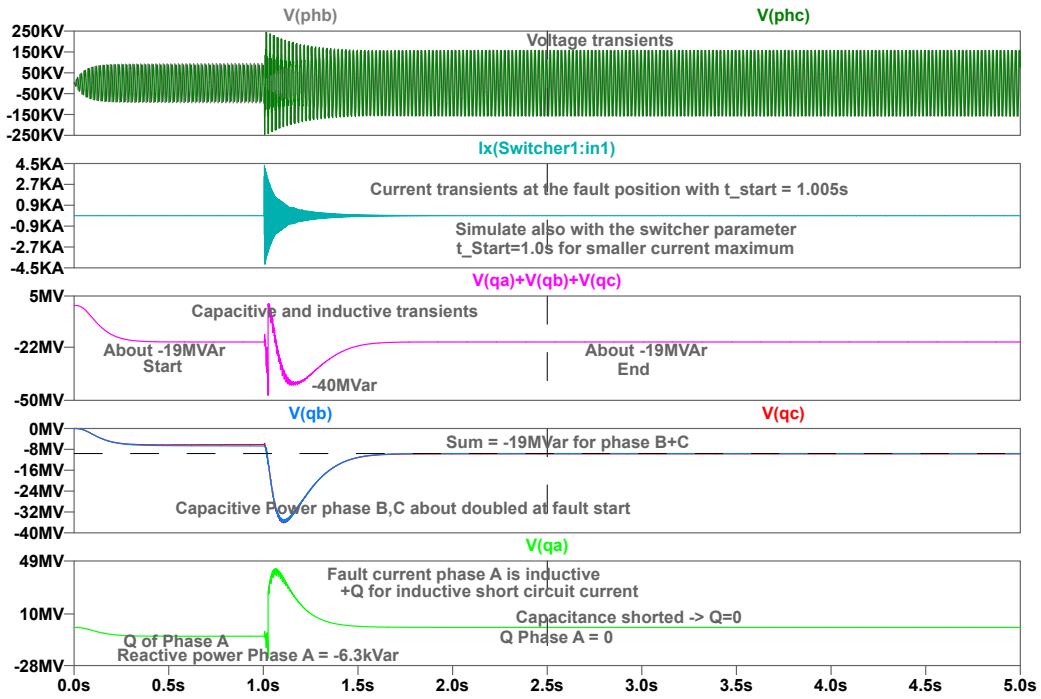


Figure 5.7: Petersen with fault TRAN - waveform

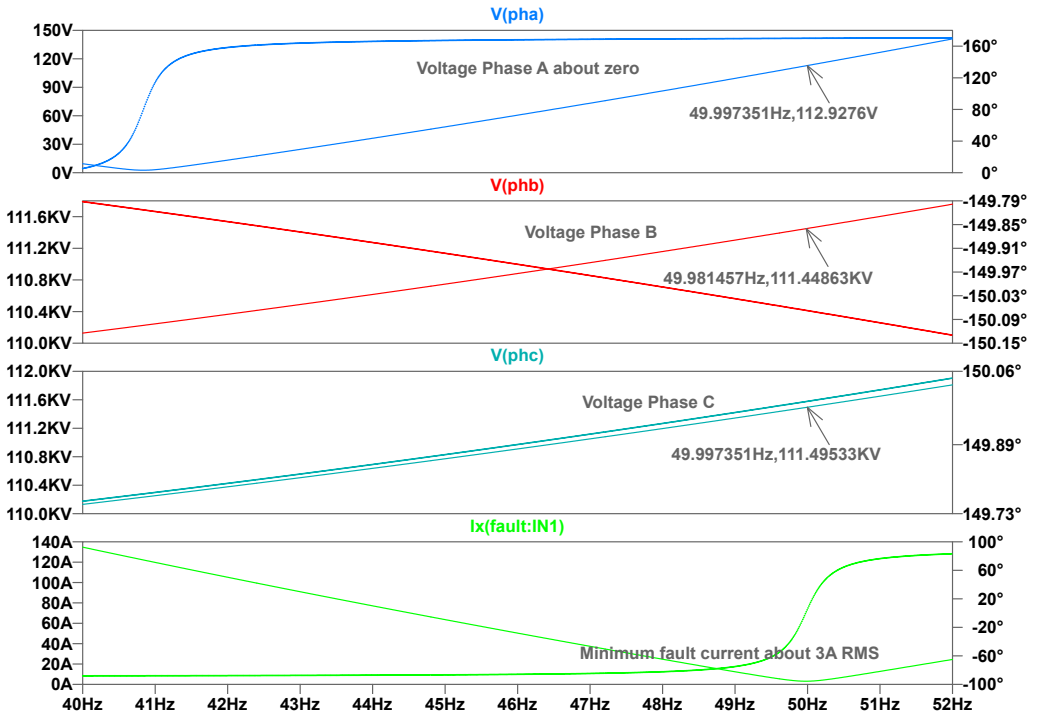


Figure 5.8: Petersen with fault AC - waveform

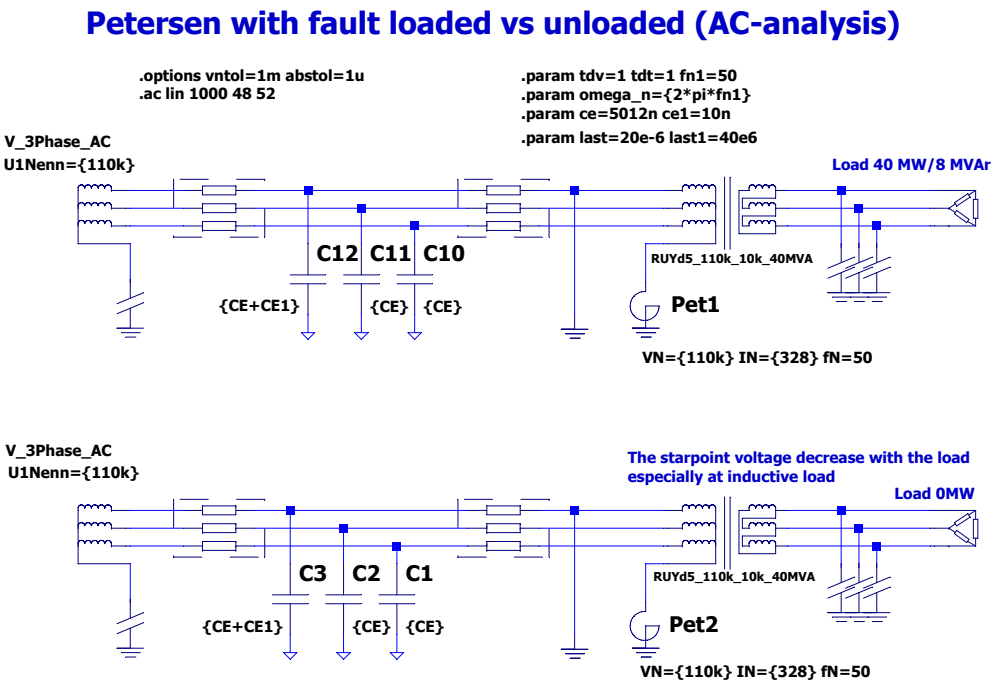


Figure 5.9: Petersen with fault and load AC - schematic

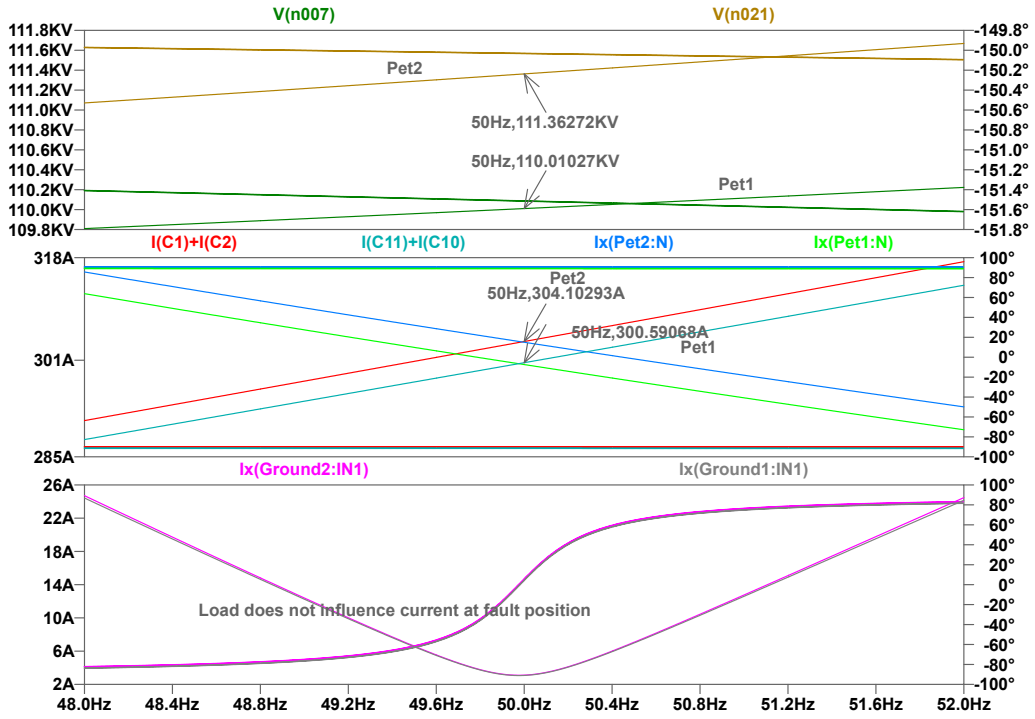


Figure 5.10: Petersen with fault and load AC - waveform

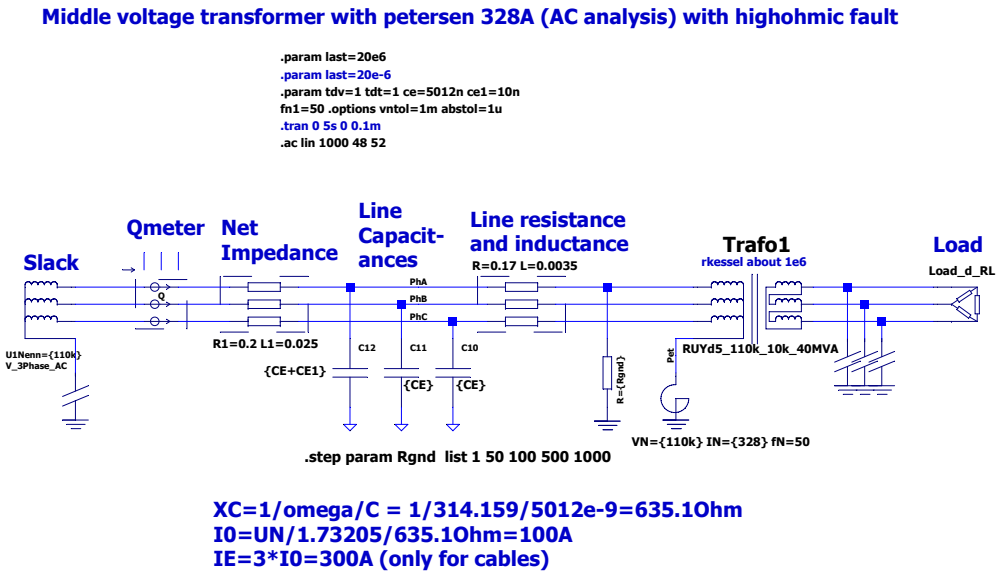


Figure 5.11: Petersen with high-resistance earth fault fault AC - schematic

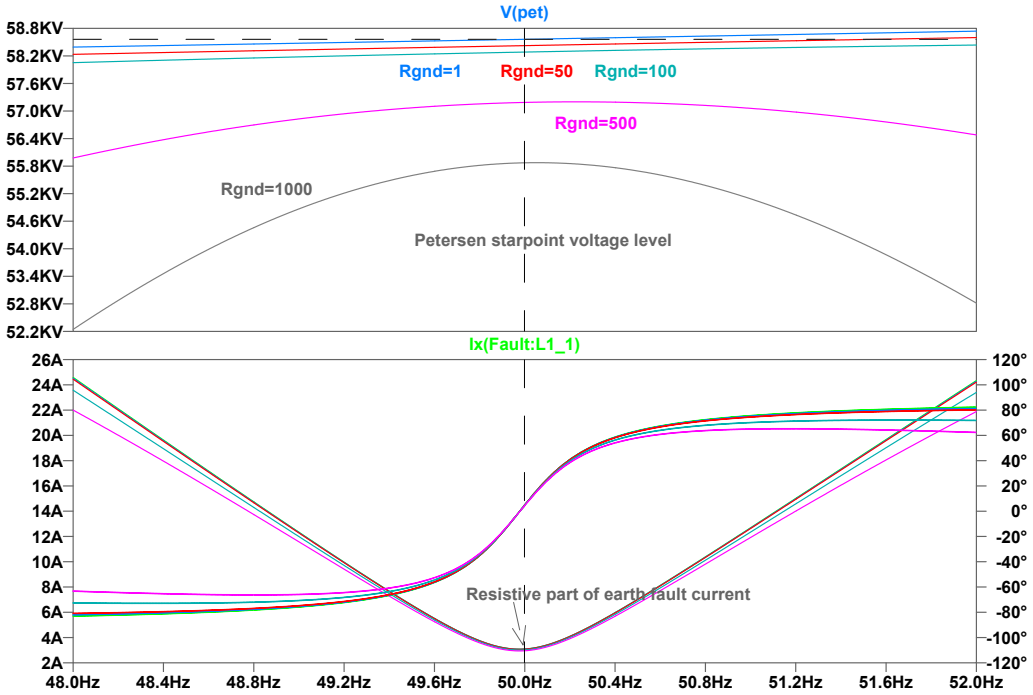


Figure 5.12: Petersen with high-resistance earth fault fault AC - waveform

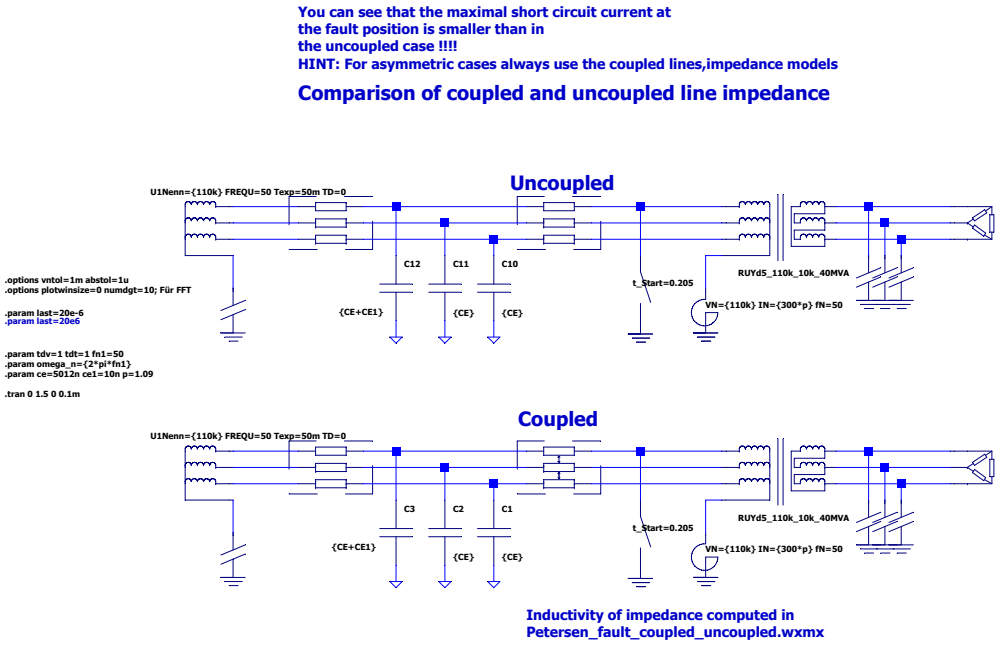


Figure 5.13: Petersen with coupled and uncoupled line impedance - schematic

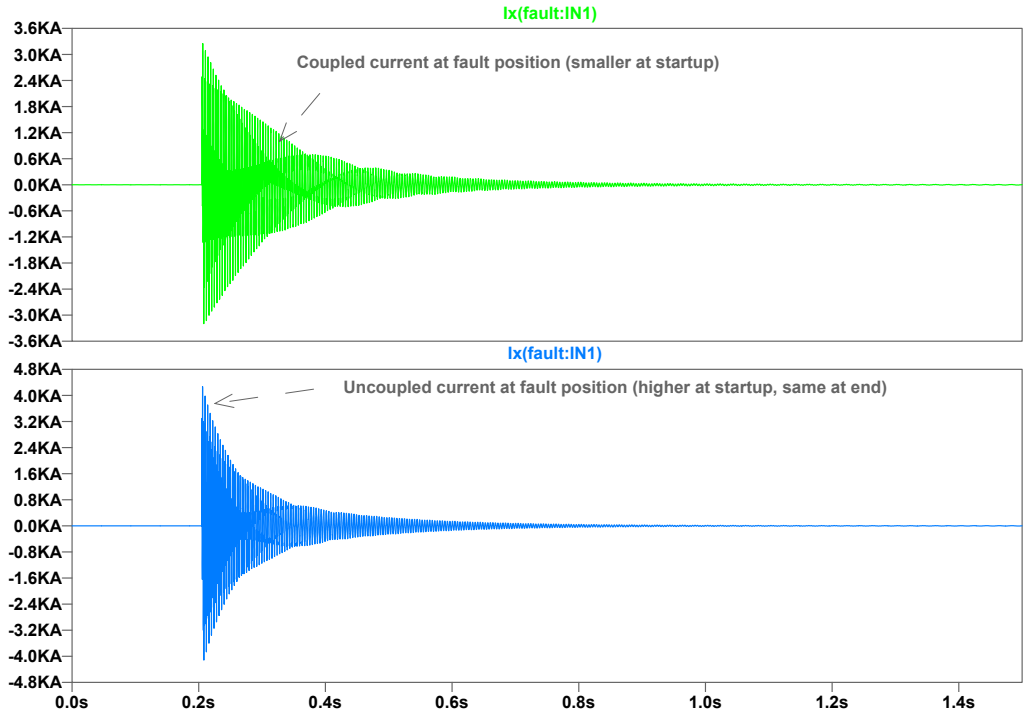


Figure 5.14: Petersen with coupled and uncoupled line impedance - waveform

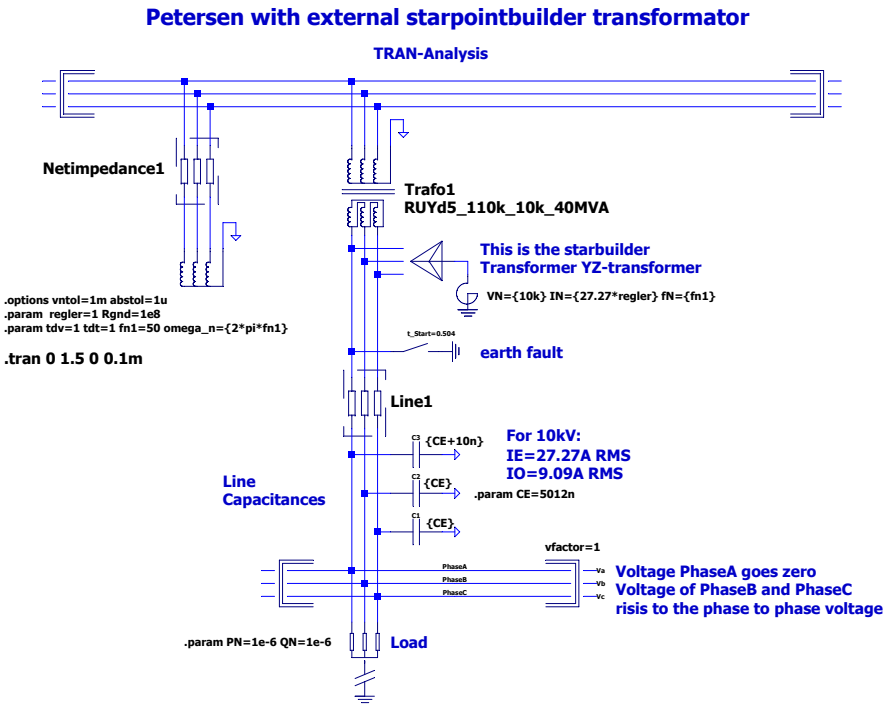


Figure 5.15: Petersen without fault and external grounding transformer TRAN - schematic

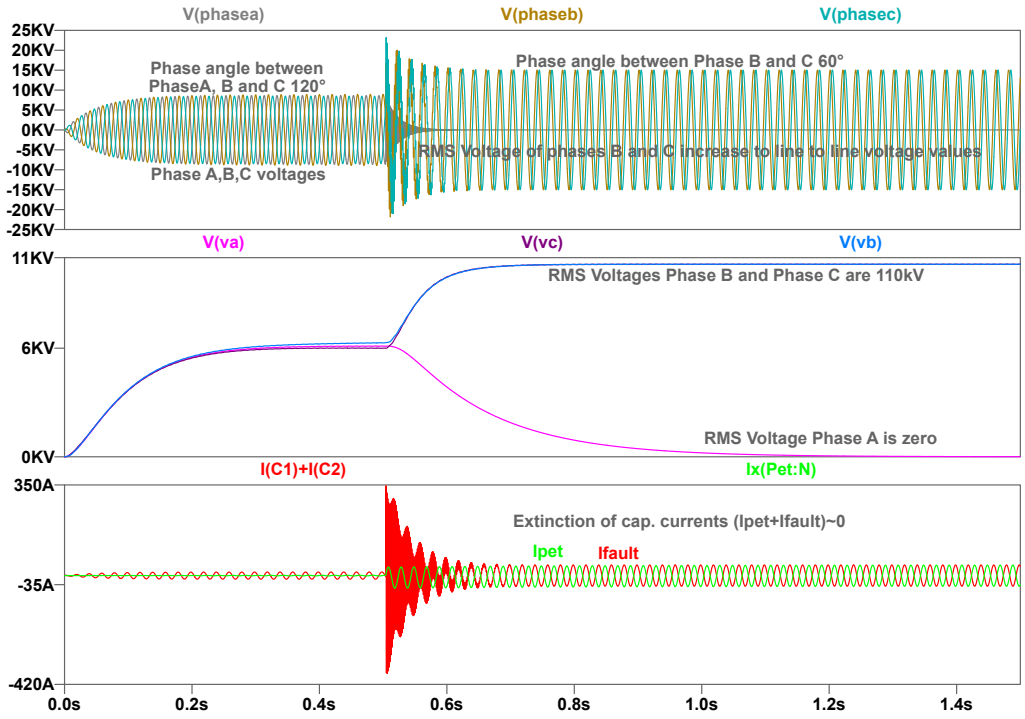


Figure 5.16: Petersen without fault and external grounding transformer TRAN - waveform

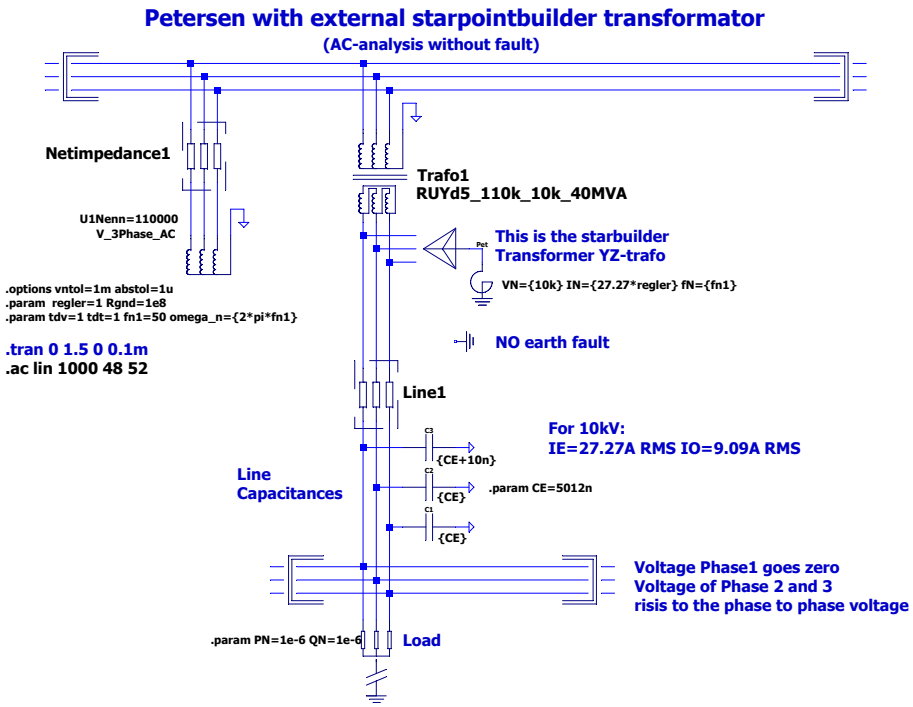


Figure 5.17: Petersen without fault and external grounding transformer AC - schematic

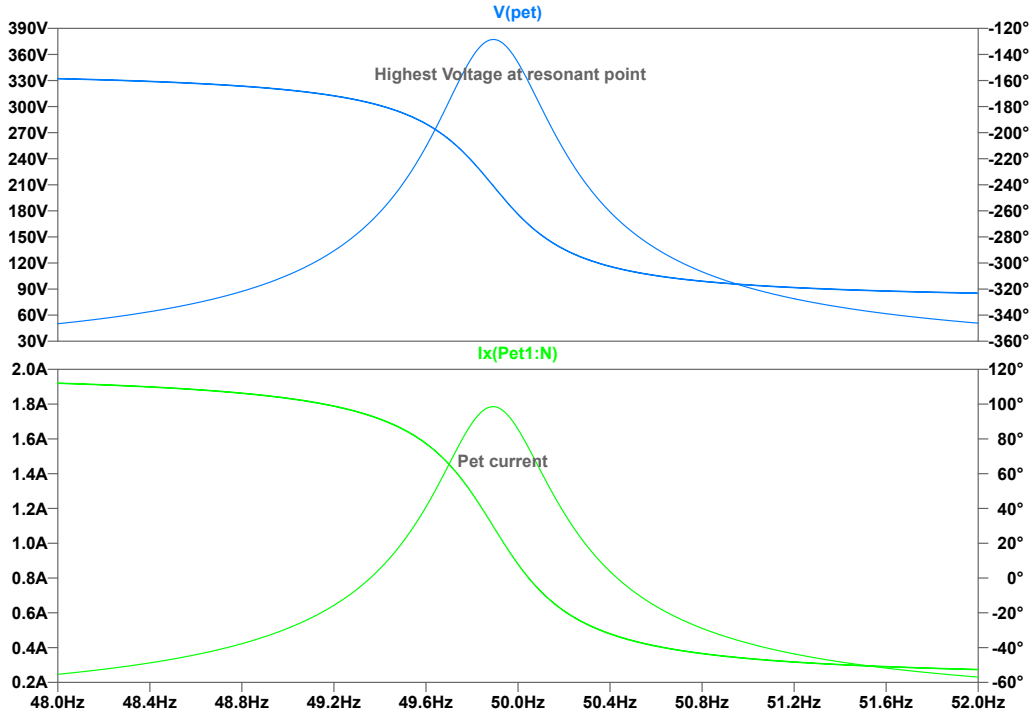


Figure 5.18: Petersen without fault and external grounding transformer AC - waveform

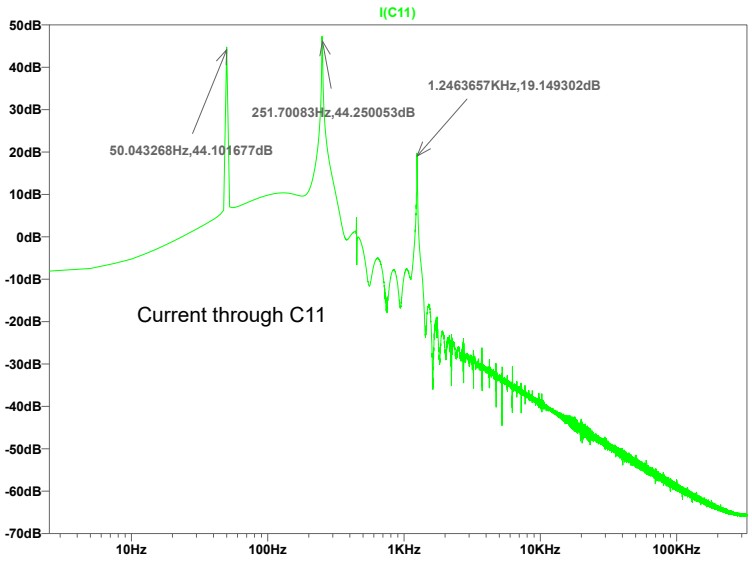


Figure 5.19: FFT of current through capacitor C11

Chapter Summary

In a resonantly earthed cable network, the Petersen coil compensates the zero-sequence capacitive earth-fault current. Network asymmetry creates a small neutral displacement voltage even without a fault. During a solid earth fault (currents like at a shorted circuit at startup), the healthy phases rise toward line-to-line voltage and the capacitive reactive power approximately doubles for the first instant. The complete tuning must include transformer and source impedance, not only the cable capacitance. Fault inception excites several natural frequencies determined by different inductive and capacitive paths. For networks without an accessible neutral, a neutral-forming transformer permits connection of the Petersen coil. High transient currents arise during the redistribution of charge in the phase-to-earth capacitances. The capacitance of the faulted phase is discharged, while the voltages across the capacitances of the two healthy phases increase to approximately $\sqrt{3}$ times their normal values. At the higher frequencies excited by the fault, capacitive reactance decreases and inductive reactance increases. These changes contribute to the high initial currents and voltage distortion. The resonant frequencies depend on the inductances and capacitances of the relevant circuit paths. During the earth fault, the total capacitive reactive power of the healthy phases is approximately twice the prefault three-phase value. As the transient decays, the Petersen coil compensates the additional zero-sequence earth-fault current, while the ordinary charging current remains supplied by the source. Note: The line inductances in this particular example are modelled without mutual coupling. Including coupling can change the effective fault-path impedance and the initial peak fault current. Compare the coupled and uncoupled models discussed in Section 5.6.

6 Switching Whole Networks

6.1 General

This chapter examines the transients that occur when an entire network (grid zone) is energized. This situation can occur during grid restoration following a blackout
The following cases are examined in detail:

- `Connecting at minimum voltage in phase R`,
- `Connecting the grid zone with phase R at maximum voltage`,
- `Connecting at minimum voltage and full remanent flux`,
- `Connecting with the main transformer set to minimum voltage`,
- `Connecting a larger grid zone`,
- `Connecting under load`.
- `Connecting with skin effect in the grid impedance`,
- `Connecting with a weak feed-in`,
- `Slowly ramping up the voltage instead of a switching operation`.

6.2 Circuit Design

Open the File `Switching_on_networks.asc`. The example is a 110 kV network with five substations. Three substations are loaded, two are unloaded or have loads that are not yet connected. The network is supplied through a 220 kV/110 kV transformer rated 200 MV A.

The main transformer is represented by a linear magnetizing characteristic using the transformer parameter `m=1`. Its voltage source initially rises exponentially so that the transformer reaches steady state before the network switch closes. The network contains three cable sections and two overhead-line sections. The cables have a total sum length of 10 km and generate approximately 15 Mvar of capacitive reactive power (see calculation 6.1). A 40 Mvar shunt reactor in bus4 overcompensates the 15 Mvar capacitive contribution, leaving a net inductive demand of approximately 25 Mvar. The energized network therefore tends to reduce rather than raise the voltage (under steady state conditions).

The cable phase-to-earth capacitances are compensated by a Petersen coil at substation 1. A small capacitance asymmetry is required to create a measurable neutral displacement voltage. In the model, this is introduced with the parameter `cuf` in phase A of line 2.

6.3 Petersen-Coil Setting

The cable capacitance is 400 nF/km per phase (see line.lib). For a total cable length of 10 km, the phase capacitance is 4 000 nF.

Calculation 6.1 – (Compensation current for the complete cable network)

At 50 Hz,

$$C = 400\,\text{nF/km} \cdot 10\,\text{km} = 4\,000\,\text{nF}, \tag{6.1}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi \cdot 50\,\text{Hz} \cdot 4\,000\,\text{nF}} \approx 796\,\Omega, \tag{6.2}$$

$$Q_C = \frac{U_{\text{LL}}^2}{X_C} = \frac{(110\,\text{kV})^2}{796\,\Omega} \approx 15.2\,\text{Mvar}. \tag{6.3}$$

For resonant earth-fault compensation,

$$X_{L,\text{P}} = \frac{X_C}{3} \approx 265\,\Omega, \tag{6.4}$$

$$I_{\text{P}} = \frac{U_{\text{LL}}/\sqrt{3}}{X_{L,\text{P}}} \approx \boxed{239.7\,\text{A}}. \tag{6.5}$$

The Petersen model uses a rounded setting of 250 A. An increase of roughly 5 % is useful because internal transformer reactance reduces the neutral voltage and prevents the ideal coil current from being reached during an earth fault. This effect was discussed in detail in Chapter 5.

6.4 Calculation of the Network Resonance

The inductances specified on the 220 kV side must be referred to the 110 kV side. With a voltage ratio of 2, impedance and inductance are divided by $2^2 = 4$ when referred from the 220 kV side to the 110 kV side. The source and transformer inductances are $L_{\text{net}} = 0.007\,7\,\text{H}$ and $L_{\text{Tr}} = 0.154\,\text{H}$. The transformer value is obtained from `xkos/omega` in `Trafo_YY0_220k_110k_200MVA_Daten.asc`. Thus, ..

Calculation 6.2 – (Dominant cable-network resonance)

$$L_{\text{eq}} = \frac{L_{\text{net}}}{4} + \frac{L_{\text{Tr}}}{4} = \frac{0.007\,7\,\text{H} + 0.154\,\text{H}}{4} = 0.04\,\text{H}, \tag{6.6}$$

$$C = 4\,000\,\text{nF}, \tag{6.7}$$

$$f_{\text{r}} = \frac{1}{2\pi\sqrt{L_{\text{eq}}C}} \approx \boxed{395.8\,\text{Hz}}. \tag{6.8}$$

This result agrees with the approximately 400 Hz peak, which is most clearly visible (after simulation) in the FFT of the main transformer current in phases B and C in Case 1. Zoom into the interval from 40 ms to 100 ms, then select View ⟶ FFT and use the current zoom extent. Additional harmonics caused by the nonlinear transformer currents are most clearly visible in phase A at 100 Hz, 150 Hz, 200 Hz and 250 Hz.

6.5 Rated current of the main transformer

The rated RMS current on the 220 kV side of the main transformer is

$$I_N = \frac{S_N}{\sqrt{3}U_N} = \frac{200\,\text{MV A}}{\sqrt{3}\cdot 220\,\text{kV}} \approx \boxed{525\,\text{A}}. \tag{6.9}$$

For further values of transformer parameters see:
Transformer_110kV_10KV.asc for the medium voltage transformer and
Transformer_220kV_110KV.asc for the high voltage transformere

6.6 Case 1: Closing at the Phase-R Voltage Zero

NOTE: The ltspice file for all cases except case9 is `Switching_on_networks.asc`. To view the results in the Waveform Viewer, you must select the appropriate plot file. After the simulation, open the Waveform Viewer window. Select "Plot Settings" / "Open Plot Settings File" and, from the available plots, select one of the corresponding case1 plots. Plots of voltages, currents, and magnetic fluxes are available. For the different simulations some parameters in the schematic are changed. With `t0=0` the network switch closes at 40 ms, when the phase A voltage crosses zero. The primary phase A current of the main transformer is then dominated by the magnetizing inrush currents of the medium-voltage transformers. Closing at voltage zero produces the largest unidirectional flux offset in Phase A (U). Even without residual flux, the transient flux can approach twice its steady-state peak starting from zero to about 0.58 Wb.

At the same instant, the phase A (U) capacitive charging current is comparatively small because the applied voltage is close to zero. In phases B (V) and C (W), the opposite situation occurs: the instantaneous voltages are high, their medium-voltage transformer flux offsets are smaller, and their initial cable-charging currents are large. The principal currents decay over approximately 500 ms. The over-voltages depends strongly on the switching instant.

All three phase currents contain high-frequency components caused by resonances between the cable capacitances and the source and transformer inductances. An FFT of a phase A medium-voltage-transformer current also contains many harmonics (100Hz,150Hz,200Hz,250Hz) of the power frequency because of the nonlinear transformer magnetizing curves. In phases B and C, a distinct network resonance appears at approximately 400 Hz.

All three medium voltage transformer currents exhibit similar but also slightly different behaviour. Since a Petersen coil is connected to the neutral point of Transformer 1, the inrush-current waveform partly extends below zero. For Transformer 2, a faster decay of the inrush current can be observed. This can be explained by the additional voltage drop caused by the shunt reactor current on Line 2. In Phase A, the shunt reactor carries a current that remains entirely positive (in phase with the transformer). This current causes a greater voltage drop across Line 2 and thus a lower voltage across the transformer. Since the flux of the transformer depends directly on the integral of the voltage, both the flux and the current of the transformer decrease. NOTE: The behaviour described above occurs only if the shunt reactor is turned on simultaneously with the transformer. If the shunt reactor is turned on 10 ms later, its current is out of phase with the transformer current, and the inrush current of the transformer will now decrease more slowly or even rise slightly at first. Try it out by inserting a switch with a switching time of 50 ms before the shunt reactor.

Lines 3 and 4 are overhead lines, so their capacitive charging transients are much lower than those of the cables. For Schematic and waveforms see figures 6.1-6.6. For measured values and signal ranges, see Table 6.1.

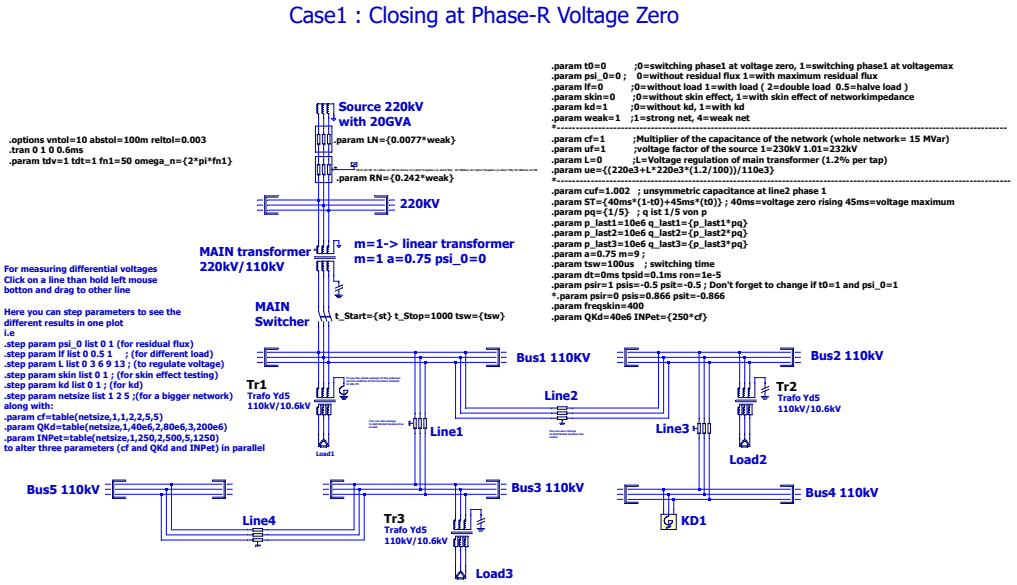


Figure 6.1: Switching on networks schematic - case1

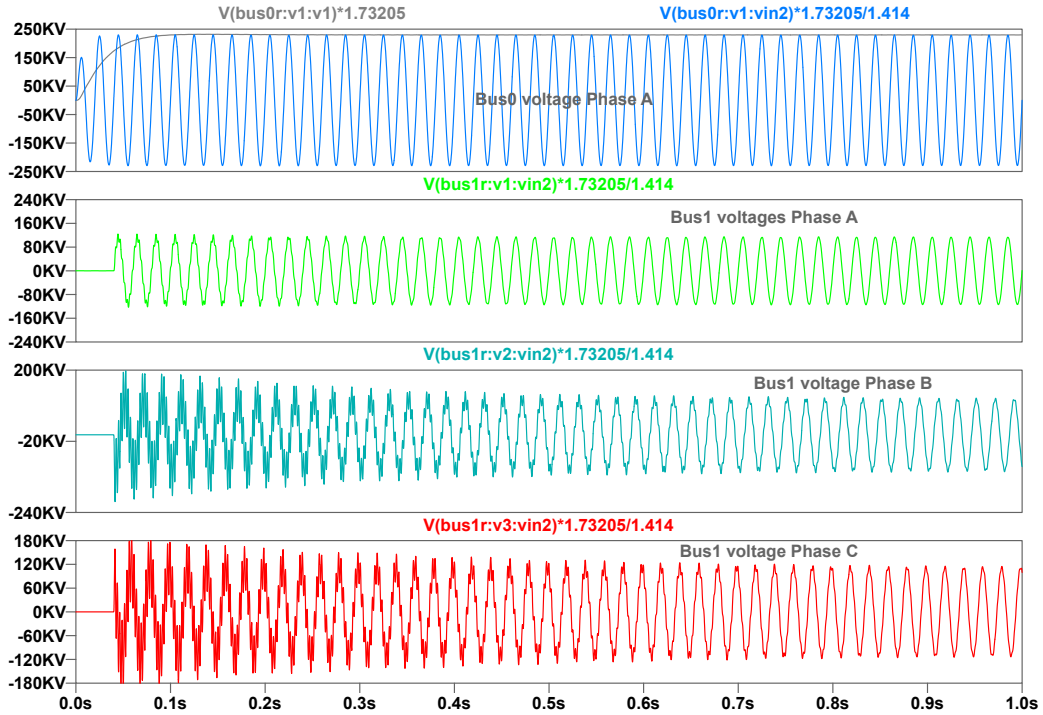


Figure 6.2: Switching on networks voltages - case1

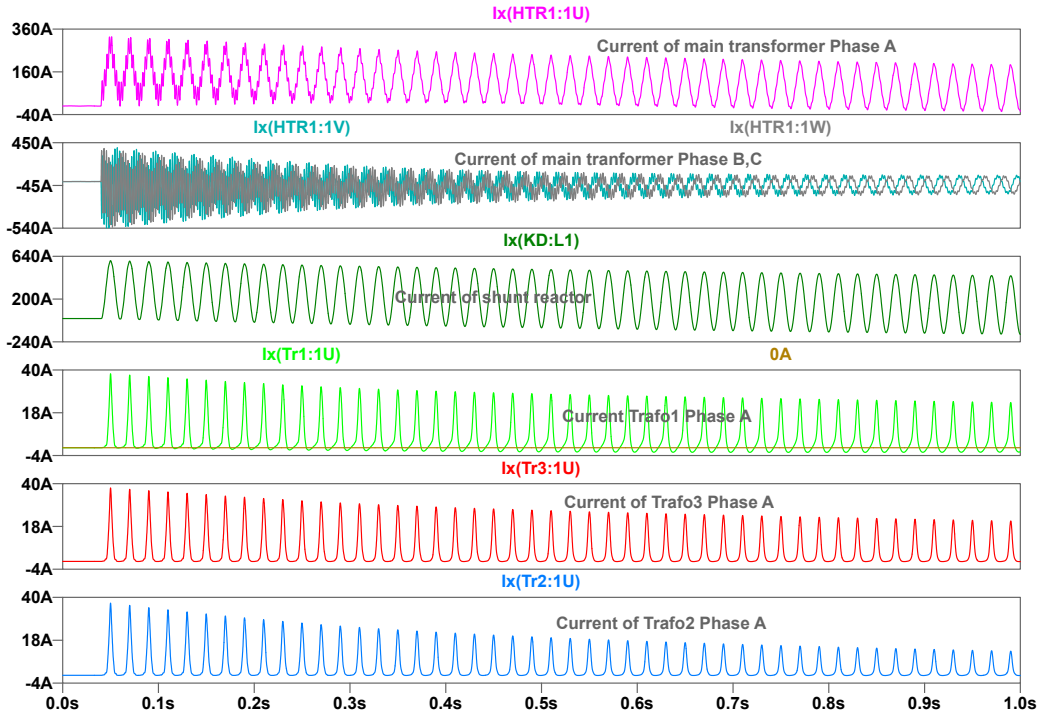


Figure 6.3: Switching on networks currents - case1

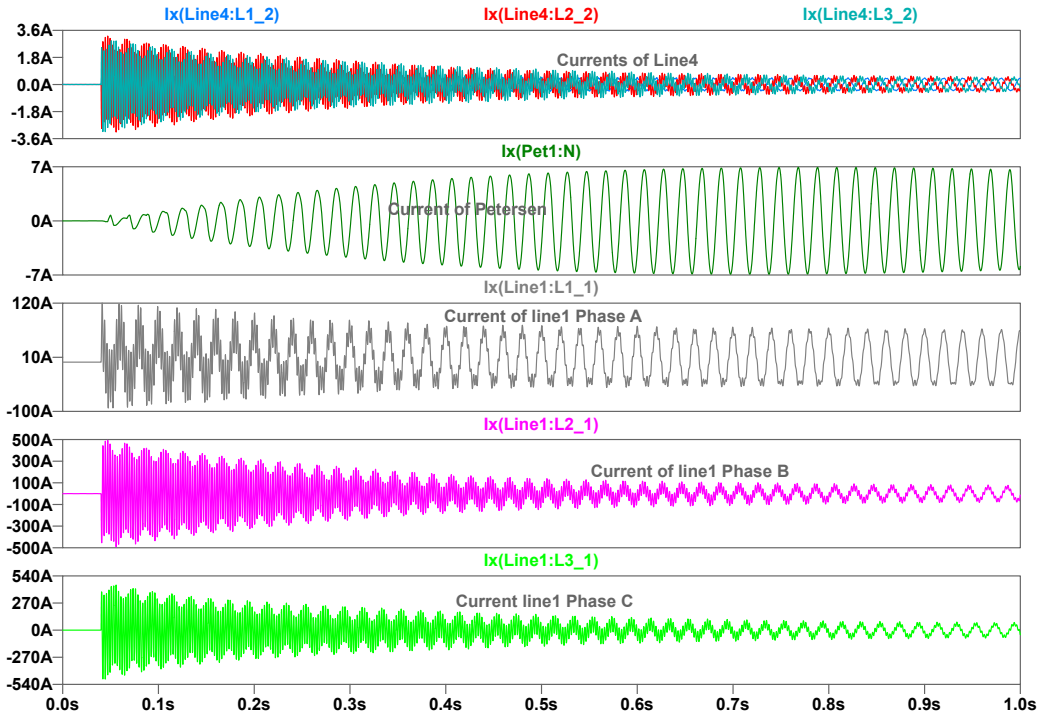


Figure 6.4: Switching on networks currents line1, line4 - case1

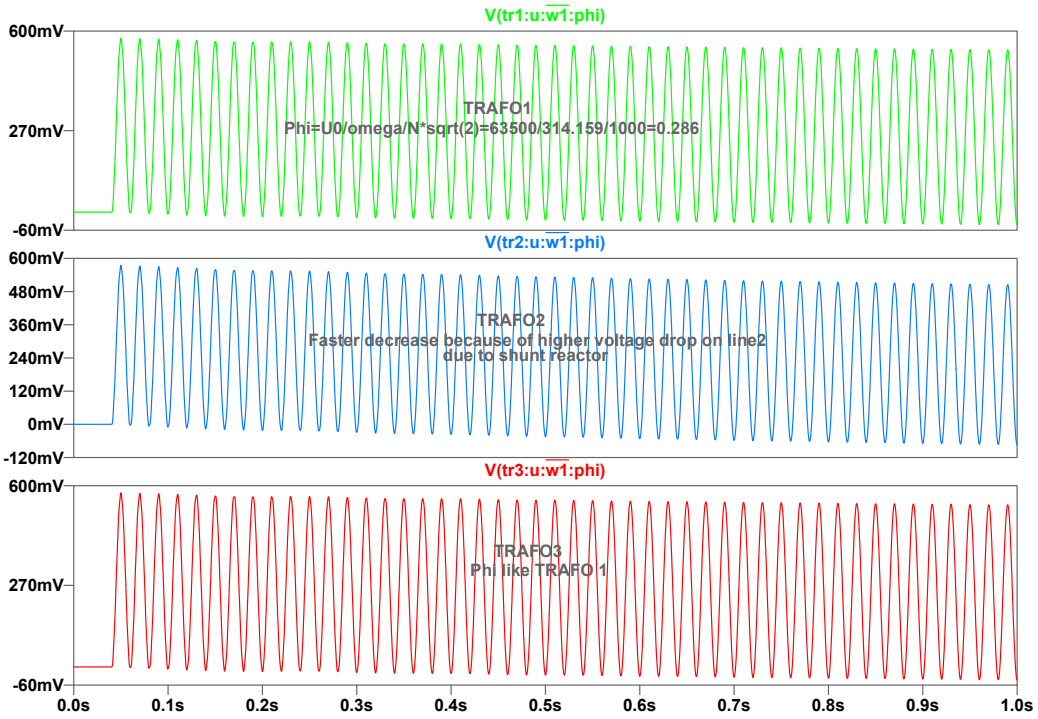


Figure 6.5: Switching on networks fluxes - case1

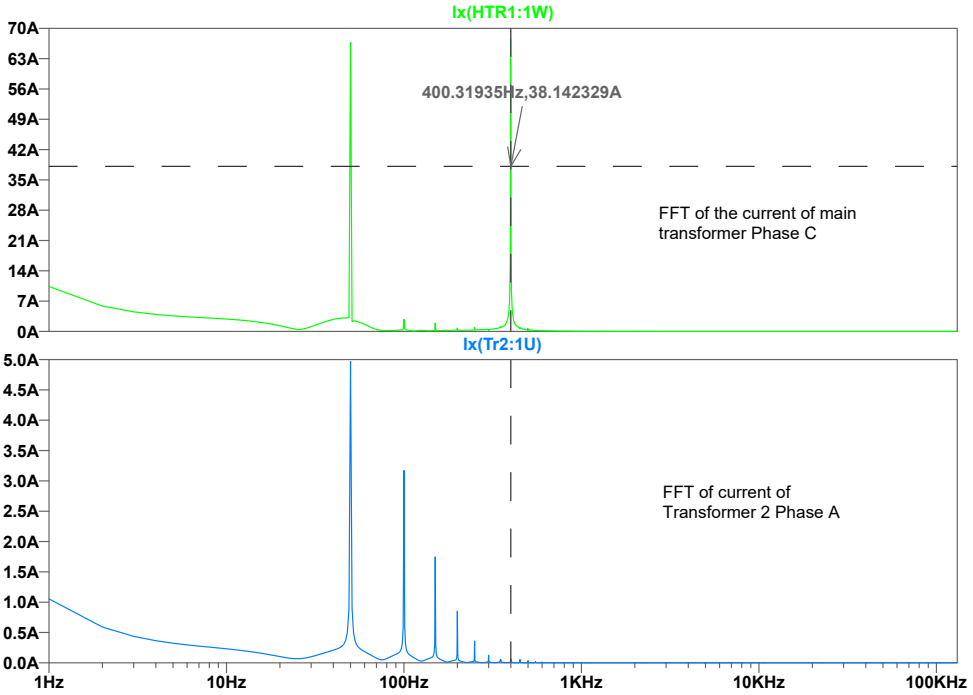


Figure 6.6: Switching on networks FFT - case1

Table 6.1: Signal ranges for Case 1. All values are peak values, bus values are multiplied by 1.73/1.414

Meas.	Grid element	Phase	Voltage level	Min.	Max.	Notes
V	Bus0	Ph A	220 kV	−229 kV	229 kV	–
V	Bus1	Ph A	110 kV	−100 kV	100 kV	–
V	Bus1	Ph B, C	110 kV	−200 kV	200 kV	–
I	HTr1	Ph A	220 kV	0 A	330 A	–
I	HTr1	Ph B, C	220 kV	−540 A	360 A	–
I	KD1	Ph A	110 kV	0 A	600 A	Steady state: 300 A
I	Tr1	Ph A	110 kV	−2 A	38 A	Due to Petersen values under zero line
I	Tr2	Ph A	110 kV	0 A	37 A	Due to KD1: faster decrease of current
I	Tr3	Ph A	110 kV	0 A	37 A	–
I	Line4	Ph B, C	110 kV	−3 A	3 A	Small capacitive inrush current due to overhead line
I	Line1	Ph B, C	110 kV	−500 A	500 A	High capacitive inrush current due to cable
I	Line1	Ph A	110 kV	−100 A	120 A	–
I	Pet1	Ph N	110 kV	−7 A	7 A	Longtime value very small
V	Pet1	Ph N	110 kV	−1.8 kV	1.6 kV	Longtime value very small
Φ	Tr1	Ph A	110 kV	0 Wb	0.576 Wb	Nominal flux: 0.288 Wb
FFT	HTr1	Ph C	220 kV	50 Hz	400 Hz	Only 50 Hz and 400 Hz
FFT	Tr2	Ph A	110 kV	50 Hz	250 Hz	50 Hz, 100 Hz, 150 Hz, 200 Hz, and 250 Hz

6.7 Case 2: Closing at the Phase-R Voltage Maximum

Set `t0=1`. Phase R now closes near its voltage maximum. Its initial capacitive charging current measured at the primary side of the main transformer is at its maximum, while the magnetising inrush currents of transformers 1, 2 and 3 - because of the corresponding flux - begins close to its natural steady-state trajectory. Phases S and T contain mixtures of cable-charging and transformer-magnetizing currents. The currents of the main transformer decay toward steady state within approximately 500 ms. The largest voltage peak occurs in phase R, while phases S and T show similar, smaller peaks.

The primary currents of the main transformer remain approximately at the level of Case 1. The inrush currents of medium-voltage transformers 1, 2, and 3 are now minimal in Phase A and higher and roughly equal in Phases B and C for transformers 2 and 3, but remain smaller than the inrush currents in Phase A at the switching time t0 = 0. Since the Petsersen current of Transformer 1 becomes quite high at approximately 16 A when t0 = 1, the currents in phases B and C of Transformer 1 differ. The current through the shunt reactor is now symmetrical about zero from the start and is approximately 300 A. The current in Line 1 Phase A is approximately 500 A, and that in Phases B and C is approximately 300 A. The magnetic flux in phases B and C is now at its highest, but not as high as it was in Phase A in Case 1. Therefore the inrush currents of Phase B and C are smaller smaller than those in phase A of Case 1. NOTE: Due to the highly nonlinear magnetic characteristic curve, small changes in the flux values cause large changes in the magnetization currents. For Waveforms see figures 6.7-6.10. For Measurement values and signal ranges see table: 6.2.

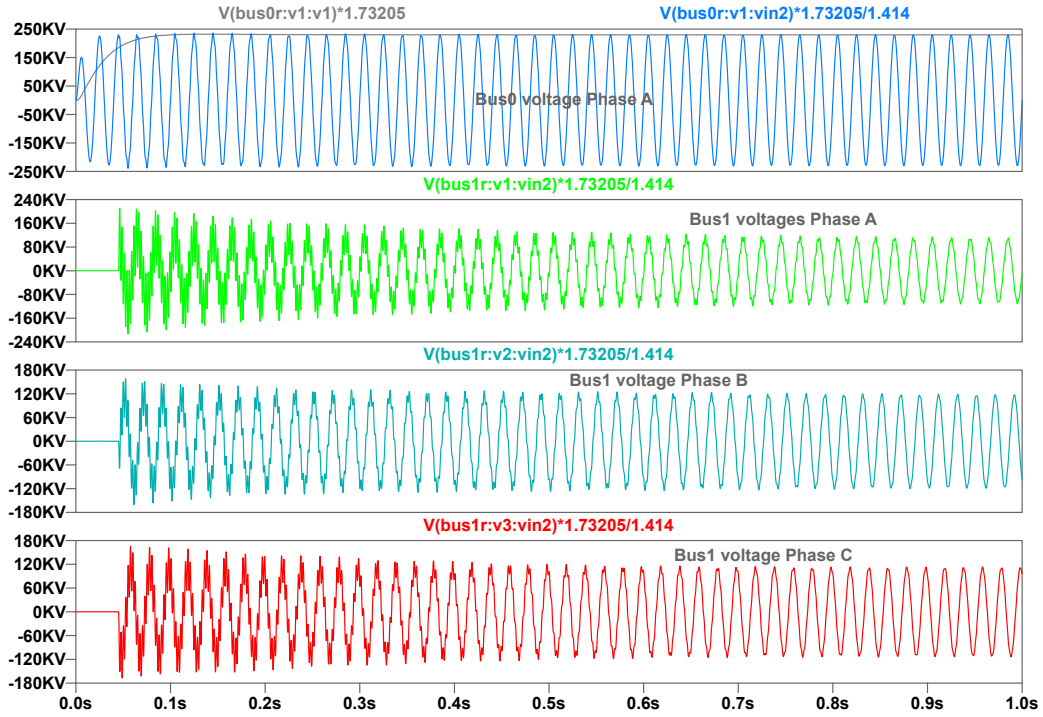


Figure 6.7: Switching on networks voltages - case2

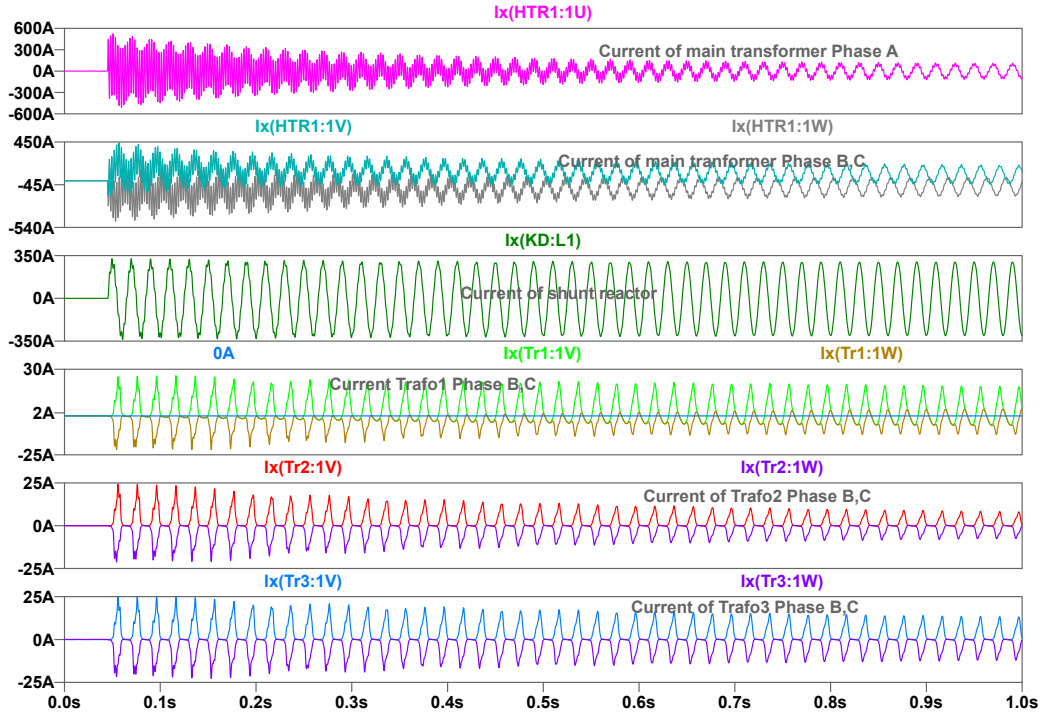


Figure 6.8: Switching on networks currents - case2

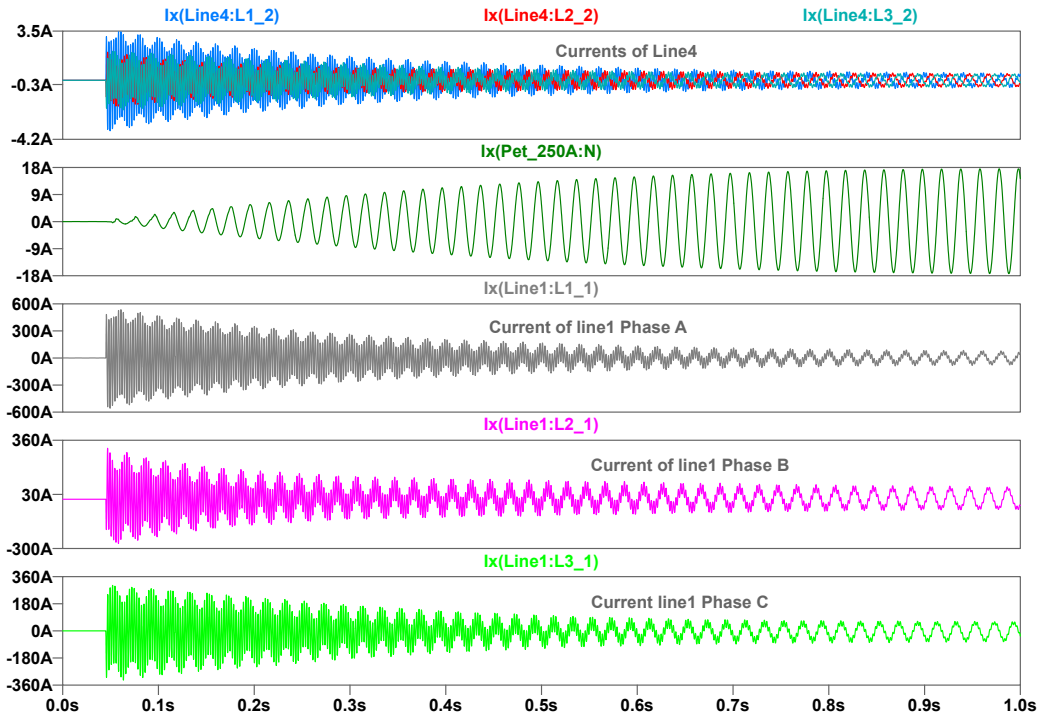


Figure 6.9: Switching on networks currents line1, line4 - case2

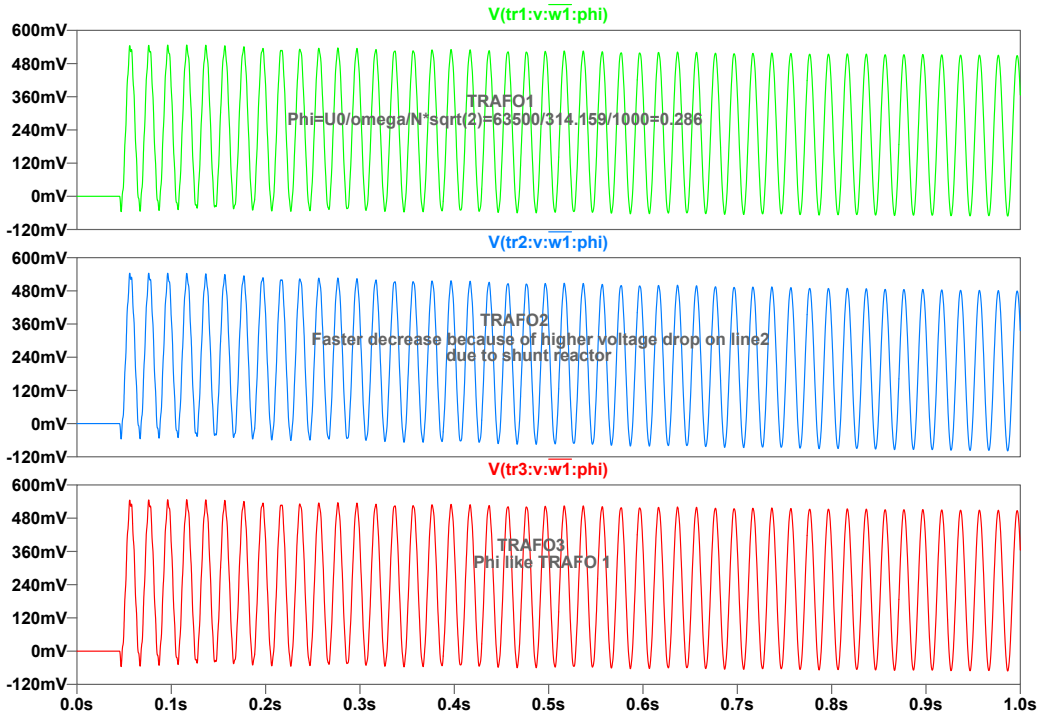


Figure 6.10: Switching on networks fluxes - case2

Table 6.2: Signal ranges for Case 2. All values are peak values, bus values are multiplied by 1.73/1.414

Meas.	Grid element	Phase	Voltage level	Min.	Max.	Notes
V	Bus0	Ph A	220 kV	−229 kV	229 kV	–
V	Bus1	Ph A, B, C	110 kV	−200 kV	200 kV	–
I	HTr1	Ph A	220 kV	−520 A	510 A	–
I	HTr1	Ph B	220 kV	−170 A	460 A	–
I	HTr1	Ph C	220 kV	−470 A	180 A	–
I	KD1	Ph A	110 kV	−300 A	300 A	Steady state: 300 A
I	Tr1	Ph B	110 kV	−4 A	26 A	Due to Petersen values under zero line
I	Tr2	Ph B	110 kV	0 A	25 A	Due to KD: faster decrease of current
I	Tr3	Ph A	110 kV	0 A	25 A	–
I	Line4	Ph A	110 kV	−3.5 A	3.5 A	Small capacitive inrush current due to overhead line
I	Line1	Ph A	110 kV	−500 A	500 A	High capacitive inrush current due to cable
I	Line1	Ph B, C	110 kV	−300 A	300 A	High capacitive inrush current due to cable
I	Pet1	Ph N	110 kV	−18 A	18 A	Longtime value very small
V	Pet1	Ph N	110 kV	−4.5 kV	4.5 kV	Longtime value very small
Φ	Tr1	Ph B, C	110 kV	0 Wb	0.54 Wb	Nominal flux: 0.288 Wb

6.8 Case 3: Closing at Voltage Zero with Full Residual Flux

Set `psi_0=1`. Phase R then carries a very large combined magnetizing current from the medium-voltage transformers. Measured on the 220 kV side of the main transformer, the peak is approximately 920 A. This exceeds the rated peak current of approximately $\sqrt{2} \cdot 525\,\text{A} = 742\,\text{A}$.

With unfavourable full residual flux, the Phase A flux can theoretically approach three times its normal peak. The theoretical peak would therefore be approximately 8.64 Wb. This theoretical peak cannot be achieved in practice. The reason is, on the one hand, the reduced voltage caused by the high current and external and internal voltage drops. The attainable flux is reduced by voltage drops caused by the high current, as well as by secondary-winding currents and leakage flux. The fast decrease of the currents is due to the nonlinear characteristic of the magnetizing curve.

The neutral-point voltage across the Petersen coil of transformer 1 initially rises to a high value (about nominal voltage) and then decays slowly. The cause is the large asymmetry of the phase magnetizing currents, which also makes the phase-to-earth voltages asymmetric. The line-to-line voltages are much less affected. If the simulation is extended to 60 s, the magnetizing currents approach their small steady-state values and the Petersen-coil voltage settles to approximately 1 kV to 2 kV.

The very high neutral displacement voltage, which reaches approximately the nominal phase-to-earth voltage, which also slowly builds up at the neutral point of the main transformer, causes a highly asymmetrical voltage in Phase B of the grid. This voltage first rises, then drops to nearly half the rated voltage, and then rises again.

For Waveforms see figures 6.11-6.14. For Measurement values and signal ranges see table: 6.3.

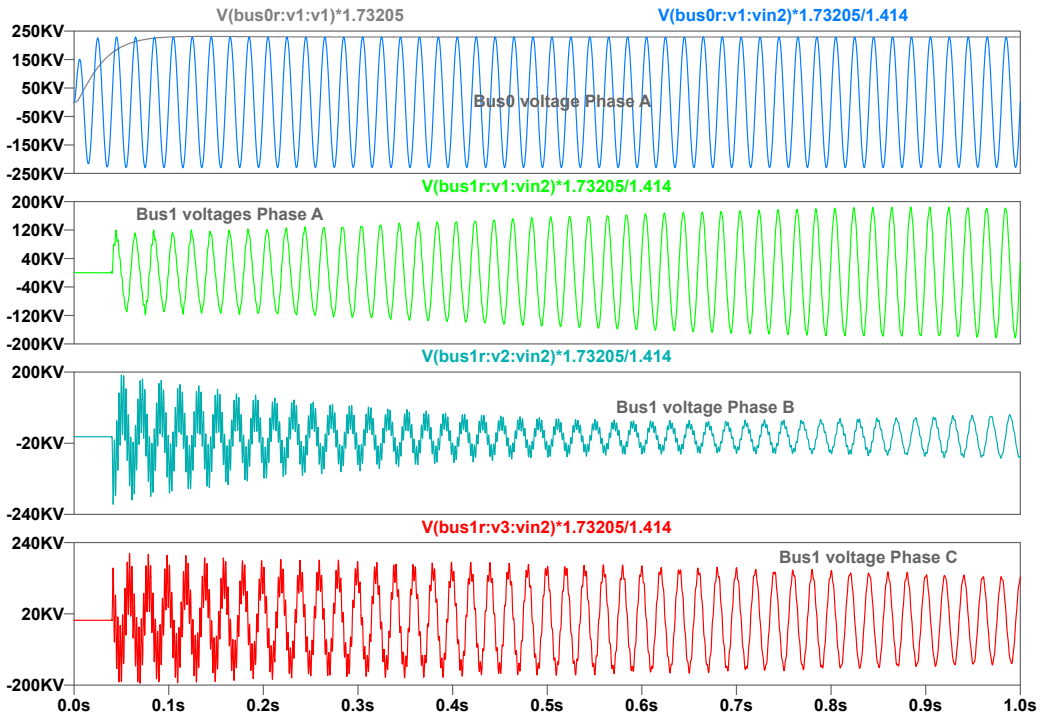


Figure 6.11: Switching on networks voltages - case3

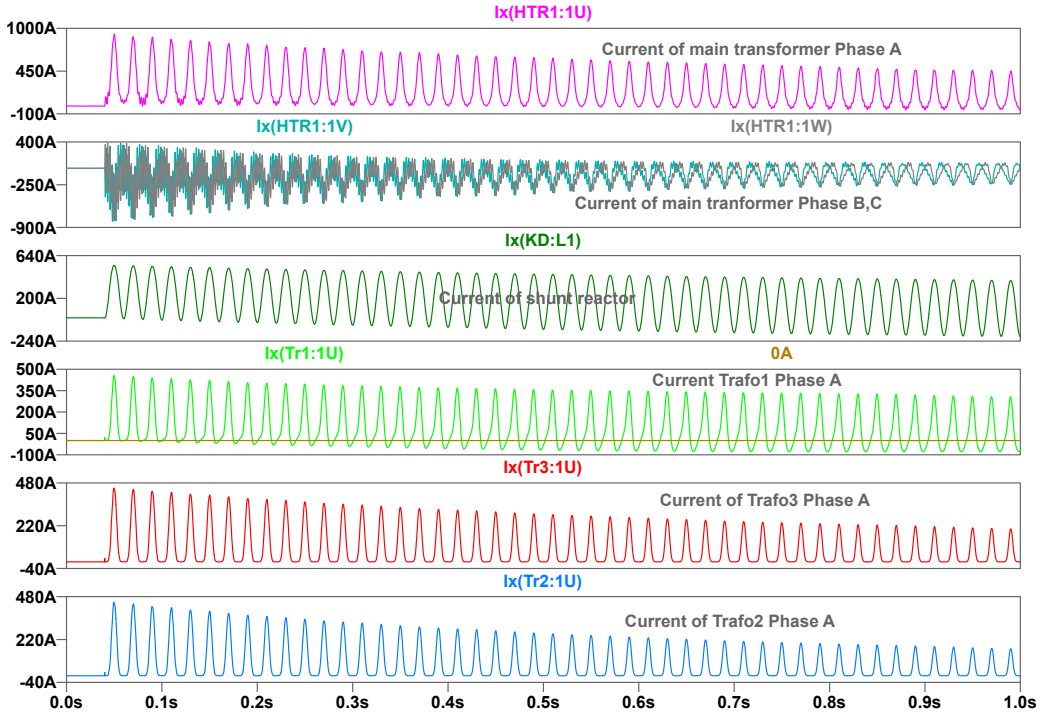


Figure 6.12: Switching on networks currents - case3

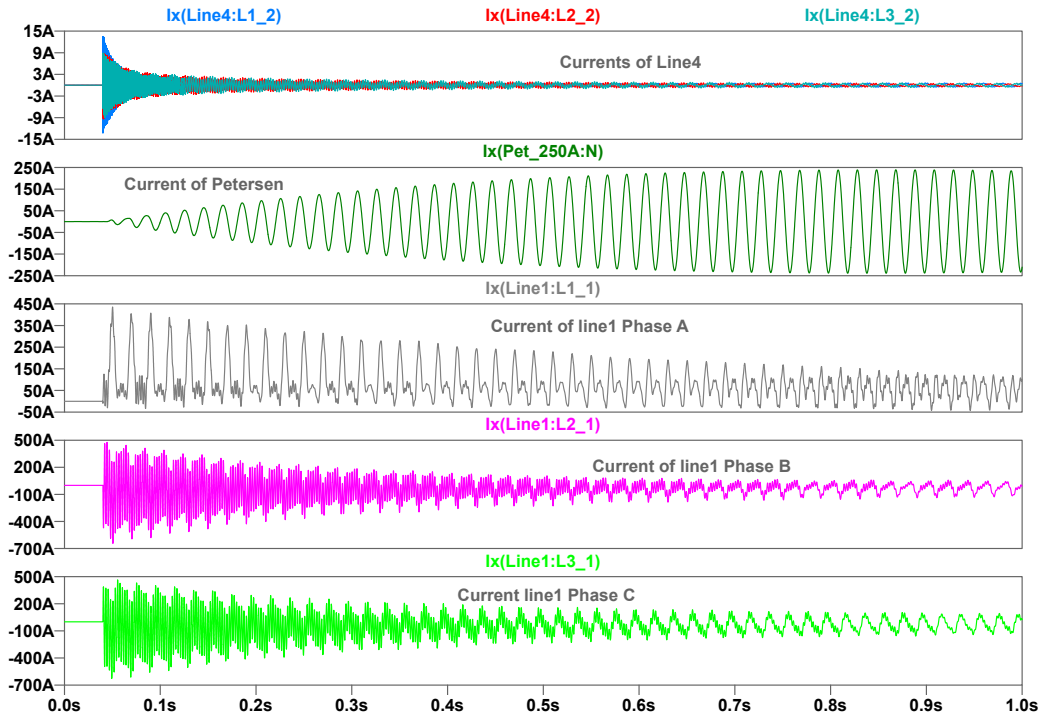


Figure 6.13: Switching on networks currents line1, line4 - case3

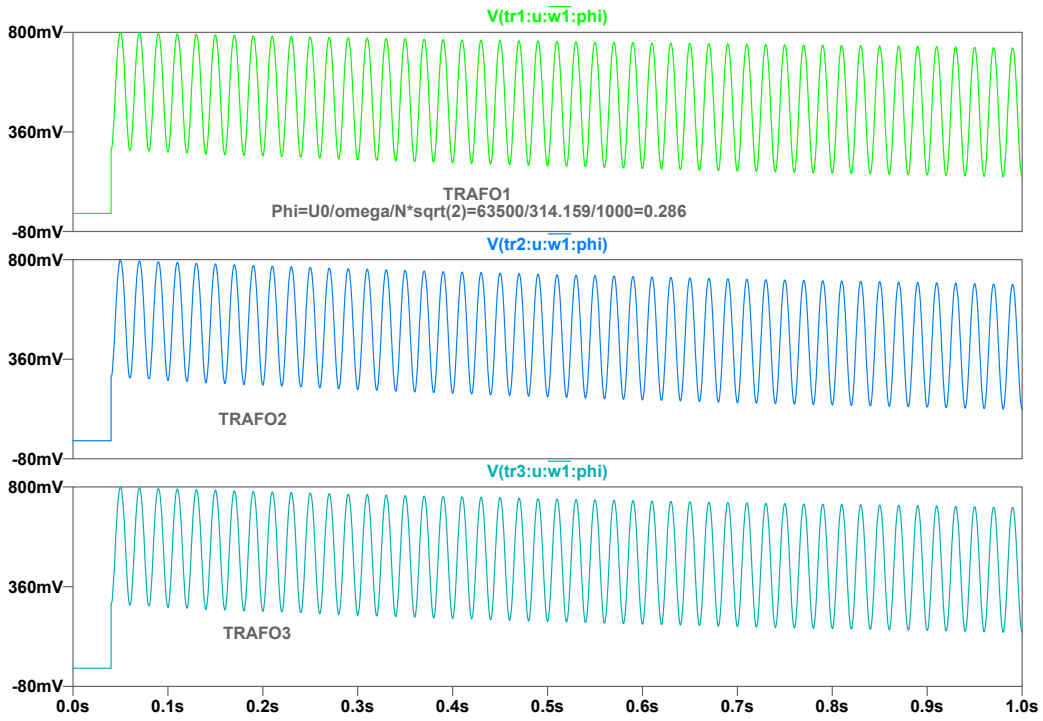


Figure 6.14: Switching on networks fluxes - case3

Table 6.3: Signal ranges for Case 3. All values are peak values, bus values are multiplied by 1.73/1.414

Meas.	Grid element	Phase	Voltage level	Min.	Max.	Notes
V	Bus0	Ph A	220 kV	−229 kV	229 kV	–
V	Bus1	Ph A,	110 kV	−120 kV	120 kV	Startup values low and later higher
V	Bus1	Ph B	110 kV	−200 kV	200 kV	Startup values high and later much smaller than higher again, Steady state: 114 kV
V	Bus1	Ph C	110 kV	−200 kV	200 kV	Steady state: 114 kV RMS
I	HTr1	Ph A	220 kV	0 A	920 A	–
I	HTr1	Ph B	220 kV	−800 A	400 A	–
I	HTr1	Ph C	220 kV	−800 A	400 A	–
I	KD1	Ph A	110 kV	0 A	540 A	Steady state: 300 A
I	Tr1	Ph A	110 kV	−80 A	450 A	Due to Petersen values under zero line
I	Tr2	Ph A	110 kV	0 A	450 A	Due to KD1: faster decrease of current
I	Tr3	Ph A	110 kV	0 A	450 A	–
I	Line4	Ph A	110 kV	−13 A	13 A	Small capacitive inrush current
I	Line1	Ph A	110 kV	−20 A	440 A	High capacitive inrush current due to cable
I	Line1	Ph B, C	110 kV	−500 A	500 A	High capacitive inrush current due to cable
I	Pet1	Ph N	110 kV	−240 A	240 A	Longtime value very small
V	Pet1	Ph N	110 kV	−60 kV	60 kV	Longtime value very small
Φ	Tr1	Ph A, B, C	110 kV	0 Wb	0.8 Wb	Nominal flux: 0.288 Wb

6.9 Case 4: Main-Transformer Tap Position L+13

Set L=13. The changed tap position reduces the secondary voltage of the main transformer. The currents and maximum flux values at 500 mWb are also lower than in Case 1. The Phase A voltage at bus1 reaches approximately 100 kV peak and contains almost no 400 Hz component associated with the capacitive charging transient. The peak voltages in phases B and C are approximately 180 kV. The Petersen-coil current is approximately 2.5 A, and the current in line 4 is similar to that in Case 1. In line 1, the peak currents are approximately 500 A in phases B and C and 100 A in phase A. The peak transformer inrush currents are approximately 11.5 A, substantially lower than in Case 1 because the maximum core flux is reduced. For Schematic and waveforms see figures 6.15- 6.19. For measured values and signal ranges, see Table 6.4. An FFT shows that the dominant resonance shifts from approximately 400 Hz to 450 Hz. With voltage ratio = 220kV/97kV=2.268 and t=2.268*2.268=5.14. Compared to calculation 6.2 the resonant frequency is

Calculation 6.3 – (Dominant cable-network resonance)

$$L_{\text{eq}} = \frac{L_{\text{net}}}{5.14} + \frac{L_{\text{Tr}}}{5.14} = \frac{0.007\,7\,\text{H} + 0.154\,\text{H}}{5.14} = 0.031\,4\,\text{H}, \tag{6.10}$$

$$C = 4\,000\,\text{nF}, \tag{6.11}$$

$$f_{\text{r}} = \frac{1}{2\pi\sqrt{L_{\text{eq}}C}} \approx \boxed{448.8\,\text{Hz}}. \tag{6.12}$$

For Waveforms see figures 6.15-6.19. For Measurement values and signal ranges see table 6.4

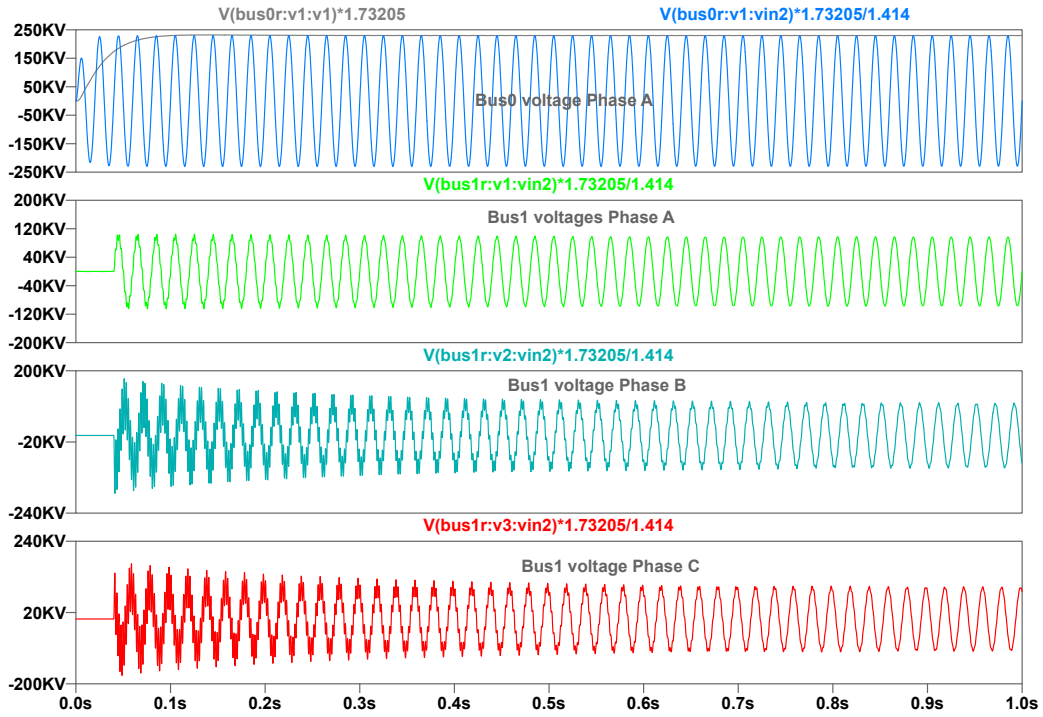


Figure 6.15: Switching on networks voltages - case4

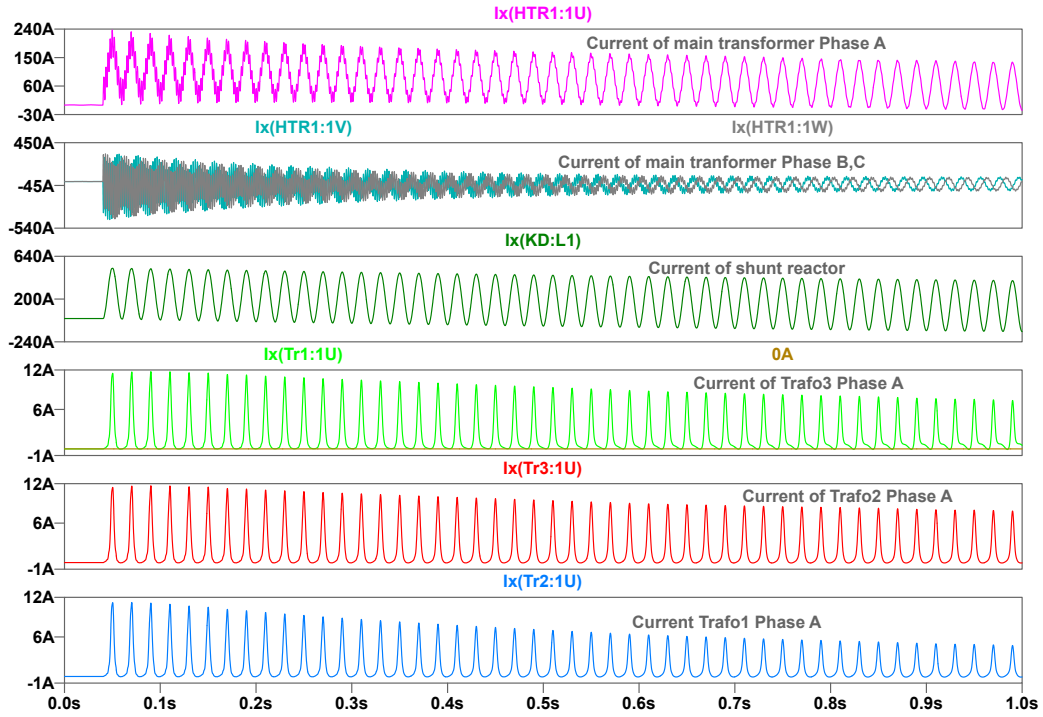


Figure 6.16: Switching on networks currents - case4

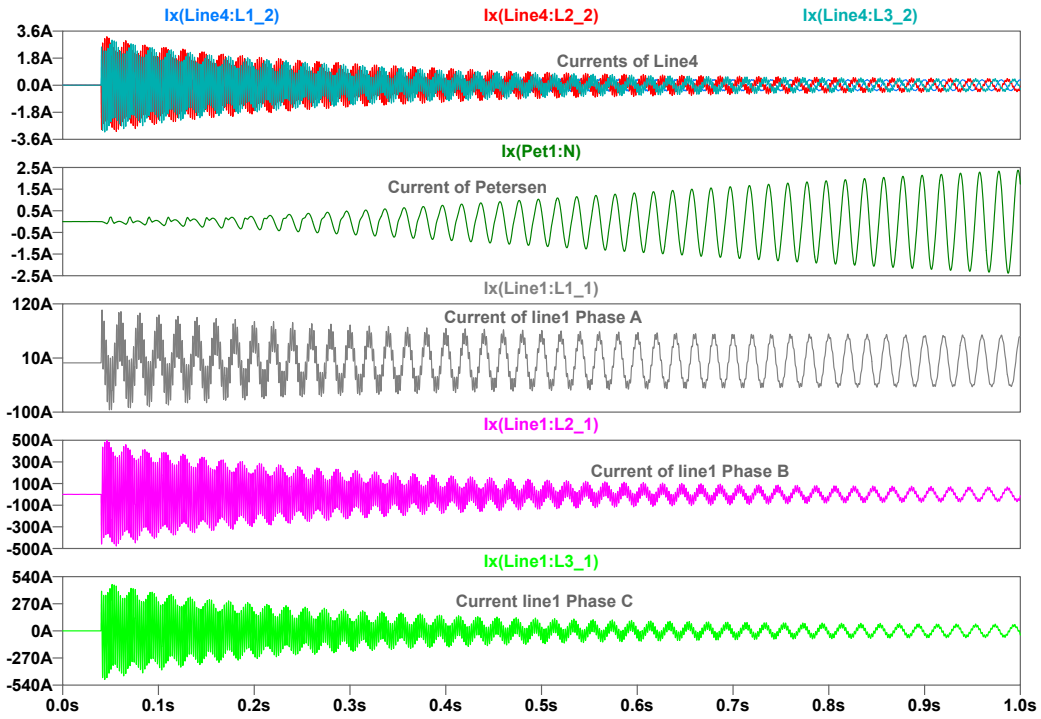


Figure 6.17: Switching on networks currents line1, line4 - case4

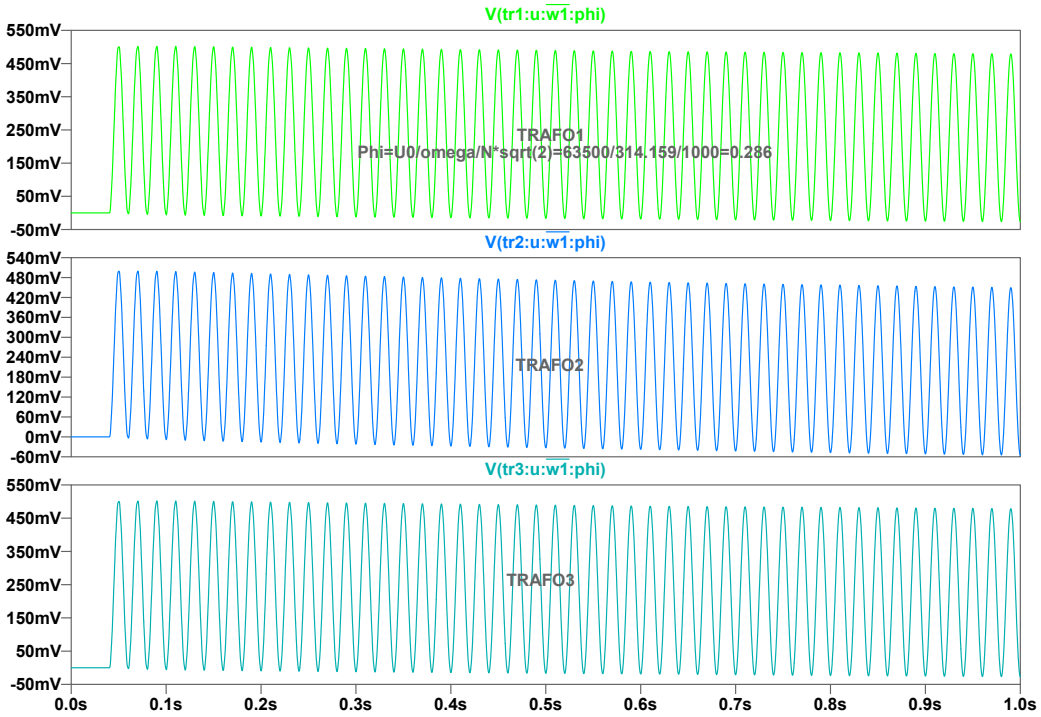


Figure 6.18: Switching on networks fluxes - case4

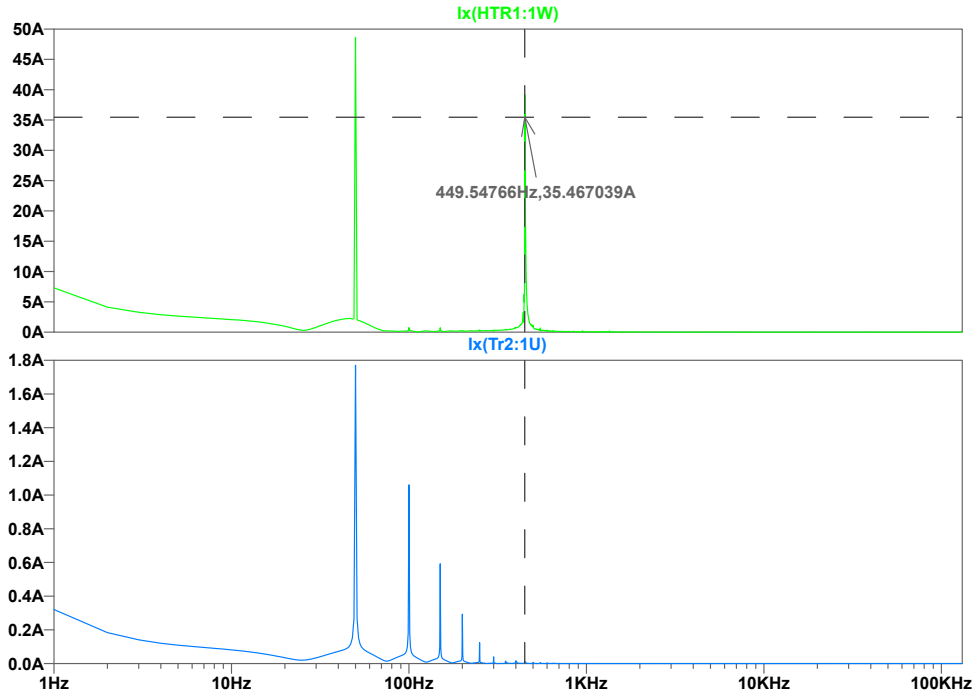


Figure 6.19: Switching on networks FFT - case4

Table 6.4: Signal ranges for Case 4. All values are peak values, bus values are multiplied by 1.73/1.414

Meas.	Grid element	Phase	Voltage level	Min.	Max.	Notes
V	Bus0	Ph A	220 kV	−229 kV	229 kV	–
V	Bus1	Ph A,	110 kV	−100 kV	100 kV	–
V	Bus1	Ph B,C	110 kV	−180 kV	180 kV	–
I	HTr1	Ph A	220 kV	0 A	230 A	–
I	HTr1	Ph B, C	220 kV	−350 A	300 A	–
I	KD1	Ph A	110 kV	0 A	520 A	Steady state: 300 A
I	Tr1	Ph A	110 kV	0 A	11 A	
I	Tr2	Ph A	110 kV	0 A	11 A	Due to KD1: faster decrease of current
I	Tr3	Ph A	110 kV	0 A	11 A	–
I	Line4	Ph A	110 kV	−3 A	3 A	Small capacitive inrush current
I	Line1	Ph A	110 kV	−100 A	100 A	High capacitive inrush current due to cable
I	Line1	Ph B, C	110 kV	−500 A	500 A	High capacitive inrush current due to cable
I	Pet1	Ph N	110 kV	−2.5 A	2.5 A	Longtime value very small
V	Pet1	Ph N	110 kV	−600 V	600 V	Longtime value very small
Φ	Tr1	Ph A, B, C	110 kV	0 Wb	0.5 Wb	Nominal flux: 0.288 Wb
FFT	HTr1	Ph C	220 kV	50 Hz	450 Hz	Only 50 Hz and 450 Hz
FFT	Tr2	Ph A	110 kV	50 Hz	250 Hz	50 Hz, 100 Hz, 150 Hz, 200 Hz, and 250 Hz

6.10 Case 5: Energizing a Larger Network

Set `cf=5` to multiply the cable-network capacitance by five. The capacitive currents are approximately twice, rather than five times, those of Section 6.6. The larger capacitance lowers the dominant resonance from approximately 400 Hz to approximately 180 Hz.

As can be seen from the low bias voltage and the low Petersen-coil current, the Petersen circuit is not yet optimally tuned. If we look at the AC analysis, we can see that the resonance point is at approximately 45 Hz. You can see this by determining the beat frequency in the transient analysis. So the Petersen circuit's tuning still needs to be adjusted (see Switching_Networks_AC.asc). NOTE: At 1 250 A, the Petersen current is, of course, much too high and would place too heavy a load on the transformer. The transformer current of Transformer 1 starts at approximately 50 A and then fluctuates up and down. This is due to the voltage containing 180Hz+50Hz components, which also fluctuates up and down at the beginning and, consequently, the integral of the voltage which gives the flux. The voltage of the 110 kV Phase A starts at 120 kV, while the voltage of the other phases initially reaches between 180 kV and 200 kV. The current through the main transformer in Phase A is approximately 530 A to −160 A, while the current in Phases B and C is approximately 1 kA. These currents are primarily caused by capacitive inrush currents and are due to the now larger capacitive network. The current through the KD ranges from 0 A to 620 A and decreases rapidly. The current in Line 1 is approximately 600 A in Phase A and approximately 1.3 kA in the other two phases, which is much higher than in Case 1. The current in Line 4 is very low, at 1 A to 2 A, since the charging current of an overhead line is very small. The FFT of the current in phase C on the primary side of the main transformer shows a peak at approximately 180 Hz, as shown in the associated calculation. Here, the 180 Hz component is even higher than the 50 Hz component. The maximum flux of the medium-voltage transformers is initially about 600 mWb, which is slightly higher than in Case 1.

For calculation of the dominant frequency of the circuit like in calculation 6.2 the inductances specified on the 220 kV side must be referred to the 110 kV side. With a voltage ratio of 2, impedance and inductance are divided by $2^2 = 4$ when referred from the 220 kV side to the 110 kV side. The source and transformer inductances are $L_{\text{net}} = 0.007\,7\,\text{H}$ and $L_{\text{Tr}} = 0.154\,\text{H}$. The sum of capacitance is now 5*4000nF=20000nF.
We obtained from the file `Trafo_YY0_220k_110k_200MVA_Daten.asc` the data for the transformer.

Calculation 6.4 – (Dominant cable-network resonance)

$$L_{\text{eq}} = \frac{L_{\text{net}}}{4} + \frac{L_{\text{Tr}}}{4} = \frac{0.007\,7\,\text{H} + 0.154\,\text{H}}{4} = 0.04\,\text{H}, \tag{6.13}$$

$$C = 20\,000\,\text{nF}, \tag{6.14}$$

$$f_{\text{r}} = \frac{1}{2\pi\sqrt{L_{\text{eq}}C}} \approx \boxed{178\,\text{Hz}}. \tag{6.15}$$

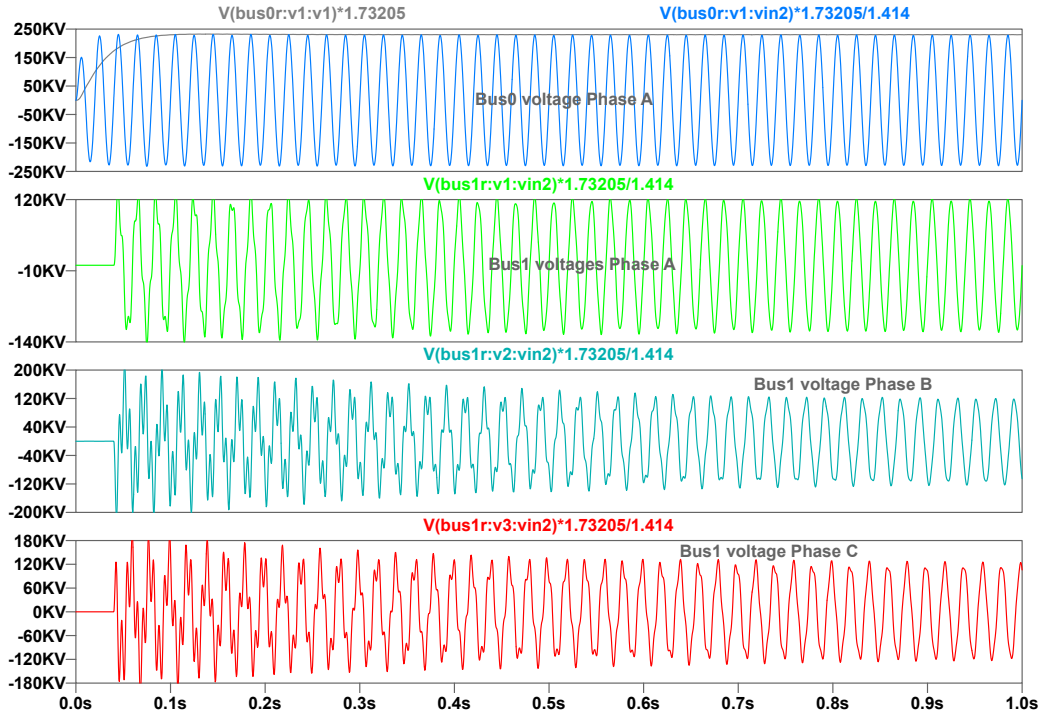


Figure 6.20: Switching on networks voltages - case5

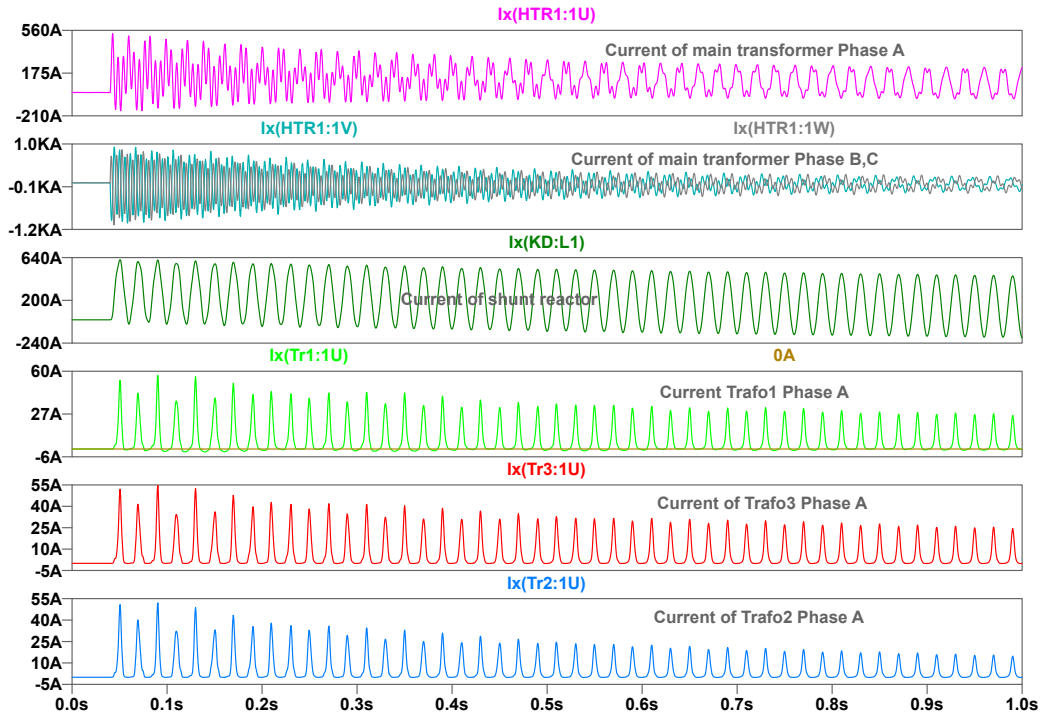


Figure 6.21: Switching on networks currents - case5

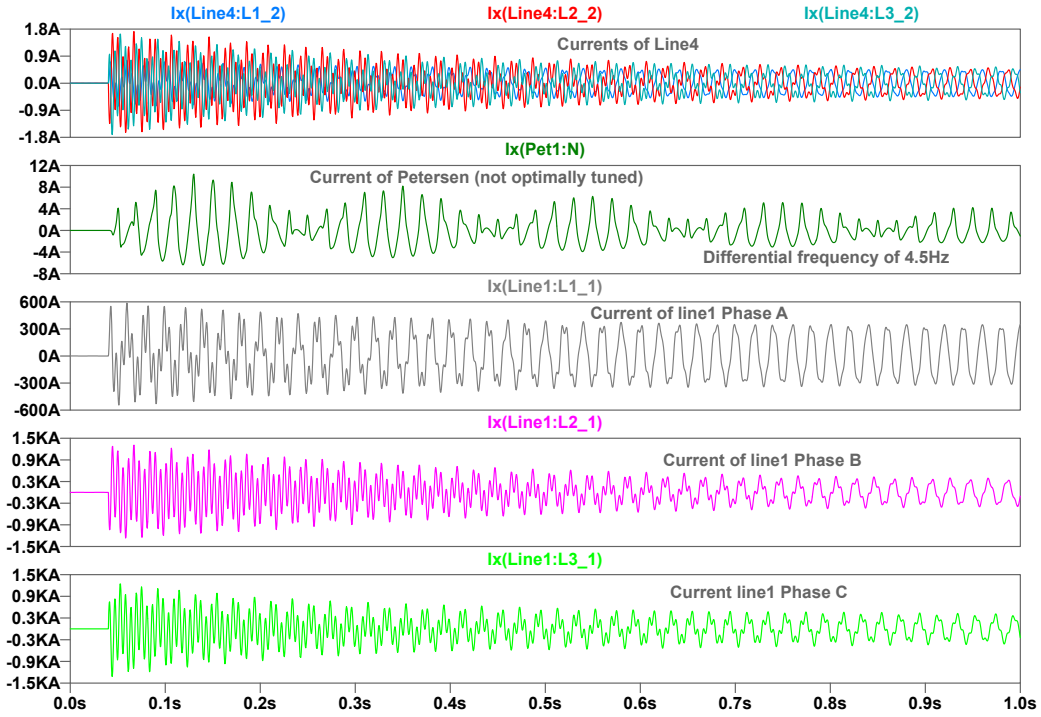


Figure 6.22: Switching on networks currents line1, line4 - case5

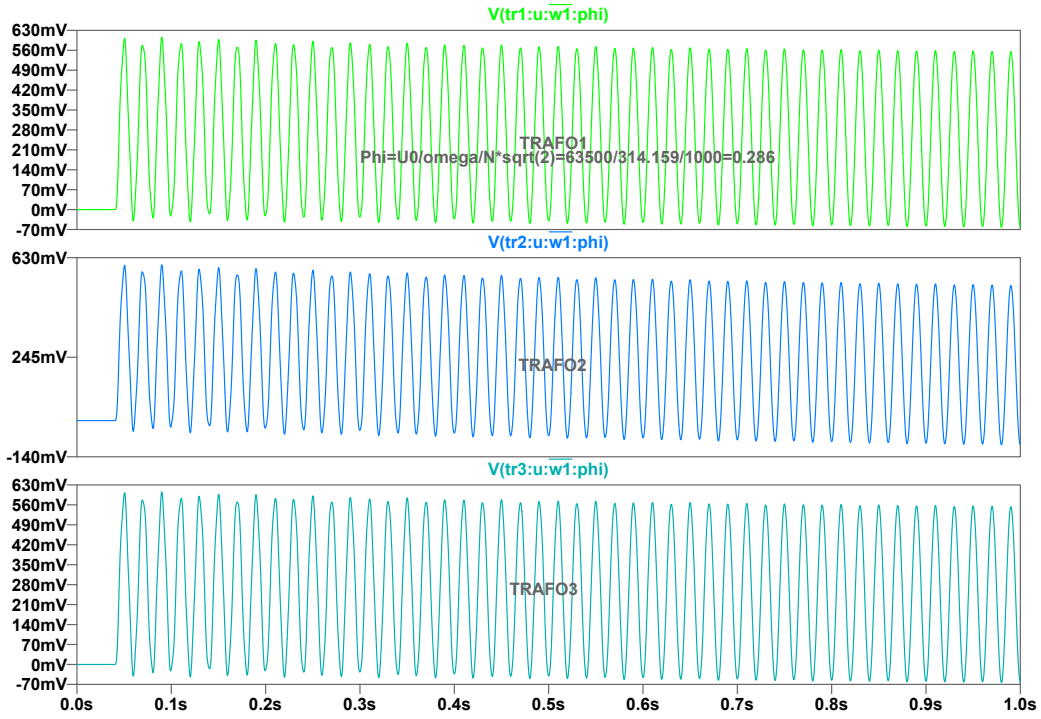


Figure 6.23: Switching on networks fluxes - case5

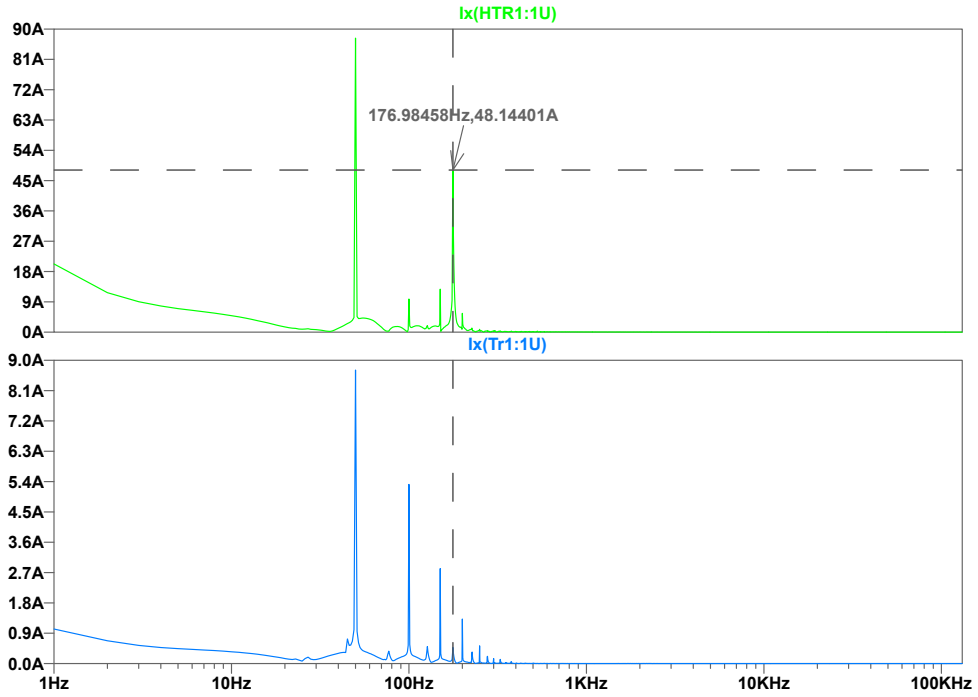


Figure 6.24: Switching on networks FFT - case5

Table 6.5: Signal ranges for Case 5. All values are peak values, bus values are multiplied by 1.73/1.414

Meas.	Grid element	Phase	Voltage level	Min.	Max.	Notes
V	Bus0	Ph A	220 kV	−229 kV	229 kV	–
V	Bus1	Ph A,	110 kV	−140 kV	120 kV	–
V	Bus1	Ph B,C	110 kV	−180 kV	180 kV	–
I	HTr1	Ph A	220 kV	−160 A	530 A	–
I	HTr1	Ph B, C	220 kV	−1 kA	1 kA	–
I	KD1	Ph A	110 kV	0 A	620 A	Steady state: 300 A
I	Tr1	Ph A	110 kV	0 A	54 A	Fluctuating
I	Tr2	Ph A	110 kV	0 A	52 A	Fluctuating
I	Tr3	Ph A	110 kV	0 A	52 A	Fluctuating
I	Line4	Ph A	110 kV	−1.6 A	1.6 A	Small capacitive inrush current
I	Line1	Ph A	110 kV	−530 A	600 A	High capacitive inrush current due to cable
I	Line1	Ph B, C	110 kV	−1.3 kA	1.3 kA	High capacitive inrush current due to cable
I	Pet1	Ph N	110 kV	−6 A	10 A	Fluctuating, Longtime value very small
V	Pet1	Ph N	110 kV	−800 V	600 V	Fluctuating, Longtime value very small
Φ	Tr1	Ph A, B, C	110 kV	0 Wb	0.60 Wb	Nominal flux: 0.288 Wb
FFT	HTr1	Ph C	220 kV	50 Hz	178 Hz	mostly 50 Hz and 178 Hz
FFT	Tr2	Ph A	110 kV	50 Hz	250 Hz	50 Hz, 100 Hz, 150 Hz, 200 Hz, and 250 Hz

6.11 Case 6: Energization with Load

Set `lf=0.5`. The connected load strongly dampens the overvoltages caused by the high-frequency capacitive charging currents, and these oscillations decay rapidly. Transformer inrush currents remain present because they are caused by the transient core flux. Among the direct-switching cases considered so far, this case produces the smallest overvoltages.

The current on the 10.6 kV side of the load is 360 A, which is approximately 36 A on the 110 kV side. The currents on the 110 kV side are a mixture of inrush currents and load currents. It is often claimed in the literature that the load current reduces the inrush of the magnetizing current. This is not really the case. This can be easily demonstrated by simultaneously simulating three identical transformers: a) Set m=1 for Transformer 1 and apply a load to it. This measures the load current on the primary side. b) Set m = 9 for Transformer 2 and set the load to zero. c) For Transformer 3, set m=9 and connect the same load. The resulting currents can then be compared. Now you can see that when adding the primary currents I(Transformer 1) + I(Transformer 2), the resulting current is approximately equal to I(Transformer 3). Any reduction in inrush current is very small in this example. The transformer inrush is simply "hidden" in the total current but is always present. See ltspice file `Hidden_inrush.asc`

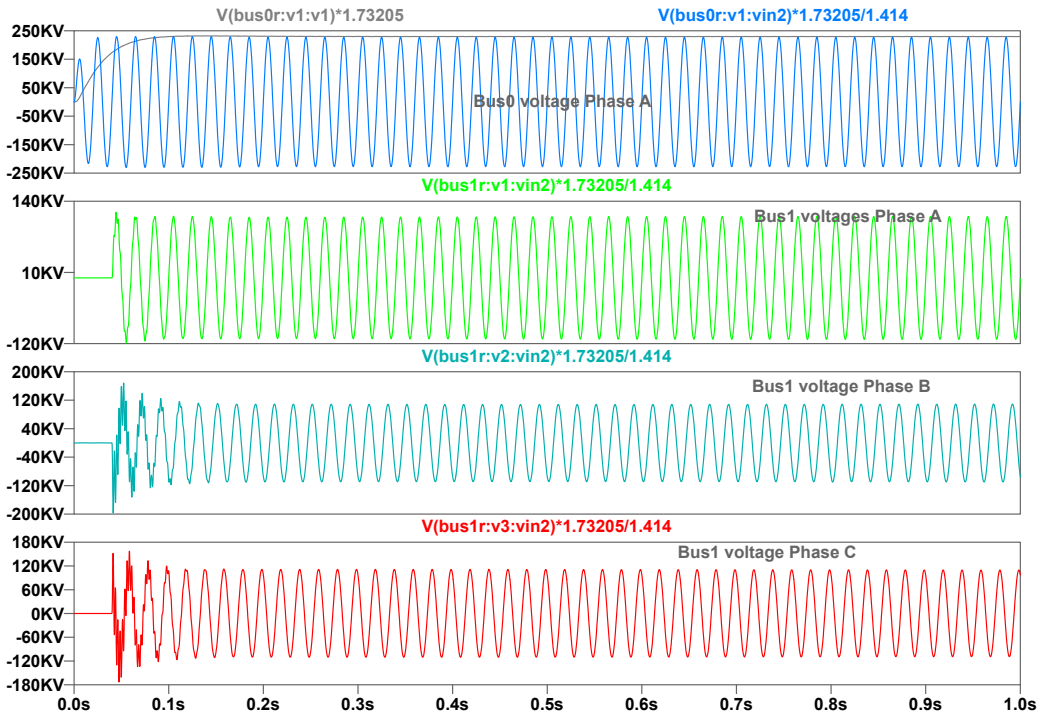


Figure 6.25: Switching on networks voltages - case6

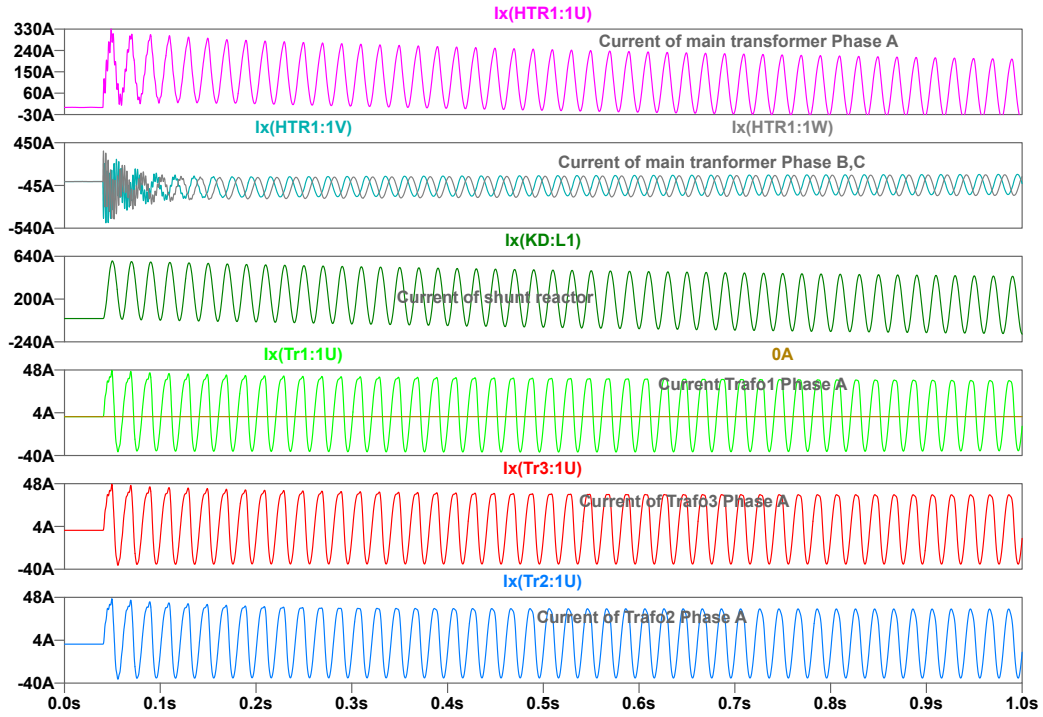


Figure 6.26: Switching on networks currents - case6

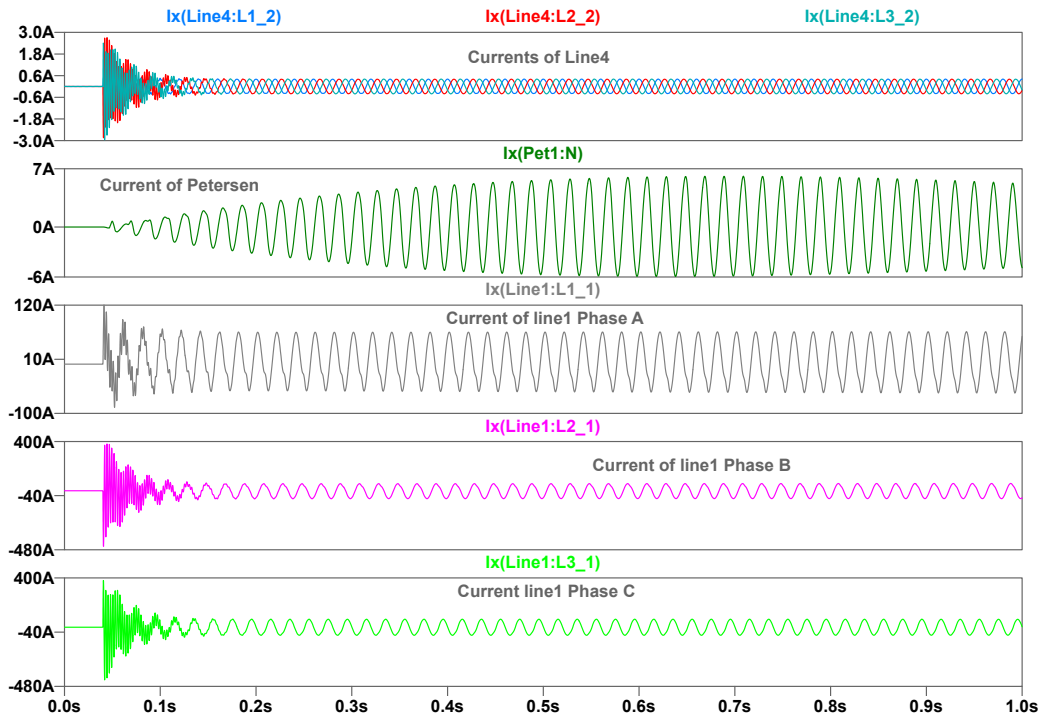


Figure 6.27: Switching on networks currents line1, line4 - case6

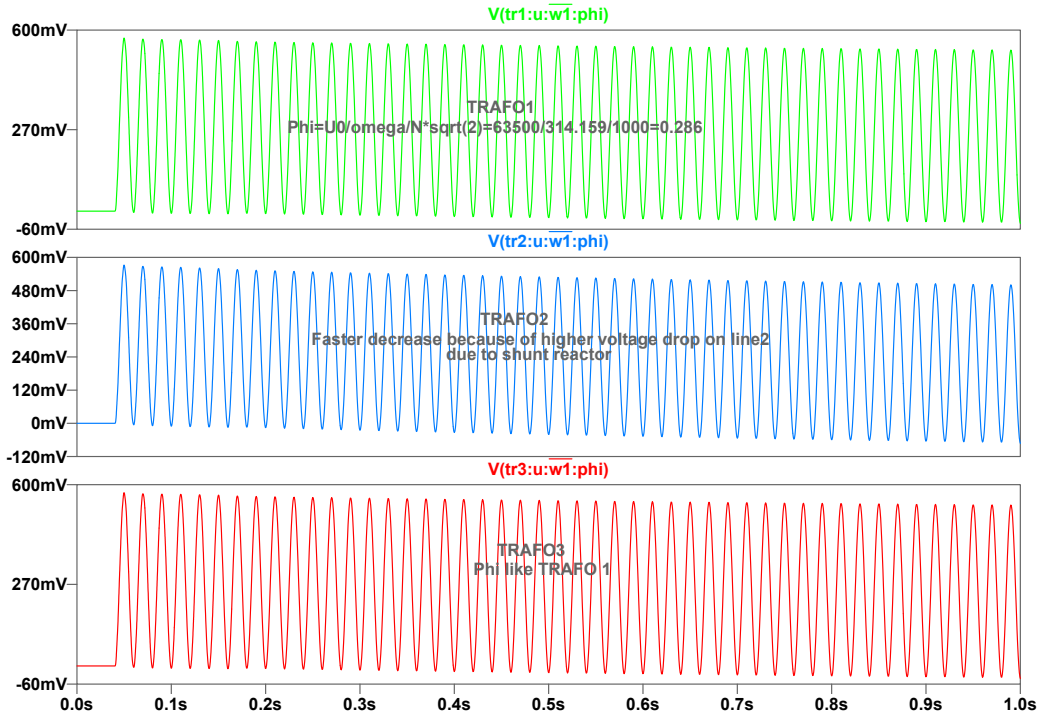


Figure 6.28: Switching on networks fluxes - case6

Table 6.6: Signal ranges for Case 6. All values are peak values, bus values are multiplied by 1.73/1.414

Meas.	Grid element	Phase	Voltage level	Min.	Max.	Notes
V	Bus0	Ph A	220 kV	−229 kV	229 kV	–
V	Bus1	Ph A,	110 kV	−120 kV	120 kV	–
V	Bus1	Ph B,C	110 kV	−170 kV	160 kV	–
I	HTr1	Ph A	220 kV	0 A	330 A	–
I	HTr1	Ph B, C	220 kV	−480 A	360 A	–
I	KD1	Ph A	110 kV	0 A	590 A	Steady state: 300 A
I	Tr1	Ph A	110 kV	−36 A	54 A	under load
I	Tr2	Ph A	110 kV	−36 A	50 A	under load
I	Tr3	Ph A	110 kV	−36 A	47 A	under load
I	Line4	Ph A	110 kV	−3 A	2.5 A	Small capacitive inrush current
I	Line1	Ph A	110 kV	−90 A	120 A	High capacitive inrush current due to cable
I	Line1	Ph B, C	110 kV	−480 A	380 A	High capacitive inrush current due to cable
I	Pet1	Ph N	110 kV	−6 A	6 A	Longtime value very small
V	Pet1	Ph N	110 kV	−1.5 kV	1.5 kV	Longtime value very small
Φ	Tr1	Ph A, B, C	110 kV	0 Wb	0.575 Wb	Nominal flux: 0.288 Wb

6.12 Case 7: Skin-Effect Model

Set `skin=1`. To apply the skin effect in a rough way, we make the simplified assumption that up to 40 ms, only the 50 Hz component determines the resistance, and from 40 ms onward, only the higher frequency of 400 Hz determines the resistance. (Normally, the 50 Hz and 400 Hz components contribute equally). This allows us to roughly determine the influence of the higher frequency. We see that the additional damping is present but small. Even smaller when split between 50 Hz and 400 Hz. The effect is expected to be more pronounced in smaller networks with higher natural frequencies, particularly when individual lines are energized.

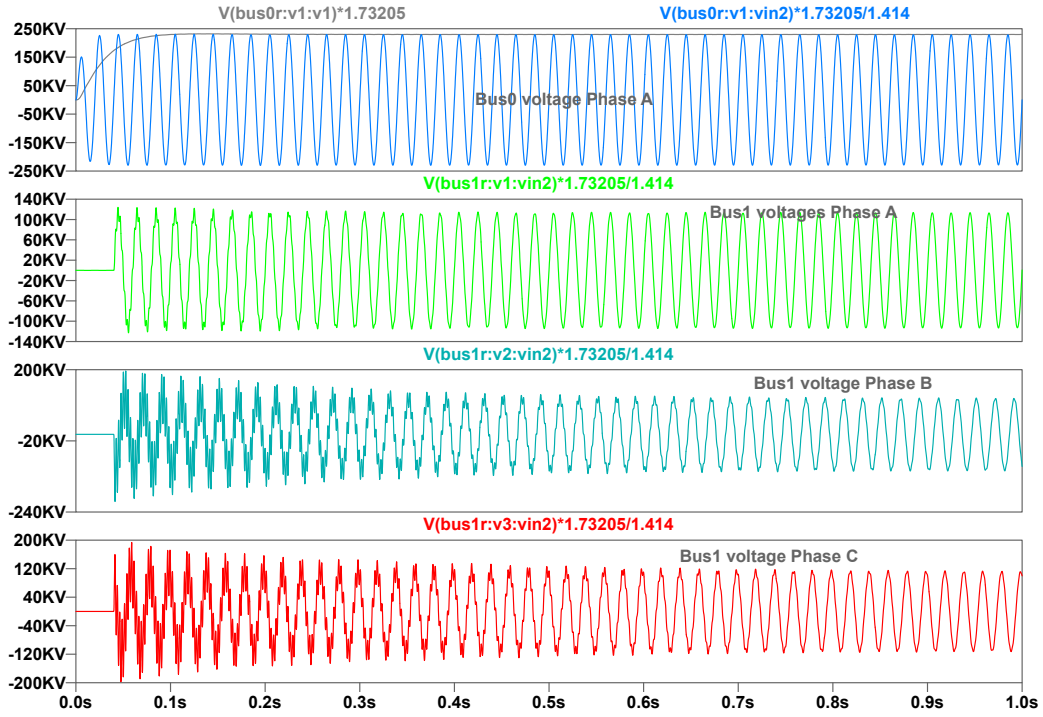


Figure 6.29: Switching on networks voltages - case7

6.13 Case 8: Energization from a Weak Network

Set `weak=40` to increase the source impedance (This setting represents an additional transformer in series, as may occur during restoration after a blackout). The larger source impedance limits the inrush current, although it may simultaneously increase voltage sensitivity elsewhere in the network. Due to the higher voltage volatility, it is not advisable to use a shunt reactor in this circuit. It is better to regulate the voltage by tap-changing the transformer. Try it out and observe the steady-state value of the 110kV voltages. The 110kV steady state voltage is about 106kV with shunt reactor and 120kV without shunt reactor. There is a voltage drop of about 15kV !! To see the difference in one plot add the line `.step param kd list 0 1` to the schematic !!

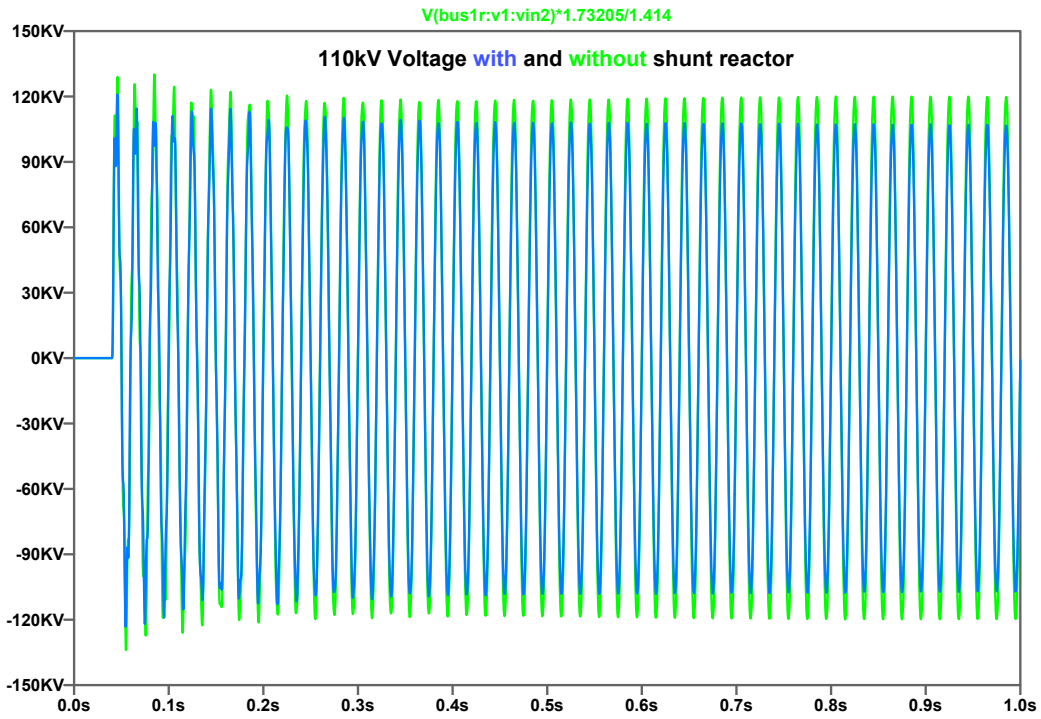


Figure 6.30: Switching on networks voltages - case8

6.14 Case 9: Network Startup by a Voltage Ramp

The simulation file is `Switching_on_networks_Voltagetrip.asc`. Set the voltage-source parameter `tpwl` to 400 ms and remove the network switch. Set the parameter ST to 10s to disable it. The source voltage then increases gradually from zero to its nominal value within 400ms. NOTE: This is not really done in practice but in simulation to keep the simulation time short.

This procedure produces practically no switching transients: there are no significant overvoltages, high-frequency charging-current oscillations, or transformer magnetizing inrush currents. It is therefore the preferred method for energizing the complete network in this study. Loads should be connected only after the voltage has reached its nominal level. The currents of this network are only the charging currents of the lines and the current of the shunt reactor. The magnetizing currents of the transformers 2 and 3 are below 1A. The current of Transformer1 is a little bit higher due to the current needed by the Petersen. The currents in all three phases of line 1 have the same amplitude. The Petersen current is 7A, the Petersen voltage is approximately 1.7kV (measured at time=100s) and the Line4 current is about 400mA. The flux of transformer 1,2 and 3 have equal amplitudes and about the nominal flux (a little bit higher since the operating voltage is not 110kV). The 110kV voltage is at about 110kV in phase A 113.7kV in Phase B and 112.3kV in Phase C (slightly different because of the neutral point displacement voltage) which is the steady-state value. The FFT shows only the nominal frequency of 50Hz for the whole circuit !!

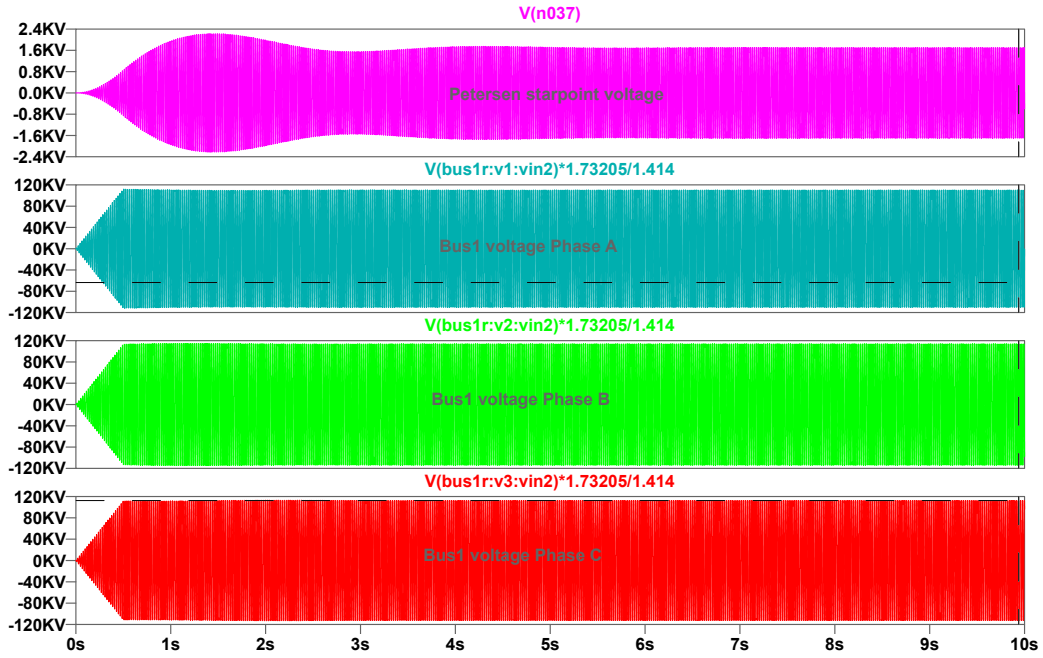


Figure 6.31: Switching on networks voltage-trip, voltages - case9

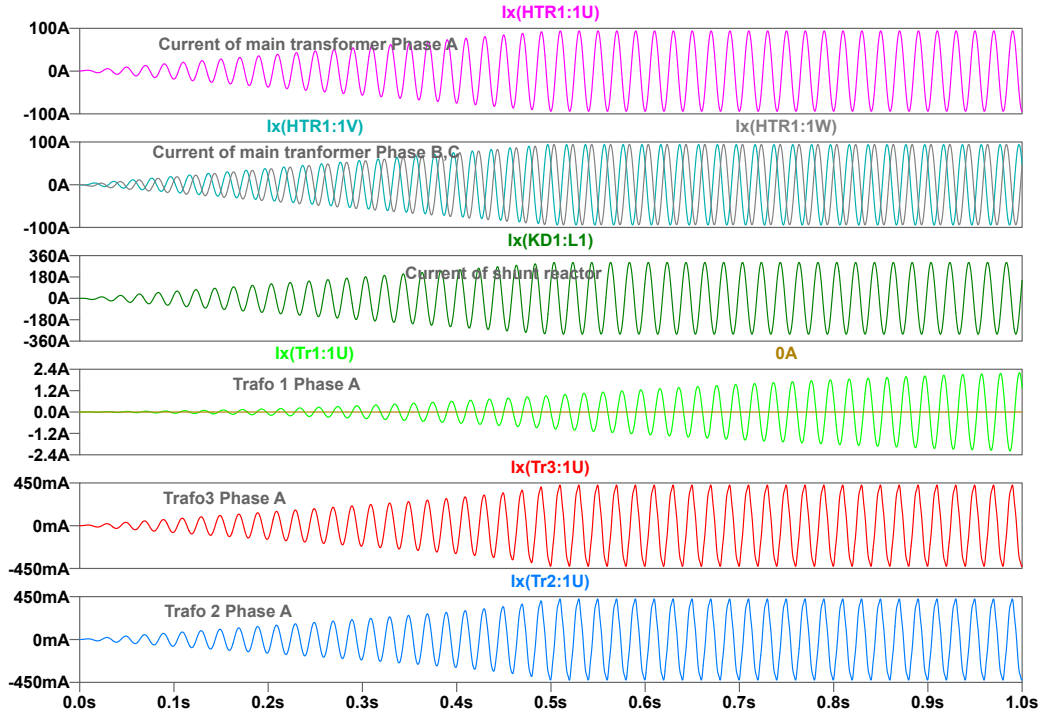


Figure 6.32: Switching on networks voltage-trip, currents - case9

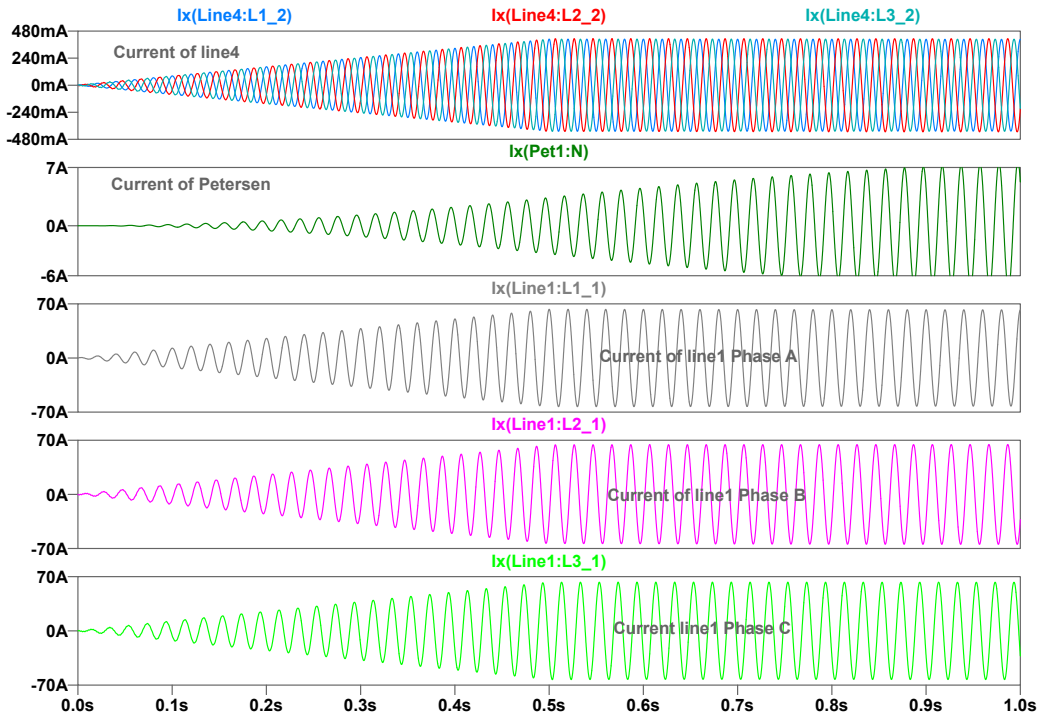


Figure 6.33: Switching on networks voltage-trip, currents line1, line4 - case9

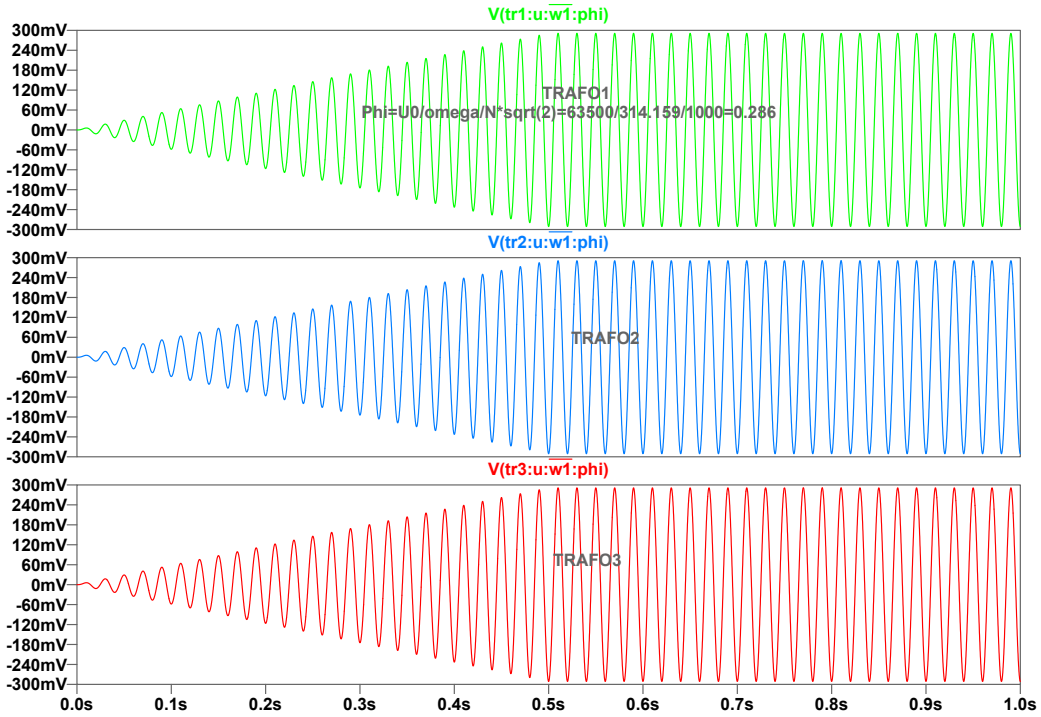


Figure 6.34: Switching on networks voltage-trip, fluxes - case9

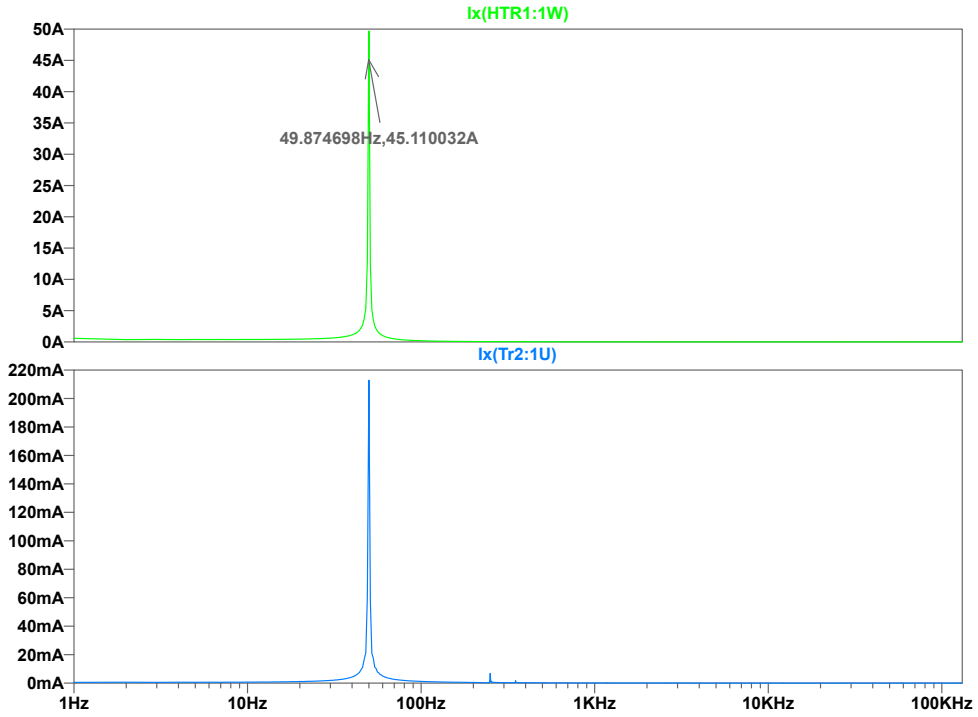


Figure 6.35: Switching on networks voltage-trip, FFT - case9

Table 6.7: Signal ranges for Case 9. All values are peak values, bus values are multiplied by 1.73/1.414

Meas.	Grid element	Phase	Voltage level	Min.	Max.	Notes
V	Bus0	Ph A	220 kV	−229 kV	229 kV	Steady state RMS
V	Bus1	Ph A	110 kV	−110 kV	110 kV	Steady state RMS
V	Bus1	Ph B, C	110 kV	−113 kV	113 kV	Steady state RMS
I	HTr1	Ph A	220 kV	−94 A	94 A	Steady state peak
I	HTr1	Ph B, C	220 kV	−94 A	94 A	Steady state peak
I	KD1	Ph A	110 kV	−300 A	300 A	Steady state peak
I	Tr1	Ph A	110 kV	−2 A	2 A	Due to Petersen higher than Tr2,3
I	Tr2	Ph A	110 kV	−450 mA	450 mA	Steady state zero load peak current
I	Tr3	Ph A	110 kV	−450 mA	450 mA	Steady state zero load peak current
I	Line4	Ph A	110 kV	−410 mA	410 mA	Steady state peak
I	Line1	Ph A, B, C	110 kV	−62 A	62 A	Steady state peak
I	Pet1	Ph N	110 kV	−7 A	7 A	Steady state peak
V	Pet1	Ph N	110 kV	−1.7 kV	1.7 kV	Steady state peak
Φ	Tr1	Ph A, B, C	110 kV	0.291 Wb	0.291 Wb	Nominal flux: 0.288 Wb peak
FFT	HTr1	Ph C	220 kV	50 Hz	50 Hz	only 50 Hz nominal frequ.
FFT	Tr2	Ph A	110 kV	50 Hz	50 Hz	50 Hz nominal frequ.

Important 6.1 – (Preferred energization method)

A controlled voltage ramp avoids the abrupt change in applied voltage that excites cable resonances and transformer flux offsets. The ramp time must be long compared with the dominant electrical oscillation periods, and load connection must be delayed until the voltage is established.

Chapter Summary

Energizing an entire cable-dominated network excites both transformer inrush and cable-charging transients. The switching instant determines which phases are dominated by magnetizing current and which by capacitive current. Cable capacitance, referred transformer inductance, tap position, load, and network strength determine the natural frequencies and transient amplitudes. Residual flux can raise transformer currents above rated peak current, while connected loads add effective damping. In this study, a gradual source-voltage ramp is the most effective energization method because it suppresses both cable resonances and transformer inrush.

7 Coupled Lines with Resonance

7.1 General

This chapter investigates two electromagnetically coupled 400 kV overhead lines on a common Danube-type tower. The capacitive reactive power of the two lines is compensated by an inductive choke. We will examine the case where the compensation is near ideal—that is, where the inductive reactive power of the choke exactly compensates for the capacitive reactive power of the line. The following cases are simulated (Line 1 is always energized):

- `Line 2 is open (lines are untransposed),`
- `Line 2 is open (lines are transposed),`
- `Line 2 is initially grounded and then ungrounded,`
- `Line 2 is initially energized and then switched off,`
- `Line 2 is initially energized and then switched off without a compensating reactor`

7.2 Circuit design

The total line length is approximately 9.75 km. Mutual inductances and capacitances couple the circuits even when one line is disconnected at both ends.

The per-kilometre inductance and capacitance matrices of the coupled line were calculated with ATP and then stored in `line_coupled.lib` The Excel file Lines_and_cables.xlsx assists with conversion to LTspice library code.

Each line is almost fully compensated by an shunt reactor element, `l_batt`. Near-exact compensation creates an important operating condition: if line 1 is energized while line 2 is open at both ends, the mutual coupling can excite a resonance and produce very high voltages on the disconnected line.

7.3 Determining the Compensation Rating

Use `Coupled_lines_resonance_TRAN.asc`. Save a working copy as `test.asc`, remove the compensation elements, and run the transient simulation. Measure the three phase reactive powers with `qmeter_uvw_sgn`. The results are approximately

$$Q_R = -2.02\,\text{Mvar}, \qquad Q_S = -2.40\,\text{Mvar}, \qquad Q_T = -2.37\,\text{Mvar}. \tag{7.1}$$

The total capacitive reactive power is therefore approximately −6.8 Mvar. A compensation rating of 6 Mvar is selected for each line, leaving the system close to, but not exactly at, resonance.

At 400 kV, the corresponding balanced three-phase current is

Calculation 7.1 – (Current of the 6 Mvar compensation reactor)

$$I_{\text{rms}} = \frac{Q}{\sqrt{3}U_{\text{LL}}} = \frac{6\,\text{Mvar}}{\sqrt{3}\cdot 400\,\text{kV}} \approx 8.66\,\text{A}, \tag{7.2}$$

$$\hat{I} = \sqrt{2}\,I_{\text{rms}} \approx \boxed{12.2\,\text{A}}. \tag{7.3}$$

7.4 Line 1 Energized and Line 2 Open: Untransposed Lines

figures: 7.2-7.3 (simulation-file: same name as the heading)

Line 1 is supplied by a source with an exponential voltage startup, while line 2 remains open at both ends. The open-line voltages exhibit a slow beat and approach their final values after approximately 2 s.

Phase R of line 2 has the highest voltage, with transient peaks near 500 kV, corresponding to approximately 350 kV RMS. The other two phase peaks remain below 300 kV. For comparison, the normal phase-voltage peak of the energized 400 kV line is

$$\widehat{U}_{\text{ph}} = \sqrt{2}\frac{U_{\text{LL}}}{\sqrt{3}} = \sqrt{2}\frac{400\,\text{kV}}{\sqrt{3}} \approx 326.6\,\text{kV}\,\textit{peak}. \tag{7.4}$$

The settled phase-R peak on line 2 is approximately 350 kV, exceeding the normal phase-voltage peak of 327 kV on line 1 with 327 kV . The slow envelope is caused by the small difference between the 50 Hz source frequency and a nearby natural frequency formed by the line capacitances and the compensating `1_batt` reactors.

The reactor-current peaks on line 1 are approximately 12 A in all three phases, consistent with calculation 7.1. On line 2, the peak currents are approximately 13.3 A, 6 A, and 7 A in phases R, S, and T, respectively. The source currents are close to zero because the line charging current is almost compensated.

The untransposed line has strongly unequal mutual capacitances and inductances. Consequently, the phase magnitudes and phase angles on open line 2 are highly asymmetric, and the sum of its three phase-to-earth voltages is nonzero, with a residual sum of approximately 46 kV.

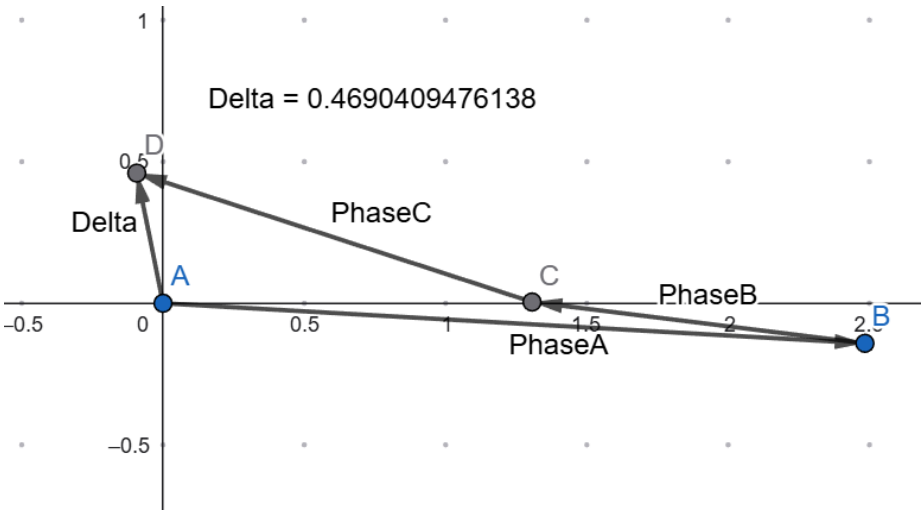


Figure 7.1: Coupled lines resonance - Sum of voltages not zero - scale multiplied by 1e5

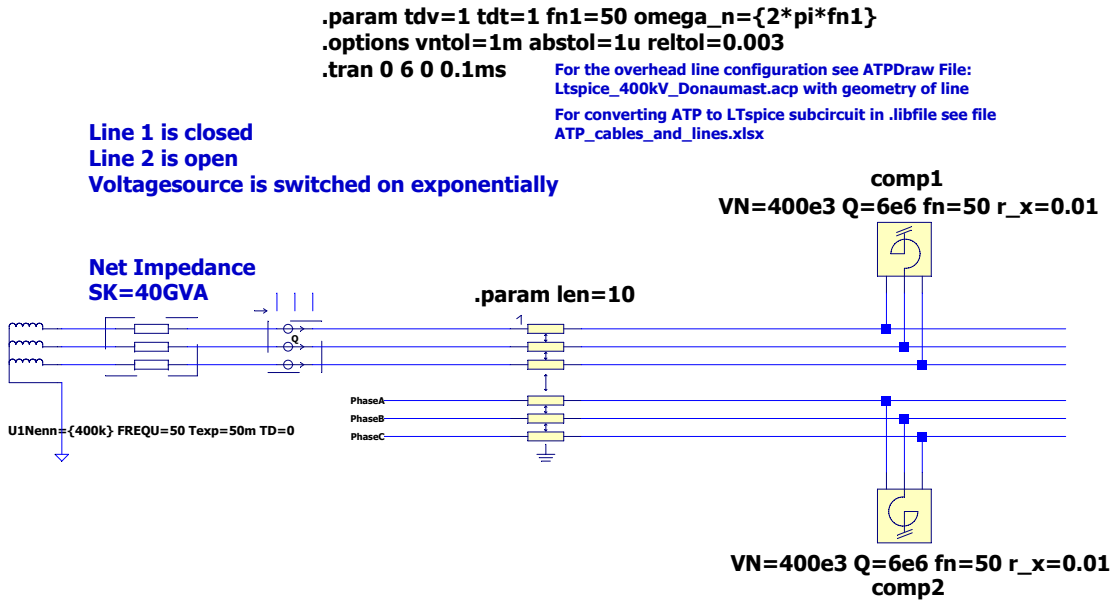


Figure 7.2: Coupled lines resonance schematic - line2 is open untransposed TRAN analysis

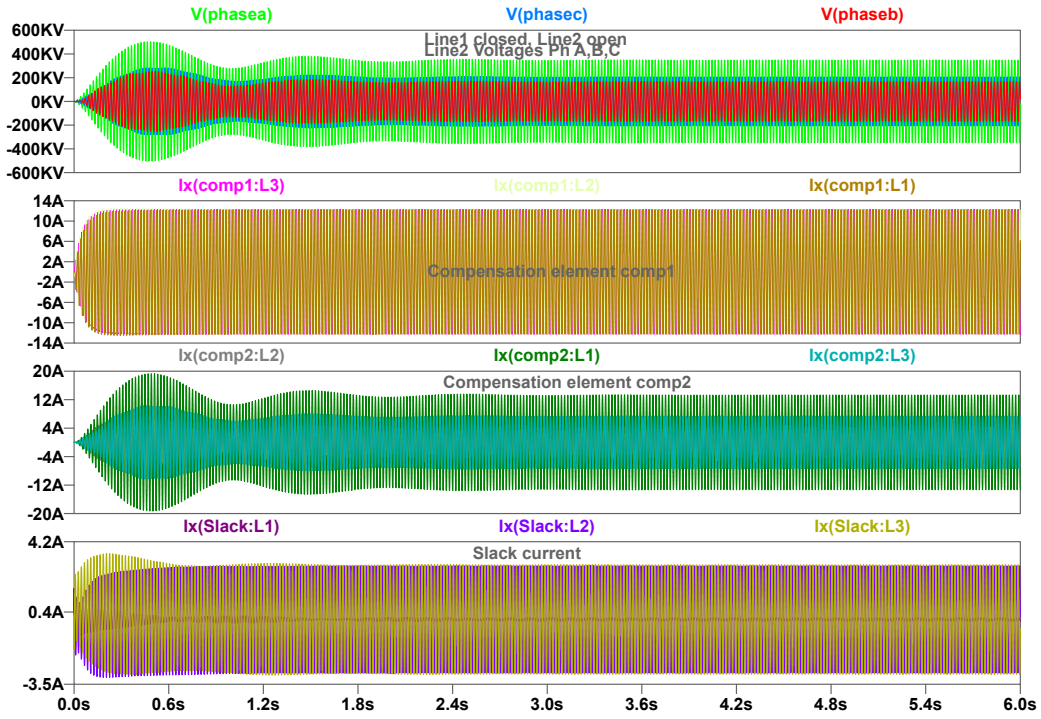


Figure 7.3: Coupled lines resonance TRAN - line2 is open untransposed TRAN analysis

7.5 AC Analysis of the Untransposed Open Line

figures: 7.4-7.5 (simulation-file: same name as the heading)

Set the vertical scale to linear by right-clicking the left axis. The principal resonance of the open-line voltage lies near 49 Hz. At 50 Hz, the phase-R voltage is approximately 250 kV RMS, compared with the normal phase voltage of approximately 231 kV RMS. This corresponds to the settled transient value of approximately 353 kV peak.

A second resonance occurs near 44 Hz. At the main resonant peak near 49 Hz, the phase-R voltage can approach approximately 1 MV peak. This demonstrates the sensitivity of the open coupled line to small changes in shunt-reactor tuning.

Important 7.1 – (Near-resonant open-line overvoltage)

An open line is not necessarily de-energized when it shares a tower with an energized line. Mutual capacitances and inductances can sustain a substantial voltage, and near-resonant shunt compensation can amplify it far above the normal phase voltage.

7.5.1 Exercises

1. Save the model as `test.asc` and increase the `l_batt` rating until the resonant peak lies at 50 Hz. Observe the resulting voltage amplification.
2. Reduce the reactor rating to 5 Mvar. Compare the resonant frequency and peak magnitude with the original case.
3. Change the reactor parameter `r_x`, which represents its ohmic component. Observe how increasing damping broadens and lowers the resonant peak.

7.6 Line 1 Energized and Line 2 Open: Transposed Lines

figures: 7.6-7.7 (simulation-file: same name as the heading)

Transposition makes the phase coupling approximately symmetrical. The three open-line voltages therefore have equal magnitudes, and the relevant resonance moves farther away from the power frequency.

The initial open-line peaks are approximately 100 kV and decay to about 70 kV after 2 s. This is a substantially more favourable operating condition than the untransposed case in Section 7.4.

Important 7.2 – (Exercise)

Save the model as `test.asc`, remove the `l_batt` elements from both lines, and compare the phase currents and reactive powers of lines 1 and 2. The transposed model should produce nearly symmetrical phase results.

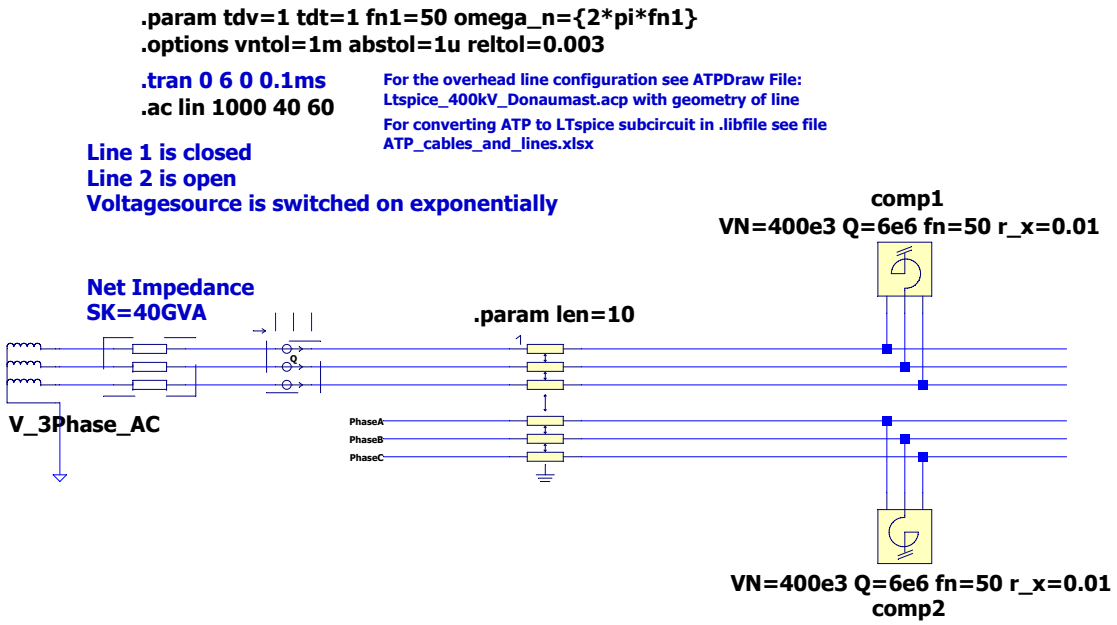


Figure 7.4: Coupled lines resonance schematic - Lines2 is open untransposed AC analysis

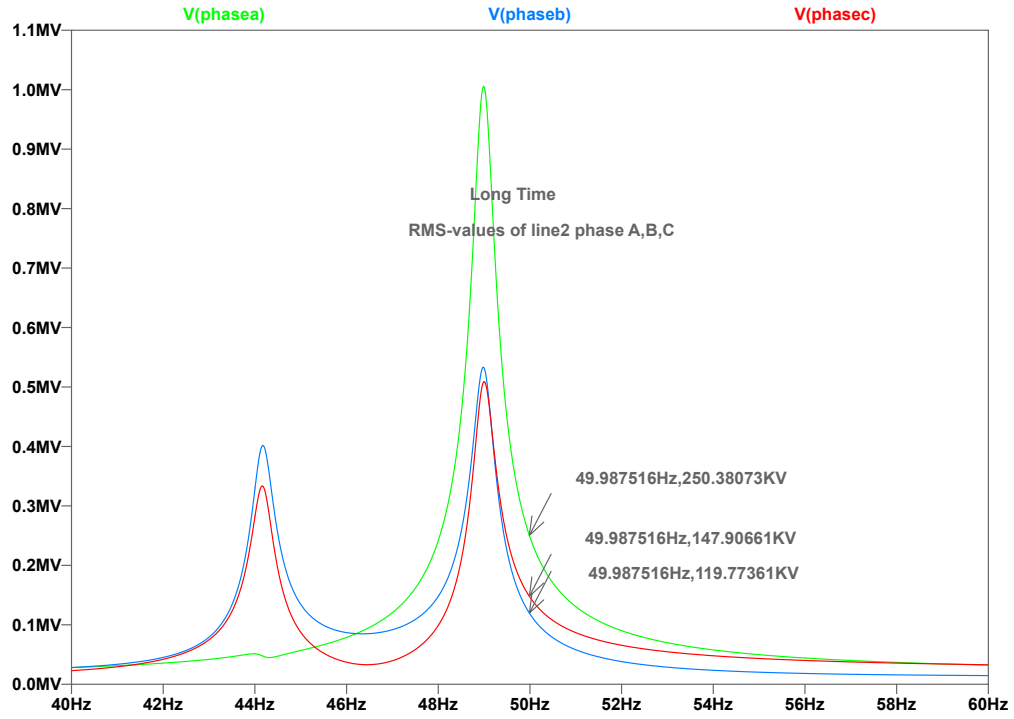


Figure 7.5: Coupled lines resonance AC - Lines2 is open untransposed

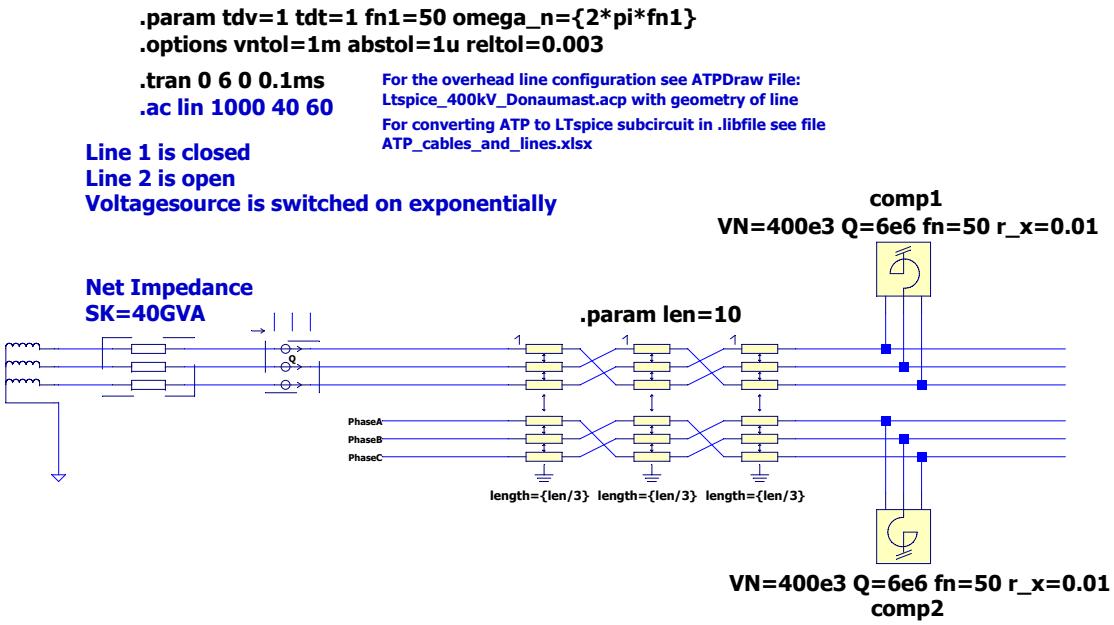


Figure 7.6: Coupled lines resonance schematic - Lines2 is open transposed - TRAN analysis

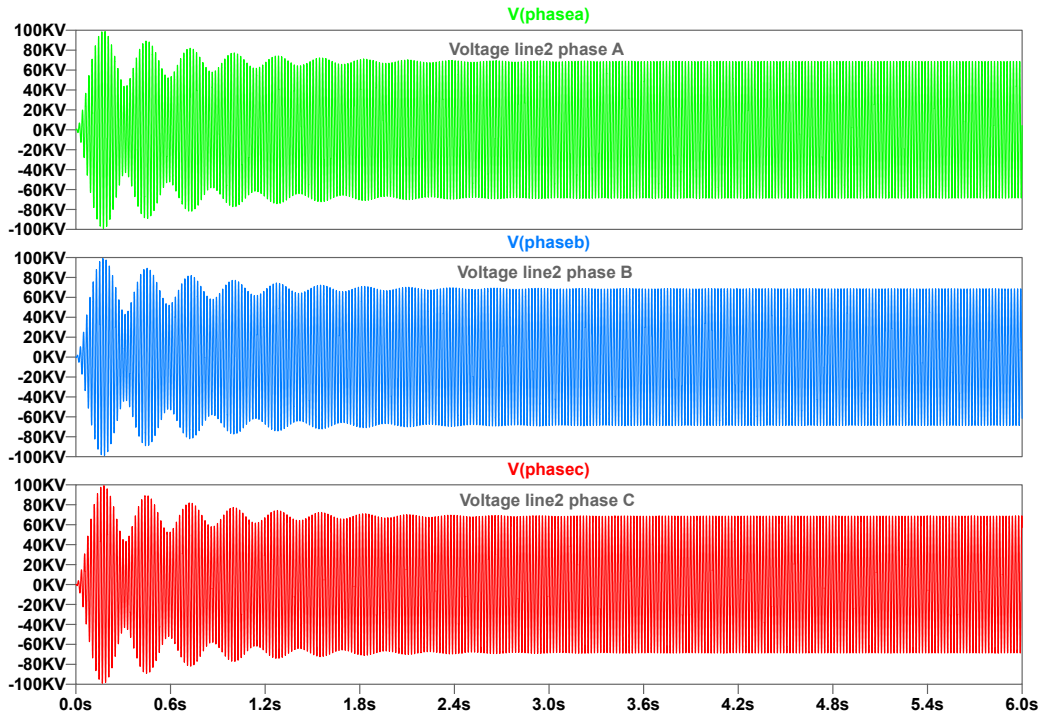


Figure 7.7: Coupled lines resonance TRAN - Lines2 is open transposed

7.7 AC Analysis of the Transposed Open Line

figures: 7.8-7.9 (simulation-file: same name as the heading)

The principal resonant peak shifts to approximately 46.3 Hz. The additional lower-frequency maxima visible in the untransposed model disappear because the phase coupling has been symmetrized. At 50 Hz, all three phase voltages are approximately 49 kV RMS, or 70 kV peak, in agreement with the final values of the transient simulation.

7.8 Line 2 Initially Grounded and Then Ungrounded

figures: 7.10-7.11 (simulation-file: same name as the heading)

Line 1 remains energized, while line 2 is initially grounded and subsequently opened. The final response is similar to the permanently open case in Section 7.4. A second beat is particularly visible in phases S and T; it is associated with the second resonance near 44 Hz identified by the AC analysis.

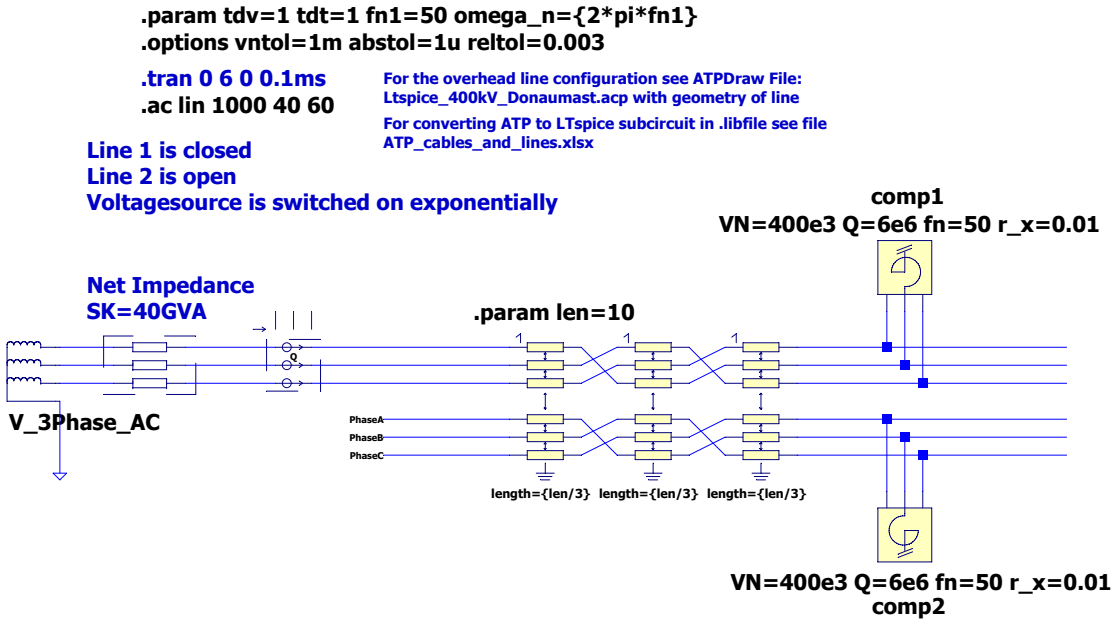


Figure 7.8: Coupled lines resonance schematic Lines2 is open transposed

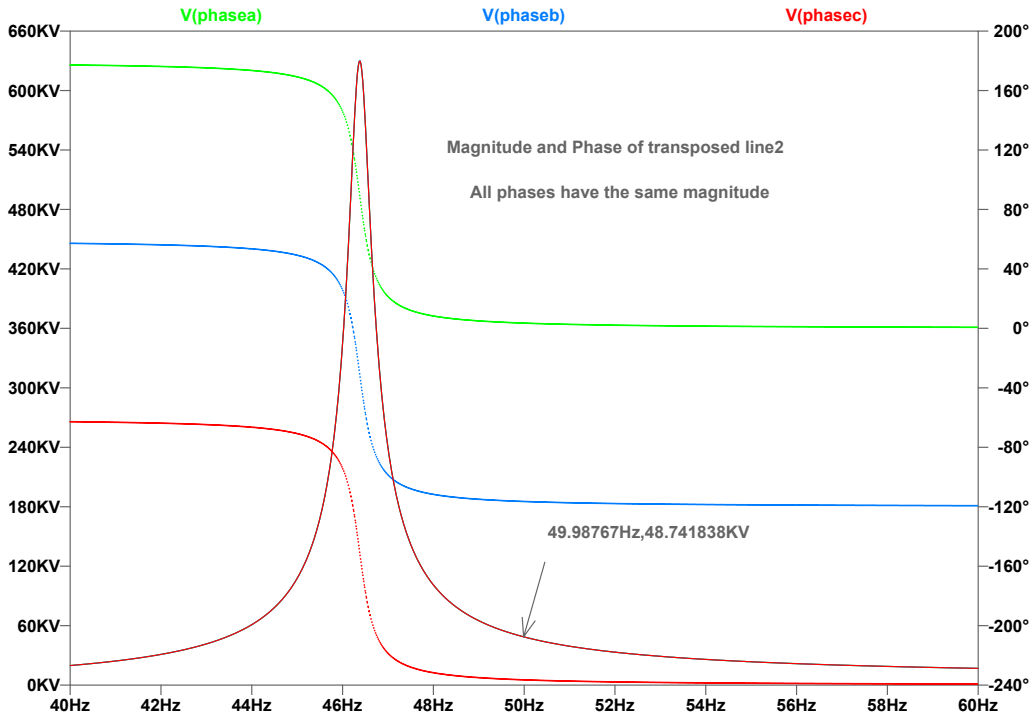
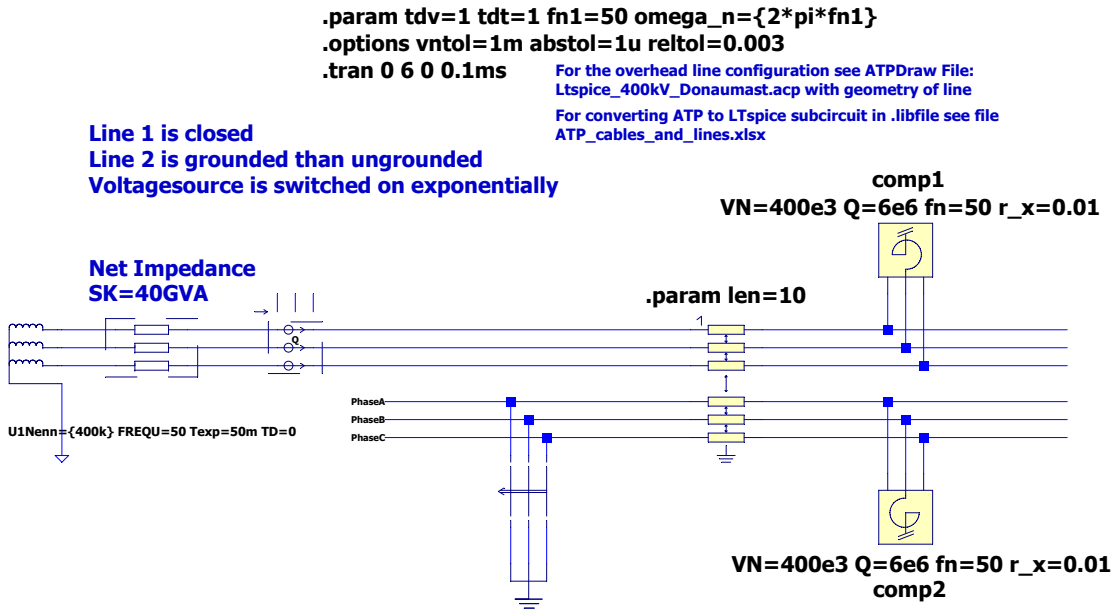


Figure 7.9: Coupled lines resonance AC - Lines2 is open transposed



e

Figure 7.10: Coupled lines resonance TRAN line2 grounded then ungrounded

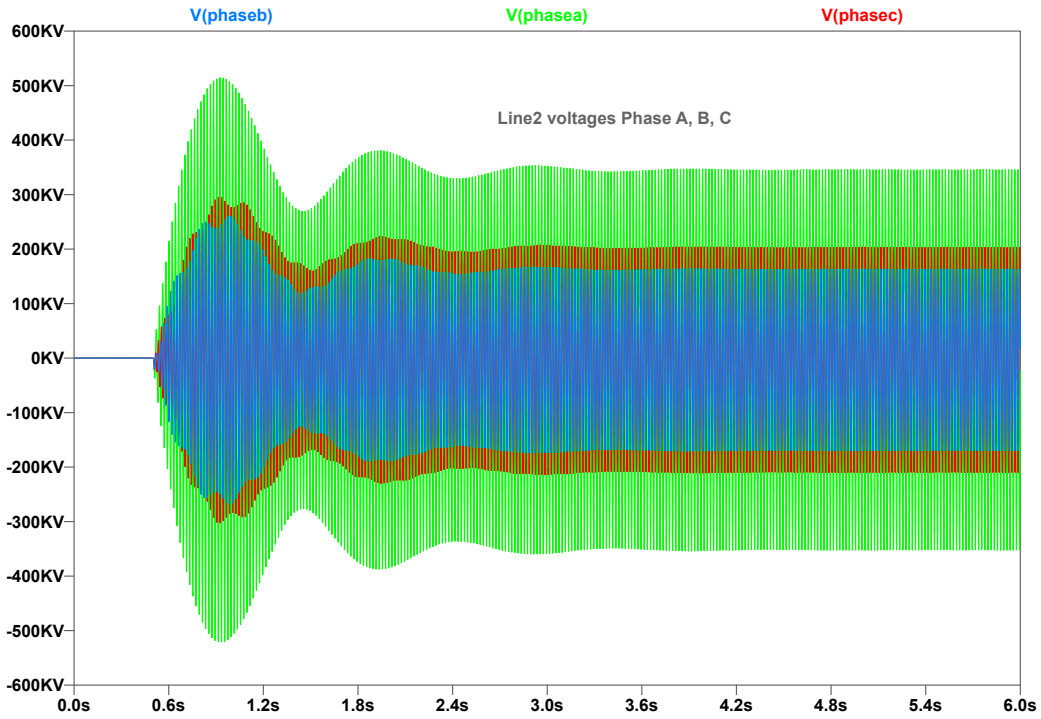


Figure 7.11: Coupled lines resonance TRAN line2 grounded then ungrounded

7.9 Line 2 Energized and Subsequently Switched Off

figures: 7.12-7.13 (simulation-file: same name as the heading)

Line 2 is disconnected at 500 ms. Immediately after opening, its voltage swings to approximately −430 kV. Phase R shows a long-period beat, while phases S and T show a shorter-period envelope associated with the 44 Hz resonance.

Select the phase-T voltage and draw a zoom window between two adjacent envelope maxima. The LTspice status line indicates approximately 5.7 Hz. This is close to the difference between the source frequency and the second natural frequency:

$$f_{\text{beat}} \approx |f_s - f_{r,2}| \approx |50\,\text{Hz} - 44\,\text{Hz}| \approx 6\,\text{Hz}. \tag{7.5}$$

Important 7.3 – (Exercise)

Save the model as `test.asc`, change the opening time to 0.510 s, and compare the phase-voltage peaks. The trapped initial conditions at the switching instant strongly influence the subsequent free oscillation.

7.10 Line 2 Switched Off Without a Shunt-Reactor

figures: 7.14-7.15 (simulation-file: same name as the heading)

After line 2 is switched off, its induced phase-voltage peaks are only approximately 20 kV to 30 kV. Without the `l_batt` element, the near-power-frequency resonance is absent.

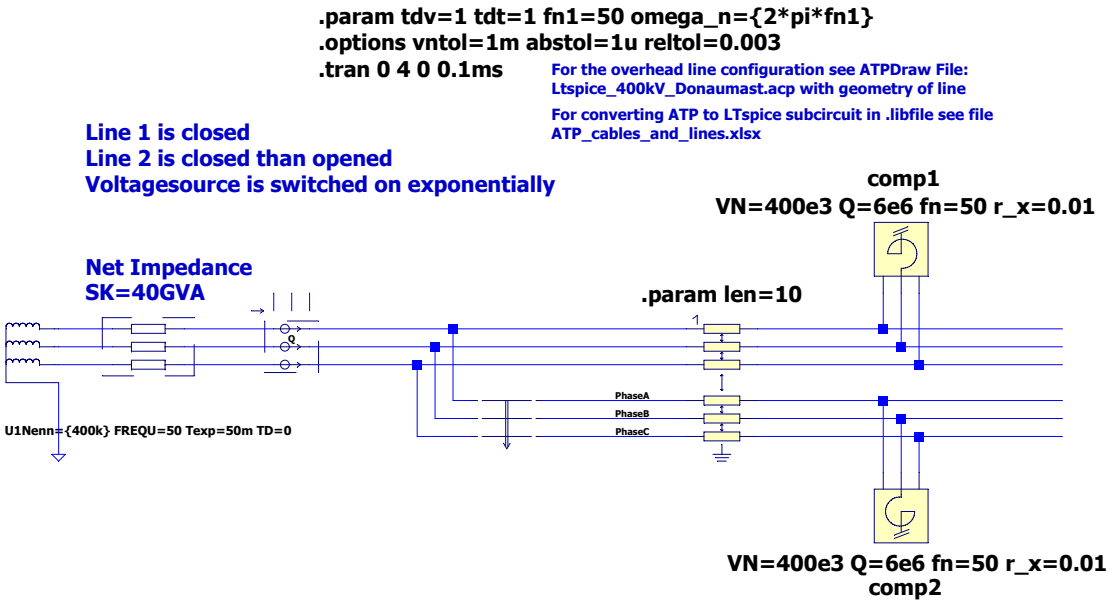


Figure 7.12: Coupled lines resonance TRAN line2 switched off schematic

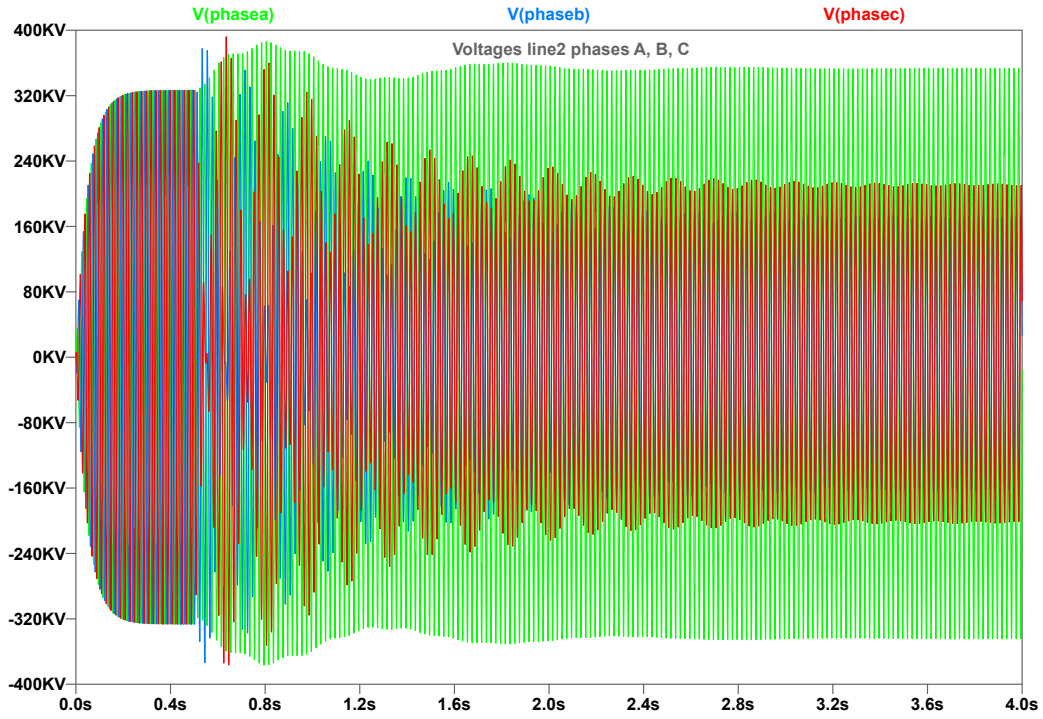


Figure 7.13: Coupled lines resonance TRAN line2 switched off

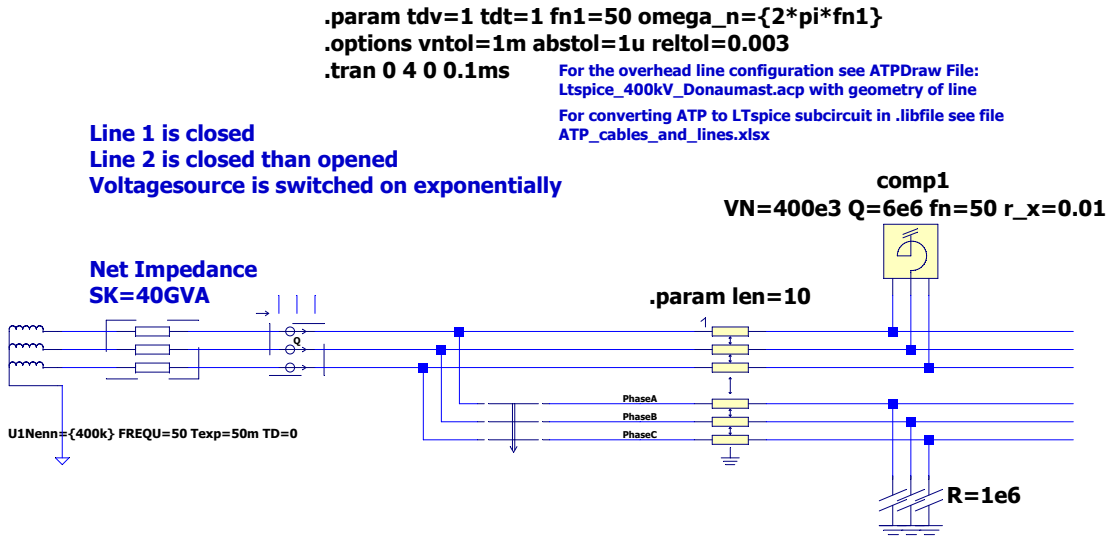


Figure 7.14: Coupled lines resonance TRAN line2 switched off without shunt reactor schematic

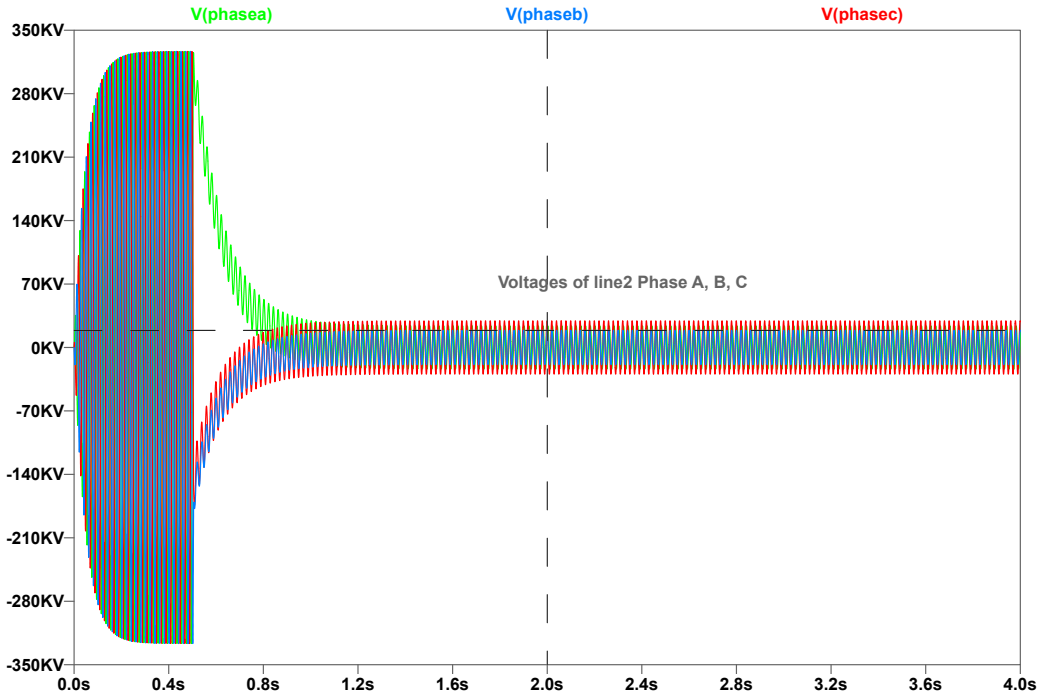


Figure 7.15: Coupled lines resonance TRAN line2 switched off without shunt reactor

7.11 Stored Energy on the Open Line

figure: 7.16 717 (simulation-file: same name as the heading)

Line 1 is initially energized and then disconnected; line 2 is subsequently grounded. To measure the dissipated energy, hold Alt and click the grounding element, then hold Ctrl and click the generated power trace. LTspice integrates the power and displays the corresponding energy. NOTE: In this case, the energy should be measured from 1 s to 2 s. Zoom into the interval from 1 s to 2 s. During this interval, the stored energy decreases substantially after the line is switched off After two seconds, only the energy transfer from Line 1 to Line 2 via capacitive coupling between both lines remains. This (small) energy should not be included in the measurement.

The energy released from line 2 is approximately 2.2 kJ. The stored energy is therefore comparatively small. With a grounding resistance of 100 Ω, the open-line voltage collapses almost immediately. It also collapses with 10 kΩ, showing that the mutual coupling path to line 1 is relatively high-impedance at 50 Hz.

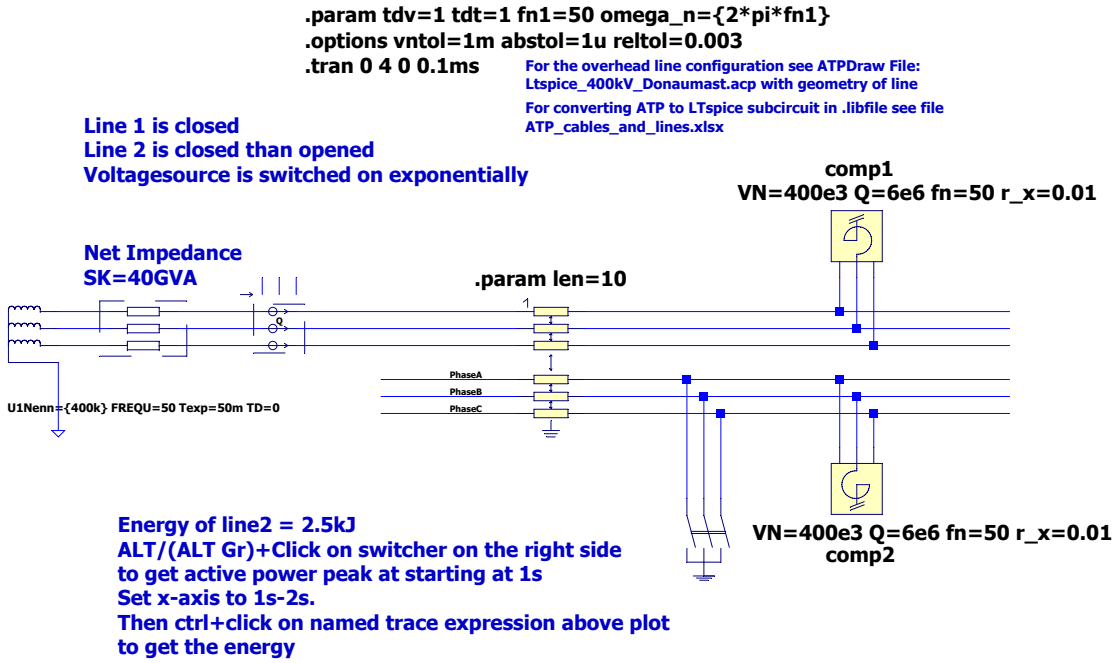


Figure 7.16: Coupled lines resonance TRAN line2 stored energy schematic

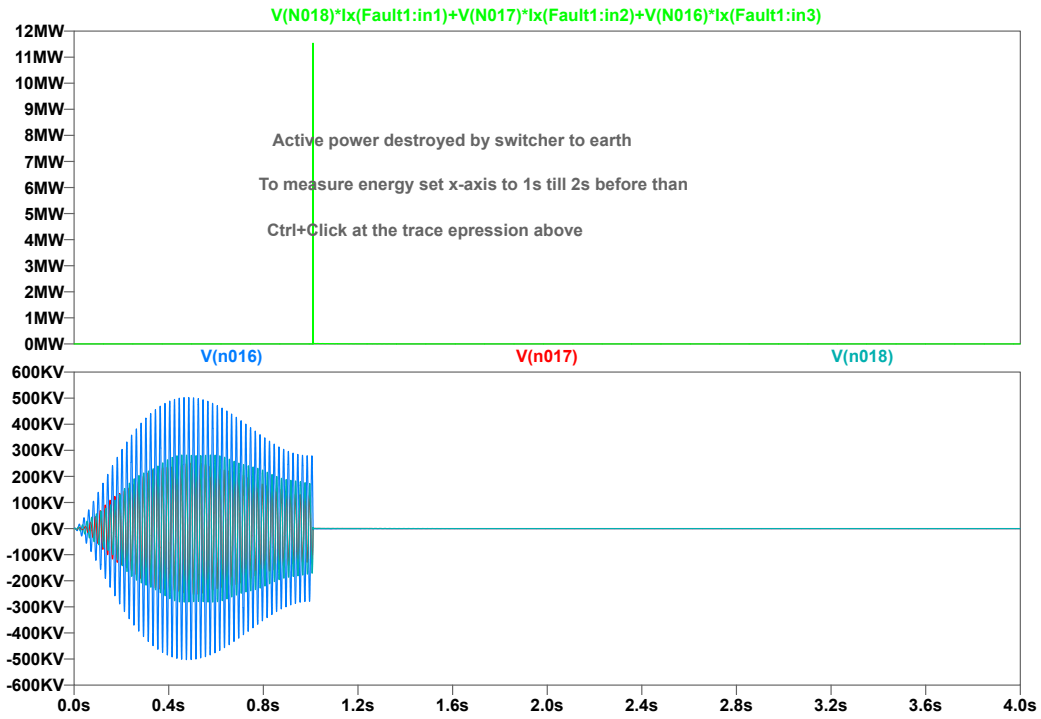


Figure 7.17: Coupled lines resonance TRAN line2 stored energy

Chapter Summary

Mutual inductive and capacitive coupling can impose substantial voltages on an open overhead-line circuit. Near-exact shunt compensation may place a natural frequency close to 50 Hz and amplify the open-line voltage to a dangerous level. Phase transposition symmetrizes the coupling and, in this example, shifts the resonance away from the operating frequency. Removing the compensation reactor eliminates the near-power-frequency resonance, while its loss parameter controls damping.

8 Operating Modes of Synchronous Generators

This chapter demonstrates three important operating modes of a synchronous generator:

- isolated or island operation,
- operation connected to a stiff power network, and
- operation at fixed mechanical speed.

The examples illustrate voltage and speed control, synchronization, active- and reactive-power control, load-angle stability, and the behaviour of the machine during symmetrical and asymmetrical faults.

8.1 Isolated Operation

8.1.1 General

In isolated operation, no external voltage source maintains the network voltage or frequency. An automatic voltage regulator (AVR) is therefore required to regulate the terminal voltage, while a speed governor must balance the mechanical input power against the electrical load. The example also demonstrates how a second synchronous generator can be synchronized with the island network.

Adding generators can increase the total inertia and improve the response of the island system. Stable load sharing, however, requires suitable governor and excitation-control characteristics.

Important 8.1 – (Control requirements in island operation)

An island network requires at least one unit that regulates frequency and at least one unit that regulates voltage. In practical parallel operation, governor droop and AVR reactive-power or voltage droop are normally used to share active and reactive power between the generators.

8.1.2 Circuit Design

The circuit contains a round-rotor synchronous generator rated at 400 MW and 20 kV. It is connected through a transformer rated at 20 kV on the generator side and 400 kV on the network side. A switch connects the generator to the initially unloaded network at 1 s.

The load increases linearly from zero to 50 MW between 2 s and 5 s. The AVR is enabled at 6 s, whereas the speed governor remains active throughout the simulation. During the first second, the main field winding is supplied by a temporary excitation ramp from 600 V to 1 400 V and back to 600 V. This accelerates the buildup of the rated terminal voltage.

The AVR measures the voltage on the high-voltage side and uses a reference of 400 kV. The governor reference is 1 500 rpm. Its switching time is placed outside the simulated interval, for example at 500 s, so that the controller remains active. Both external controllers are implemented as simple PI controllers to illustrate the fundamental behaviour of the controlled machine. Their dynamics are intentionally fast in order to shorten the simulation.

Simulation 8.1 – (AVR switching sequence)

The AVR switch must close shortly before the switch supplying the fixed field voltage opens. The field voltage must not be interrupted, because this would cause the terminal voltage to collapse. Since the excitation circuit may carry a high current, the switch on-resistance must be sufficiently small to avoid an unwanted field-voltage drop.
The torque-control changeover is less critical. Owing to the large moment of inertia J, the rotor speed cannot change abruptly even if the electromagnetic or mechanical torque is briefly disturbed.

8.1.3 Simulation Start

see Figures: 8.1 and 8.2 (simulation-file: same name as the heading)

The machine starts with zero terminal voltage. The initial field-voltage boost to 1 400 V allows the rated voltage to be reached quickly. For this demonstration, the stator copper-resistance parameter `Rcu` is set to 0.01 to accelerate voltage buildup and thereby reduce simulation time.

8.1.4 Voltage Behaviour

see Figures: 8.1 and 8.2

After approximately 1 s, the network voltage reaches 400 kV, corresponding to 20 kV at the generator terminals. As the load rises from zero to 50 MW, the network voltage falls slightly to approximately 397 kV. When the AVR is enabled at 6 s, it rapidly restores the voltage to 400 kV.

In an island network, a voltage change can also change the active and reactive power absorbed by voltage-dependent loads. For an ideal constant-power load, by contrast, the specified active and reactive powers remain constant and the current changes as the voltage changes.

8.1.5 Speed Behaviour

see Figures: 8.1 and 8.2

During the load ramp, the speed initially falls below 1 500 rpm. Once the final load has been reached and the governor has increased the mechanical torque, the speed returns to its reference value. When the AVR is enabled, the rising voltage temporarily increases the load current and electromagnetic torque. Consequently, the frequency briefly decreases before returning to its steady-state value.

8.1.6 Load Angle and Load Steps

see Figures: 8.3-8.6 (simulation-file: same names as the heading)

During the first load ramp, the load angle increases and settles at approximately 12°. A 30 MW load step can be introduced by setting `Vfact=3` at 2 s and closing an additional switch at 3 s. Further load steps at 5 s and 7 s increase the generator output from 30 MW to 60 MW and finally to 90 MW. The maximum load angle is approximately 21°.

Each load step produces a transient change in frequency and load angle. The governor subsequently establishes a new torque balance.

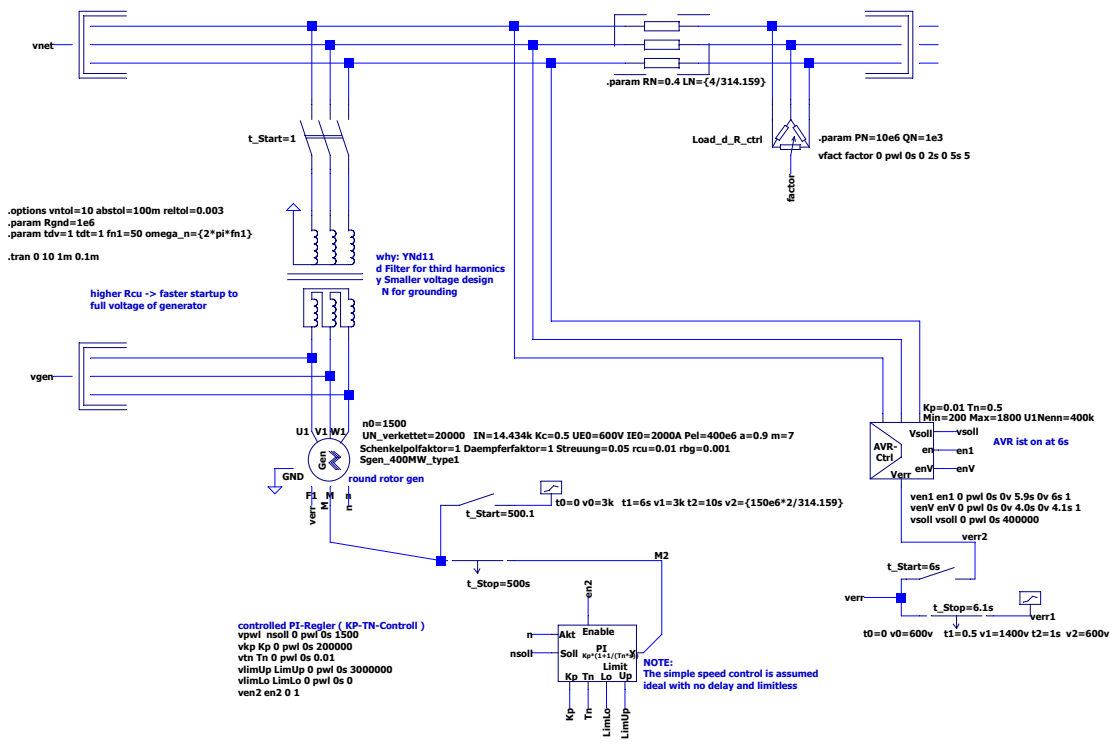


Figure 8.1: Simulation start - Voltage behaviour - Speed behaviour - schematic

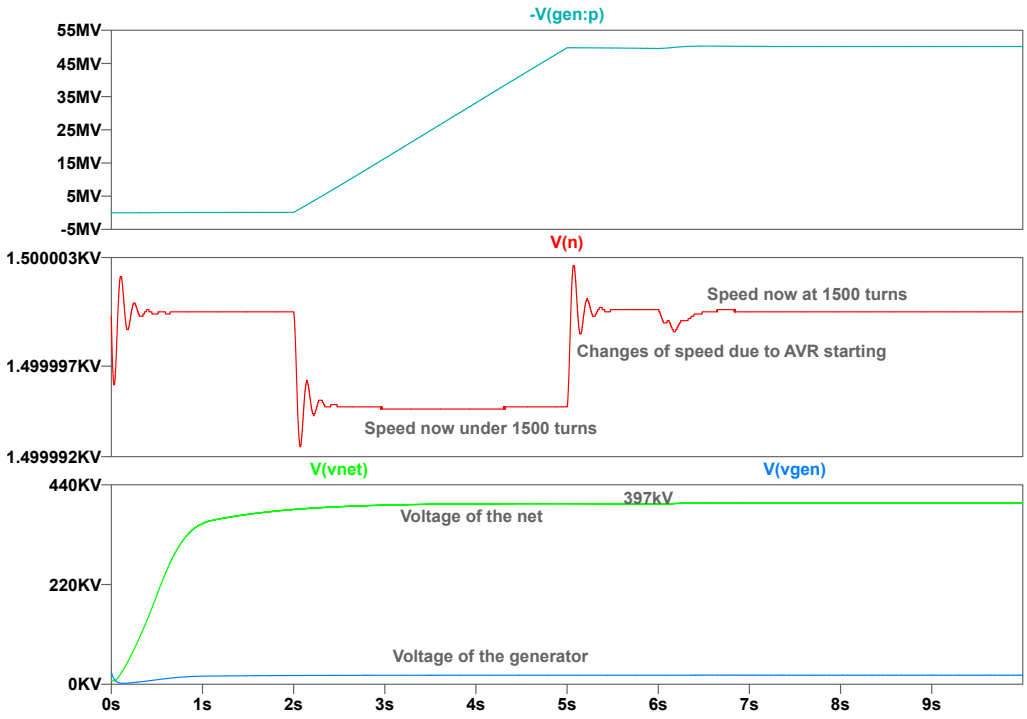


Figure 8.2: Simulaton start - Voltage behaviour - Speed behaviour - Waveform

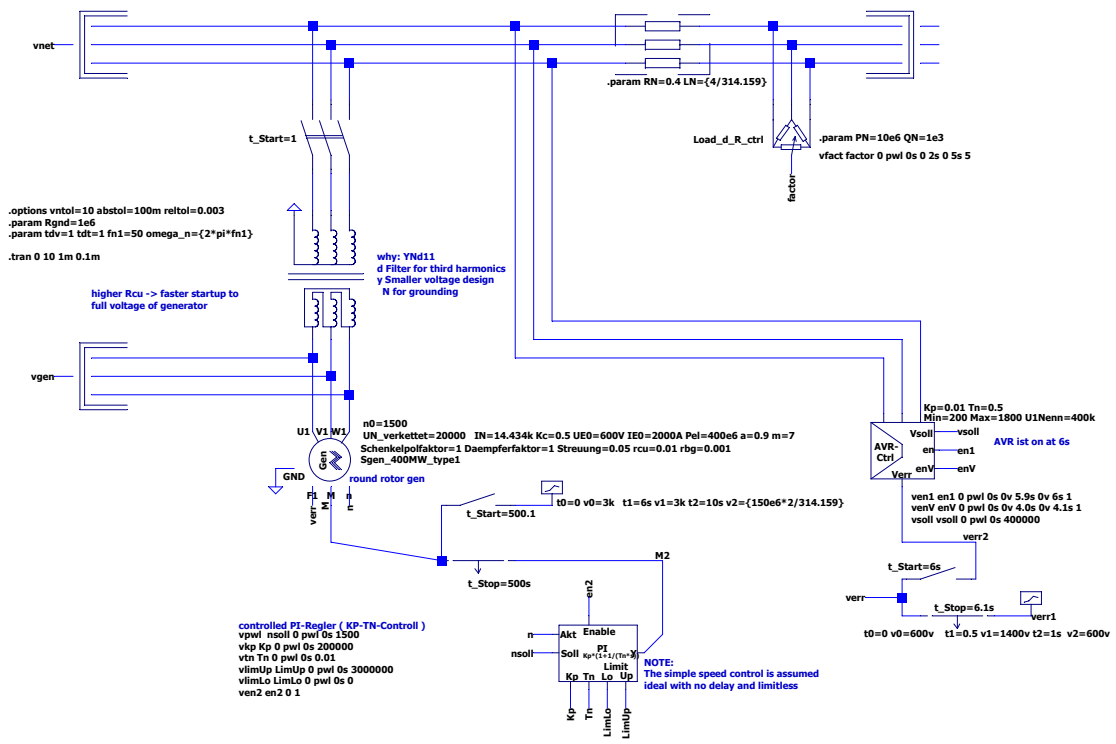


Figure 8.3: Load angle - schematic

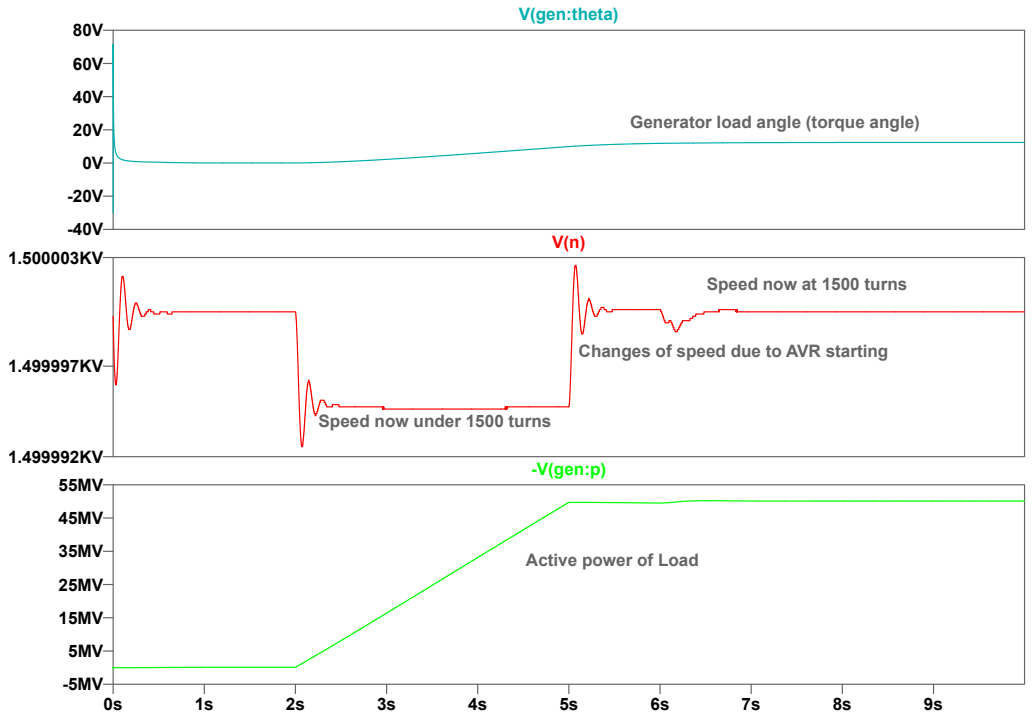


Figure 8.4: Load angle - Waveform

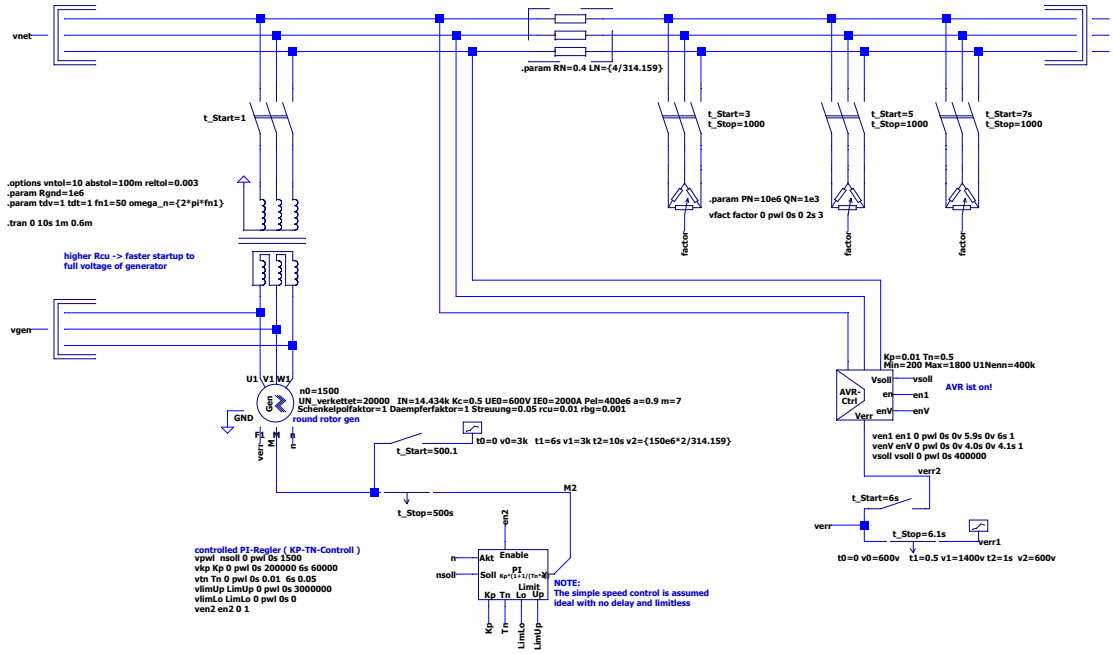


Figure 8.5: Load steps - schematic

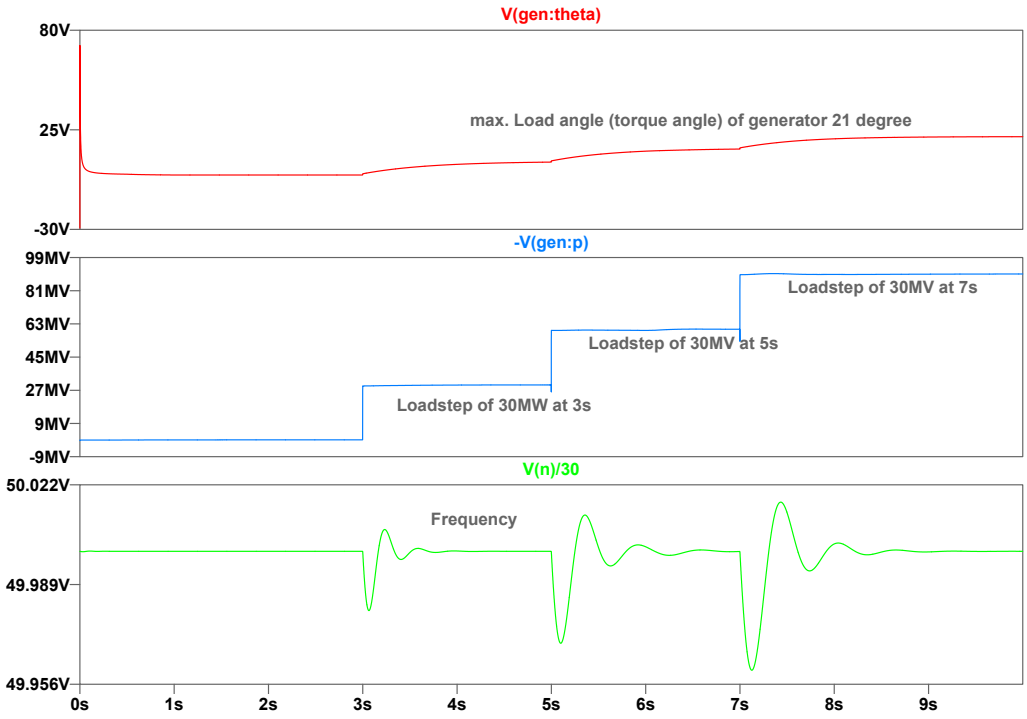


Figure 8.6: Load steps - Waveform

8.1.7 Comparison with a Salient-Pole Machine

see figures: 8.7 and 8.8 (simulation-file: same name as the heading)

The parameter settings `SchenkelPoleFactor=0.5` and `KC=0.8` represent a salient-pole machine in this model. After restarting the simulation, the load angle is approximately 7°, compared with about 21° for the round-rotor case. The difference results from the changed direct- and quadrature-axis reactances and the additional reluctance-torque component.

8.1.8 Reactive Power in Island Operation

In an isolated network, the total active and reactive powers are determined by the connected loads and network losses. Excitation primarily controls the terminal voltage and the sharing of reactive power between parallel sources. With only one generator, changing the field voltage cannot independently specify the reactive power demanded by a constant-power load.

Remark 8.1 – (Voltage dependence of the load)

The influence of the AVR on load power depends on the load model. Constant-impedance loads change their power approximately with the square of the voltage, whereas ideal constant-power loads maintain their specified P and Q by changing their current.

8.1.9 Switching Off the Loads

see figures: 8.9 and 8.10 (simulation-file: same name as the heading)

The three loads are disconnected at 9 s, 11 s, and 13 s. The load angle decreases after each switching operation. When the final load is removed, the frequency overshoots above its nominal value, which is the opposite of its response to a positive load step. In the chosen model, the frequency takes approximately 3.5 s to return to 50 Hz, because only the governor action and the comparatively small friction torque decelerate the rotor.

8.1.10 Scaling Quantities in the Waveform Viewer

Dimensionless internal signals can be displayed in engineering units by using the following scale factors in LTspice:

Quantity	Scale factor	Displayed unit
Load angle	1 deg/V	degree
Frequency	1 Hz/V	hertz
Active power	1 W/V	watt

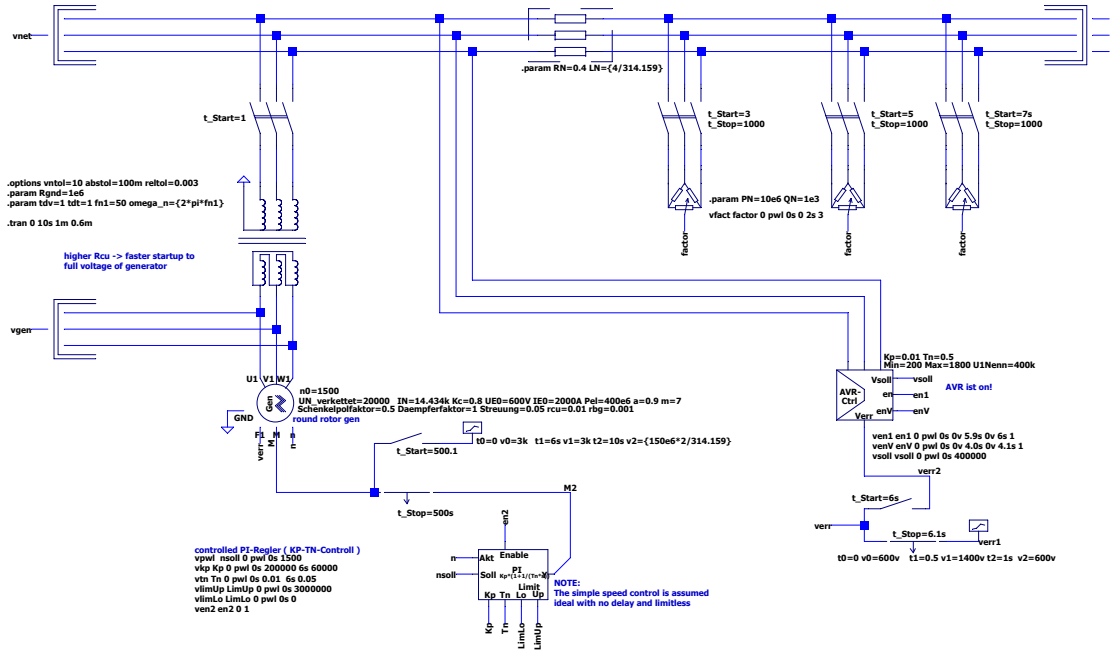


Figure 8.7: Comparison with a salient-pole machine - schematic

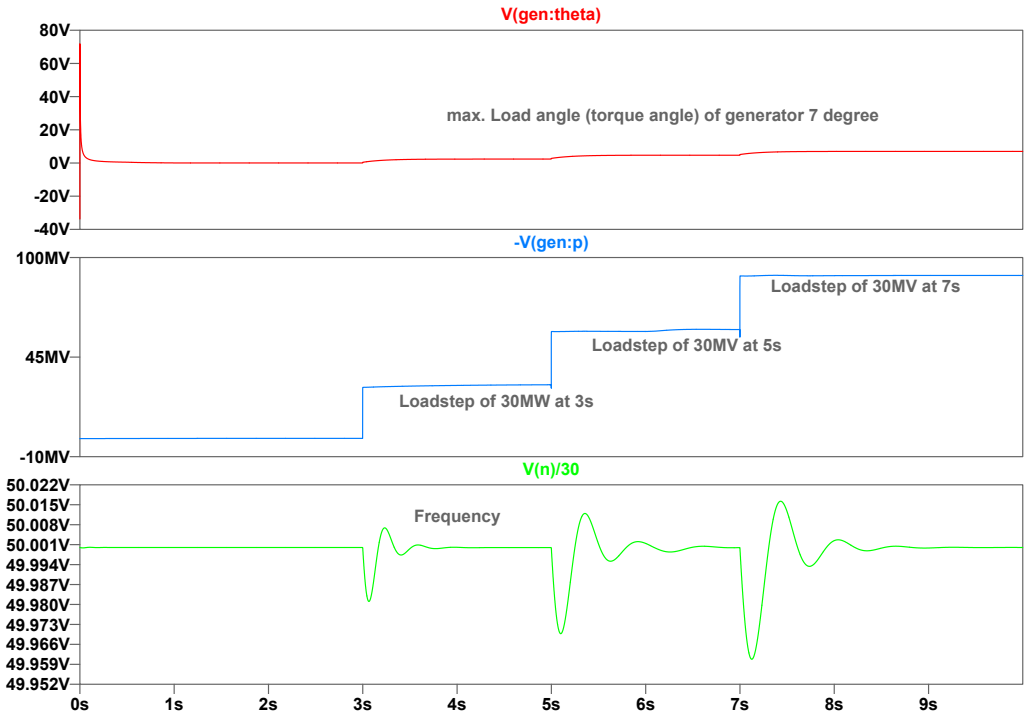


Figure 8.8: Comparison with a salient-pole machine - Waveform

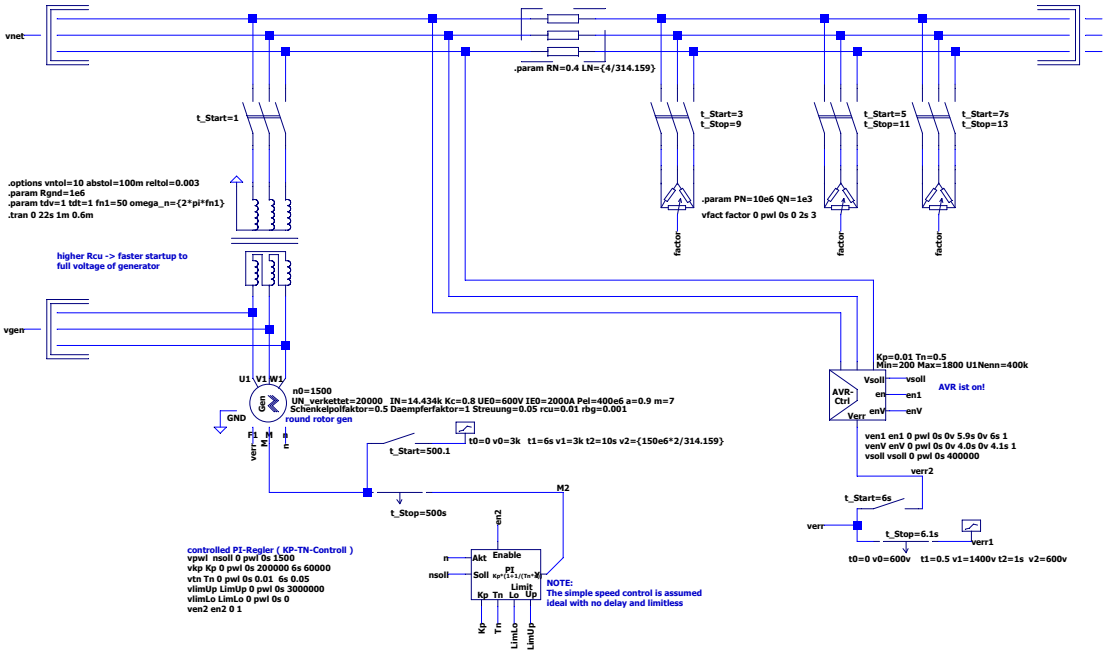


Figure 8.9: Switching off loads - schematic

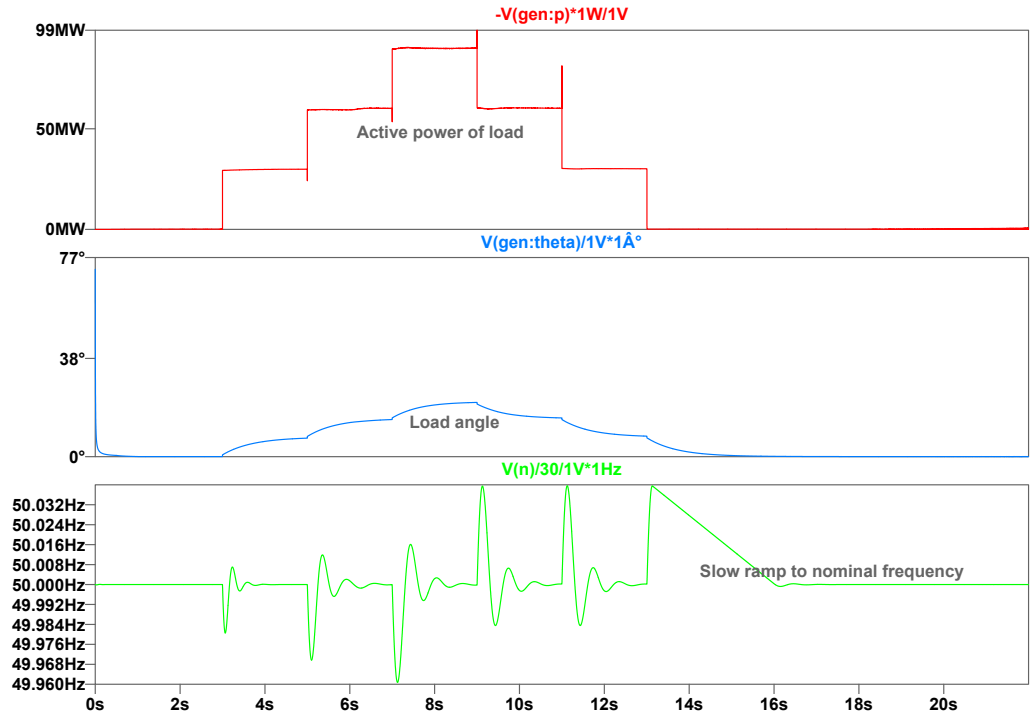


Figure 8.10: Switching off loads - Waveform

8.1.11 Synchronizing a Second Generator

see figures 8.11-8.13 (simulation-file: same name as the heading)

A second generator is added on the right-hand side together with its speed controller, but without an active AVR in this particular demonstration. The simulation end time is increased to 30 s. Generator 2 is connected at 12 s; its speed controller is disabled at that instant, and a constant-torque input becomes active at 12.1 s. A ramp from zero to 20 MW between 14 s and 18 s establishes its active-power output. A third load is switched at 20 s to examine the response after synchronization.

8.1.11.1 Synchronization Conditions

see figures 8.11 and 8.12 and 8.13

Between approximately 9.8 s and 10.2 s, the generator and network voltages have nearly equal magnitudes, although Generator 2 leads slightly in phase. Its speed reference can be reduced from 50 Hz to 49.98 Hz, corresponding to a small negative slip. At 12 s, the two voltages are approximately in phase and the switch can be closed.

Important 8.2 – (Conditions before synchronization)

Before connecting a synchronous generator to an energized network, verify the phase sequence, voltage magnitude, frequency, and phase angle. Closing the switch with an excessive mismatch can produce high currents and severe shaft-torque transients.

8.1.11.2 Power Sharing

see figures 8.11 and 8.12 and 8.13

Generator 2 supplies 20 MW, so Generator 1 reduces its output to approximately 40 MW. A subsequent 30 MW load step is supplied almost entirely by Generator 1, because the active-power controller of Generator 2 has been disabled. The final outputs are therefore approximately 70 MW for Generator 1 and 20 MW for Generator 2. Generator 2 exhibits larger transient speed deviations because it has no active governor in this example.

Remark 8.2 – (Regulation of parallel generators)

Disabling the AVR and governor of Generator 2 is a deliberate simplification of this example, not a general operating rule. In practical island systems, parallel generators normally retain coordinated voltage and speed control with appropriate droop settings.

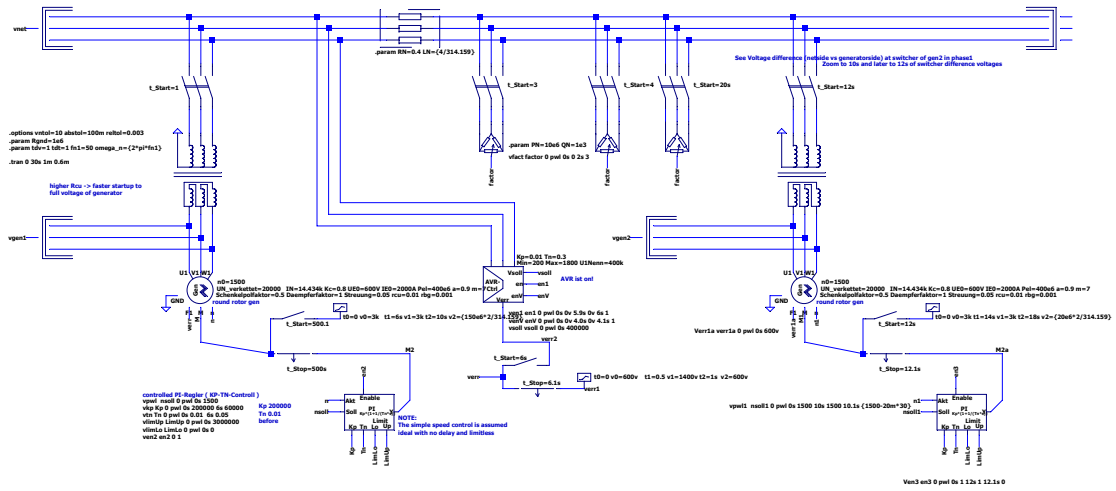


Figure 8.11: Synchronizing a second generator - schematic

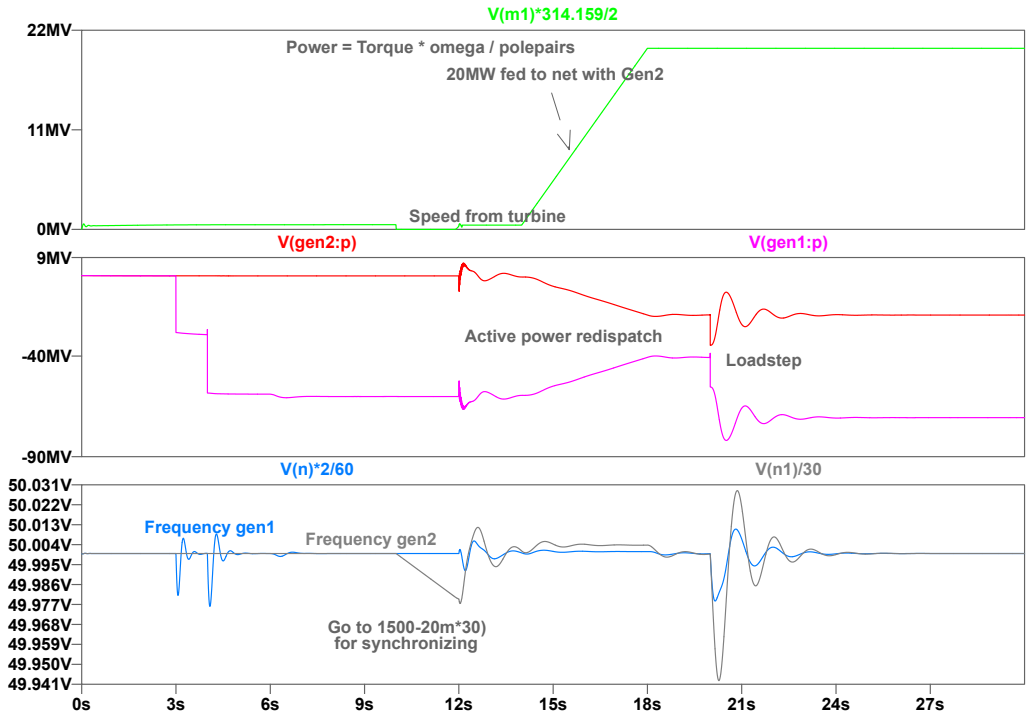


Figure 8.12: Synchronizing a second generator - Waveform

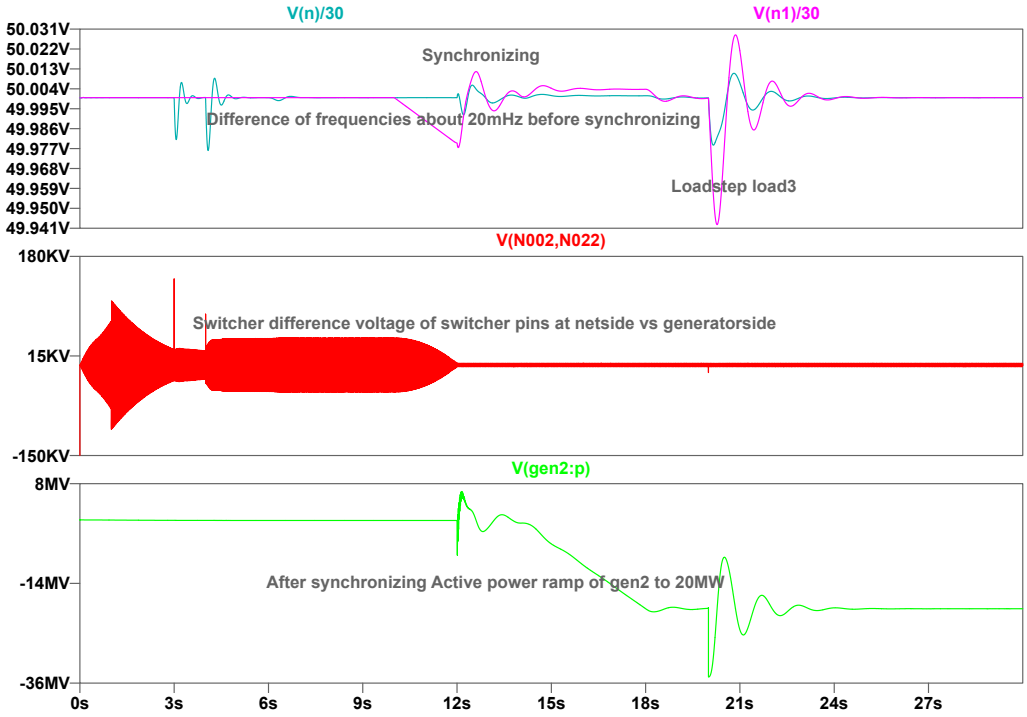


Figure 8.13: Synchronizing a second generator - Waveform2

8.2 Grid-Connected and Constant-Speed Operation

8.2.1 General Principle

The following simulations examine synchronization with a stiff grid and the subsequent delivery of active and reactive power. They also cover faults near and far from the generator, a single-phase-to-ground fault, operation at fixed speed, and the loss of synchronism when the transferable active power is exceeded.

8.2.2 Circuit Design

From left to right, the circuit contains:

- a 250 MV A, 13.8 kV, 60 Hz synchronous generator rated at 3 600 rpm;
- a speed governor and a 5 MW auxiliary load;
- an AVR measuring the 13.8 kV generator bus;
- a Yd11 step-up transformer;
- a synchronization switch and a 10 MW load;
- a switch for a three-phase-to-ground fault; and
- a grid impedance and a 230 kV stiff voltage source.

The high-voltage transformer winding is star-connected, while the generator-side winding is delta-connected. The delta provides a path for triplen harmonic currents. For the synchronization study, the transformer is made linear with `M=1`; the model parameter `err0=1` is also used to suppress nonlinear energization transients when full voltage is applied.

The rated excitation voltage is 200 V. The AVR is enabled at 0.1 s and provides an excitation range from 100 V to 400 V, allowing both underexcited and overexcited operation. The speed governor starts with a reference of 3 600 rpm.

The generator is synchronized at 1 s. Immediately afterwards, the governor is disabled and the mechanical input power is ramped from zero to 150 MW between 2 s and 10 s. For a machine with p pole pairs,

Calculation 8.1 – (Mechanical power and shaft torque)

$$P_{\mathrm{m}} = T_{\mathrm{m}}\omega_{\mathrm{m}}, \tag{8.1}$$

$$\omega_{\mathrm{e}} = p\,\omega_{\mathrm{m}}, \tag{8.2}$$

$$T_{\mathrm{m}} = \frac{pP_{\mathrm{m}}}{\omega_{\mathrm{e}}}. \tag{8.3}$$

For the two-pole turbo-generator, $p = 1$ and $\omega_{\mathrm{m}} = \omega_{\mathrm{e}}$.

To compensate the 30° displacement of the Yd11 transformer during the initial synchronization study, the grid-source phase parameter is set to `DPHI=330`. The exact sign depends on the voltage reference directions used in the schematic and should be confirmed from the voltage difference across the open synchronization switch.

8.2.3 Synchronization and Active-Power Output

see figures 8.14 and 8.15 (simulation-file: same name as the heading)

In a simulation from zero to 12 s, the synchronization condition at 1 s can be checked from the voltage difference across the open switch. In this example, its instantaneous magnitude is approximately 4 kV, and the frequencies on both sides are close to 60 Hz.

Closing the switch produces transient active-power and frequency oscillations. While the active power is being increased, the rotor speed can remain slightly above synchronous speed. Once the input-power ramp becomes constant, the frequency returns to 60 Hz. The power ramp is generated by applying the corresponding torque ramp to the generator input pin `M-PIN`.

Remark 8.3 – (Accelerated simulation)

An active-power ramp of approximately 19 MW/s is faster than would normally be used for a large generating unit. It is selected here only to keep the simulation time short.

8.2.4 Three-Phase Fault in the Grid

see figures 8.16-8.18 (simulation-file: same name as the heading)

A three-phase-to-ground fault is applied in the grid from 12 s to 12.1 s. For a restarted simulation using a saved operating point, the corresponding local time interval may appear as 0.5 s to 0.6 s. The AVR partly supports the voltage by increasing the excitation.

During the fault, the exported active power decreases and the load angle moves towards zero. Because the mechanical input power initially remains unchanged, the rotor accelerates and its kinetic energy increases until the fault is cleared. With an end time of 13 s and a saved state at 11.5 s, the peak fault current is approximately 58 kA; the high current persists for the 0.1 s fault duration.

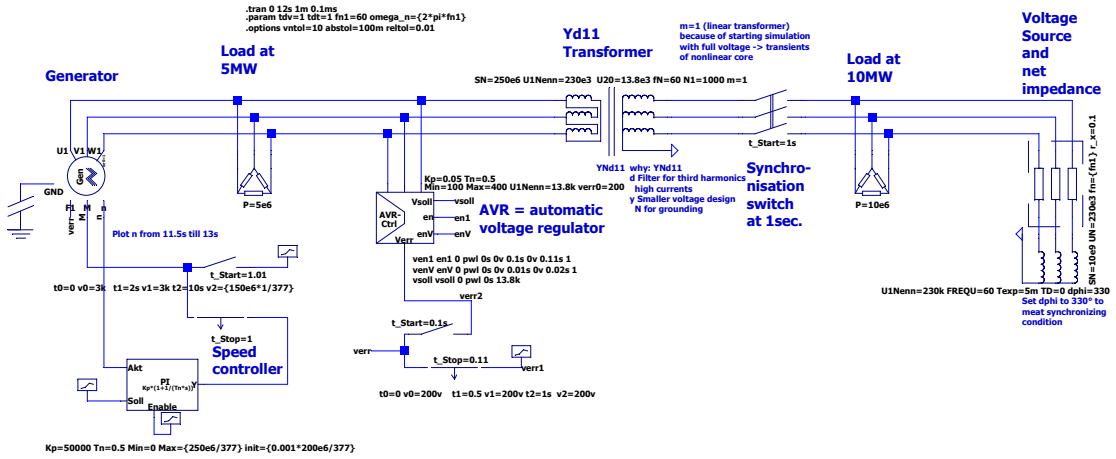


Figure 8.14: Synchronization and active power output - schematic

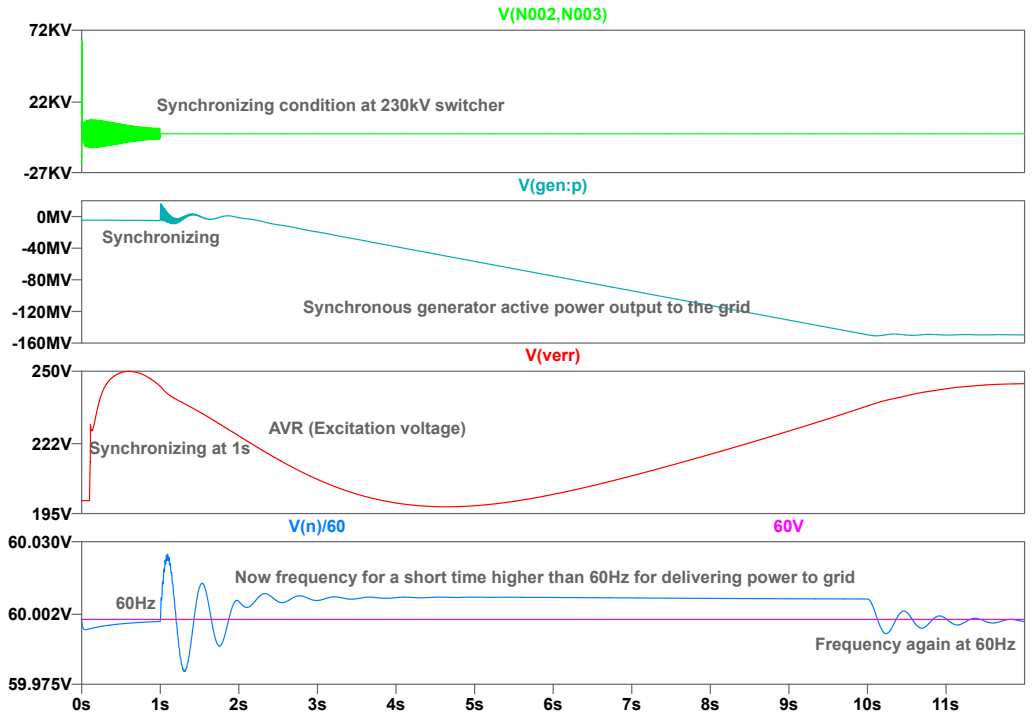


Figure 8.15: Synchronization and active power output - Waveform

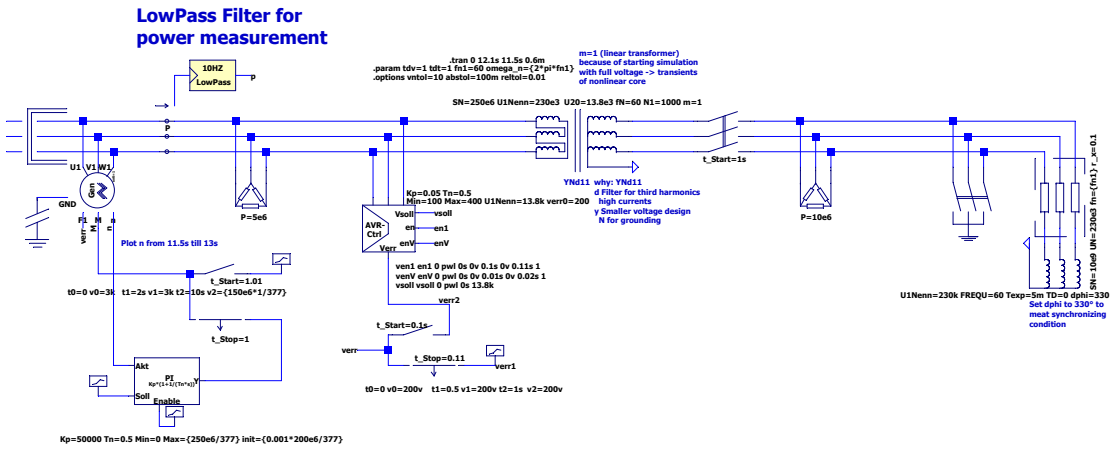


Figure 8.16: Three-phase fault in the grid - schematic

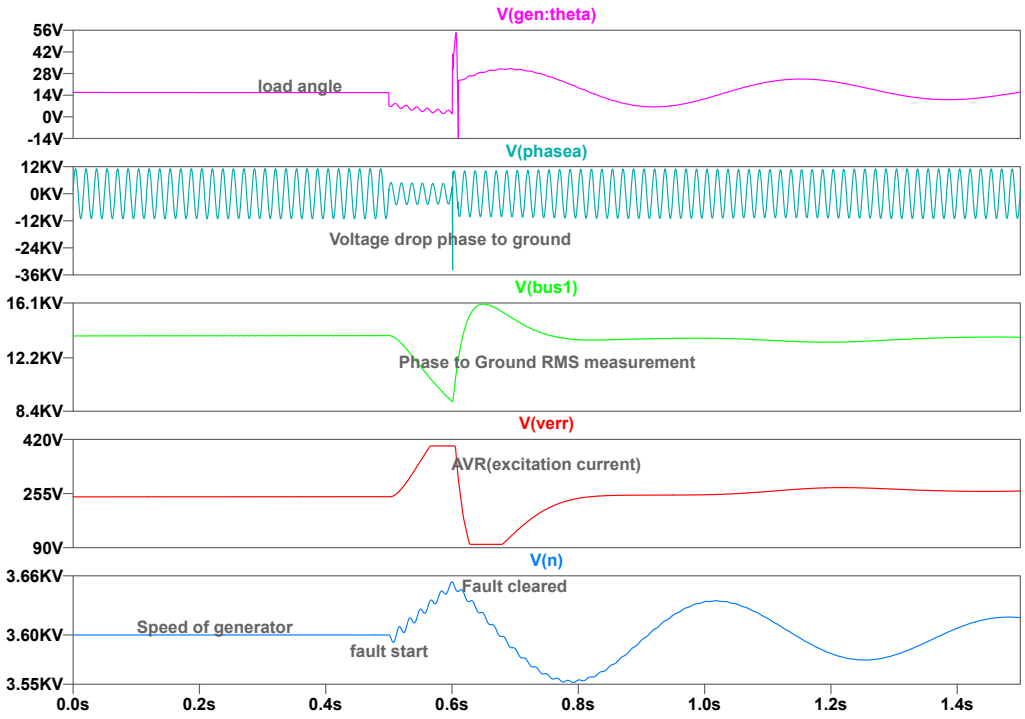


Figure 8.17: Three-phase fault in the grid - Waveform

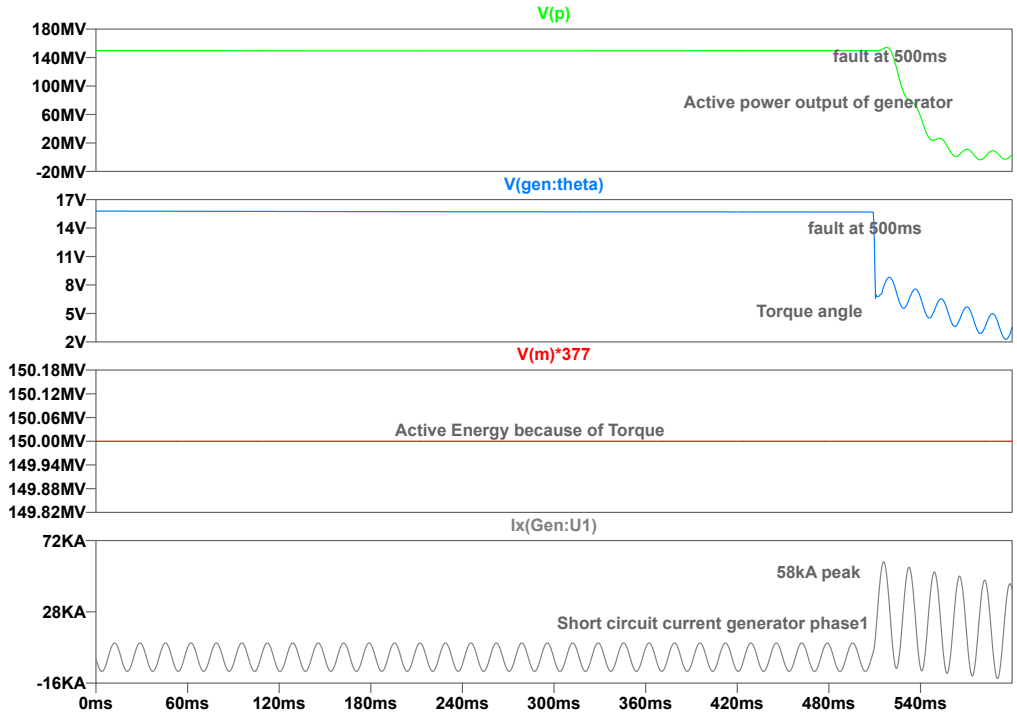


Figure 8.18: Three-phase fault in the grid - Waveform2

8.2.5 Three-Phase Fault Near the Generator

see figures 8.19 and 8.20 (simulation-file: same name as the heading)

The fault is now moved directly to the generator busbar. The terminal voltage drops more strongly, the peak current increases to approximately 80 kA, and the load angle rises beyond 90° in the reported simulation.

Although a terminal short circuit is often described as unloading the external network, the generator can still deliver substantial electrical power into the fault resistance and the internal machine resistances. Here, the power dissipated in the fault path is approximately 200 MW, about 50 MW greater than the applied mechanical input. The deficit is supplied by the rotor's *kinetic* energy, causing the speed to decrease. After fault clearing, the current and speed exhibit an approximately 2 Hz electromechanical oscillation before returning to their pre-fault values.

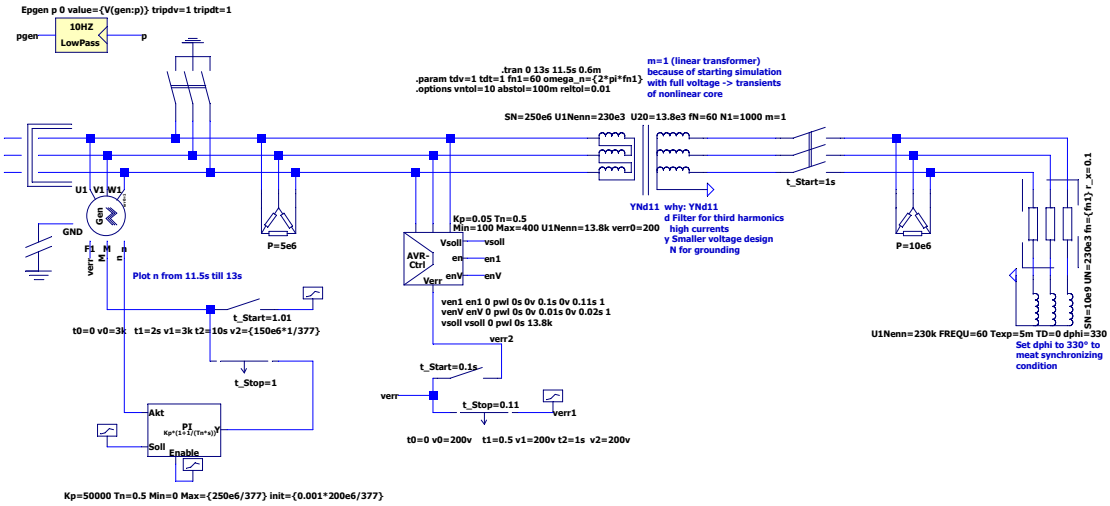


Figure 8.19: Three phase fault near the generator - Schematic

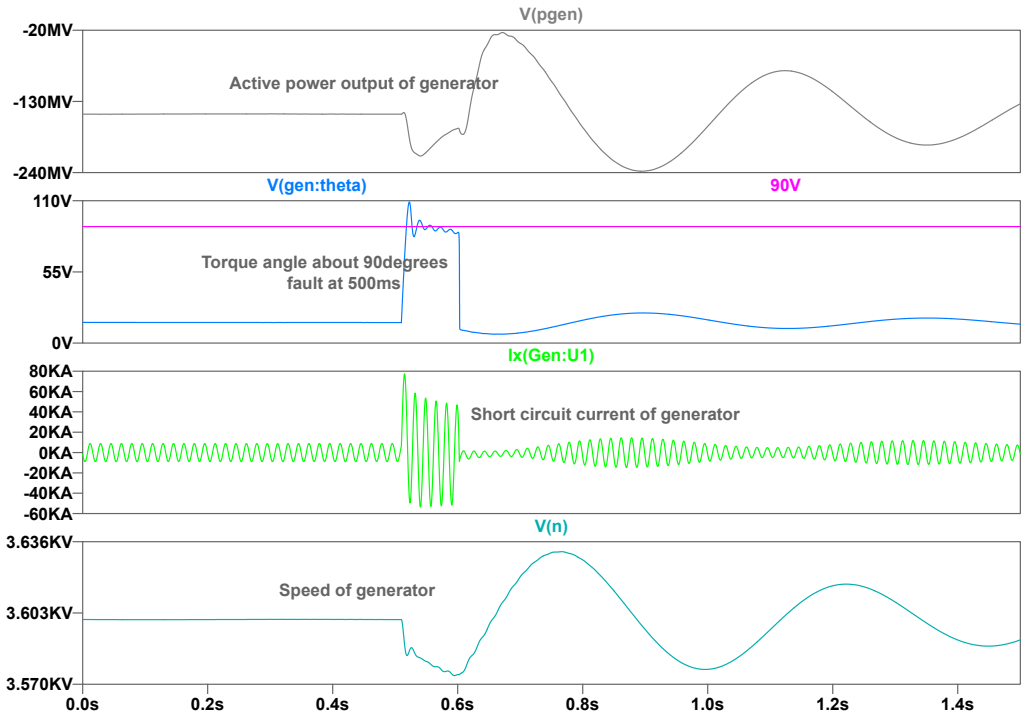


Figure 8.20: Three phase fault near the generator - Waveform

8.2.6 Uncleared Generator-Side Fault

see figures 8.21 and 8.22 (simulation-file: same name as the heading)

For this study, the generator is initially operated without an external load; only the small friction-torque input remains. Setting the fault switch parameter `Tstop` to 100 s keeps the fault closed throughout the relevant simulation interval.

The calculated electrical power initially rises to approximately 450 MW, producing strong braking torque. The rotor angle decreases, crosses zero after approximately 750 ms, and continues towards −180°. The machine then loses synchronism and passes periodically between generating and motoring operation. Rotor acceleration, deceleration, power reversal, and pole slipping repeat as long as the fault and the stiff grid remain connected.

Unstable operating condition

An uncleared generator-side short circuit leads to extreme electrical and mechanical stress and loss of synchronism. Real protection systems must clear such a fault rapidly and isolate the affected unit.

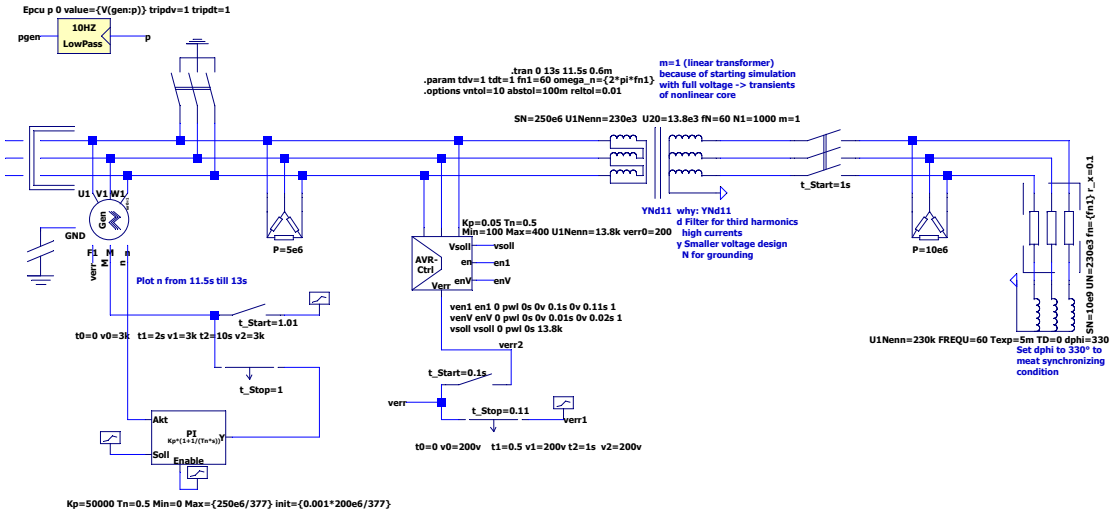


Figure 8.21: Uncleared Generator-side Fault - Schematic

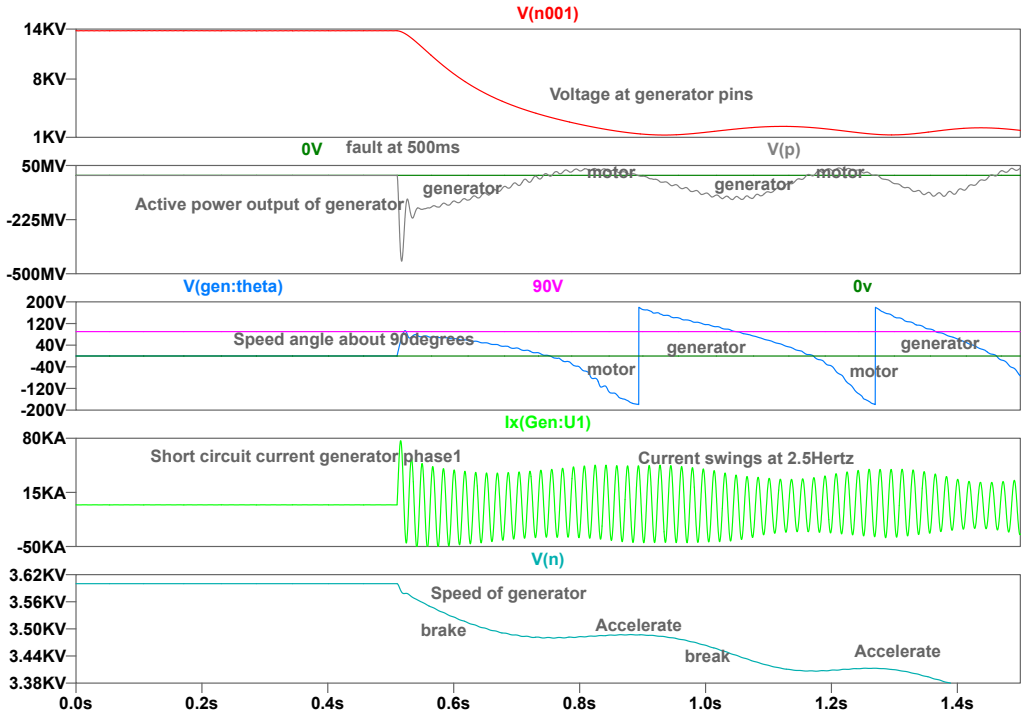


Figure 8.22: Uncleared Generator-side Fault - Waveform

8.2.7 Generator-Side Fault at Constant Speed

see figures 8.23 and 8.24 (simulation-file: same name as the heading)

When the generator is constrained to constant speed, the calculated peak short-circuit current rises to approximately 95 kA. The instantaneous electromagnetic torque, expressed as an equivalent power at the fixed speed, exceeds 900 MW at fault inception.

The current waveform shows the short subtransient interval, followed by the longer transient interval and finally the steady-state short-circuit current. Although the applied excitation voltage remains constant, transformer action between the stator and rotor circuits causes the field current to rise to about 4.4 kA before it decays towards its nominal value of approximately 200 A. The generator terminal voltage approaches zero, and the calculated load angle is approximately 90°.

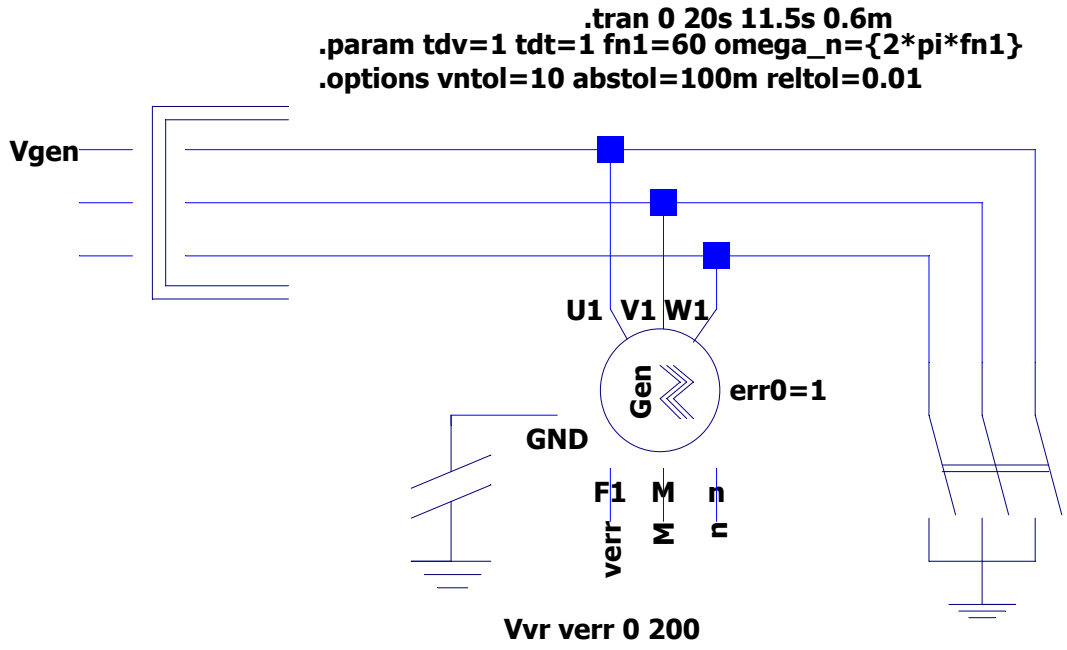


Figure 8.23: Generator-side fault at constant speed - Schematic

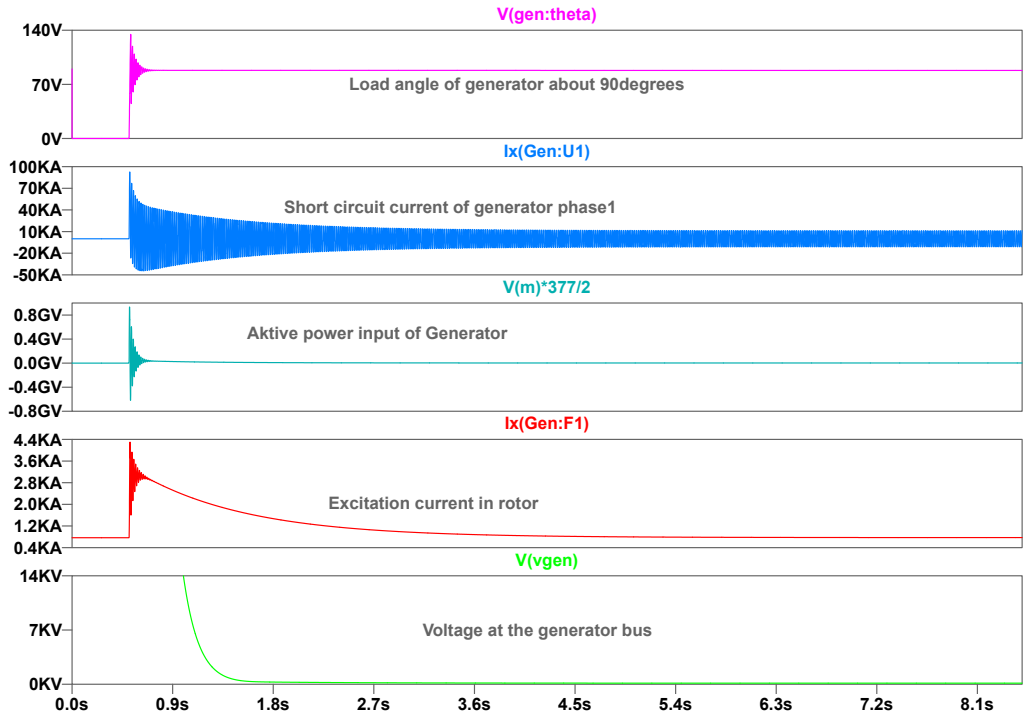


Figure 8.24: Generator-side fault at constant speed - Waveform

8.2.8 Single-Phase-to-Ground Fault Near the Generator

see figures 8.25 and 8.26 (simulation-file: same name as the heading)

For the single-phase-to-ground fault, the generator neutral point is grounded. The simulated peak fault current is approximately 110 kA, which is higher than the three-phase peak in this particular network and grounding configuration.

Whereas the principal transients of the three-phase fault decay within about 100 ms, the asymmetrical ground fault continues to excite negative-sequence components. The system can be decomposed into positive-sequence (+omega) and negative-sequence (-omega) components. The negative system rotates in the opposite direction to the rotor and induces voltages at twice the fundamental frequency in the field and damper windings. These fields can now also be divided into two rotating fields. One rotates at +omega and the other at –omega. The field with +omega compensates for the stator field, and the field with -omega now has 3*omega relative to the stator and induces voltages there at triple the frequency. This continues to repeat, but only the fields with 2x to 4x the frequency are significant.

An FFT evaluated between approximately 600 ms and 800 ms shows components at 60 Hz, 120 Hz, and 240 Hz in the excitation current. The stator current contains prominent components at 60 Hz and 180 Hz. Similar effects can occur, with smaller magnitudes, when a synchronous generator supplies an asymmetrical load.

Remark 8.4 – (Interpretation of FFT components)

The exact harmonic spectrum depends on the machine model, saturation, reference-frame definitions, fault inception angle, grounding impedance, and the selected FFT window. The listed frequency components are results of this simulation and should not be interpreted as universal amplitudes for every synchronous machine.

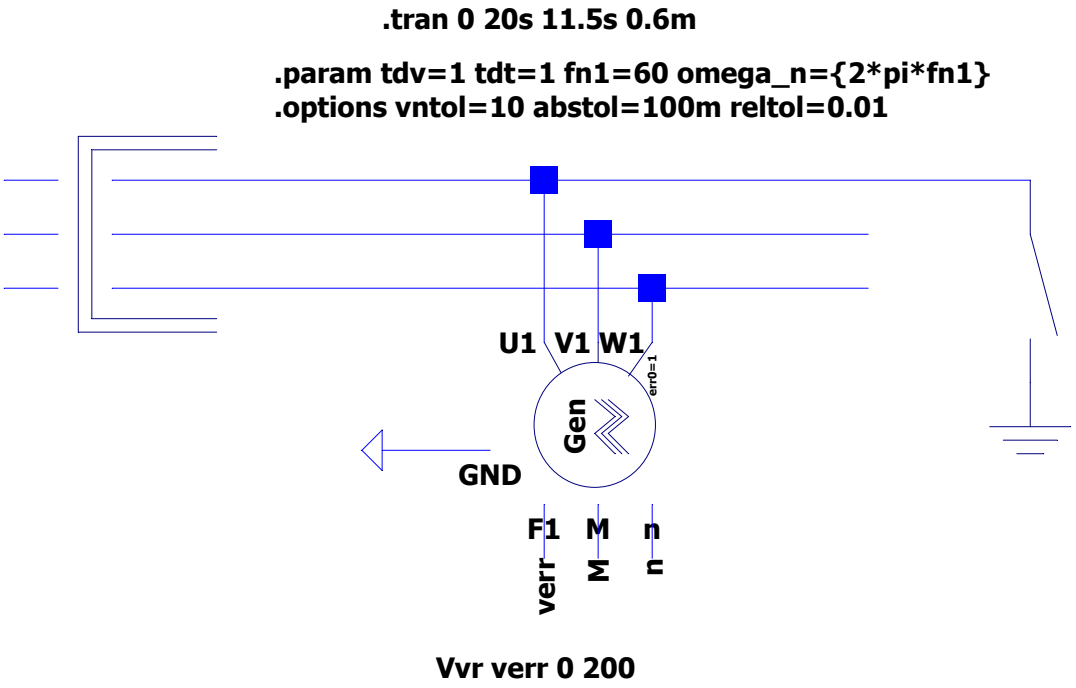


Figure 8.25: Single-phase to ground fault near the generator - Schematic

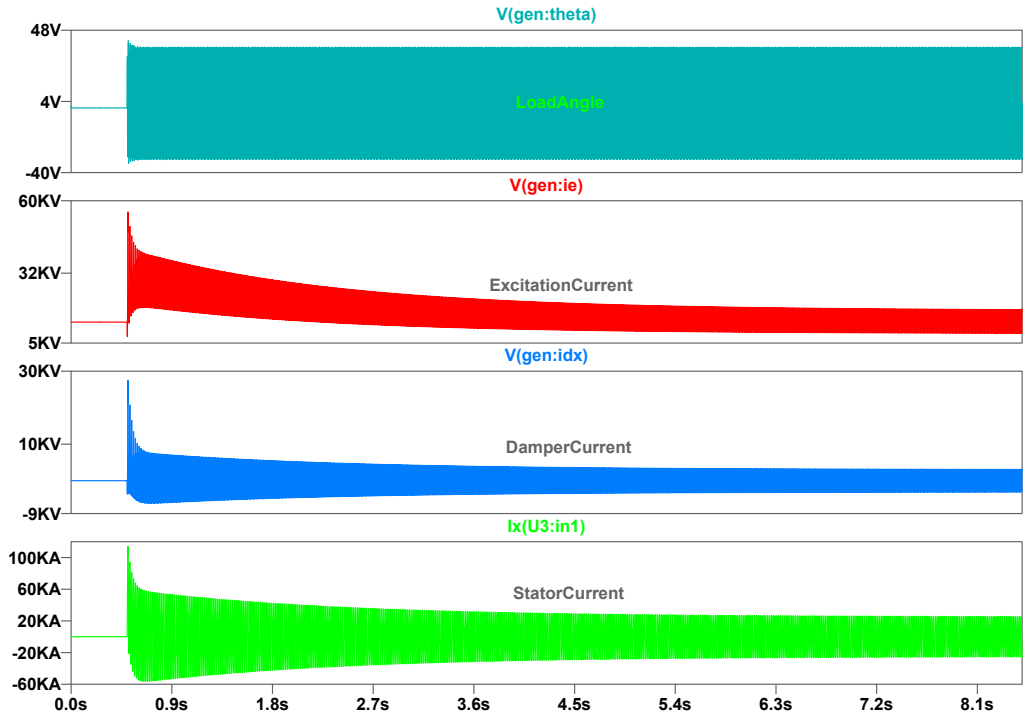


Figure 8.26: Single-phase to ground fault near the generator - Waveform

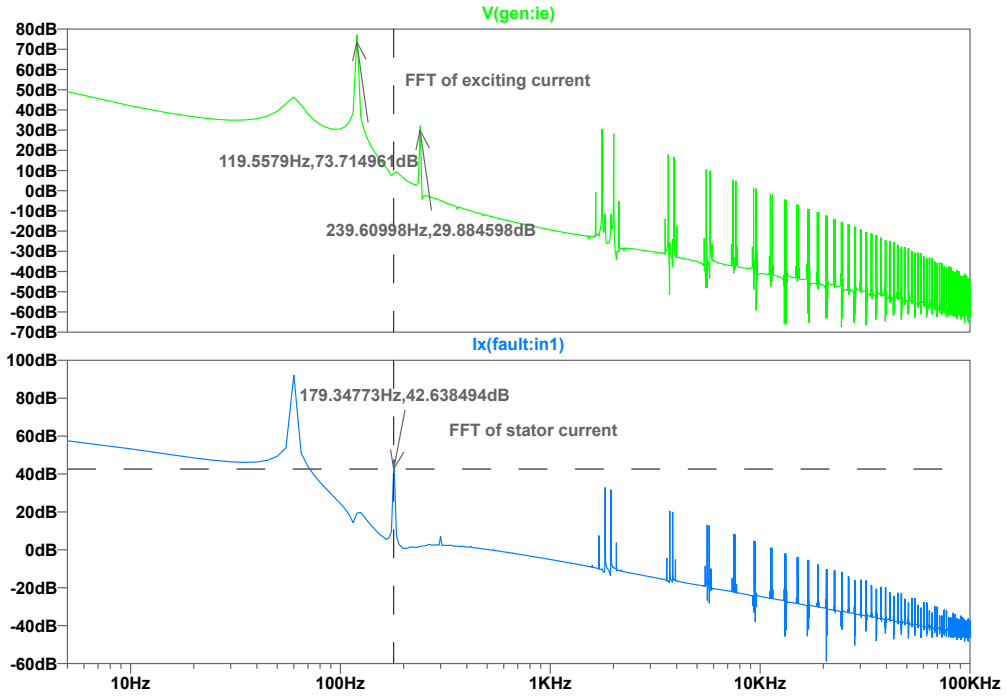


Figure 8.27: Single-phase to ground fault near the generator - Waveform

8.2.9 Phasor Diagram During a Continuous Fault

see figures 8.28-8.33 (simulation-file: same name as the heading)

For a steady-state phasor diagram at constant speed, read the values `Usx`, `Usy`, `Isx`, and `Isy` from the LTspice waveform viewer at 8.5 s. Enter these quantities in the wxMaxima file

`synchrongenerator.wxmx`

The script scales the values and calculates the additional quantities required by the phasor diagram. The values of X_d and X_q are taken from `synchrongeneratordata.wxmx`. Finally, transfer the calculated vectors to the GeoGebra file `synchrongenerator.ggb`.

Because `Usx` and `Usy` are very small, their vector sum, the terminal voltage U, is almost zero. The internal excitation voltage is large and lies on the quadrature axis. The stator current lies approximately on the negative real axis, whereas the small terminal voltage lies on the positive real axis. Under the sign convention used here, the machine therefore delivers mainly active power into the resistive fault path. The corresponding loss is approximately 2 MW. For a subtransient phasor diagram shortly after fault inception, zoom into the interval from 500 ms to 520 ms and read the values at approximately 515.95 ms. In addition to the stator quantities, record the damper-current components `IDx` and `IDy`. Scale these values in

`synchrongeneratorwithdamper.wxmx`

before transferring them to GeoGebra. The resulting diagram shows the strong initial influence of the damper winding.

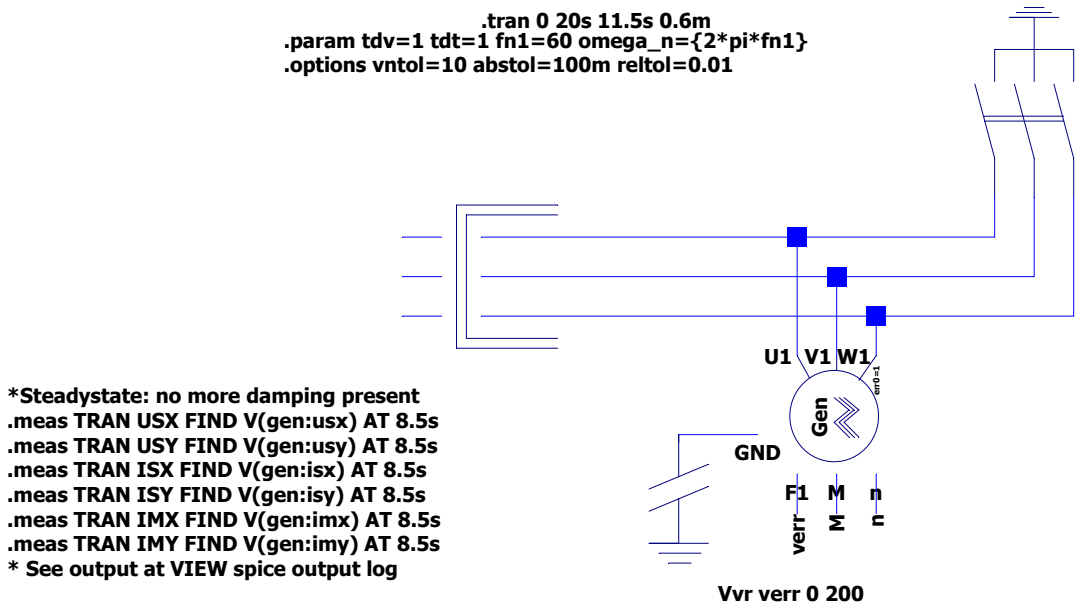


Figure 8.28: Phasor diagram during a sustained fault - Schematic

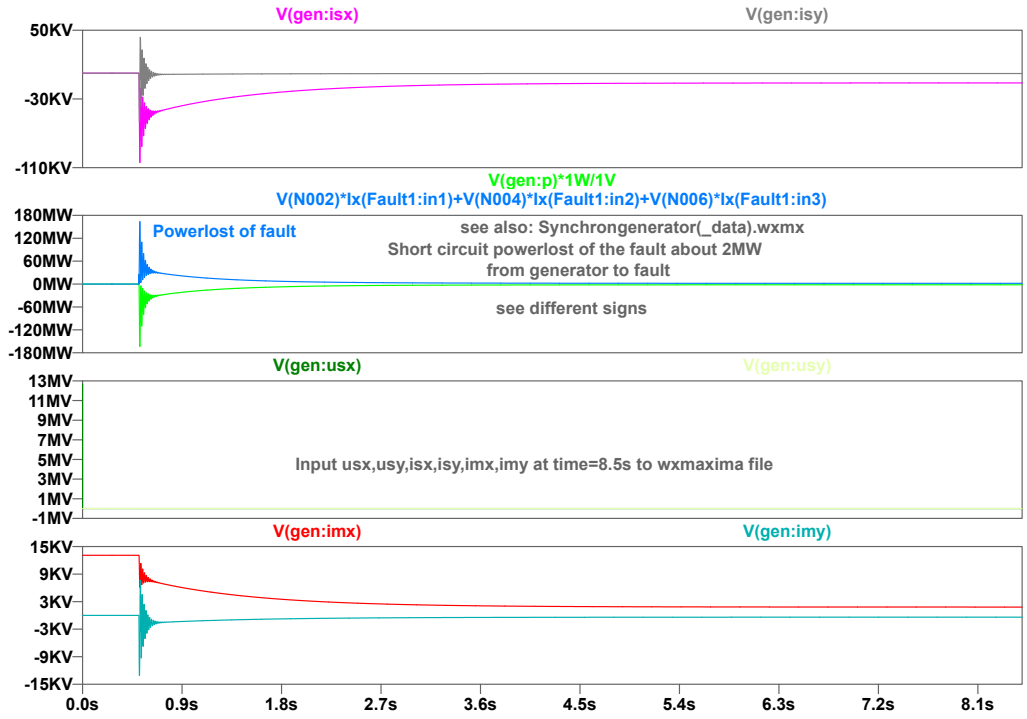


Figure 8.29: Phasor diagram during a sustained fault - Waveform

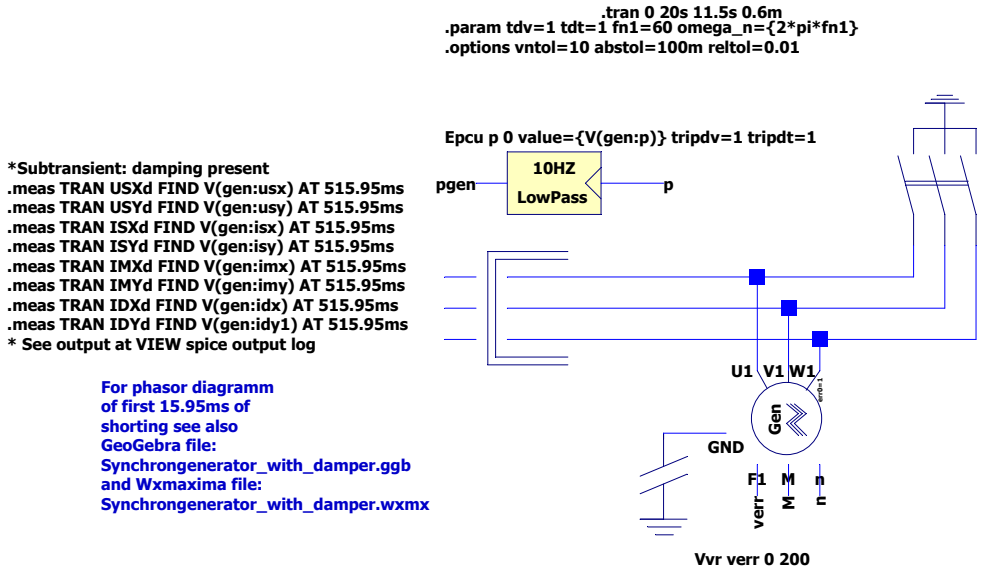


Figure 8.30: Phasor diagram during a sustained fault with damper - Schematic

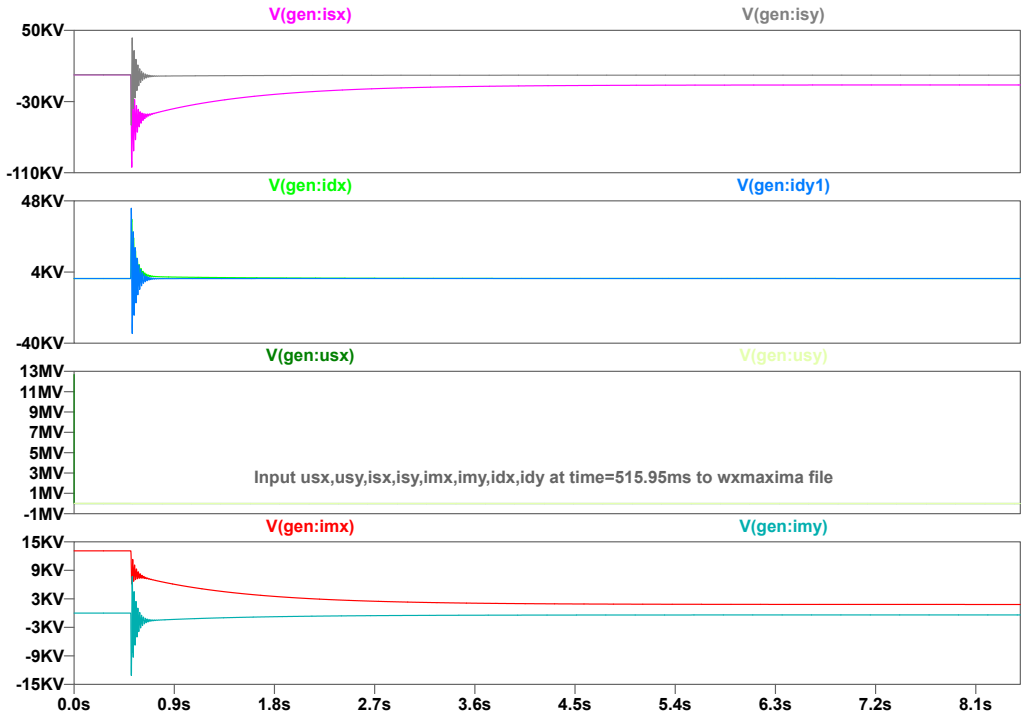


Figure 8.31: Phasor diagram during a sustained fault with damper - Waveform

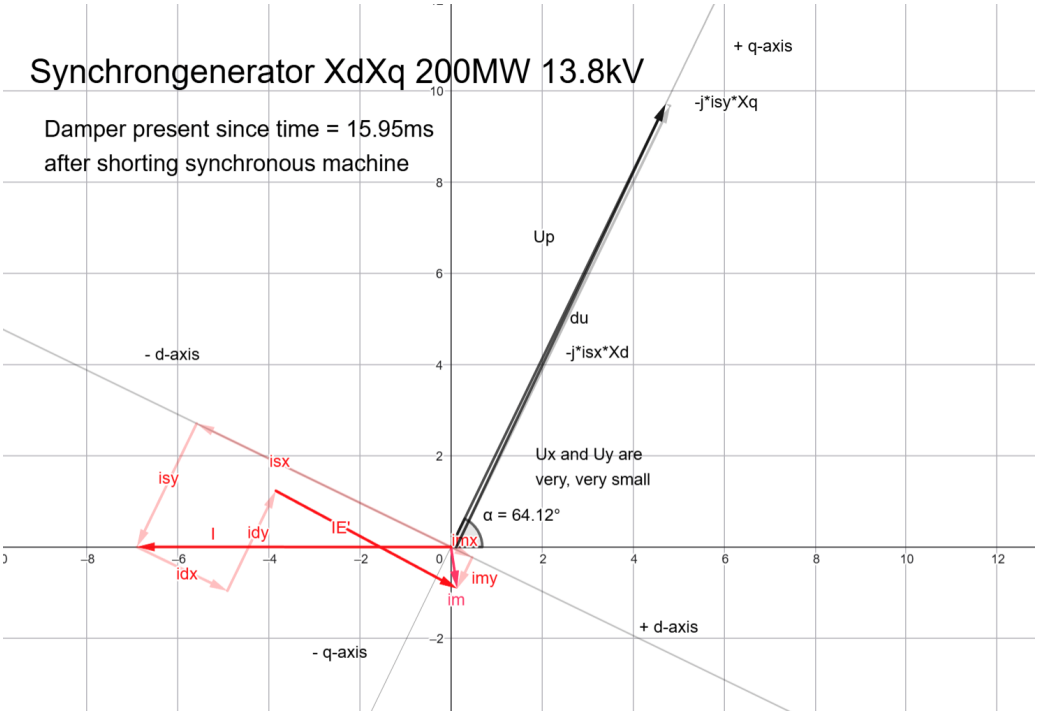


Figure 8.32: Phasor Diagramm without damper subtransient - Waveform

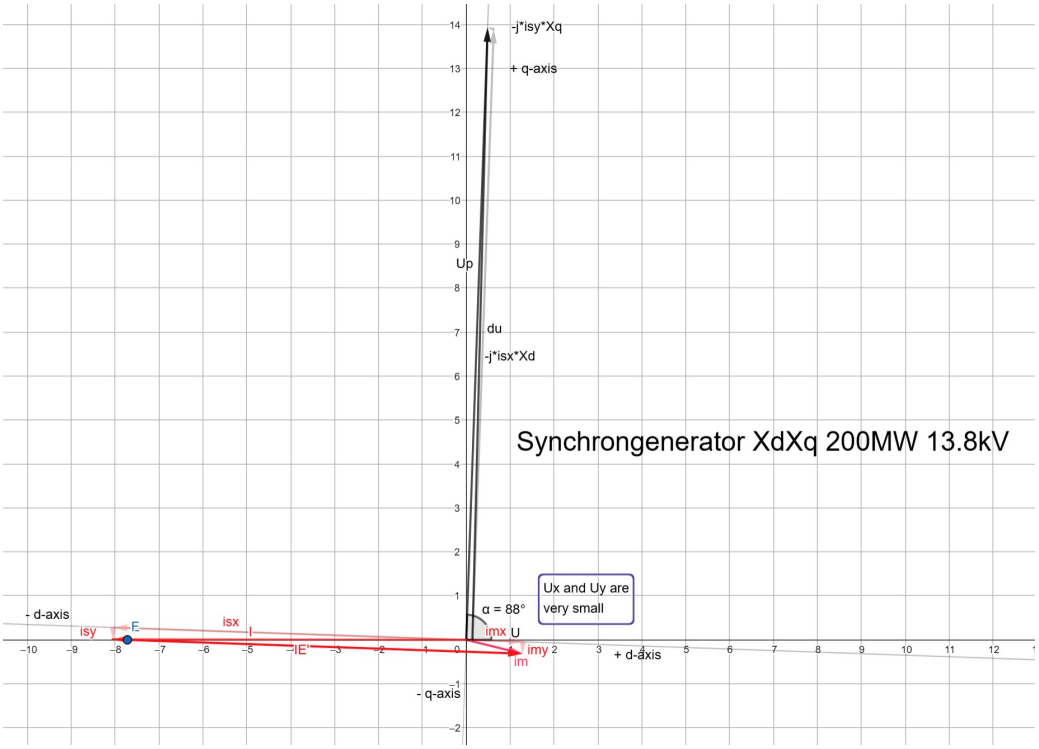


Figure 8.33: Phasor Diagramm with damper steady state - Waveform

8.2.10 Active- and Reactive-Power Control

see figures 8.34-8.37 (simulation-file: same name as the heading)

Active power is controlled through the mechanical torque applied at the generator's `M` input. The speed governor must first be changed from speed-regulating operation to the intended grid-connected power-control mode. In the example, the machine is ramped to 100 MW, during which the load angle rises to approximately 10°. The fault switch is disabled by placing its stop time outside the simulated interval.

Reactive power is varied by changing the field supply at `Verr` from 200 V to 100 V between 11 s and 13 s, and then from 100 V to 400 V between 13 s and 15 s. Increasing the excitation reduces the load angle for the chosen active power and improves the steady-state stability margin. Because the stiff grid largely fixes the terminal voltage, excitation mainly changes reactive power. This differs from island operation, where excitation has a much stronger influence on voltage.

For a deliberately fast demonstration, the field-resistance parameter can be set to `RE=1e4`. At 200 V, the reactive-power exchange is approximately zero. Reducing the excitation to 100 V makes the underexcited generator absorb reactive power; increasing it to 400 V makes the overexcited generator supply reactive power.

Remark 8.5 – (Artificial field resistance)

Changing `RE` to accelerate the response is useful for demonstrating the qualitative relationship between excitation and reactive power, but it no longer represents the original physical field time constant.

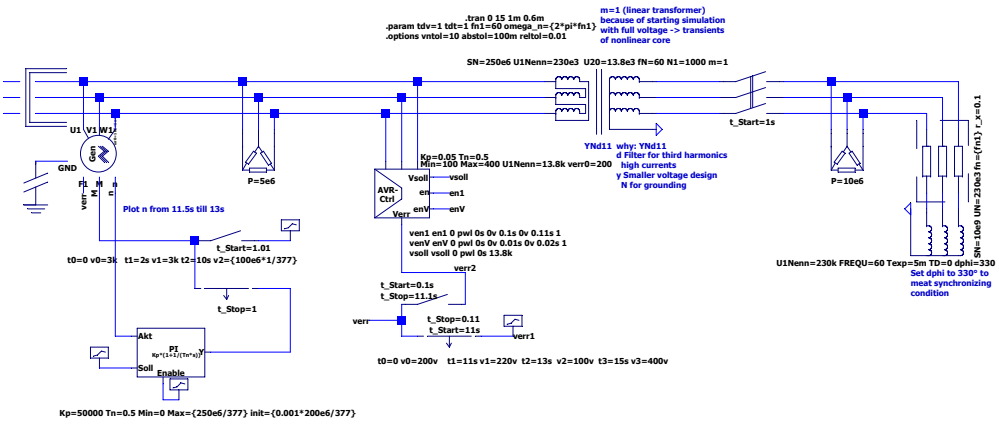


Figure 8.34: Active and reactive power control - Schematic

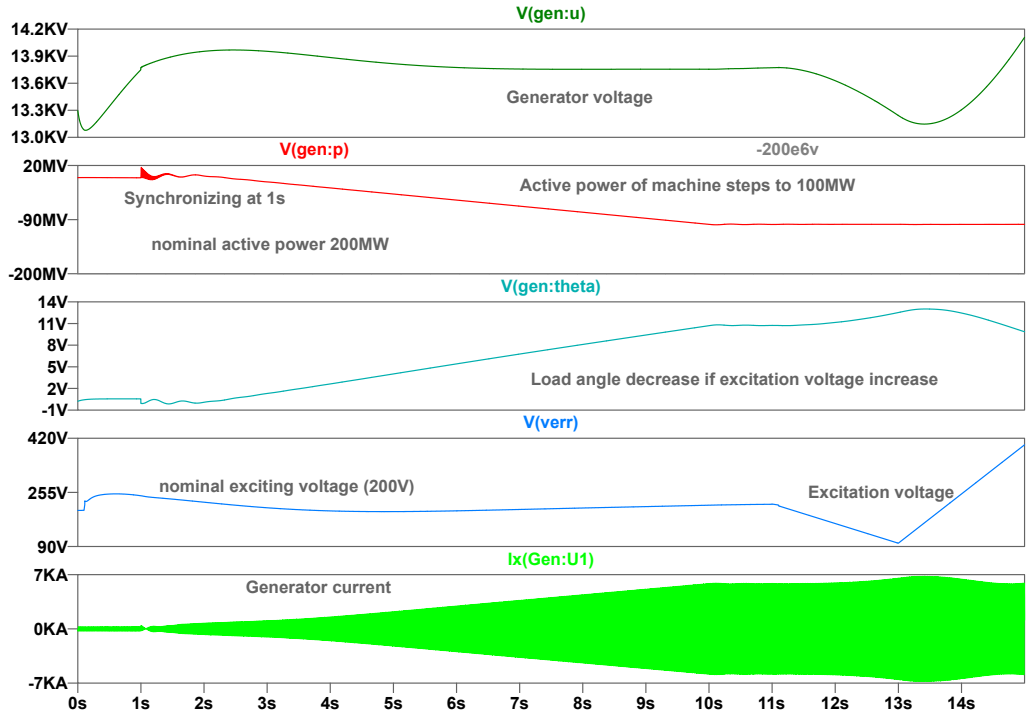


Figure 8.35: Active and reactive power control - Waveform

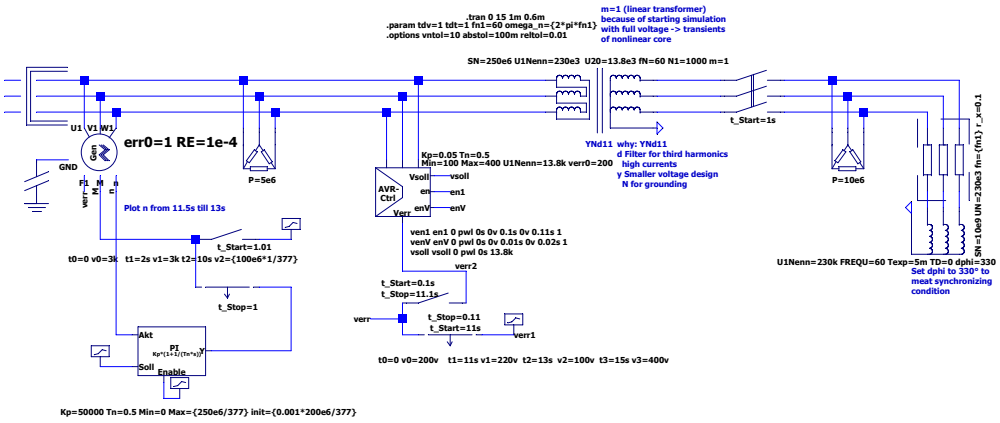


Figure 8.36: Active and reactive power control zoomed - Schematic

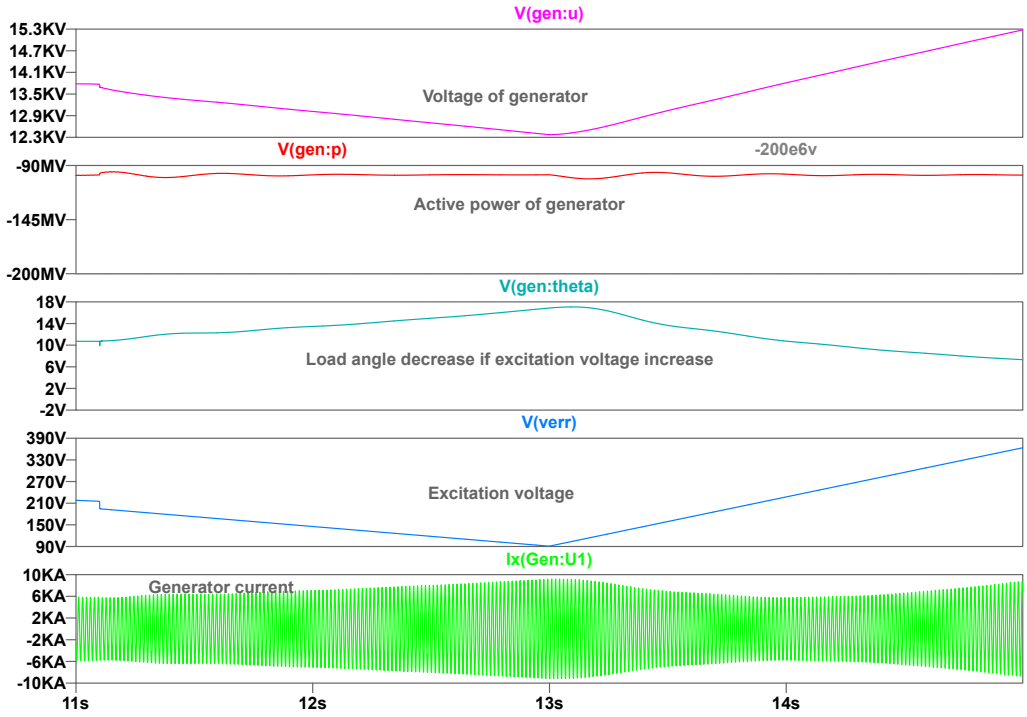


Figure 8.37: Active and reactive power control zoomed - Waveform

8.2.11 Phasor Diagrams for Overexcited and Underexcited Operation

see figures 8.38-8.40 (simulation-file: same name as the heading)

An overexcited synchronous generator operates with a field current higher than the value required for approximately unity power factor. The stronger rotor magnetic field causes the generator to supply reactive power to the electrical network. Its stator current is typically lagging with respect to the terminal voltage. Overexcited generators therefore support the grid voltage and can be used for voltage regulation and reactive-power compensation. An underexcited synchronous generator operates with a reduced field current. In this condition, the generator absorbs reactive power from the network. The stator current is typically leading with respect to the terminal voltage. Underexcited operation can therefore be used to reduce excessive grid voltage. For a generator delivering approximately constant active power, changing the excitation mainly changes its reactive power and power factor:

- Overexcited: supplies reactive power ⟶ lagging power factor,
- Underexcited: absorbs reactive power, and
- Normal excitation: reactive power is approximately zero ⟶ power factor close to unity

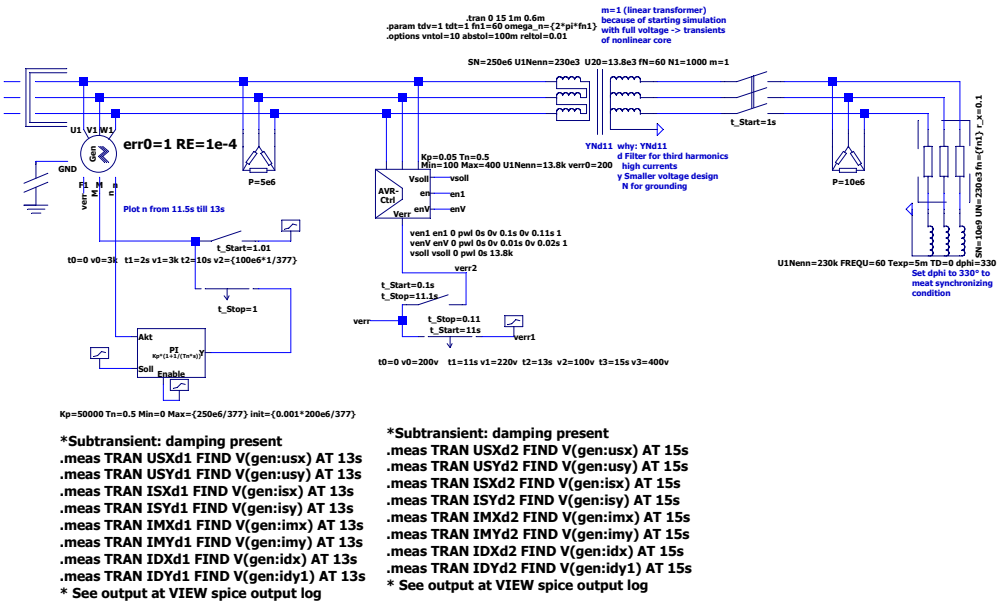


Figure 8.38: synchronous generator underexcited/overexcited - schematic

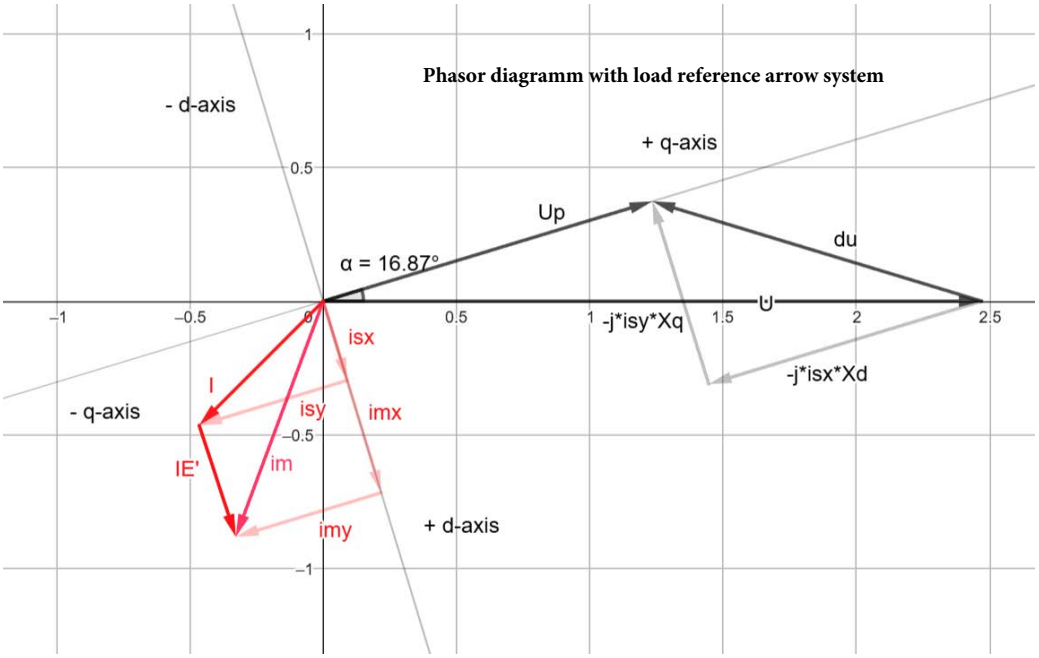


Figure 8.39: synchronous generator underexcited (plotted with GeoGebra) - phasor diagramm

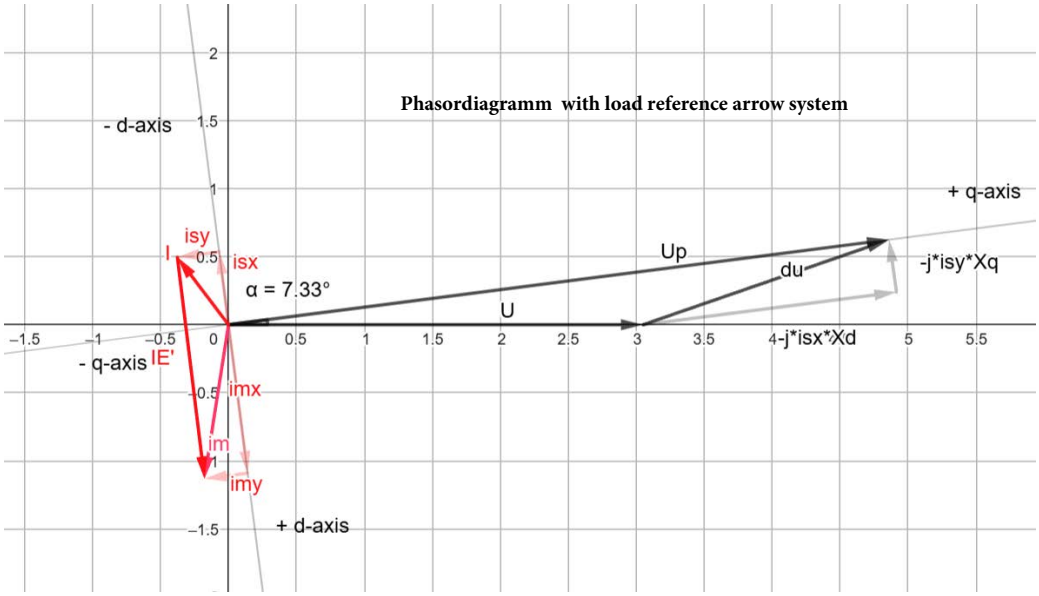


Figure 8.40: synchronous generator overexcited (plotted with GeoGebra) - phasor diagramm

8.2.12 Maximum Transferable Active Power

see figures 8.41-8.42 (simulation-file: same name as the heading)

If the mechanical input power is increased continuously, or if excitation is reduced excessively, the load angle rises towards the static stability limit. Its exact location depends on the machine type, excitation, network impedance, and saliency. A round-rotor approximation has its theoretical maximum near 90°; the reluctance-torque term of a salient-pole machine changes the power-angle characteristic and can shift this maximum.

For secure operation, a substantial angular margin must be maintained. In this example, the load angle should remain below approximately 60°. If the stable operating range is exceeded, the rotor slips poles, active power oscillates, and the terminal voltage can collapse severely.

Loss of synchronism

Operation near or beyond the maximum of the power-angle characteristic must be avoided. The simulation is intended to demonstrate instability, not an allowable continuous operating condition.

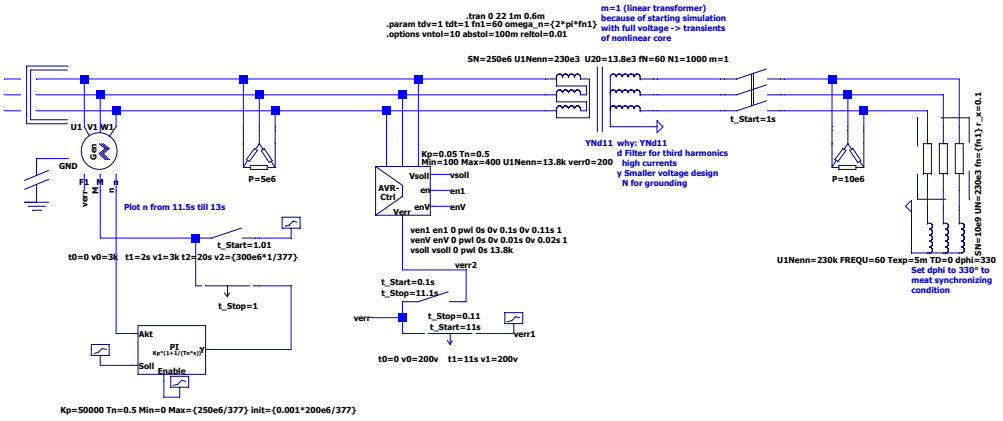


Figure 8.41: Maximum Transferable Active Power - Schematic

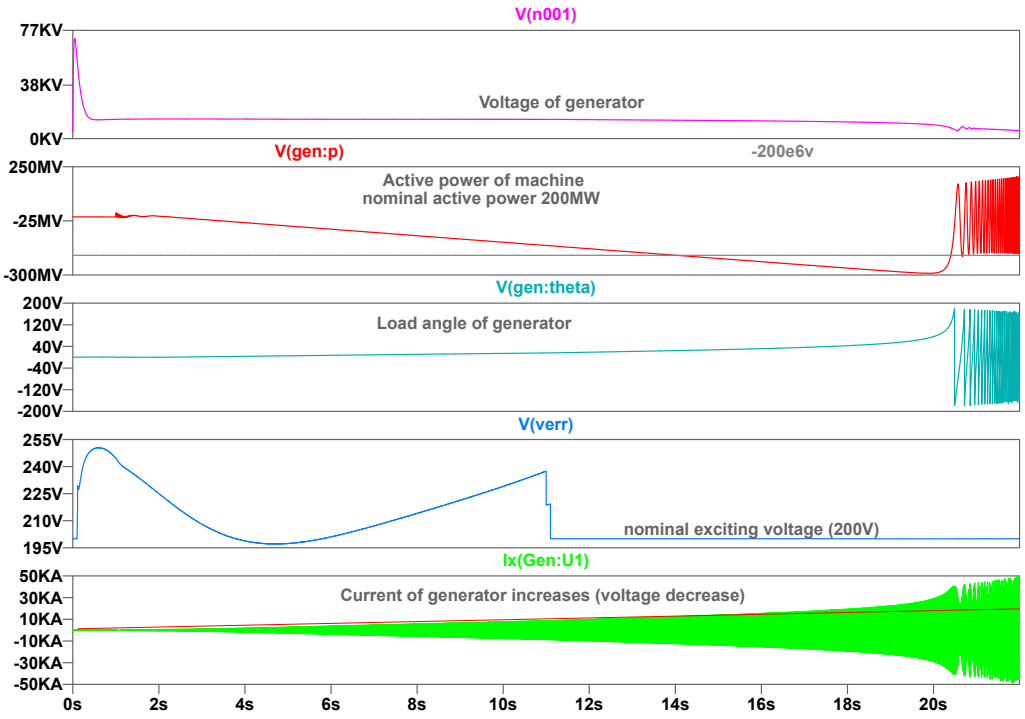


Figure 8.42: Maximum Transferable Active Power - Waveform

Chapter Summary

In island operation, voltage and frequency must be established by the generating units. The AVR regulates voltage, while the governor balances mechanical power and load demand. Parallel island generators require coordinated control and careful synchronization.

When connected to a stiff grid, mechanical torque mainly controls active power, whereas excitation mainly controls reactive power. Fault location, fault symmetry, grounding, rotor dynamics, and the selected mechanical-speed model strongly influence the short-circuit current and electromechanical response. Excessive load angle leads to loss of synchronism and must be prevented by suitable operating limits and protection.

9 Characteristic Curves of a Synchronous Generator

The steady-state behaviour of a synchronous generator can be described by a set of characteristic curves. Each characteristic is obtained by keeping certain operating quantities constant while one selected quantity is varied. Together, the curves describe excitation requirements, magnetic saturation, voltage regulation, short-circuit behaviour, reactive-power exchange, and steady-state operating limits.

The most important characteristics are:

- the open-circuit characteristic,
- the short-circuit characteristic,
- the load or external characteristic,
- the regulation characteristic, and
- the V-curves.

For the conventional tests, the generator is normally driven at constant rated speed so that its electrical frequency remains constant. The test machine used to record the characteristic curves has a phase-to-phase voltage of 400 V, a rated current of 21 A, and a rated power of 12 kW. The exact values can be found in the WxMaxima file `Synchrongenerator_400V.wxmx`.

9.1 Open-Circuit Characteristic

see figure 9.1 and figure 9.2 (simulation-file: same name as the heading)

The open-circuit characteristic (OCC) describes the relationship between the no-load terminal voltage and the rotor field current:

$$U_0 = f(I_f), \qquad I_s = 0, \qquad n = n_N. \tag{9.1}$$

Here, U_0 is the open-circuit terminal voltage, I_f is the field current, I_s is the stator current, and n_N is the rated mechanical speed.

9.1.1 Test Procedure

The generator is driven at rated speed while its stator terminals remain open. No electrical load is connected, and the stator current is therefore practically zero. The field current is increased gradually from zero, and the corresponding terminal voltage is recorded.

Because there is no stator current, the voltage drops across the stator resistance and synchronous reactance are negligible. The terminal voltage is therefore approximately equal to the internally generated voltage:

$$\underline{U}_0 \approx \underline{E}. \tag{9.2}$$

The internally generated phase voltage can be expressed approximately as

$$E = \omega N k_w \Phi, \tag{9.3}$$

where ω is the electrical angular frequency, N is the effective number of series turns per phase, k_w is the winding factor, and Φ is the magnetic flux per pole. At constant speed and frequency,

$$E \propto \Phi. \tag{9.4}$$

The OCC therefore also represents the magnetic relationship between field current and air-gap flux.

9.1.2 Shape of the Curve

At low field currents, the magnetic circuit is unsaturated. The flux is approximately proportional to the field current, and the open-circuit voltage increases almost linearly:

$$\Phi \propto I_f, \qquad U_0 \propto I_f. \tag{9.5}$$

The straight-line extension of this initial region is called the *air-gap line*. At higher excitation levels, the iron core begins to saturate. A further increase in field current then produces a progressively smaller increase in flux and terminal voltage. The measured OCC consequently bends below the air-gap line.

Even when $I_f = 0$, a small voltage may remain at the stator terminals because of remanent magnetism in the rotor and stator iron.

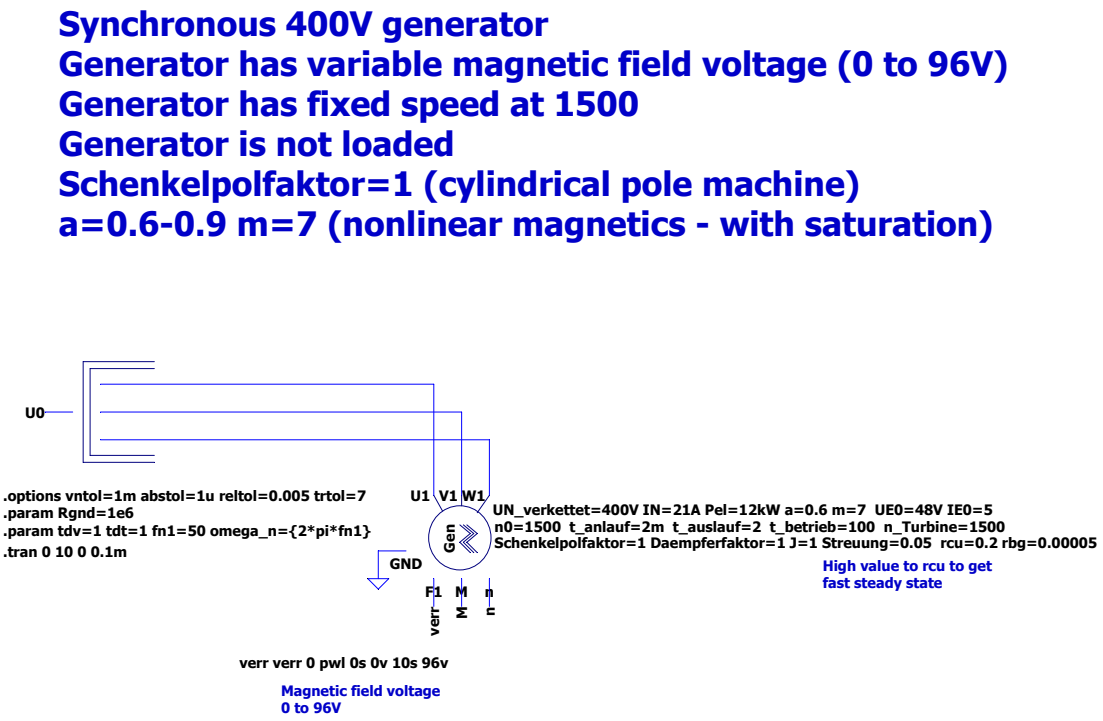


Figure 9.1: Open circuit characteristic - Schematic

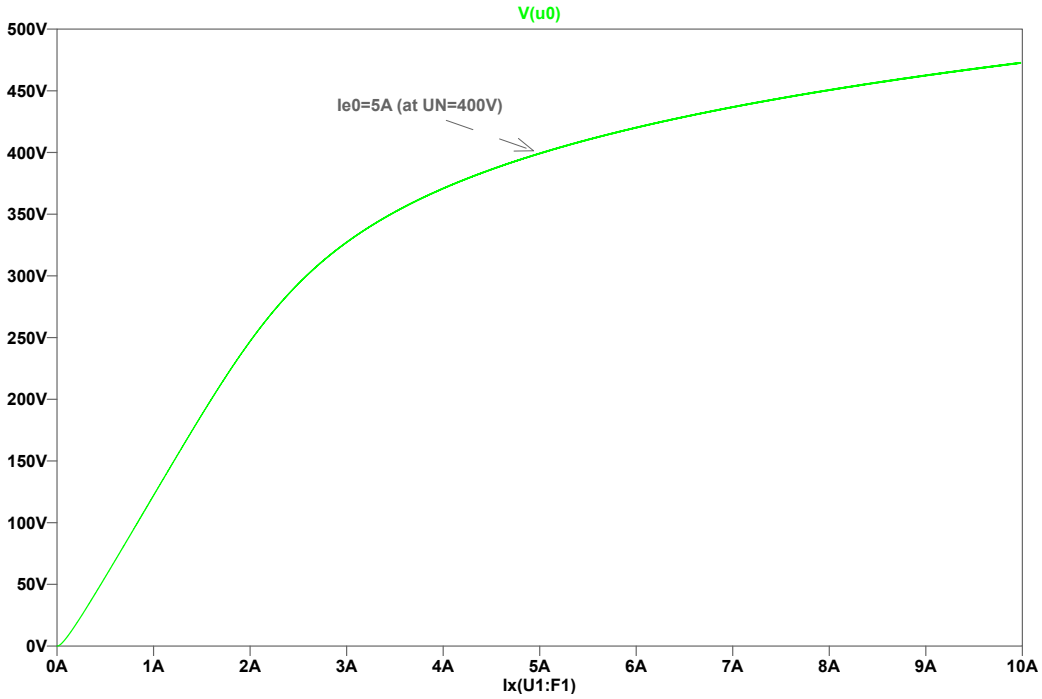


Figure 9.2: Open circuit characteristic - Waveform

- determine the field current required for rated no-load voltage,
- investigate magnetic saturation and remanence,
- establish the air-gap line,
- estimate synchronous reactance and voltage regulation,
- determine the short-circuit ratio,
- select the operating range of the excitation system and AVR, and
- validate the saturation curve of a simulation model.

9.2 Short-Circuit Characteristic

see figure 9.3 and figure 9.4 (simulation-file: same name as the heading)

The short-circuit characteristic (SCC) describes the relationship between the steady-state stator short-circuit current and the field current:

$$I_k = f(I_f), \qquad U_s = 0, \qquad n = n_N. \tag{9.6}$$

9.2.1 Test Procedure

The three stator terminals are short-circuited symmetrically through suitable measuring equipment. The generator is driven at rated speed, and the excitation is increased slowly from a small initial value. The corresponding steady-state stator current is measured at each field-current setting.

Important 9.1 – (Current limitation during the test)

The field current must be increased carefully. Rated stator current can be reached at a field current considerably lower than that required for rated open-circuit voltage. The conventional SCC test is normally stopped when the rated stator current has been obtained.

9.2.2 Shape of the Curve

The short-circuit characteristic is approximately linear:

$$I_k \propto I_f. \tag{9.7}$$

During a symmetrical terminal short circuit, the terminal voltage is nearly zero. The internal generated voltage is therefore balanced mainly by the voltage drop across the synchronous impedance:

$$\underline{E} \approx (R_s + \mathrm{j}X_s)\,\underline{I}_k. \tag{9.8}$$

Since normally $X_s \gg R_s$, the magnitude can be approximated by

$$E \approx X_s I_k. \tag{9.9}$$

The armature reaction is strongly demagnetizing during the symmetrical short circuit. The resulting air-gap flux therefore remains comparatively small, even when the stator current is high. The magnetic circuit operates mainly in its unsaturated region, which explains the almost straight SCC.

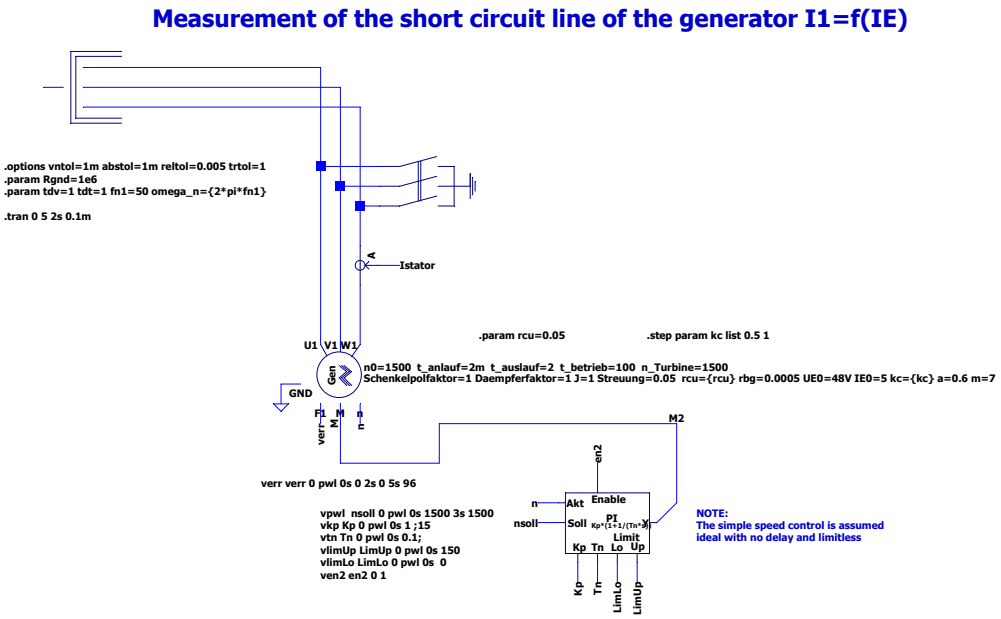


Figure 9.3: Short circuit characteristic - Schematic

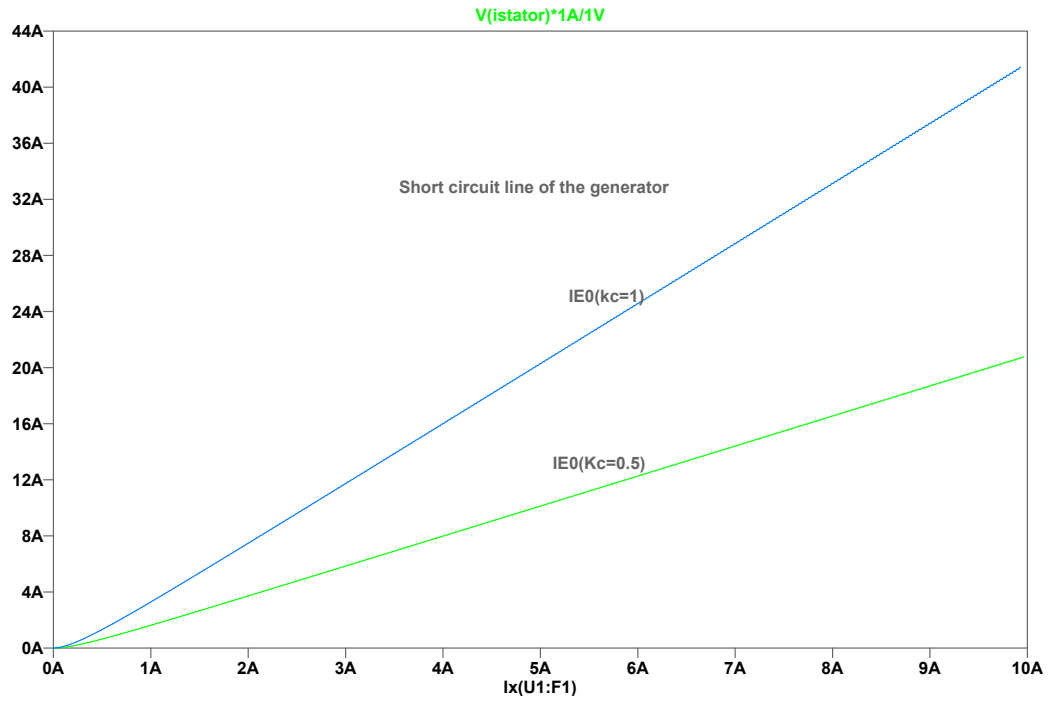


Figure 9.4: Short circuit characteristic - Waveform

The OCC is used to:

9.2.3 Steady-State and Transient Short-Circuit Currents

The conventional SCC represents the *steady-state* short-circuit current. Immediately after a sudden fault, however, the current is considerably higher and passes through subtransient, transient, and steady-state intervals.

Neglecting resistance and using the appropriate direct-axis reactances, the symmetrical AC components can be estimated as

$$I_k'' \approx \frac{E}{X_d''}, \qquad \text{subtransient current,} \tag{9.10}$$

$$I_k' \approx \frac{E}{X_d'}, \qquad \text{transient current,} \tag{9.11}$$

$$I_k \approx \frac{E}{X_d}, \qquad \text{steady-state current.} \tag{9.12}$$

The first interval is dominated by the stator and damper-winding flux constraints. After the damper currents decay, the transient reactance becomes effective. Finally, after the field transient has decayed, the current is determined mainly by the synchronous reactance.

Remark 9.1 – (Sudden-fault waveform)

The actual phase current following a sudden short circuit may also contain a decaying DC component. Its magnitude depends on the fault inception angle and the pre-fault operating point. The peak phase current can therefore be much larger than the symmetrical RMS current represented by the SCC.

9.2.4 Synchronous-Impedance Estimate

The OCC and SCC can be combined to estimate the synchronous impedance. The open-circuit voltage and short-circuit current must be evaluated at the same field current.

Calculation 9.1 – (Synchronous impedance from OCC and SCC)

Using phase voltage and phase current,

$$Z_s = \frac{U_{0,\mathrm{ph}}}{I_k}, \tag{9.13}$$

$$X_s = \sqrt{Z_s^2 - R_s^2}. \tag{9.14}$$

If the stator resistance is small compared with the reactance,

$$X_s \approx Z_s. \tag{9.15}$$

The OCC may already be saturated at the selected field current, whereas the SCC remains almost unsaturated. The synchronous-impedance method therefore usually gives a conservative, relatively high estimate of voltage regulation.

9.2.5 Short-Circuit Ratio

The short-circuit ratio is commonly defined as

$$\mathrm{SCR} = \frac{I_{f,U_N}}{I_{f,I_N}}, \tag{9.16}$$

where I_{f,U_N} is the field current required to produce rated voltage on open circuit and I_{f,I_N} is the field current required to produce rated stator current under steady-state short-circuit conditions. Approximately,

$$\mathrm{SCR} \approx \frac{1}{x_d}, \tag{9.17}$$

when x_d is expressed in per unit. Saturation and the exact test definition affect this relationship.

A comparatively large SCR generally indicates a smaller synchronous reactance, improved steady-state voltage regulation, and a larger stability margin. It also leads to higher short-circuit current and usually requires a physically larger and more expensive machine.

9.3 Load or External Characteristic

see figure 9.5-9.10 (simulation-files: same names as the headings)

The load characteristic describes how the generator terminal voltage changes with stator current when speed, field current, and load power factor remain constant:

$$U = f(I_s), \qquad n = \text{constant}, \qquad I_f = \text{constant}, \qquad \cos\varphi = \text{constant}. \tag{9.18}$$

It is therefore also called the *external characteristic* of the generator.

For a simplified round-rotor generator, the steady-state phasor equation is

$$\underline{E} = \underline{U} + (R_s + \mathrm{j}X_s)\,\underline{I}_s. \tag{9.19}$$

If field current and speed are constant, the magnitude of E is approximately constant. The terminal voltage must consequently change when the stator current changes. The direction and magnitude of this voltage change depend strongly on the load power factor.

9.3.1 Inductive Load

For a lagging load, the stator current lags behind the terminal voltage. The reactive voltage drop $\mathrm{j}X_s\underline{I}_s$ and the demagnetizing armature reaction reduce the terminal voltage. Consequently, the voltage falls significantly as the load current rises, and the external characteristic has a pronounced negative slope.

9.3.2 Unity-Power-Factor Load

At unity power factor, stator current and terminal voltage are in phase. The terminal voltage still decreases as the current increases, but usually less strongly than for an inductive load. The reduction results from stator resistance, synchronous reactance, armature reaction, and magnetic saturation.

9.3.3 Capacitive Load

For a leading load, the stator current leads the terminal voltage. The armature reaction can then have a magnetizing effect. As the current increases, the terminal voltage may remain nearly constant or even rise. The external characteristic can therefore have a positive slope.

This behaviour explains why capacitive loads and overexcited synchronous machines can support the voltage of a power system.

9.3.4 Voltage Regulation

The voltage regulation of a synchronous generator is commonly expressed as

$$\varepsilon_U = \frac{U_0 - U_N}{U_N} \cdot 100\%, \tag{9.20}$$

where U_0 is the no-load voltage obtained after removing the load without changing speed or field current, and U_N is the terminal voltage at the specified load.

For an inductive load, the regulation is normally positive because $U_0 > U_N$. For a sufficiently capacitive load, it may become negative because the loaded terminal voltage can exceed the corresponding no-load voltage.

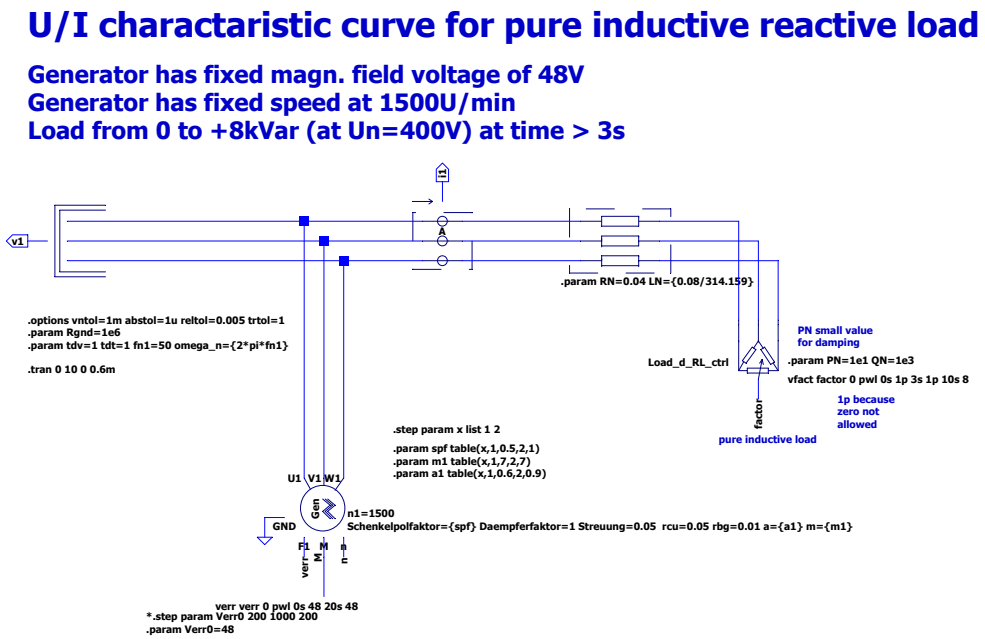


Figure 9.5: Inductive characteristic - Schematic

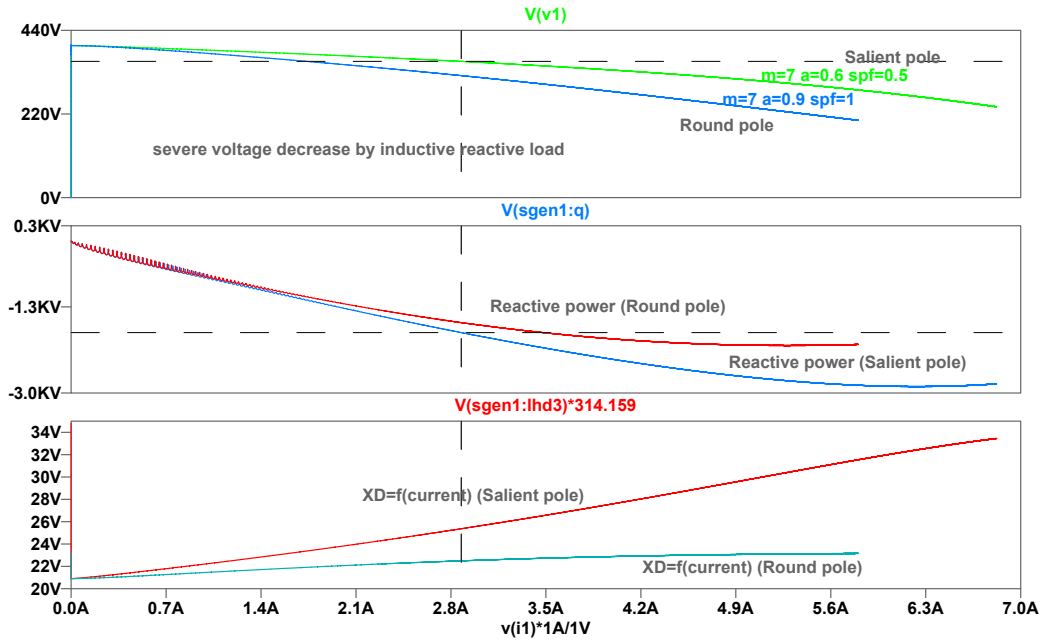


Figure 9.6: Inductive characteristic - Waveform

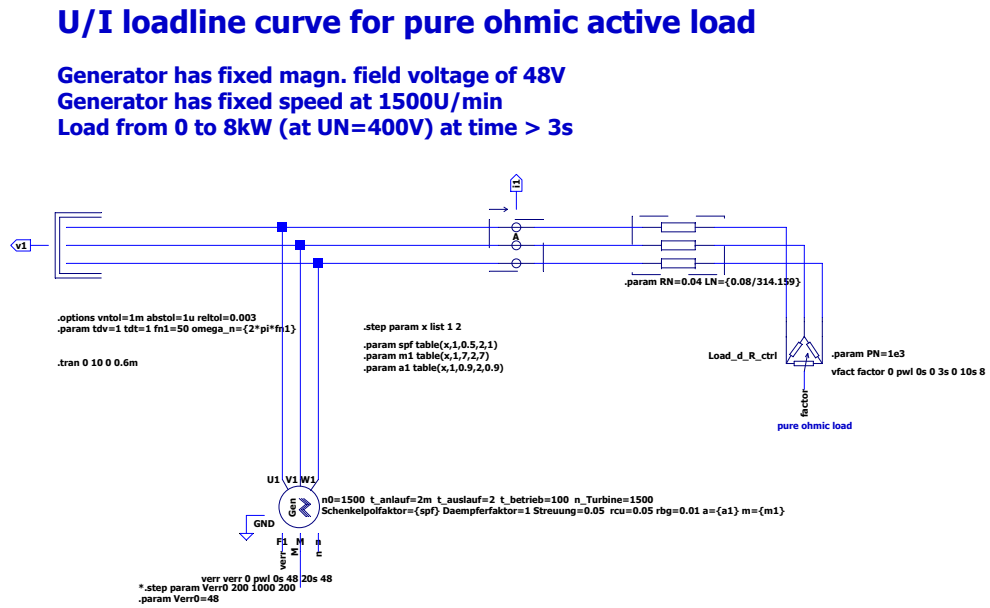


Figure 9.7: Resistive characteristic - Schematic

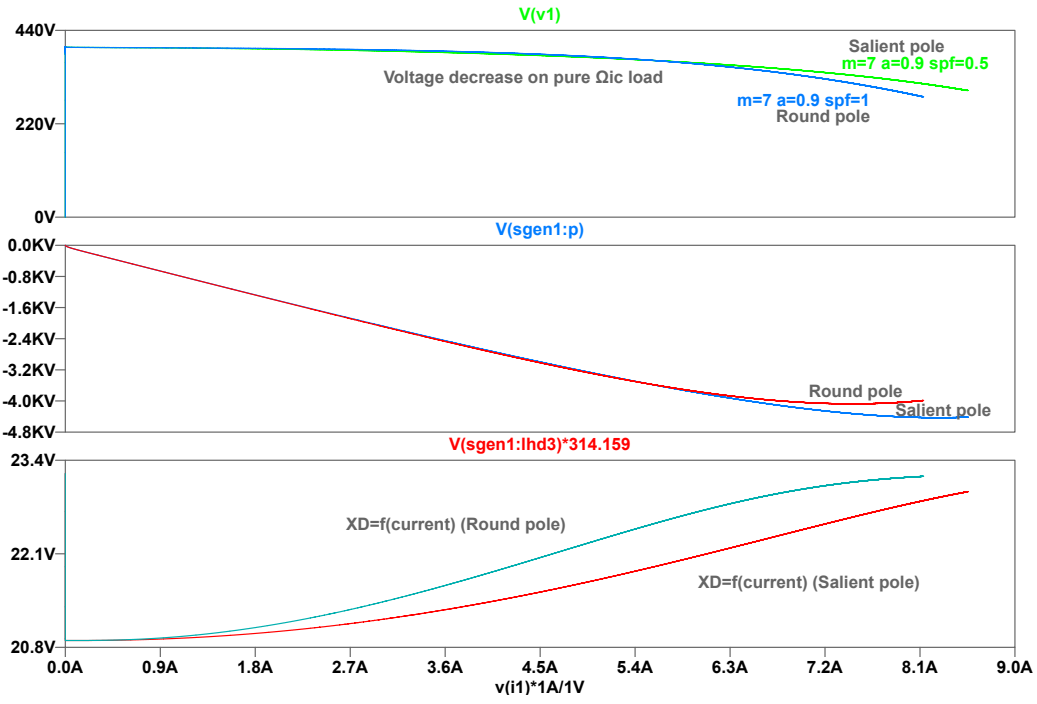


Figure 9.8: Resistive characteristic - Waveform

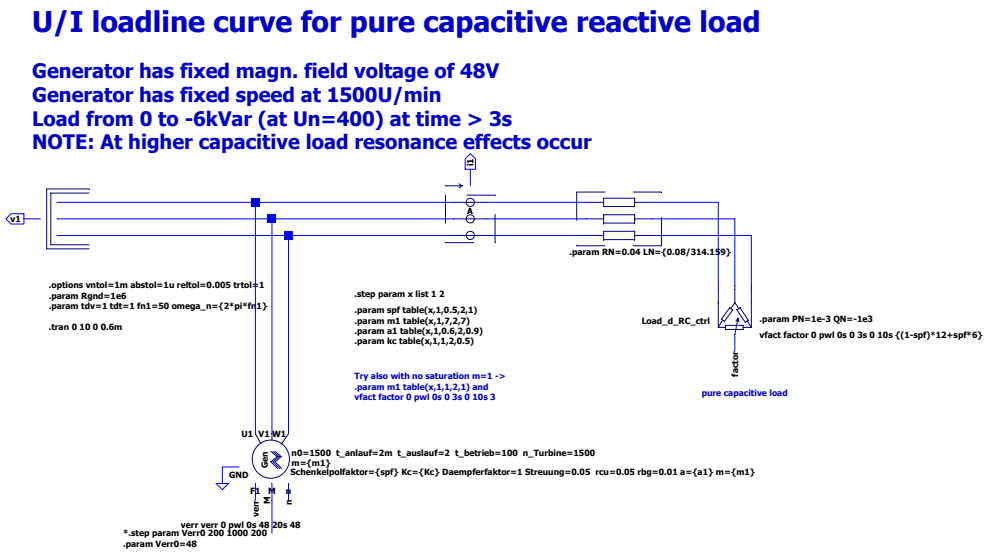


Figure 9.9: Capacitive characteristic - Schematic

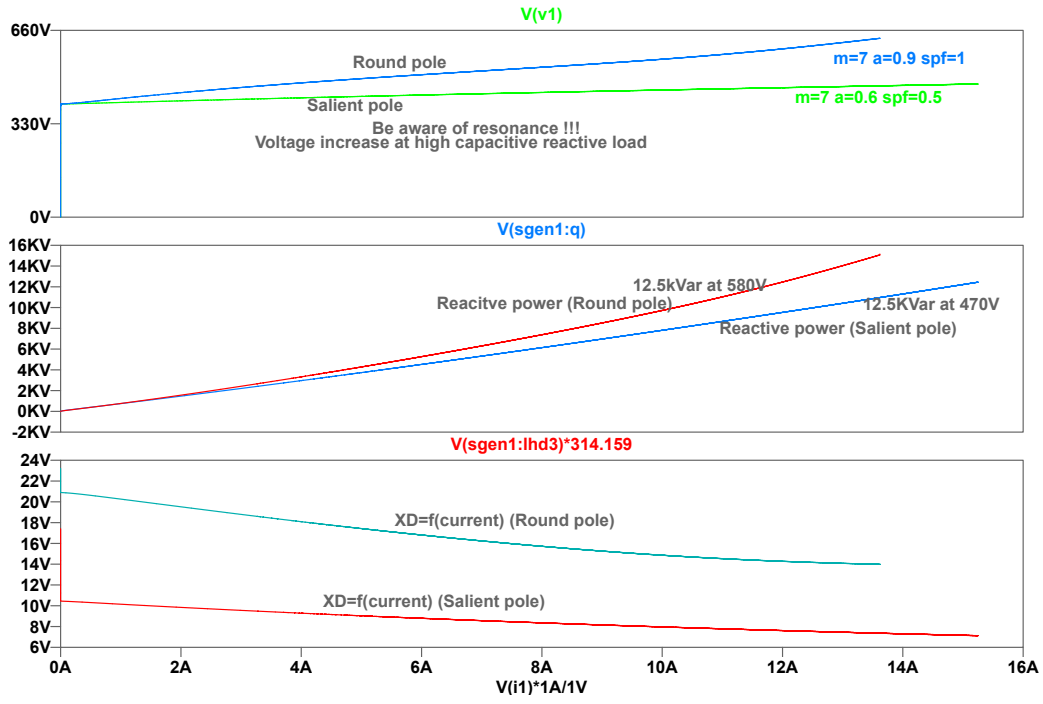


Figure 9.10: Capacitive characteristic - Waveform

9.4 Regulation Characteristic

see figures 9.11-9.16 (simulation-file: same names as the headings)

The regulation characteristic describes the field current required to maintain constant terminal voltage while the stator current changes:

$$I_f = f(I_s), \qquad U = U_N, \qquad n = n_N, \qquad \cos\varphi = \text{constant}. \tag{9.21}$$

It is sometimes called the excitation characteristic under load.

Important 9.2 – (Difference between load and regulation characteristics)

The load characteristic gives $U = f(I_s)$ at constant excitation. The regulation characteristic gives $I_f = f(I_s)$ while the terminal voltage is kept constant. The two characteristics therefore describe complementary operating conditions.

9.4.1 Inductive Load

For a lagging load, armature reaction is predominantly demagnetizing, and the synchronous-reactance voltage drop increases with stator current. The excitation must therefore be increased as the load rises:

$$I_s \uparrow \quad \Longrightarrow \quad I_f \uparrow . \tag{9.22}$$

The regulation curve consequently rises relatively steeply.

9.4.2 Unity-Power-Factor Load

At unity power factor, the required field current normally increases with stator current, but less strongly than for an inductive load.

9.4.3 Capacitive Load

For a leading load, armature reaction is partly magnetizing. Less field current is therefore required to maintain rated terminal voltage. Over part of the operating range,

$$I_s \uparrow \quad \Longrightarrow \quad I_f \downarrow . \tag{9.23}$$

For strongly capacitive operating conditions, the generator may therefore have to operate underexcited.

9.4.4 Practical Importance

The regulation characteristic is directly related to the task of the AVR. When load current and power factor change, the AVR adjusts field voltage and field current so that the terminal voltage remains at its reference value.

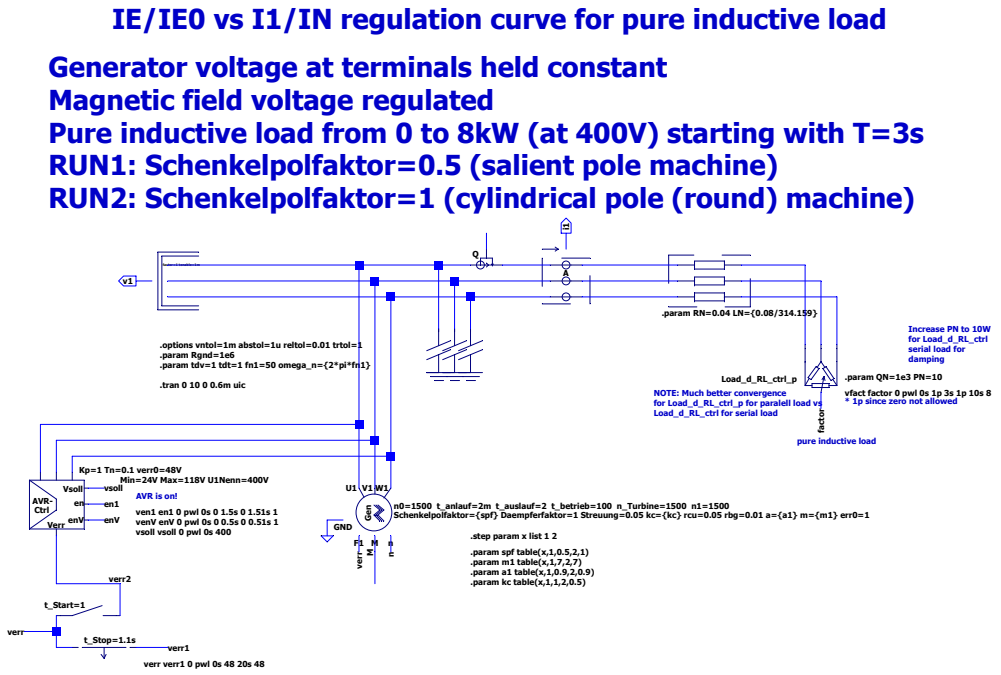


Figure 9.11: Inductive Regulation curve - Schematic

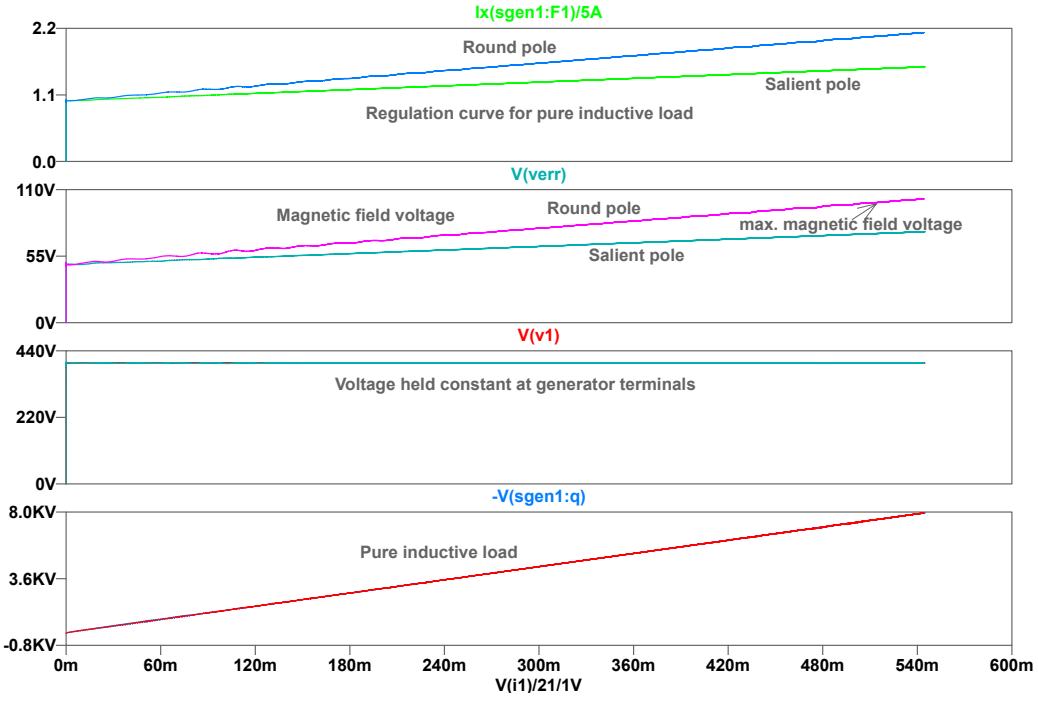


Figure 9.12: Inductive Regulation curve - Waveform

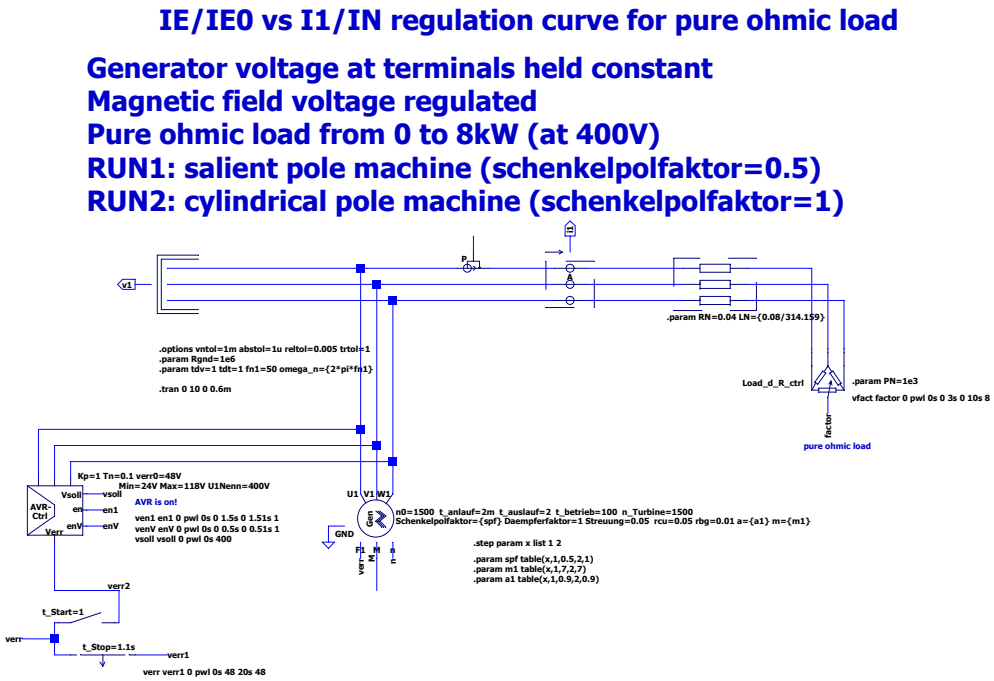


Figure 9.13: Ohmic Regulation curve - Schematic

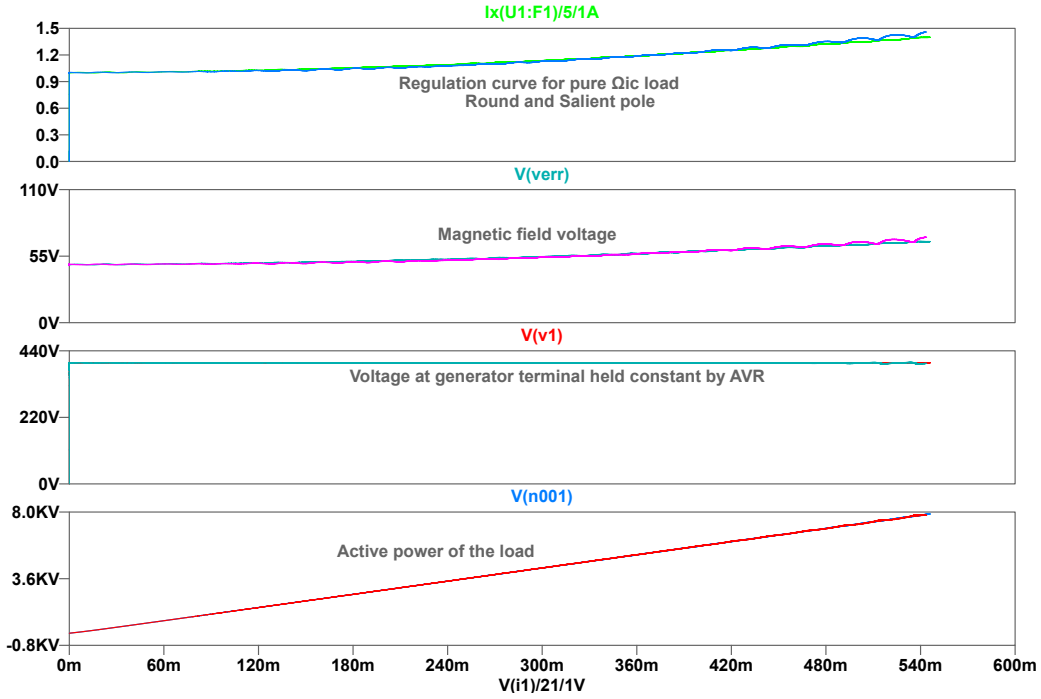


Figure 9.14: Ohmic Regulation curve - Waveform

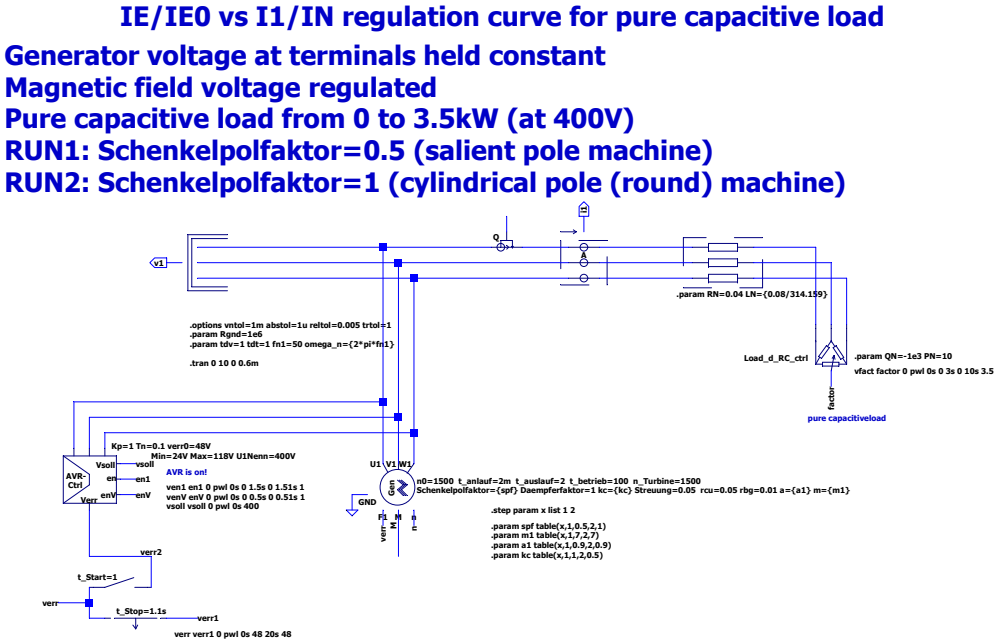


Figure 9.15: Capacitive Regulation curve - Schematic

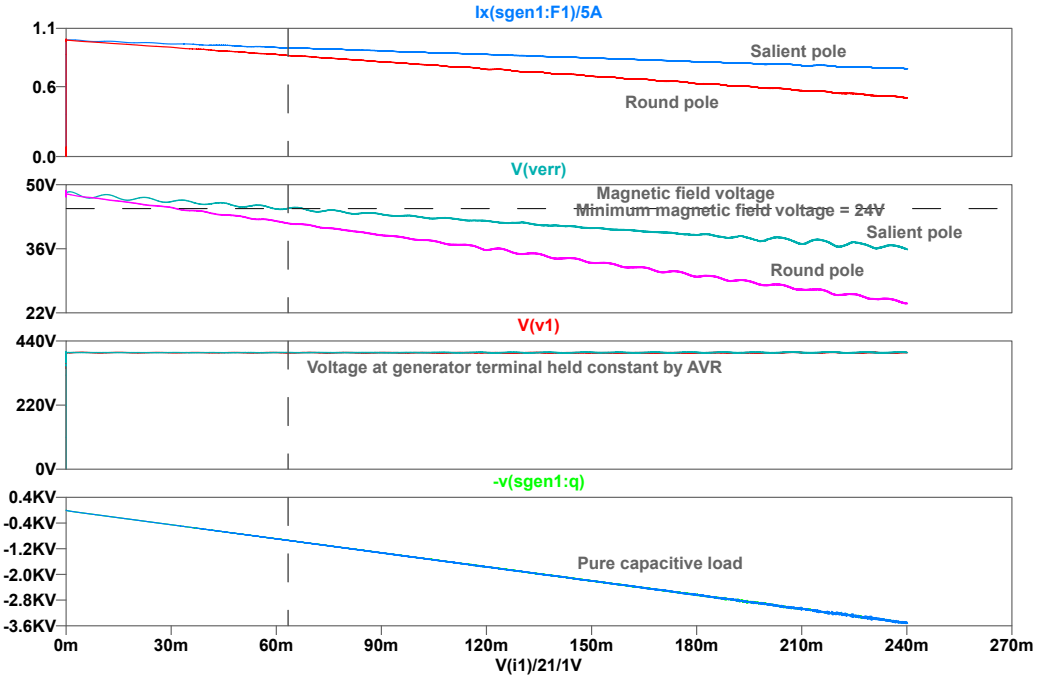


Figure 9.16: Capacitive Regulation curve - Waveform

9.5 V-Curves

see figure 9.17 and figure 9.18 (simulation-file: same name as the heading)

The V-curves of a synchronous machine show the relationship between stator current and field current at constant active power and terminal voltage:

$$I_s = f(I_f), \qquad P = \text{constant}, \qquad U = \text{constant}, \qquad f = \text{constant}. \tag{9.24}$$

The curves are named after their characteristic V-shaped appearance.

9.5.1 Test Procedure

The generator is connected to a stiff grid, which maintains approximately constant voltage and frequency. The mechanical torque is adjusted to establish a specified active-power output. The field current is then varied while the mechanical power remains constant. For each excitation value, the stator current, power factor, and reactive power are recorded.

9.5.2 Minimum Stator Current

For a balanced three-phase generator,

$$S = \sqrt{3}\, U_{\mathrm{LL}} I_s, \tag{9.25}$$

$$P = \sqrt{3}\, U_{\mathrm{LL}} I_s \cos\varphi. \tag{9.26}$$

At constant active power and terminal voltage,

$$I_s = \frac{P}{\sqrt{3}\, U_{\mathrm{LL}} \cos\varphi}. \tag{9.27}$$

The stator current reaches its minimum when

$$\cos\varphi = 1. \tag{9.28}$$

At this operating point, $Q \approx 0$, and the generator exchanges almost no reactive power with the grid.

9.5.3 Underexcited Generator

If the field current is reduced below the value corresponding to the minimum stator current, the generator becomes underexcited. Under the usual generator sign convention, it absorbs reactive power from the grid.

The stator current increases because it contains both an active-current and a reactive-current component:

$$I_s = \sqrt{I_P^2 + I_Q^2}. \tag{9.29}$$

Although the active power remains constant, the increasing reactive-current component increases the total stator current. Excessive underexcitation can be limited by stator heating, end-region heating, reduced steady-state stability, and the risk of losing synchronism.

9.5.4 Overexcited Generator

If the field current is increased above the value corresponding to unity power factor, the generator becomes overexcited and normally supplies reactive power to the grid. The stator current again rises because the magnitude of the reactive-current component increases.

Excessive overexcitation is limited mainly by rotor field-winding heating, stator-current heating, magnetic saturation, and the maximum capability of the excitation system.

Remark 9.2 – (Reactive-power sign convention)

The statements "supplies reactive power" and "absorbs reactive power" use the generator convention. Some simulation programs define positive power as power entering a component. In that convention, the displayed sign of Q is reversed and must be checked against the model reference directions.

9.5.5 Families of V-Curves

A separate V-curve can be recorded for every active-power level, for example,

$$P = 0, \quad 0.25P_N, \quad 0.5P_N, \quad 0.75P_N, \quad P_N. \tag{9.30}$$

As active power increases, the minimum possible stator current also increases:

$$I_{s,\min} = \frac{P}{\sqrt{3}\,U_{\mathrm{LL}}}. \tag{9.31}$$

The position of the minimum can shift slightly because of magnetic saturation, stator resistance, and the synchronous-reactance voltage drop.

9.5.6 Inverted V-Curves

An inverted V-curve represents power factor as a function of field current:

$$\cos\varphi = f(I_f). \tag{9.32}$$

Its maximum occurs near unity power factor. On one side lies underexcited operation; on the other lies overexcited operation. Moving away from the maximum in either direction reduces the magnitude of the power factor and increases stator current for the same active power.

9.5.7 Relationship to Power Control

The V-curves illustrate the approximate separation of active- and reactive-power control when a synchronous generator is connected to a stiff grid:

- mechanical torque primarily controls active power P,
- field excitation primarily controls reactive power Q,
- the grid frequency fixes the average rotor speed, and
- the grid voltage largely fixes the terminal voltage.

This separation is not exact because excitation changes also influence the load angle, saturation, stator current, and machine losses.

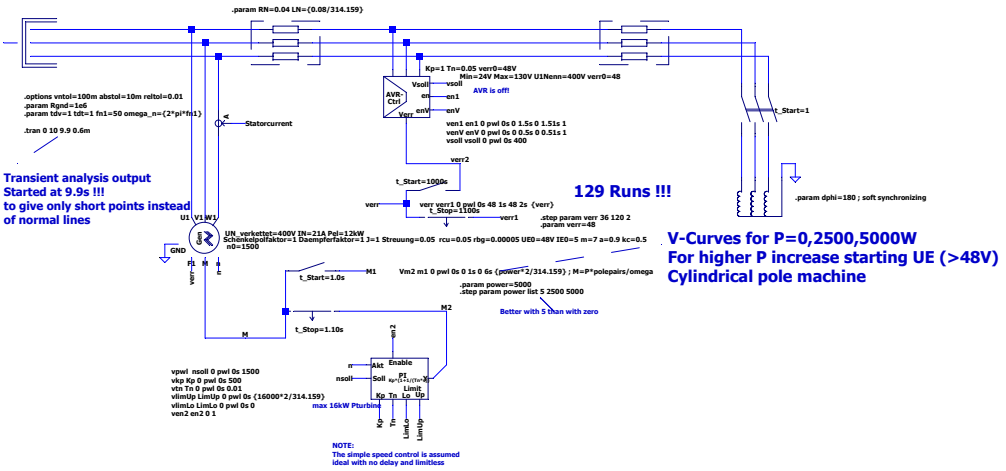


Figure 9.17: V-curve - Schematic

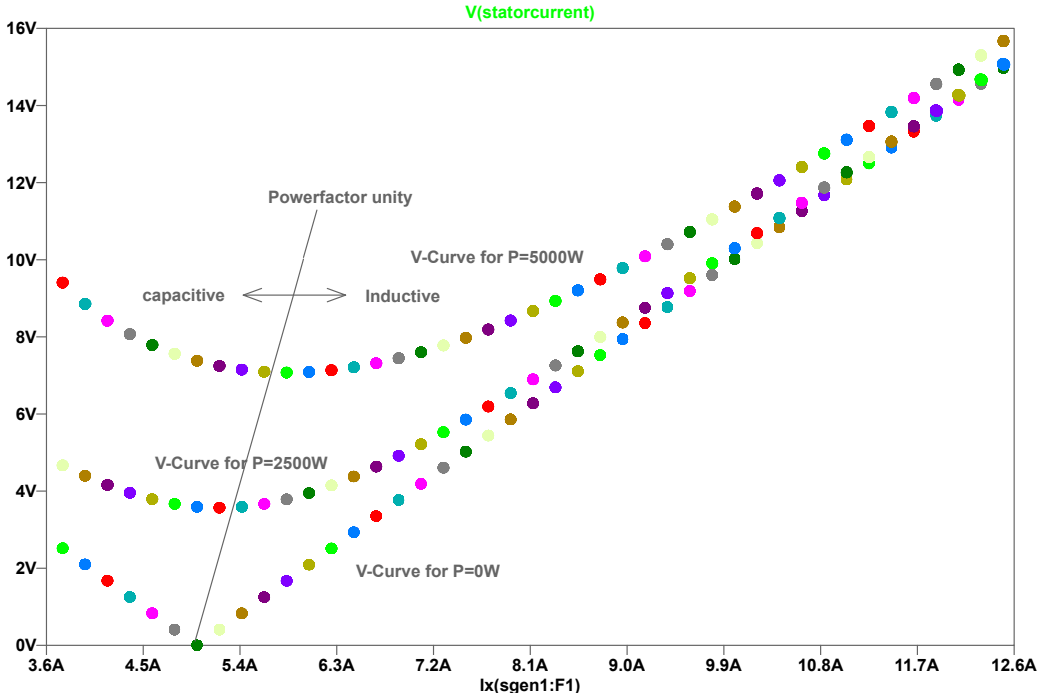


Figure 9.18: V-curve - Waveform

9.6 Comparison of the Characteristics

Table 9.1: Comparison of synchronous-generator characteristic curves

Characteristic	Horizontal axis	Vertical axis	Constant quantities	Main information
Open circuit	I_f	U_0	$I_s = 0, n = n_N$	Saturation and generated voltage
Short circuit	I_f	I_k	$U = 0, n = n_N$	Steady-state fault current and synchronous impedance
Load/external	I_s	U	$I_f, n, \cos\varphi$	Natural terminal-voltage variation
Regulation	I_s	I_f	$U, n, \cos\varphi$	Excitation required for constant voltage
V-curve	I_f	I_s	P, U, f	Reactive-power and power-factor behaviour

The open-circuit and short-circuit characteristics are principally machine test characteristics. Together, they provide information about saturation, synchronous impedance, voltage regulation, and short-circuit ratio.

The load and regulation characteristics describe complementary operating conditions. The load characteristic shows what happens when excitation remains unchanged, whereas the regulation characteristic shows how excitation must be adjusted to prevent the terminal voltage from changing.

The V-curves are mainly operating characteristics. They show the relationship between excitation, stator current, power factor, and reactive power during grid-connected operation.

Chapter Summary

The open-circuit characteristic describes generated voltage and magnetic saturation. The short-circuit characteristic describes the steady-state fault current and, together with the OCC, permits an approximate determination of synchronous impedance and short-circuit ratio.

The load characteristic gives the terminal-voltage variation at constant excitation, whereas the regulation characteristic gives the field current required to keep voltage constant. V-curves show how underexcitation and overexcitation increase stator current and determine whether a grid-connected generator absorbs or supplies reactive power.

Synchronous Generator Model

Formula sheet for the $xy0$ machine model

Notation. Subscripts S, E, DX, and DY denote the stator, excitation winding, and the damper windings on the x- and y-axes. The dot denotes a time derivative, e.g. $\dot{\psi} = \mathrm{d}\psi/\mathrm{d}t$. The symbols x and y correspond to the rotating machine axes used in the source model.

Clarke transformation: three-phase to $xy0$

$$u_0 = \frac{u_1 + u_2 + u_3}{3} = R_0 i_0 + L_0 \dot{i}_0, \tag{A.1}$$

$$u_{SXS} = \frac{2u_1 - (u_2 + u_3)}{3}, \tag{A.2}$$

$$u_{SYS} = \frac{u_2 - u_3}{\sqrt{3}}. \tag{A.3}$$

The stationary components are rotated through the electrical angle α:

$$u_{SX} = u_{SXS}\cos\alpha + u_{SYS}\sin\alpha, \tag{A.4}$$

$$u_{SY} = u_{SYS}\cos\alpha - u_{SXS}\sin\alpha. \tag{A.5}$$

Electrical equations in the rotating reference frame

$$u_{SX} = R_S i_{SX} + \dot{\psi}_{SX} - p\omega\psi_{SY}, \tag{A.6}$$

$$g u_E = g^2 \frac{2}{3} R_E i_E + \dot{\psi}_E, \tag{A.7}$$

$$0 = R_{DX} i_{DX} + \dot{\psi}_{DX}, \tag{A.8}$$

$$u_{SY} = R_S i_{SY} + \dot{\psi}_{SY} + p\omega\psi_{SX}, \tag{A.9}$$

$$0 = R_{DY} i_{DY} + \dot{\psi}_{DY}. \tag{A.10}$$

Flux linkages and current components

The common magnetizing flux linkages are

$$\psi_{hX} = \psi_{SX} - L_S i_{SX} = \psi_{DX} - L_{DX} i_{DX} = \psi_E - L_E \frac{i_E}{g}, \tag{A.11}$$

$$\psi_{hY} = \psi_{SY} - L_S i_{SY} = \psi_{DY} - L_{DY} i_{DY}. \tag{A.12}$$

The associated magnetizing currents are

$$i_{\mu X} = f(\psi_{hX}) = i_{SX} + i_{DX} + \frac{i_E}{g}, \tag{A.13}$$

$$i_{\mu Y} = f(\psi_{hY}) = i_{SY} + i_{DY}. \tag{A.14}$$

For the nonlinear magnetization characteristic shown in the source model,

$$i_\mu(\psi) = i_{\mu N} \left[0.9 \frac{\psi}{\psi_N} + 0.1 \left(\frac{\psi}{\psi_N} \right)^7 \right]. \tag{A.15}$$

Mechanical equations

$$M = M_{RBG} - M_i, \tag{A.16}$$

$$M_i = \frac{3}{2} p \left(i_{SY} \psi_{hX} - i_{SX} \psi_{hY} \right), \tag{A.17}$$

$$\alpha = p \int \omega \, \mathrm{d}t, \tag{A.18}$$

$$M_{RBG} = M_{\text{Reibung}} \, \operatorname{sign}(n). \tag{A.19}$$

Here, p is the number of pole pairs and n is the mechanical speed.

Inverse transformation: $xy0$ to three-phase

First rotate from the machine frame back to stationary coordinates:

$$i_{SXS} = i_{SX} \cos\alpha - i_{SY} \sin\alpha, \tag{A.20}$$

$$i_{SYS} = i_{SX} \sin\alpha + i_{SY} \cos\alpha. \tag{A.21}$$

Then recover the phase currents:

$$i_1 = i_{SXS} + i_0, \tag{A.22}$$

$$i_2 = \frac{-i_{SXS} + \sqrt{3}\, i_{SYS}}{2} + i_0, \tag{A.23}$$

$$i_3 = \frac{-i_{SXS} - \sqrt{3}\, i_{SYS}}{2} + i_0. \tag{A.24}$$

Default model simplifications (if not otherwise supported by the parameter line):

$$\frac{i_E}{g} = i'_E, \qquad L_E = L_S, \quad L_0 = L_S, \quad L_{DX} = L_S, \quad L_{DY} = L_S,$$

$$R_0 = R_S, \qquad R_{DX} = R_D, \qquad R_{DY} = R_D.$$

Constant machine parameters

The rated input data used for the 400 MW synchronous generator are

$$U_{N,\text{verkettet}} = 20\,\text{kV}, \quad I_N = 14.434\,\text{kA}, \quad P_{EL} = 400\,\text{MW}, \tag{A.25}$$

$$f_N = 50\,\text{Hz}, \quad K_C = 0.5, \quad p = 2, \tag{A.26}$$

$$U_{E0} = 600\,\text{V}, \quad I_{E0} = 2\,000\,\text{A}, \quad \text{Streuung} = 0.05, \tag{A.27}$$

$$r_{cu} = 0.001, \quad r_{bg} = 0.001, \quad \text{Schenkelpolfaktor} = 1, \tag{A.28}$$

$$\text{Daempferfaktor} = 1. \tag{A.29}$$

The apparent power and power factor obtained from the rated voltage and current are

$$S_N = \sqrt{3}\,U_{N,\text{verkettet}} I_N \tag{A.30}$$

$$= \sqrt{3} \cdot 20\,\text{kV} \cdot 14.434\,\text{kA} = 500.008\,\text{MV A} \approx 500\,\text{MV A}, \tag{A.31}$$

$$\cos\varphi_N = \frac{P_{EL}}{S_N} = \frac{400\,\text{MW}}{500.008\,\text{MV A}} = 0.799987 \approx 0.8. \tag{A.32}$$

The constant machine parameters are calculated from

$$U_N = \frac{U_{N,\text{verkettet}}}{\sqrt{3}}, \qquad Z_N = \frac{U_N}{I_N}, \tag{A.33}$$

$$\omega_{EL} = 2\pi f_N, \qquad R_S = Z_N\, r_{cu}, \tag{A.34}$$

$$L_S = \frac{\text{Streuung}\, Z_N}{K_C \omega_{EL}}, \qquad \Psi_N = \frac{\sqrt{2}\,U_N}{\omega_{EL}}, \tag{A.35}$$

$$g = \frac{I_{E0}(1 - \text{Streuung})}{K_C \sqrt{2}\, I_N}, \qquad R_D = \frac{R_S}{\text{Daempferfaktor}}, \tag{A.36}$$

$$I_{MNX} = \frac{I_{E0}}{g}, \qquad I_{MNY} = \frac{I_{MNX}}{\text{Schenkelpolfaktor}}, \tag{A.37}$$

$$M_N = \frac{pP_{EL}(1 - r_{bg} - r_{cu})}{\omega_{EL}}. \tag{A.38}$$

Inserting the rated data gives

$$U_N = \frac{20\,\text{kV}}{\sqrt{3}} = \boxed{11.547\,0\,\text{kV}}, \tag{A.39}$$

$$Z_N = \frac{11.547\,0\,\text{kV}}{14.434\,\text{kA}} = \boxed{0.799\,987\,\Omega}, \tag{A.40}$$

$$\omega_{EL} = 2\pi \cdot 50\,\text{Hz} = \boxed{314.159\,\text{rad/s}}, \tag{A.41}$$

$$R_S = 0.799\,987\,\Omega \cdot 0.001 = \boxed{7.999\,87 \cdot 10^{-4}\,\Omega}, \tag{A.42}$$

$$L_S = \frac{0.05 \cdot 0.799\,987\,\Omega}{0.5 \cdot 314.159\,\text{rad/s}} = \boxed{2.546\,44 \cdot 10^{-4}\,\text{H}}, \tag{A.43}$$

$$\Psi_N = \frac{\sqrt{2} \cdot 11.547\,0\,\text{kV}}{314.159\,\text{rad/s}} = \boxed{51.979\,8\,\text{Wb}}. \tag{A.44}$$

For the excitation-current transformation,

$$g = \frac{2\,000\,\text{A}(1 - 0.05)}{0.5\sqrt{2} \cdot 14.434\,\text{kA}} = \boxed{0.186158}, \tag{A.45}$$

$$I_{MNX} = \frac{2\,000\,\text{A}}{0.186158} = \boxed{10.743\,6\,\text{kA}}, \tag{A.46}$$

$$I_{MNY} = \frac{10.743\,6\,\text{kA}}{1} = \boxed{10.743\,6\,\text{kA}}. \tag{A.47}$$

Because Daempferfaktor = 1, the damper resistance becomes

$$R_D = \frac{R_S}{1} = \boxed{7.999\,87 \cdot 10^{-4}\,\Omega}. \tag{A.48}$$

The rated electromagnetic torque, including the specified relative friction and copper losses, is

$$M_N = \frac{2 \cdot 400\,\text{MW}(1 - 0.001 - 0.001)}{314.159\,\text{rad/s}} \tag{A.49}$$

$$= \boxed{2.541\,39 \cdot 10^{6}\,\text{N m}}. \tag{A.50}$$

The excitation-winding resistance and its resistance referred to the stator side equation are

$$R_E = \frac{U_{E0}}{I_{E0}} = \frac{600\,\text{V}}{2\,000\,\text{A}} = \boxed{0.300\,000\,\Omega}, \tag{A.51}$$

$$R'_E = g^2 \frac{2}{3} R_E = (0.186158)^2 \frac{2}{3} \cdot 0.3\,\Omega = \boxed{6.930\,97 \cdot 10^{-3}\,\Omega}, \tag{A.52}$$

$$U'_{E0} = g U_{E0} = 0.186158 \cdot 600\,\text{V} = \boxed{111.695\,\text{V}}. \tag{A.53}$$

For $p = 2$, the synchronous mechanical speed is

$$n_N = \frac{60 f_N}{p} = \boxed{1\,500\,\text{rpm}}, \qquad \omega_{mN} = \frac{\omega_{EL}}{p} = \boxed{157.080\,\text{rad/s}}. \tag{A.54}$$

Interpretation of Ψ_N. The value $\Psi_N = 51.979\,8$ Wb is the rated peak stator flux linkage in volt-seconds, not the physical air-gap flux in webers per pole. A physical flux value would additionally require the effective number of stator turns and the winding factor.

Parameter definitions

Symbol	Meaning
Streuung	Leakage factor
Daempferfaktor	Damping factor
Schenkelpolfaktor	Salient-pole factor
r_{bg}	Friction-loss factor
r_{cu}	Copper-loss (resistance) factor
p	Number of pole pairs
K_C	Model scaling or correction factor
$U_{N,\text{verkettet}}$	Rated line-to-line RMS voltage
$U_N,\ I_N,\ Z_N$	Rated phase voltage, rated current, and base impedance
$f_N,\ \omega_{EL}$	Rated electrical frequency and angular frequency
I_{E0}	Reference excitation current
P_{EL}	Electrical power

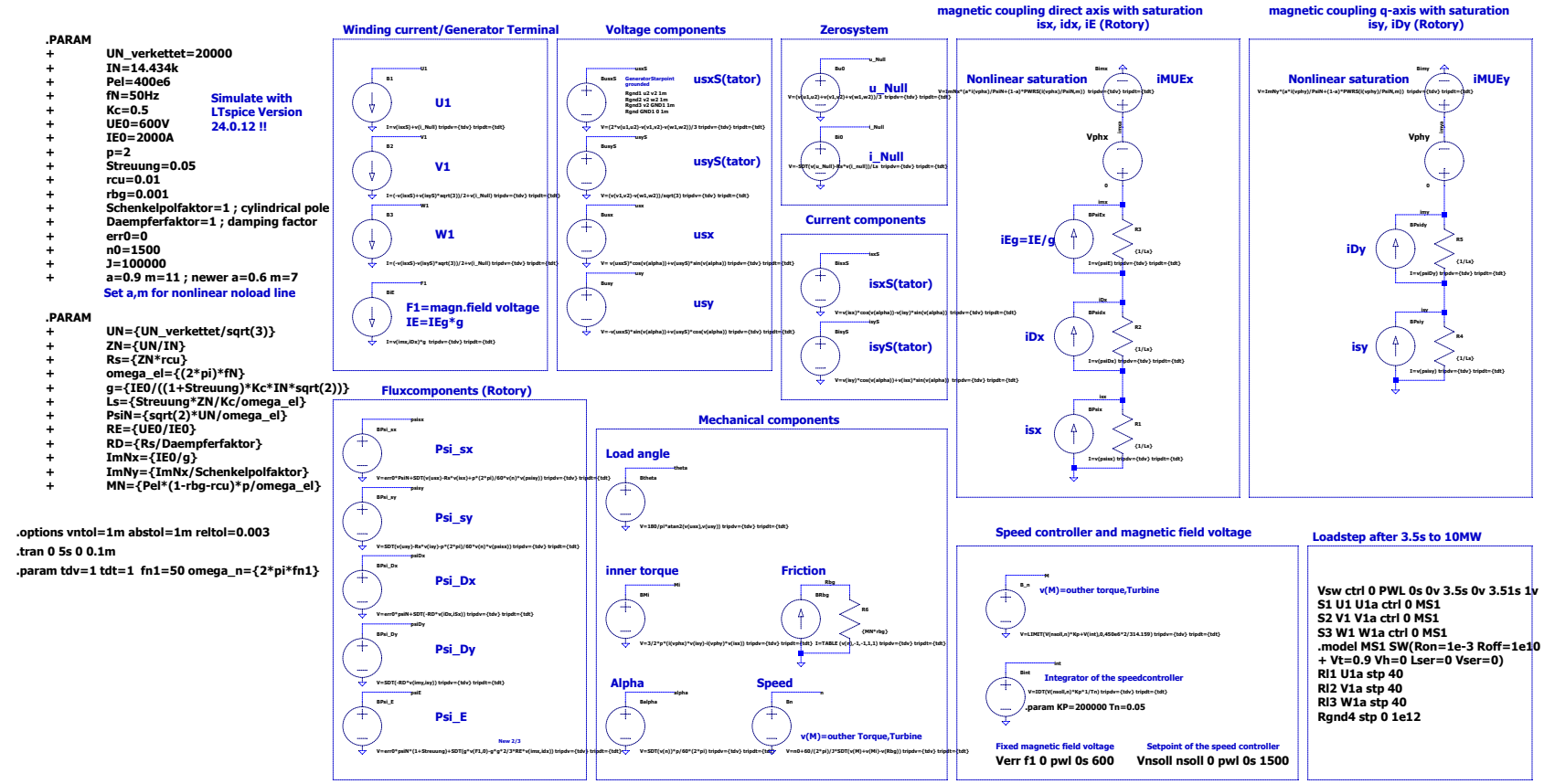


Figure A.1: Synchronous machine schematic model

```
1 * NOTE: idt() = integration function
2 .SUBCKT Block1_fixe_Drehzahl U1 V1 W1 F1 M n GND1
3 + PARAMS:
4 + UN_verkettet=20000
5 + IN=14.434k
6 + Pel=400e6 ;UN_verkettet*IN*sqrt(3)*cosphi of 0.8
7 + fn=50
8 + Kc=0.5
9 + UE0=600V ; Power of exciter = 0.25-5% * SN machine (smaller values for
    higher SN)
10 + IE0=2000A
11 + p=2
12 + Streuung=0.05 ;
13 + rcu=0.001
14 + rbg=0.001
15 + Schenkelpolfaktor=1 ;(1=cylindrical pole 0.5=Salient pole)
16 + Daempferfaktor=1
17 + err0=0
18 + RE_factor=0.6666
19 + n0=1500
20 + n1={n0} ;starts with 1500U/min
21 + alpha_0=0
22 + a=0.9 ; Parameter for nonlinear noload line (should be about 0.6-0.9)
23 + m=7
24 .PARAM
25 + a1={a}; Saturation parameter d-axis
26 + m1={m}; Saturation parameter d-axis
27 + a2={a1}; Saturation parameter q-axis
28 + m2={m1}; Saturation parameter q-axis
29 + UN={UN_verkettet/sqrt(3)}
30 + ZN={UN/IN}
31 + Rs={ZN*rcu}
32 + omega_el={(2*pi)*fN}
33 + Ls={Streuung*ZN/Kc/omega_el}
34 + PsiN={sqrt(2)*UN/omega_el}
35 + RE={UE0/IE0}
36 + RD={Rs/Daempferfaktor}
37 + RDx={RD} ; can be overwritten by setting this parameters on modelline
38 + RDy={RD} ; can be overwritten by setting this parameters on modelline
39 + LDx={Ls} ; can be overwritten by setting this parameters on modelline
40 + LDy={Ls} ; can be overwritten by setting this parameters on modelline
41 + LE={Ls} ; can be overwritten by setting this parameters on modelline
42 + R0={Rs} ; can be overwritten by setting this parameters on modelline
43 + L0={Ls} ; can be overwritten by setting this parameters on modelline (X0
     is 0.03-0.1 p.u)
44 + Lsd={Ls}
45 + Lsq={Ls}
46 + XD={UN/IN/Kc}
47 + XQ={XD*Schenkelpolfaktor}
48 + g={UE0/(UN*sqrt(2)*RE/(XD-omega_el*Lsd))} ; updated formula for g
    parameter
49 + g1={UE0/(UN*sqrt(2)*RE/(XQ-omega_el*Lsq))} ; updated formula for salient
```

```
    pole machine
50 + ImNx={IE0/g} ; d-axis magnetizing current
51 + ImNy={IE0/g1} ;q-axis magnetizing current
52 + MN={Pel*(1-rbg-rcu)*p/omega_el}
53 *
54 * currents of windings
55 G1 U1 U2 VALUE={v(isxS)+v(i_Null)}
56 G2 V1 V2 VALUE={(-v(isxS)+v(isyS)*sqrt(3))/2+v(i_Null)}
57 G3 W1 W2 VALUE={(-v(isxS)-v(isyS)*sqrt(3))/2+v(i_Null)}
Synchrongenerator_spicecode.txt Gedruckt am 15. August 2026
Seite 2
58 GiE F1 0 VALUE={v(ie)*g}
59 * voltage components
60 EuxS usxS 0 VALUE={(2*v(u1,u2)-v(v1,v2)-v(w1,w2))/3}
61 EuyS usyS 0 VALUE={(v(v1,v2)-v(w1,w2))/sqrt(3)}
62 Eux usx 0 VALUE={ v(usxS)*cos(v(alpha))+v(usyS)*sin(v(alpha))}
63 Euy usy 0 VALUE={-v(usxS)*sin(v(alpha))+v(usyS)*cos(v(alpha))}
64 * Starpoint
65 R2 v2 GND1 1u
66 R12 u2 GND1 1u
67 R23 w2 GND1 1u
68 RGND1 GND1 0 1e6
69 * Zerosystem
70 Eu0 u_Null 0 VALUE={(v(u1,u2)+v(v1,v2)+v(w1,w2))/3}
71 Ei0 i_Null 0 VALUE={idt(v(u_Null)-R0*v(i_null))/L0}
72 * current components
73 EisxS isxS 0 VALUE={v(isx)*cos(v(alpha))-v(isy)*sin(v(alpha))}
74 EisyS isyS 0 VALUE={v(isy)*cos(v(alpha))+v(isx)*sin(v(alpha))}
75 * magnetic coupling of direct axis with saturation
76 Gpsix 0 isx VALUE={v(psisx)}
77 Rsx isx 0 {1/Lsd}
78 Gpsidx isx iDx1 VALUE={v(psiDx)} ;
79 Rsdx iDx1 isx {1/LDx}
80 GpsiEx iDx1 imx1 VALUE={v(psiE)}
81 RsEx imx1 iDx1 {1/LE}
82 vphx imx1 imxa1 0
83 Eimx imxa1 0 VALUE={ImNx*(a1*i(vphx)/PsiN+(1-a1)*PWRS(i(vphx)/PsiN,m1))}
84 * magnetic coupling of quadrature axis with saturation
85 Gpsiy 0 isy VALUE={v(psisy)}
86 Rsy isy 0 {1/Lsq}
87 Gpsidy isy imy1 VALUE={v(psiDy)}
88 Rsdy imy1 isy {1/LDy}
89 vphy imy1 imya1 0
90 Eimy imya1 0 VALUE={ImNy*(a2*i(vphy)/PsiN+(1-a2)*PWRS(i(vphy)/PsiN,m2))}
91 * damper currents
92 Eidx idx 0 VALUE={v(idx1)-v(isx)}
93 Eidy idy 0 VALUE={v(imy1)-v(isy)}
94 * exciter current
95 Eie ie 0 VALUE={v(imx1)-v(idx1)}
96 * Flux components
97 Epsi_sx psisx 0 VALUE={err0*PsiN+idt(v(usx)-Rs*v(isx)+p*(2*pi)/60*v(n)*v(
    psisy))}
98 Epsi_sy psisy 0 VALUE={idt(v(usy)-Rs*v(isy)-p*(2*pi)/60*v(n)*v(psisx))}
```

```
99 Epsi_Dx psiDx 0 VALUE={err0*psiN+idt(-RDx*v(idx))}
100 Epsi_Dy psiDy 0 VALUE={idt(-RDy*v(idy))}
101 Epsi_E psiE 0 VALUE={err0*psiN*(1+Streuung)+idt(v(F1,0)*g*RE_factor-RE*
    RE_factor*g*g*v(ie))}
102 * Load angle
103 Etheta 0 theta VALUE={idt(180/pi*atan2(v(usx),v(usy)))}
104 * inner torque
105 Emi Mi 0 VALUE={1.5*p*(i(vphx)*v(isy)-i(vphy)*v(isx))}
106 * outher torque
107 EMoment M 0 VALUE={-v(Mi)+v(Rbg)}
108 Vn n 0 {n1}
109 * friction
110 GRbg 0 Rbg TABLE {v(n)} (-1,-1) (1,1)
111 RRbg Rbg 0 {MN*rbg}
112 * Compute angle alpha
113 Ealpha alpha 0 VALUE={alpha_0*pi/180+idt(v(n))*p/60*(2*pi)}
114 .ENDS
```

Listing A.1: Behavioral implementation of a synchronous generator.

```
%%%
```

```
xrq=xmq^2/(xq-xq2)
sigmae=1-xmd/xe
sigmaxd=1-xmd/xd
sigmard=1-xmd/xrd
% Computation of Fundamental parameters for LTSpice model
% Delete Formulas and overwrite with the values of
% Rs,Ls,RE,RDx,RDy,LDx,LDy,LE,R0,L0 below
Rs=rcu*ZBase % Rs
Ls=xl*ZBase/omega % Ls
Lmd=xmd*ZBase/omega % (Lhd3)
Lmq=xmq*ZBase/omega % (Lhq3)
Lfd=Lmd*sigmae % Le
Lkd=(xrd-xmd)*ZBase/omega % LDx
Lkq=(xrq-xmq)*ZBase/omega % LDy
Rkd=(xrd-xmd^2/xe)*ZBase/omega/Td02 % RDx
Rkq=(xrq*ZBase/omega/Tq02) % RDy
Re=Ue0/Ie0*2/3 % RE
Re0=Ue0/Ie0
g=Ue0/(UN*sqrt(2)*Re0/(Xd-omega*Ls))
Res=Re*g*g
R0=Rs % R0
L0=Ls % L0
Td01=Xd/Res/omega % Should be same as above
% If not change Ie0 to meet Td01


```

Figure A.3: MATLAB code for converting standard machine parameters to fundamental model parameters

Transformer Model

Formula sheet for a three-phase two-winding transformer

Notation. Indices $k = 1, 2, 3$ identify the transformer legs or phases; p and s denote the primary and secondary windings. Uppercase U and I denote electrical quantities, Φ the magnetic flux, $\Psi = N\Phi$ the flux linkage, and $\Theta = NI$ the magnetomotive force. The example uses vector group Yy0.

Transformer-leg equations

For each leg $k \in \{1, 2, 3\}$, the reluctance associated with the iron-loss resistance is

$$\mathscr{R}_{R\mathrm{Fe}} = \frac{N_1^2}{R_{\mathrm{Fe}}}. \tag{B.1}$$

The nonlinear magnetizing characteristic is represented by

$$\Theta_{\mu k} = N_1 I_{\mu N} \left[a \frac{\Phi_k}{\Phi_N} + (1 - a) \left(\frac{\Phi_k}{\Phi_N} \right)^m \right], \qquad k = 1, 2, 3. \tag{B.2}$$

Equivalently, the magnetizing current of one leg is

$$I_{\mu k} = I_{\mu N} \left[a \frac{\Phi_k}{\Phi_N} + (1 - a) \left(\frac{\Phi_k}{\Phi_N} \right)^m \right]. \tag{B.3}$$

Primary-winding equations

For every phase k,

$$\Phi_{kp} = \frac{\Psi_{kp}}{N_1} = \frac{1}{N_1} \int \left(U_{kp} - R_1 I_{kp} \right) \mathrm{d}t, \qquad I_{kp} = \frac{\Theta_k}{N_1}. \tag{B.4}$$

Written separately for the three phases, this relation applies for $k = 1$, 2, and 3 with the corresponding phase quantities.

Secondary-winding equations

For every phase k,

$$\Phi_{ks} = \frac{\Psi_{ks}}{N_2} = \frac{1}{N_2} \int (U_{ks} - R_2 I_{ks}) \, \mathrm{d}t, \qquad I_{ks} = \frac{\Theta_k}{N_2}. \tag{B.5}$$

The voltage and current reference directions must be chosen consistently with the winding polarities used in the circuit model.

Common core flux and winding flux linkages

For ideal magnetic coupling of the windings on each leg,

$$\Phi_{kp} = \Phi_{ks} = \Phi_k, \qquad k = 1, 2, 3. \tag{B.6}$$

The corresponding flux linkages are

$$\Psi_{kp} = N_1 \Phi_k, \qquad \Psi_{ks} = N_2 \Phi_k. \tag{B.7}$$

Thus, in general,

$$\Psi_{kp} \neq \Psi_{ks} \quad \text{when} \quad N_1 \neq N_2, \tag{B.8}$$

even though both windings link the same core flux Φ_k.

Leakage-flux branch

With $\mathscr{R}_{\Phi\sigma}$ denoting the equivalent magnetic reluctance of the leakage path,

$$\Phi_{k\sigma} = \frac{\Theta_k}{\mathscr{R}_{\Phi\sigma}}, \qquad k = 1, 2, 3. \tag{B.9}$$

For a leakage inductance $L_{1\sigma}$ referred to the primary winding,

$$L_{1\sigma} = \frac{N_1^2}{\mathscr{R}_{\Phi\sigma}} \qquad \Longleftrightarrow \qquad \mathscr{R}_{\Phi\sigma} = \frac{N_1^2}{L_{1\sigma}}. \tag{B.10}$$

Tank representation

The tank is represented by an additional magnetic reluctance $\mathscr{R}_{\text{tank}}$. The source text recommends the following approximate choices:

$$\mathscr{R}_{\text{tank}} \approx (0.1 \ldots 0.01)\frac{N_1^2}{L_{1\sigma}}, \quad \text{rough initial estimate: 10–100 times the leakage inductance,} \tag{B.11}$$

$$\mathscr{R}_{\text{tank}} \approx 10^5 \ldots 10^6, \quad \text{tank magnetic reluctance in the model units,} \tag{B.12}$$

$$\mathscr{R}_{\text{tank}} \approx 10^{20}, \quad \text{tank branch absent (practically open).} \tag{B.13}$$

Transformer main data:

Symbol	Meaning
S_N	Three-phase rated apparent power
$U_{1N,LL}$	Primary rated line-to-line voltage
$U_{2N,LL}$	Secondary rated line-to-line voltage
f_N	Rated frequency
N_1	Primary number of turns
u_R	Resistive part of relative short-circuit voltage
u_X	Inductive part of relative short-circuit voltage
i_0	Relative no-load current
p_0	Relative no-load loss, with $p_0 < i_0$
a	Linear weighting in the magnetizing characteristic
m	Exponent of the nonlinear magnetizing term

Per-leg quantities for vector group Yy0

The three-phase values are first converted to one transformer leg:

$$S_{N,\mathrm{leg}} = \frac{S_N}{3}, \qquad U_{1N} = \frac{U_{1N,LL}}{\sqrt{3}}, \qquad U_{2N} = \frac{U_{2N,LL}}{\sqrt{3}}, \tag{B.14}$$

$$I_{1N} = \frac{S_{N,\mathrm{leg}}}{U_{1N}}, \qquad Z_{1N} = \frac{U_{1N}}{I_{1N}}, \qquad \omega_N = 2\pi f_N. \tag{B.15}$$

Winding and iron-loss parameters

The short-circuit impedance is split equally between the two windings and referred initially to the primary side:

$$R_1 = \frac{Z_{1N} u_R}{2}, \qquad L_{1\sigma} = \frac{Z_{1N} u_X}{2\omega_N}, \tag{B.16}$$

$$R_{\mathrm{Fe}} = \frac{U_{1N}^2}{p_0 S_{N,\mathrm{leg}}}, \qquad \mathscr{R}_{R\mathrm{Fe}} = \frac{N_1^2}{R_{\mathrm{Fe}}}, \tag{B.17}$$

$$N_2 = N_1 \frac{U_{2N}}{U_{1N}}, \qquad R_2 = R_1 \left(\frac{N_2}{N_1}\right)^2, \tag{B.18}$$

$$\mathscr{R}_{\Phi\sigma} = \frac{N_1^2}{L_{1\sigma}}. \tag{B.19}$$

Magnetizing branch and rated flux

From the no-load current and no-load losses,

$$X_h = \frac{U_{1N}^2}{S_{N,\mathrm{leg}}\sqrt{i_0^2 - p_0^2}}, \tag{B.20}$$

$$I_\mu = \frac{U_{1N}}{X_h}, \tag{B.21}$$

$$I_{\mu N} = \sqrt{2}\, I_\mu, \tag{B.22}$$

$$\Phi_N = \frac{\sqrt{2}\, U_{1N}}{\omega_N N_1}. \tag{B.23}$$

The nonlinear characteristic then becomes

$$I_\mu(\Phi) = I_{\mu N}\left[a\frac{\Phi}{\Phi_N} + (1-a)\left(\frac{\Phi}{\Phi_N}\right)^m\right]. \tag{B.24}$$

Parameter definitions

Symbol	Meaning
Θ_k	Magnetomotive force of leg k
$\Phi_k,\ \Phi_{k\sigma}$	Main and leakage fluxes of leg k
$\Psi_{kp},\ \Psi_{ks}$	Primary and secondary flux linkages
$\mathscr{R}_{R\mathrm{Fe}}$	Magnetic reluctance associated with the iron-loss branch
$\mathscr{R}_{\Phi\sigma}$	Equivalent leakage-path reluctance
$\mathscr{R}_{\mathrm{tank}}$	Tank-path magnetic reluctance
$R_1,\ R_2$	Primary and secondary winding resistances
R_{Fe}	Equivalent iron-loss resistance
$L_{1\sigma}$	Primary-referred leakage inductance
X_h	Main magnetizing reactance
$I_{\mu N},\ \Phi_N$	Rated peak magnetizing current and rated peak flux
$a,\ m$	Shape parameters of the nonlinear magnetization curve

Implementation notes. The time integrals may be implemented with behavioural sources in LTspice. When forming the winding equations, verify the chosen voltage polarities, current reference directions, initial flux, and the sign of each magnetomotive-force contribution. The numerical tank-reluctance range is model- and unit-dependent and should be validated against the intended zero-sequence behaviour.

Rated data: 300 MVA, 400 kV / 120 kV, 50 Hz

Symbol	Value	Meaning
S_N	300×10^6 VA	Three-phase rated apparent power
$U_{1N,LL}$	400×10^3 V	Primary rated line-to-line voltage
$U_{2N,LL}$	120×10^3 V	Secondary rated line-to-line voltage
f_N	50 Hz	Rated frequency
N_1	1000	Primary number of turns
u_R	0.002	Resistive part of relative short-circuit voltage
u_X	0.2	Inductive part of relative short-circuit voltage
i_0	0.0006	Relative no-load current
p_0	0.0004	Relative no-load loss, with $p_0 < i_0$
a	0.75	Linear weighting in the magnetizing characteristic
m	9	Exponent of the nonlinear magnetizing term

Assumptions. The calculation uses a balanced per-phase equivalent circuit for vector group Yy0. The total short-circuit resistance and leakage reactance are split equally between the primary winding and the secondary winding referred to the primary side. All magnetizing-branch quantities are initially referred to the 400 kV side.

Per-leg rated quantities

The three-phase rated values are converted to one transformer leg:

$$S_{N,\mathrm{leg}} = \frac{S_N}{3} = \frac{300\,\mathrm{MV\,A}}{3} = 100\,\mathrm{MV\,A}, \tag{B.25}$$

$$U_{1N} = \frac{U_{1N,LL}}{\sqrt{3}} = \frac{400\,\mathrm{kV}}{\sqrt{3}} = 230.940\,\mathrm{kV}, \tag{B.26}$$

$$U_{2N} = \frac{U_{2N,LL}}{\sqrt{3}} = \frac{120\,\mathrm{kV}}{\sqrt{3}} = 69.282\,\mathrm{kV}, \tag{B.27}$$

$$I_{1N} = \frac{S_{N,\mathrm{leg}}}{U_{1N}} = 433.013\,\mathrm{A}, \tag{B.28}$$

$$Z_{1N} = \frac{U_{1N}}{I_{1N}} = \frac{U_{1N,LL}^2}{S_N} = 533.333\,\Omega, \tag{B.29}$$

$$\omega_N = 2\pi f_N = 314.159\,\mathrm{rad/s}. \tag{B.30}$$

The turns ratio and secondary number of turns are

$$u = \frac{N_1}{N_2} = \frac{U_{1N}}{U_{2N}} = \frac{400}{120} = 3.33333, \tag{B.31}$$

$$N_2 = N_1 \frac{U_{2N}}{U_{1N}} = 1000\,\frac{120}{400} = \boxed{300}. \tag{B.32}$$

Winding parameters

The total short-circuit resistance and leakage reactance referred to the primary side are

$$R'_k = u_R Z_{1N} = 0.002 \cdot 533.333\,\Omega = 1.066\,67\,\Omega, \tag{B.33}$$

$$X'_{k\sigma} = u_X Z_{1N} = 0.2 \cdot 533.333\,\Omega = 106.667\,\Omega, \tag{B.34}$$

$$L'_{k\sigma} = \frac{X'_{k\sigma}}{\omega_N} = 0.339\,531\,\mathrm{H}. \tag{B.35}$$

With equal division between the two windings,

$$R_1 = R'_2 = \frac{R'_k}{2} = \boxed{0.533\,333\,\Omega}, \tag{B.36}$$

$$L_{1\sigma} = L'_{2\sigma} = \frac{L'_{k\sigma}}{2} = \boxed{0.169\,765\,\mathrm{H}}. \tag{B.37}$$

The actual secondary-side parameters follow from $(N_2/N_1)^2 = 0.09$:

$$R_2 = R'_2 \left(\frac{N_2}{N_1}\right)^2 = 0.533\,333\,\Omega \cdot 0.09 = \boxed{0.048\,000\,\Omega}, \tag{B.38}$$

$$L_{2\sigma} = L'_{2\sigma} \left(\frac{N_2}{N_1}\right)^2 = 0.169\,765\,\mathrm{H} \cdot 0.09 = \boxed{0.015\,278\,9\,\mathrm{H}}. \tag{B.39}$$

For the magnetic-circuit leakage representation,

$$\mathscr{R}_{\Phi\sigma} = \frac{N_1^2}{L_{1\sigma}} = \frac{1000^2}{0.169\,765\,\mathrm{H}} = \boxed{5.890\,49 \cdot 10^6/\mathrm{H}}. \tag{B.40}$$

Iron-loss parameters

The total rated no-load loss is

$$P_0 = p_0 S_N = 0.0004 \cdot 300\,\mathrm{MV\,A} = 120\,\mathrm{kW}. \tag{B.41}$$

Thus, each transformer leg represents 40 kW of iron loss. The primary-referred iron-loss resistance is

$$R_{\mathrm{Fe}} = \frac{U_{1N}^2}{p_0 S_{N,\mathrm{leg}}} \tag{B.42}$$

$$= \frac{(230.940\,\mathrm{kV})^2}{0.0004 \cdot 100\,\mathrm{MV\,A}} = \boxed{1.333\,33\,\mathrm{M\Omega}}. \tag{B.43}$$

The corresponding RMS iron-loss current per phase is

$$I_{\mathrm{Fe}} = \frac{U_{1N}}{R_{\mathrm{Fe}}} = \boxed{0.173\,205\,\mathrm{A}}. \tag{B.44}$$

In the magnetic-circuit representation used by the transformer model,

$$\mathscr{R}_{R\mathrm{Fe}} = \frac{N_1^2}{R_{\mathrm{Fe}}} = \frac{1000^2}{1.333\,33 \cdot 10^6\,\Omega} = \boxed{0.750000}. \tag{B.45}$$

Magnetizing branch and rated flux

The rated RMS no-load current is

$$I_0 = i_0 I_{1N} = 0.0006 \cdot 433.013\,\mathrm{A} = 0.259\,808\,\mathrm{A}. \tag{B.46}$$

After subtracting the active iron-loss component vectorially, the RMS magnetizing current is

$$I_\mu = I_{1N}\sqrt{i_0^2 - p_0^2} \tag{B.47}$$

$$= 433.013\,\mathrm{A}\sqrt{0.0006^2 - 0.0004^2} = \boxed{0.193\,649\,\mathrm{A}}. \tag{B.48}$$

The magnetizing reactance and inductance referred to the primary side are

$$X_h = \frac{U_{1N}}{I_\mu} = \frac{U_{1N}^2}{S_{N,\mathrm{leg}}\sqrt{i_0^2 - p_0^2}} = \boxed{1.192\,57\,\mathrm{M\Omega}}, \tag{B.49}$$

$$L_h = \frac{X_h}{\omega_N} = \boxed{3.796\,07 \cdot 10^3\,\mathrm{H}}. \tag{B.50}$$

The peak magnetizing current used in the nonlinear characteristic is

$$I_{\mu N} = \sqrt{2}\, I_\mu = \boxed{0.273\,861\,\mathrm{A}}. \tag{B.51}$$

For a sinusoidal phase voltage, the rated peak core flux is

$$\Phi_N = \frac{\sqrt{2}\,U_{1N}}{\omega_N N_1} \tag{B.52}$$

$$= \frac{\sqrt{2} \cdot 230.940\,\mathrm{kV}}{314.159\,\mathrm{rad/s} \cdot 1000} = \boxed{1.039\,60\,\mathrm{Wb}}. \tag{B.53}$$

With $a = 0.75$ and $m = 9$, the numerical magnetizing characteristic becomes

$$I_\mu(\Phi) = 0.273\,861\,\mathrm{A}\left[0.75\frac{\Phi}{1.039\,60\,\mathrm{Wb}} + 0.25\left(\frac{\Phi}{1.039\,60\,\mathrm{Wb}}\right)^9\right]. \tag{B.54}$$

Calculated parameter summary

Parameter	Calculated value	**Reference**
N_2	300	Secondary number of turns
R_1	0.533 333 Ω	Primary winding resistance
$L_{1\sigma}$	0.169 765 H	Primary leakage inductance
R'_2	0.533 333 Ω	Secondary resistance referred to primary
$L'_{2\sigma}$	0.169 765 H	Secondary leakage inductance referred to primary
R_2	0.048 000 Ω	Actual secondary resistance
$L_{2\sigma}$	0.015 278 9 H	Actual secondary leakage inductance
R_{Fe}	1.333 33 MΩ	Iron-loss resistance, primary-referred
X_h	1.192 57 MΩ	Magnetizing reactance, primary-referred
L_h	$3.796\,07 \cdot 10^3$ H	Linear magnetizing inductance
$I_{\mu N}$	0.273 861 A	Rated peak magnetizing current
Φ_N	1.039 60 Wb	Rated peak core flux
$\mathscr{R}_{R\mathrm{Fe}}$	0.750000	Iron-loss magnetic-model parameter
$\mathscr{R}_{\Phi\sigma}$	$5.890\,49 \cdot 10^6$/H	Leakage-path magnetic parameter

Magnetic and electrical circuit

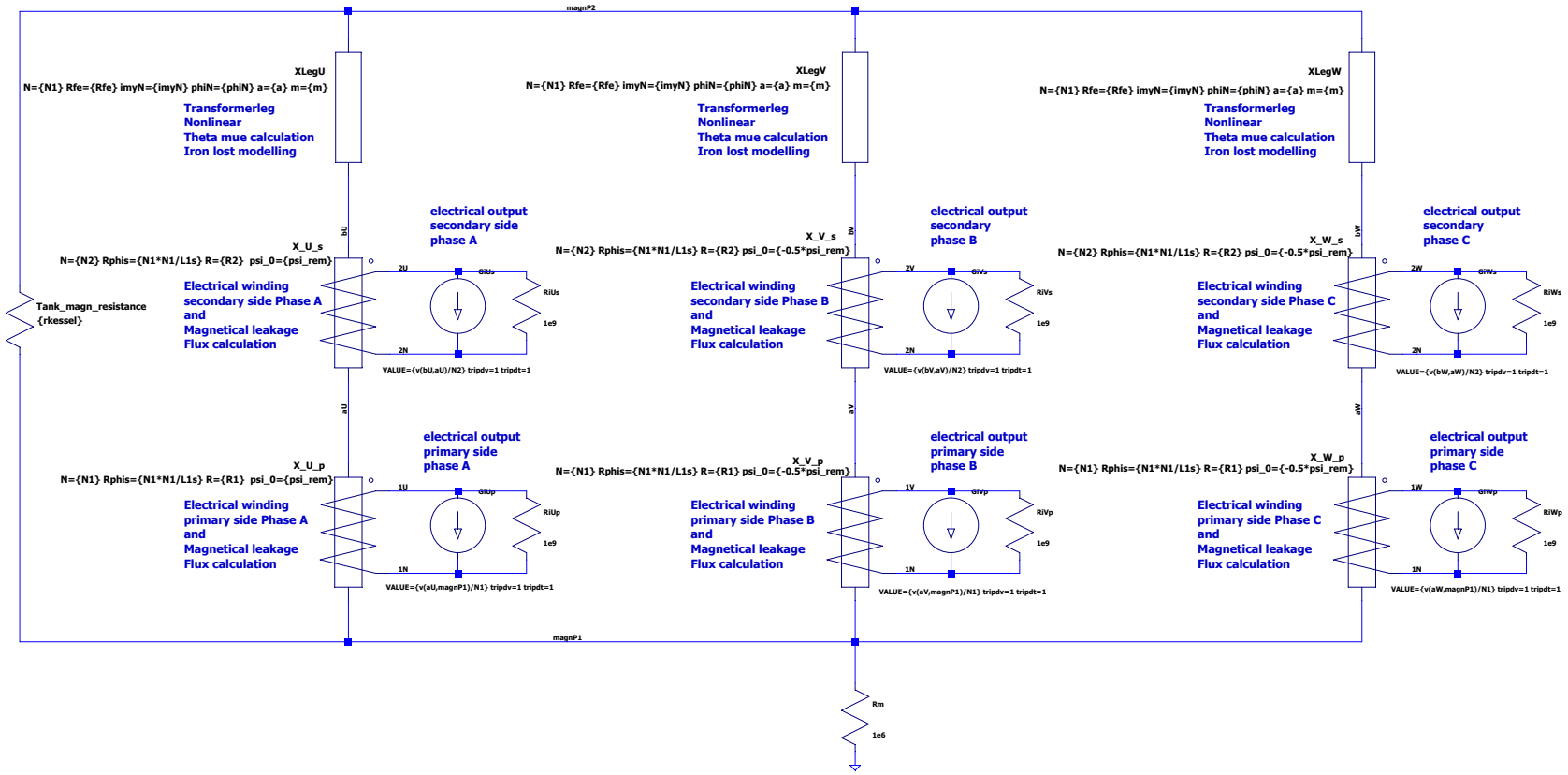


Figure B.1: Electrical and magnetic model of a three-phase Yy0 transformer with a three-leg core

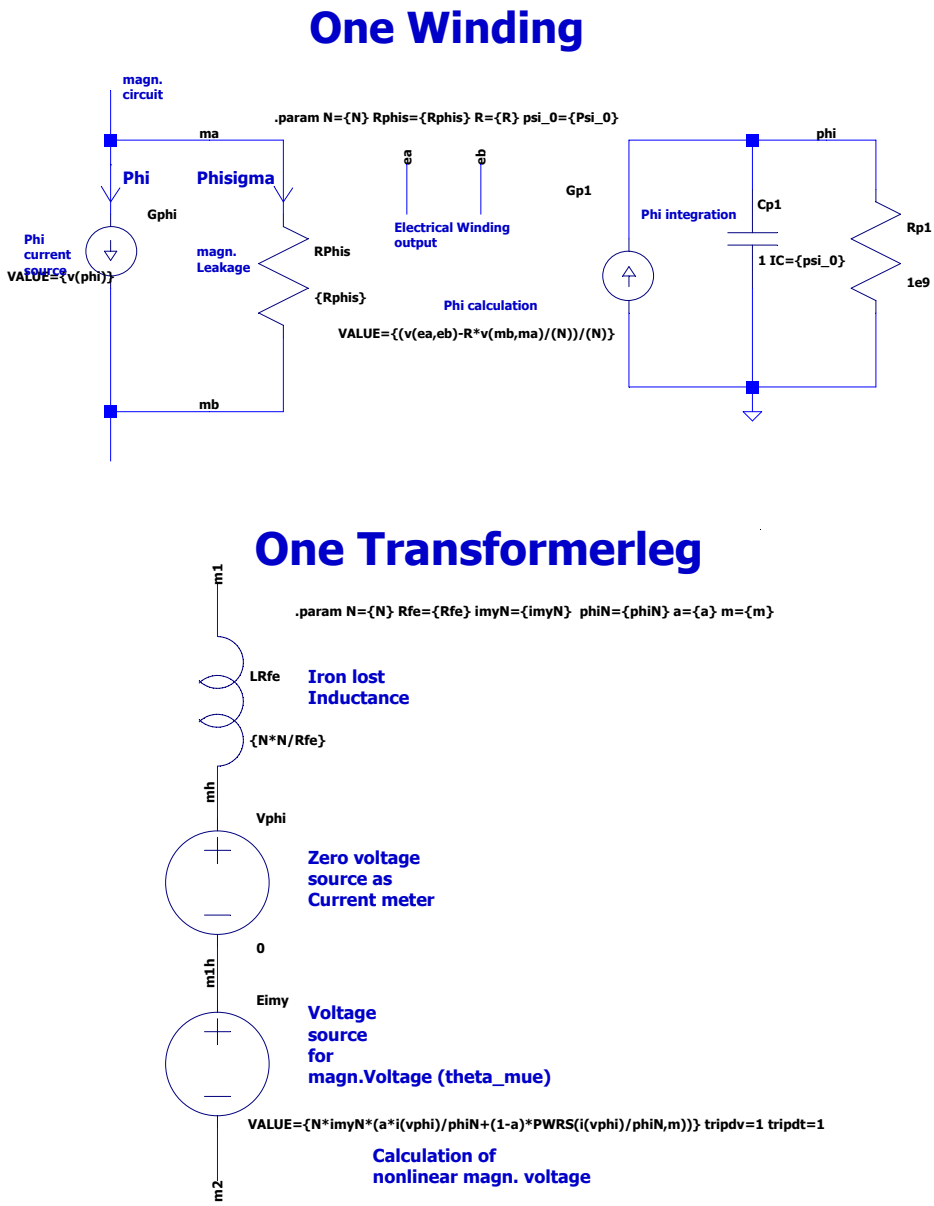


Figure B.2: One winding and one leg of the transformer model

```
1 .SUBCKT RUYy0_400k_120k_300MVA 1U 1V 1W 1N 2U 2V 2W 2N
2 + PARAMS:
3 + SN=300e6
4 + U1Nenn=400e3
5 + U20=120e3
6 + fn={fn1}
7 + N1=1000
8 + uR=0.002
9 + uX=0.2
10 + i0=0.0006
11 + p0=0.0004
12 + a=0.75 ; FS
13 + m=9;FS
14 + psi_0=0 ; Remanence-parameter 0,1,-1
15 + psir=1
16 + psis=-0.5
17 + psit=-0.5
18 + tpsi0=1000s ; Remanence-timepoint
19 + tpsid=0.1ms ; On-time of remanent flux
20 + ron=0.00001 ; On - Parameter of switcher for remanent flux
21 + bf=0 ; bf=1 -> middle leg with shorter magn. path correction factor
22 + rkessel=6e5 ; Tank resistance
23 + l_=0
24 + middle=0
25 + stproz=1.0
26 + ctrl={1+(l_-middle)*stproz/100} ; simple Tap changer
**************************************************************************
27 Xu 1U 1N 2U 2N magn1 magn2 Trafoschenkel2_Yy
28 + PARAMS:
29 + SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
30 + uX={uX} i0={i0*IF(bf==0,1,1.2)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
31 + psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
**************************************************************************
32 Xv 1V1N 2V 2N magn1 magn2 Trafoschenkel2_Yy
33 + PARAMS:
34 + SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
35 + uX={uX} i0={i0*IF(bf==0,1,0.6)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
36 + psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
**************************************************************************
37 Xw 1W 1N 2W 2N magn1 magn2 Trafoschenkel2_Yy
38 + PARAMS:
39 + SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
40 + uX={uX} i0={i0*IF(bf==0,1,1.2)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
41 + psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
**************************************************************************
42 Rm magn1 0 1MEG
43 Rkessel magn1 magn2 {rkessel}
44 RGND1U 1U 0 1e12
45 RGND1V 1V 0 1e12
46 RGND1W 1W 0 1e12
47 RGND2U 2U 0 1e12
```

```
48 RGND2V 2V 0 1e12
49 RGND2W 2W 0 1e12
50 RGND1N 1N 0 1e12
51 RGND2N 2N 0 1e12
52 .ENDS
************************************************************
54 .SUBCKT Trafoschenkel2_Yy 1_1 1_2 2_1 2_2 magnP1 magnP2
55 + PARAMS:
56 + SN=10kVA
57 + U1N=400V
58 + U20=400V
59 + fn={fn1}
60 + N1=1000
61 + uR=0.04
62 + uX=0.08
63 + i0=0.05
64 + p0=0.005
65 + a={0.3} ; FS
66 + m={7} ; FS
67 + psi_0=0 ; Remanenzparameter 0,1,-1
68 + tpsi0=40ms
69 + tpsid=0.1ms
70 + ron=0.00001
71 + ctrl=1
72 .PARAM
73 + I1N={SN/U1N}
74 + Z1N={U1N/I1N}
75 + omegaN={(2*pi)*fN}
76 + R1={Z1N*uR/2}
77 + L1s={Z1N*uX/2/omegaN}
78 + RFe={U1N*U1N/(p0*SN)}
79 + N2={N1*U20/U1N}
80 + R2={R1*N2/N1*N2/N1}
81 + Xh={(U1N*U1N)/(SN*sqrt(i0*i0-p0*p0))} ;FS
82 + iu={U1N/Xh} ;FS
83 + ImyN={sqrt(2)*iu} ;FS
84 + phiN={sqrt(2)*U1N/omegaN/N1}
85 + a={a} ; FS
86 + m={m} ; FS
*********************************************************************
87 * Windingscurrents
88 Gi1 1_1 1_2 VALUE={v(a,magnP1)/(N1*ctrl)}
89 Rpi1 1_1 1_2 1000MEG
90 Gi2 2_1 2_2 VALUE={v(b,a)/N2}
91 Rpi2 2_1 2_2 1000MEG
92 * Windings
*********************************************************************
93 X_W1 magnP1 a 1_1 1_2 winding_Yy
94 +PARAMS: R={R1} N={N1} Rphis={N1*N1/L1s}
95 +psi_0={psi_0*phiN} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
*********************************************************************
96 X_W2 a b 2_1 2_2 winding_Yy
97 +PARAMS: R={R2} N={N2} Rphis={N1*N1/L1s}
```

```
98 +psi_0={psi_0*phiN} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl=1
**********************************************************************
99 * Iron leg
100 X_SCH b magnP2Leg_Yy PARAMS:
101 + N={N1} RFe={RFe} ImyN={ImyN} ctrl={ctrl}
102 + phiN={phiN} a={a} m={m}
103 .ENDS
104 **********************************************************************
105 .SUBCKT winding_Yy ma mb ea eb
106 +PARAMS: R=1 N=1 Rphis=1 psi_0=0 tpsi0=1000s
+   tpsid=0.1ms ron=0.00001 ctrl=1
107 Gphi ma mb VALUE={v(phi)}
108 Rphis ma mb {Rphis*ctrl*ctrl}
109 ; Flux phi
110 Gp 0 phi VALUE={(v(ea,eb)-R*ctrl*v(mb,ma)/(N*ctrl))/(N*ctrl)}
111 Cp phi 0 1 IC=0
112 Rpphi 01000MEG
113 Vctrl ctrl 0 pwl 0s 0v {tpsi0+1p} 0v {tpsi0+tpsid+2p} 1V
+   v{tpsi0+2*tpsid+3p} 1v {tpsi0+3*tpsid+4p} 0v
114 S1 phi phix ctrl 0 MS1
115 .model MS1 SW(Ron={Ron} Roff=1e9 Vt=0.9 Vh=0 Lser=0 Vser=0)
116 E1 phix 0 value={psi_0}
117 .ENDS ;
118 **********************************************************************
119 * magnetic leg
120 .SUBCKT Leg_Yy m1 m2
121 +PARAMS: N=1 RFe=1 ImyN=1 a=1 phiN=1 m=9 ctrl=1
122 LRFem1 mh {N*ctrl*N*ctrl/RFe} Rser={0.001m*N*N/RFe} ; Rser=parasitic
    Value
123 vphi mh mh1 0
124 rphi mh 0 1e12
125 Eimy mh1 m2
126 +VALUE={N*imyN*(a*i(vphi)/phiN+(1-a)*PWRS(i(vphi)/phiN,m))}
127 .ENDS;
128 **********************************************************************
```

Listing B.1: Behavioral implementation of a three-phase transformer.

C Mathematical Modelling with Behavioral Sources in LTspice

C.1 Behavioral E and G Sources

Behavioral voltage and current sources make it possible to implement mathematical models directly in LTspice. Node voltages, branch currents, parameters, arithmetic operations, logical operations, and built-in functions can be combined in one expression.

Basic LTspice syntax

An E source generates a voltage between its output nodes:

```
E1 node1 node2 VALUE={mathematical_expression}
```

A G source generates a branch current between its output nodes:

```
G1 node1 node2 VALUE={mathematical_expression}
```

Typical expression operands are

```
V(node)              ; node voltage relative to ground
V(node_plus,node_minus) ; differential node voltage
I(Vsense)             ; current through a voltage source
Ix(U1:pin)            ; current into a subcircuit pin
```

Elements used in the toolboxes. The principal element designators are V, I, L, C, R, K, E, G, S, T, and O. Consult the LTspice help for the precise syntax, sign convention, and analysis restrictions of each element.

Important C.1 – (Dimensional consistency)

The value returned by an E-source expression is interpreted as volts; the value returned by a G-source expression is interpreted as amperes. LTspice does not enforce physical dimensional consistency inside a behavioral expression. The model author must therefore verify all scaling factors and units.

C.2 Functions in Behavioral Expressions

The following functions can be used in behavioral E and G sources. Optional arguments are shown in square brackets. Function availability can depend on the installed LTspice version.

Table C.1: Functions available in LTspice behavioral expressions.

Function	Description
`abs(x)`	Absolute value of x.
`absdelay(x,t[,tmax])`	Delays x by t; the optional maximum delay is `tmax`.
`acos(x)`	Real part of the arc cosine. For example, `acos(-5)` returns `3.14159`, not a complex result.
`arccos(x)`	Synonym for `acos(x)`.
`acosh(x)`	Real part of the inverse hyperbolic cosine.
`asin(x)`	Real part of the arc sine. Out-of-domain arguments are limited to the real result.
`arcsin(x)`	Synonym for `asin(x)`.
`asinh(x)`	Inverse hyperbolic sine.
`atan(x)`	Arc tangent of x.
`arctan(x)`	Synonym for `atan(x)`.
`atan2(y,x)`	Four-quadrant arc tangent of y/x.
`atanh(x)`	Inverse hyperbolic tangent.
`buf(x)`	Returns 1 if $x > 0.5$; otherwise returns 0.
`cbrt(x)`	Real cube root of x.
`ceil(x)`	Smallest integer greater than or equal to x.
`cos(x)`	Cosine of x; the argument is in radians.
`cosh(x)`	Hyperbolic cosine of x.
`ddt(x)`	Time derivative $\mathrm{d}x/\mathrm{d}t$.
`delay(x,t[,tmax])`	Same as `absdelay(x,t[,tmax])`.
`dnlim(x,A,Delta)`	Smooth approximation of `max(x,A)` with first-derivative transition width Δ.
`exp(x)`	Exponential function e^x.
`floor(x)`	Largest integer less than or equal to x.
`hypot(x,y)`	Returns $\sqrt{x^2+y^2}$ with suitable numerical scaling.
`idt(x[,ic[,a]])`	Integrates x; `ic` is the optional initial condition and the integrator is reset when `a` is true.
`idtmod(x[,ic[,m[,o]]])`	Integrates x, resets on reaching modulus `m`, and offsets the output by `o`.
`if(x,y,z)`	Returns y if $x > 0.5$; otherwise returns z.
`int(x)`	Converts x to an integer.
`inv(x)`	Returns 0 if $x > 0.5$; otherwise returns 1.
`limit(x,y,z)`	Returns the intermediate value of x, y, and z.
`ln(x)`	Natural logarithm of x.
`log(x)`	Alternate syntax for `ln(x)`.
`log10(x)`	Base-10 logarithm of x.
`max(x,y)`	Greater of x and y.
`min(x,y)`	Smaller of x and y.
`mod(x,y)`	Remainder of the division x/y.
`noise()`	Returns 1 during a `.noise` analysis; otherwise returns 0.
`pow(x,y)`	Real part of x^y; for example, `pow(-1,0.5)` returns 0 rather than an imaginary value.

Continued on next page

Table C.1: Functions available in LTspice behavioral expressions (continued).

Function	Description
`pwr(x,y)`	Returns $\|x\|^y$.
`pwrs(x,y)`	Returns $\text{sgn}(x)\|x\|^y$.
`rand(x)`	Pseudorandom value between 0 and 1 determined by the integer value of x.
`random(x)`	Similar to `rand(x)`, but transitions smoothly between values.
`round(x)`	Integer nearest to x.
`sdt(x[,ic[,assert]])`	Alternate syntax for `idt(x[,ic[,assert]])`.
`sgn(x)`	Sign of x: -1, 0, or 1.
`sin(x)`	Sine of x; the argument is in radians.
`sinh(x)`	Hyperbolic sine of x.
`smallsig()`	Returns 1 during a `.ac` or `.noise` analysis; otherwise returns 0.
`square(x)`	Returns x^2.
`sqrt(x)`	Square root of x.
`table(x,a,b,c,d,...)`	Linearly interpolates a value for x from the point pairs $(a,b),(c,d),\ldots$.
`table(x, .include ``filename'')`	Interpolates from point pairs stored as numbers in an included file.
`tan(x)`	Tangent of x.
`tanh(x)`	Hyperbolic tangent of x.
`u(x)`	Unit step: 1 if $x > 0$; otherwise 0.
`uplim(x,A,Delta)`	Smooth approximation of `min(x,A)` with first-derivative transition width Δ.
`uramp(x)`	Returns x if $x > 0$; otherwise returns 0.
`white(x)`	Pseudorandom value between -0.5 and 0.5 with smoother transitions than `random(x)`.
`xor(x,y)`	Converts x and y to Boolean values and returns their exclusive OR.
`!(x)`	Alternate syntax for `inv(x)`.
`~(x)`	Alternate syntax for `inv(x)`.

C.3 Operators and Precedence

Table C.2 lists the behavioral-expression operators in ascending order of precedence. Operators in the same group have equal precedence; parentheses should be used whenever the intended evaluation order could be ambiguous.

Table C.2: LTspice behavioral-expression operators in ascending order of precedence.

Operator	Description
`&`	Converts both operands to Boolean values and returns logical AND.
`|`	Converts both operands to Boolean values and returns logical OR.
`^`	Converts both operands to Boolean values and returns logical XOR.
`>, <`	Greater-than and less-than comparisons.
`>=, <=`	Greater-than-or-equal and less-than-or-equal comparisons.

Table C.2: LTspice behavioral-expression operators (continued).

Operator	Description
`==`	Returns true if both expressions are equal.
`!=`, `<>`	Returns true if both expressions are unequal.
`+`, `-`	Floating-point addition and subtraction.
`*`, `/`	Floating-point multiplication and division.
`%`	Floating-point remainder; equivalent to `mod(x,y)`.
`**`	Raises the left operand to the power of the right operand. Only the real part is returned.
`!`, `~`	Unary Boolean inversion of the following expression.

Boolean conversion. For the behavioral functions `buf()`, `if()`, and `inv()`, LTspice treats values greater than 0.5 as true. The exact threshold behavior of logical operators should be checked when expressions can take non-binary analog values.

C.4 Example: Single-Phase Transformer

The example implements the voltage equations by numerical integration and solves the flux-current relations as a simultaneous linear system.

C.4.1 Differential equations

The winding-voltage equations are

$$u = R_1 i_1 + \frac{\mathrm{d}\psi_1}{\mathrm{d}t}, \tag{C.1}$$

$$0 = (R_2 + R_{\mathrm{load}}) i_2 + \frac{\mathrm{d}\psi_2}{\mathrm{d}t}. \tag{C.2}$$

Consequently,

$$\frac{\mathrm{d}\psi_1}{\mathrm{d}t} = u - R_1 i_1, \tag{C.3}$$

$$\frac{\mathrm{d}\psi_2}{\mathrm{d}t} = -(R_2 + R_{\mathrm{load}}) i_2. \tag{C.4}$$

C.4.2 Flux-current equations

The coupled linear system is

$$\begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} = L_{1h} \begin{bmatrix} 1+\sigma_1 & 1 \\ 1 & 1+\sigma_2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}. \tag{C.5}$$

Equivalently, the two residual equations are

$$0 = \psi_1 - (1+\sigma_1) L_{1h} i_1 - L_{1h} i_2, \tag{C.6}$$

$$0 = \psi_2 - L_{1h} i_1 - (1+\sigma_2) L_{1h} i_2. \tag{C.7}$$

C.4.3 LTspice netlist

```
* Single-phase transformer using behavioral sources
* Parameters
.PARAM U=20 R1=10 R2=8 R_Last=17
.PARAM L1h=0.1 sigma1=0.1 sigma2=0.15

* Integrator for flux psi1.
* G-C-R realization is used because idt() can be less robust.
Gpsi1  0 psi1 VALUE={ U - R1*V(i1) }
Cpsi1  psi1 0 1 IC=0
Rpsi1  psi1 0 1E6

* Integrator for flux psi2
Gpsi2  0 psi2 VALUE={ -(R2+R_Last)*V(i2) }
Cpsi2  psi2 0 1 IC=0
Rpsi2  psi2 0 1E6

* Simultaneous linear system
Gi1  0 i1 VALUE={ V(psi1)
+ (-(1+sigma1)*L1h*V(i1)-L1h*V(i2)) }
Ri1  i1 0 1E12

Gi2  0 i2 VALUE={ V(psi2)
+ (-L1h*V(i1)-(1+sigma2)*L1h*V(i2)) }
Ri2  i2 0 1E12

.TRAN 0.001 0.04 0 0.001 UIC
```

Listing C.1: Behavioral implementation of a single-phase transformer.

Remark C.1 – (Solution of the algebraic system)

The resistances `Ri1` and `Ri2` are set to $1 \cdot 10^{12}\,\Omega$. The G-source currents can therefore produce only negligible node voltages unless their behavioral expressions are nearly zero. During each Newton iteration, LTspice consequently adjusts `V(i1)` and `V(i2)` until both residual equations are satisfied simultaneously. The node voltages thus represent the calculated currents i_1 and i_2.

Important C.2 – (Numerical interpretation of the G-C-R integrator)

With `Cpsi1=1 F`, the capacitor relation gives $\mathrm{d}V(\mathtt{psi1})/\mathrm{d}t = I(\mathtt{Gpsi1})$, so the node voltage directly represents flux linkage. The parallel 1 MΩ resistor provides a DC path but also introduces a small leakage term. Increase its value if this artificial decay is significant, while retaining a sufficiently robust DC operating point.

Appendix Summary

Behavioral E and G sources allow differential equations, algebraic equations, logical functions, and lookup tables to be implemented directly in LTspice. E sources return voltages, whereas G sources return currents. Integrators may be formed either with `idt()` or explicitly with a G-C-R network. A simultaneous algebraic system can be solved by expressing each residual with a behavioral G source and forcing it toward zero through a very large shunt resistance. Careful attention to sign conventions, units, damping, and operator precedence is essential for numerically robust models.

D Download and Installation of LTspice and the Power-System Toolboxes

D.1 Purpose and Required Downloads

The examples in this book use LTspice Version 24.0.12 together with two power-system toolboxes:

- **Transient toolbox:** models and examples for electromagnetic transient simulations of high-energy power networks;
- **Steady-state toolbox:** models and examples for load-flow and steady-state simulations.

This book concentrates on transient simulations. The steady-state toolbox is optional and is required only when the corresponding load-flow models are to be used.

Files to download

Download the following archive from the project homepage: The transient and loadflow Toolboxes and the LTspice 24.0.12 simulator installation file in one archive `LTspice.zip`; The archive is large because it also contains the now free MicroCap12 simulator with Library file for loadflow and transient simulation and with some examples! This file seems a little messy because it's the development version.

If you only need the LTspice toolbox files preferably option is to download the archive from the LTspice usergroup in the files-z_yahoo section `adventures with loadflow`. NOTE: The LTspice simulator 24.0.12 must be downloading separately

Project homepage:

```
my homepage: <www.friedrichschmidt.at>  or
LTspice usergroup files at: <https://groups.io/g/LTspice/files/z_yahoo/
    adventures%20with%20loadflow>
```

Important D.1 – (Version used by this book)

The models and screenshots described in this book were prepared for LTspice Version 24.0.12. Newer releases may have problems displaying simulation results in the waveform viewer. Retain the 24.0.12 installer if exact reproducibility is required.

D.2 Installing LTspice 24.0.12

1. Extract the downloaded LTspice archive to a temporary folder.
2. Right-click the installer and select Run as administrator.
3. Complete the installation using the default installation options.
4. Start LTspice once and close it again before copying the toolboxes. This confirms that the simulator has been installed correctly and allows it to create its normal user configuration files.

Administrator rights. Administrator rights are required only for the LTspice installation when the installer writes to protected Windows folders. The toolbox files themselves are installed in the user's Documents folder and should normally not require administrator rights.

D.3 Installing the Toolboxes

In the paths below, replace `<username>` with the actual Windows user name. Extract the transient toolbox into

```
C:\Users\<username>\Documents\LTspice
```

After extraction, the transient-toolbox folder must be

```
C:\Users\<username>\Documents\LTspice\loadflow_transient
```

If the steady-state toolbox is also required, extract it so that the following folder exists:

```
C:\Users\<username>\Documents\LTspice\loadflow_steady_state
```

Important D.2 – (Check the extracted folder level)

Avoid an additional, duplicated directory level such as `LTspice\LTspice\loadflow_transient`. The search paths configured in LTspice must point directly to the folders that contain the toolbox symbols and library files.

Updating an existing installation. If a toolbox folder already exists, make a backup before extracting a newer archive. This preserves locally modified symbols, models, plot settings, and example circuits.

D.4 Configuring Search Paths

Start LTspice and open the configuration dialog with

Tools → Control Panel.

Open the Sym. & Lib. Search Paths tab. Add the transient-toolbox path to both the **Symbol Search Path** and the **Library Search Path**:

```
C:\Users\<username>\Documents\LTspice\loadflow_transient
```

Thus, the same path is entered twice: once in each of the two search-path lists.
If the steady-state toolbox is installed, also add

```
C:\Users\<username>\Documents\LTspice\loadflow_steady_state
```

to both search-path lists.

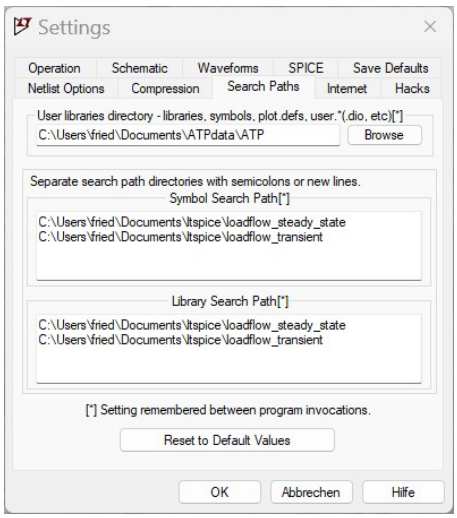


Figure D.1: Symbol and Library search path tab

Table D.1: Required LTspice symbol and library search paths.

Toolbox	Folder	Add to both lists
Transient	`C:\Users\<username>\Documents\LTspice\loadflow_transient`	Yes
Steady state	`C:\Users\<username>\Documents\LTspice\loadflow_steady_state`	Yes

D.5 Configuring the Simulator

Keep the Control Panel open and configure the following tabs.

D.5.1 SPICE tab

Set the numerical solver to

Solver: Alternate.

The alternate solver is recommended for the large networks, nonlinear magnetic models, switches, and behavioral equations used in the transient toolbox.

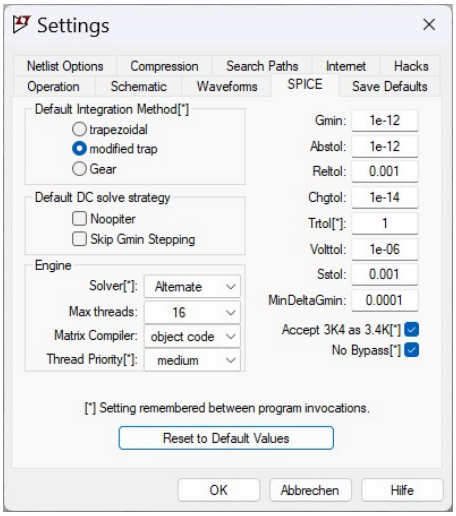


Figure D.2: Spice tab

D.5.2 Save Defaults tab

Enable the first three checkboxes on this tab. The exact labels can vary slightly between LTspice builds. These options ensure that the internal data required by many toolbox plots and hierarchical models are retained.

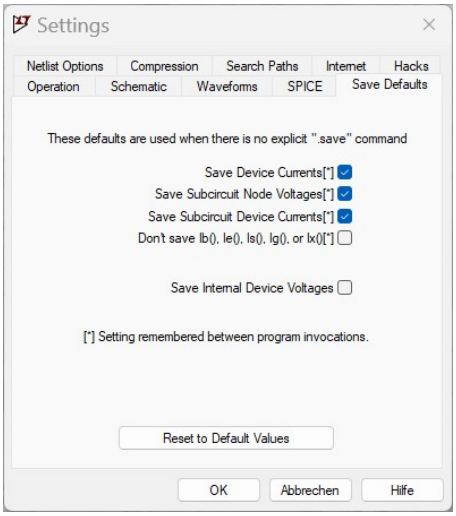


Figure D.3: Save defaults tab

> **Internal subcircuit quantities.** Several examples inspect internal transformer fluxes, winding currents, or hierarchical node voltages. If these traces are unavailable, return to SAVE DEFAULTS and verify that the relevant subcircuit node voltages and currents are saved.

D.5.3 Schematic tab

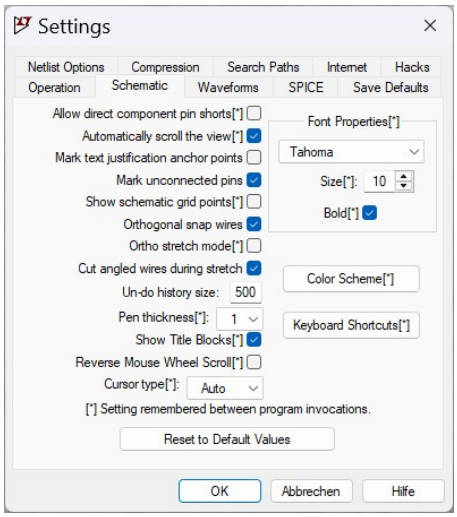


Figure D.4: Schematic tab

Set the schematic-editor font to

Arial, 10 pt, Bold.

D.5.4 Waveforms tab

Set the waveform-viewer font to

Arial, 12 pt, Data trace width=2, Cursor width=2.

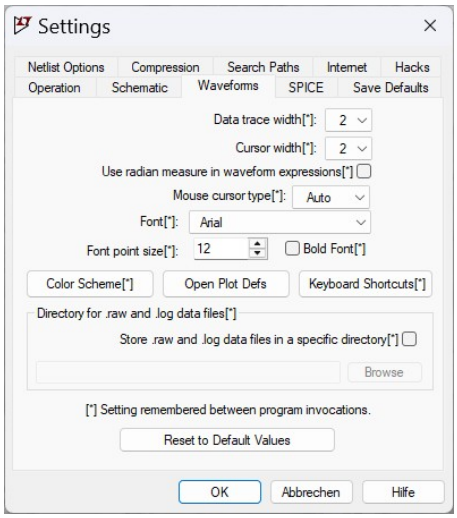


Figure D.5: Waveforms tab

D.5.5 Operation tab

Enable the following options:

- **Marching Waveforms**;
- **Generate Expanded Listing**;
- **Save all open files on start of simulation**;
- **Automatically Delete .raw Files**.

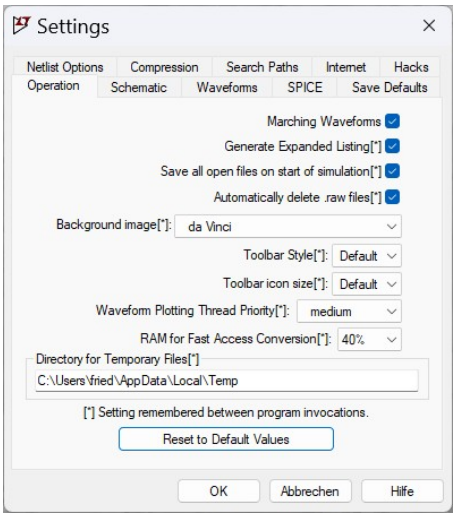


Figure D.6: Operation tab

Remark D.1 – (Automatically deleting RAW files)

RAW files from long transient simulations can become very large. Automatic deletion saves disk space after the waveform window is closed. Disable this option temporarily if a RAW file must be retained for later evaluation or shared with another user.

Table D.2: Summary of the recommended LTspice settings.

Control Panel tab	Recommended setting
Sym. & Lib. Search Paths	Add each installed toolbox folder to both the symbol and library search-path lists.
SPICE	Solver: **Alternate**.
Save Defaults	Enable the first three checkboxes.
Drafting Options	Arial, 10 pt, Bold.
Waveforms	Arial, 10 pt, Bold.
Operation	Enable Marching Waveforms, Generate Expanded Listing, and Automatically Delete .raw Files.

D.6 Verifying the Installation

After closing the Control Panel, restart LTspice and perform the following checks:

1. Open the component-selection dialog and verify that toolbox symbols can be selected.
2. Open one transient-toolbox example circuit.
3. Confirm that LTspice reports no missing symbol or library-file errors.
4. Run a short transient analysis.
5. Verify that the waveform viewer opens and that hierarchical quantities required by the example can be selected.
6. If the steady-state toolbox is installed, open and run one of its load-flow examples as a separate check.

Important D.3 – (If models or symbols are not found)

Check the spelling of the Windows user name, confirm the extracted folder level, and verify that the toolbox path was entered in both search-path lists. After changing a search path, restart LTspice before testing the installation again.

Appendix Summary

Install LTspice 24.0.12 with its default options, extract the transient toolbox to the user's Documents folder, and optionally install the steady-state load-flow toolbox. Add every installed toolbox folder to both the symbol and library search paths. Use the alternate solver, save the internal model data required by the examples, configure consistent schematic and waveform fonts, and enable the recommended operating options. A short example simulation should be run after installation to verify that all symbols and library models are available.

E Questions and Answers for Transient Simulation

This appendix collects practical questions that frequently arise when using the transient-simulation toolbox. The answers refer to the conventions and models used in this book.

E.1 Basic Simulation Setup

Which parameters should be present on every schematic?

The following global parameters are recommended:

```
.param tdv=1 tdt=1
.param fn1=50 omega_n={2*pi*fn1}
.options vntol=1m abstol=1u
```

Use `fn1=60` for a 60 Hz system. The values `vntol=1m` and `abstol=1u` correspond to 1 mV and 1 µA; they are included in the toolbox template `template.asc`.

For a 50 Hz to 60 Hz system, a useful maximum transient time step is between 0.1 ms and 0.6 ms. A smaller step improves temporal resolution but increases simulation time. The alternate solver is recommended under Tools → Control Panel → SPICE.

If a result is physically implausible, first check transformer orientation and all engineering prefixes. In LTspice, `m` means milli; use `Meg` for mega. A value of `400` means 400 V, not 400 kV.

How can steady state be reached quickly?

The voltage-source parameters `texp` and `tpwl` can be used to ramp the source voltage. Values between approximately 10 ms and 100 ms normally produce a final voltage buildup over roughly 50 ms to 500 ms.

Transformer energization can additionally be improved by coordinating the residual-flux parameter `psi_0` with the switching instant. For a phase-R voltage zero crossing followed by a positive voltage half-cycle:

- `psi_0=-1` can place the flux close to its steady-state path;
- `psi_0=0` can produce approximately twice rated peak flux;
- `psi_0=1` can produce approximately three times rated peak flux.

At a voltage maximum, `psi_0=0` normally gives the smallest flux offset. The exact polarity depends on the winding and voltage sign convention.

What can be done if the circuit does not converge?

Use the following sequence:

1. Verify units, polarities, initial conditions, floating nodes, and switch timing.
2. Reduce the maximum time step.

3. Use the alternate solver.
4. Provide realistic parasitic resistances and DC paths.
5. If necessary, relax `vntol`, `abstol`, or `reltol` moderately.

Increasing tolerances can help convergence but reduces numerical accuracy; repeat the final simulation with tighter tolerances as a check.

How can subcircuit currents and voltages be accessed?

Open

Tools → Control Panel → Save Defaults

and enable **Save Subcircuit Node Voltages** and **Save Subcircuit Device Currents**. Restart or rerun the simulation after changing these settings.

How are power, energy, and current direction displayed?

Hold Alt and click a component to plot its instantaneous power, based on the voltages and currents at its pins. Hold Ctrl and click the power trace label to integrate the trace over the displayed interval and obtain energy.

LTspice defines device-pin currents as positive into the pin. Resistors, capacitors, and inductors display a current arrow. Mirroring a symbol also mirrors this reference direction.

What must be considered for LTspice inductors?

LTspice inductors may contain a default series resistance of 1 mΩ. Right-click the inductor and set `Rser=0` if an ideal inductor is required. The global default can be changed under Tools → Control Panel → Hacks by enabling the option that defaults inductors to `Rser=0`.

Removing all loss can create an undamped resonant circuit and may worsen convergence. Use a physically meaningful resistance whenever possible.

E.2 Transformer Models and Magnetic Quantities

How is a nonlinear transformer changed to a linear model?

Right-click the transformer and set

```
m=1
```

in the value or parameter field. This removes the high-order nonlinear term in the toolbox magnetization characteristic.

Which parameters determine the magnetization curve?

The characteristic is controlled primarily by a and m:

$$i_\mu(\Phi) = I_{\mu N}\left[a\frac{\Phi}{\Phi_N} + (1-a)\left(\frac{\Phi}{\Phi_N}\right)^m\right]. \tag{E.1}$$

The parameter a determines the linear share and the small-signal slope, while m controls the sharpness of saturation. Typical large-transformer examples use $m = 7$ or $m = 9$. Example curves are available in `Magnetisierungskurven_different_xxx.asc`.

The quantities $I_{\mu N}$ and Φ_N must use the same amplitude convention. If peak flux is used, the rated peak values are

$$\Phi_N = \frac{\sqrt{2}\,U_{N,\mathrm{ph}}}{\omega N}, \qquad I_{\mu N} = \sqrt{2}\,I_{\mu N,\mathrm{rms}}. \tag{E.2}$$

How is residual flux specified?

Use the transformer parameter `psi_0` for a predefined residual-flux pattern. Where supported, phase-specific parameters should be used for an arbitrary residual-flux vector. The initialization time must match the desired switching event.

How is residual flux applied at an arbitrary time?

The model briefly connects an initialization voltage source to the 1 F flux-integrator capacitor. The switch on-resistance and duration determine the charging accuracy:

$$\tau = R_{\mathrm{on}}C. \tag{E.3}$$

For example,

$$R_{\mathrm{on}} = 10^{-5}\,\Omega, \quad C = 1\,\mathrm{F} \quad\Longrightarrow\quad \tau = 0.01\,\mathrm{ms}. \tag{E.4}$$

A closing duration of 0.1 ms corresponds to 10τ and charges the capacitor essentially to its final value. The rapid initialization can produce a narrow voltage or current spike at the charging instant.

Where can introductory LTspice tutorials be found?

An introductory tutorial collection is available at http://www.simonbramble.co.uk/lt_spice/ltspice_lt_spice.htm.

How is the JUSTUS transformer model composed?

The model combines an electrical circuit with a magnetic equivalent circuit. Iron loss, leakage, and core paths can therefore be represented with magnetic quantities. Under the magnetic-electrical duality used in the model:

- electrical series connections correspond to magnetic parallel paths;

- electrical parallel connections correspond to magnetic series paths;
- resistance corresponds to conductance under the chosen analogy;
- voltage-source and current-source roles are interchanged;
- T and Π topologies are dual representations.

For a winding with N turns,

$$\mathscr{R}_\sigma = \frac{N^2}{L_\sigma}, \tag{E.5}$$

$$\mathscr{R}_{R\mathrm{Fe}} = \frac{N^2}{R_{\mathrm{Fe}}}, \tag{E.6}$$

according to the scaling convention of the toolbox model. The placement of the corresponding branches follows the electrical-magnetic duality described above.

How can approximate data for a 300 MV A transformer be estimated?

Assume a 400 kV/120 kV Yy transformer, $N_1 = 1000$, magnetic path length $l_m = 10\,\mathrm{m}$, rated flux density $B_N = 1.8\,\mathrm{T}$, and RMS magnetizing current $I_{\mu N} = 0.5\,\mathrm{A}$.

Using RMS sinusoidal quantities,

$$\Psi_{1N,\mathrm{rms}} = \frac{U_{1N,LL}}{\sqrt{3}\,\omega} = 735.1\,\mathrm{Wb}, \tag{E.7}$$

$$\Phi_{N,\mathrm{rms}} = \frac{\Psi_{1N,\mathrm{rms}}}{N_1} = 0.735\,1\,\mathrm{Wb}, \tag{E.8}$$

$$A_{\mathrm{Fe}} = \frac{\Phi_{N,\mathrm{rms}}}{B_N} \approx 0.408\,\mathrm{m}^2, \tag{E.9}$$

$$\Theta_{\mu N} = N_1 I_{\mu N} = 500\,\mathrm{A}, \tag{E.10}$$

$$H_N = \frac{\Theta_{\mu N}}{l_m} = 50\,\mathrm{A/m}, \tag{E.11}$$

$$\mu_r = \frac{B_N}{\mu_0 H_N} \approx 2.86 \times 10^4, \tag{E.12}$$

$$\mathscr{R}_m = \frac{\Theta_{\mu N}}{\Phi_{N,\mathrm{rms}}} \approx 680/\mathrm{H}, \tag{E.13}$$

$$L_{1h} = \frac{N_1^2}{\mathscr{R}_m} \approx 1\,470\,\mathrm{H}. \tag{E.14}$$

The equivalent peak convention gives $\widehat{\Phi}_N = \sqrt{2}\Phi_{N,\mathrm{rms}} = 1.039\,6\,\mathrm{Wb}$ and $\hat{I}_{\mu N} = \sqrt{2} I_{\mu N}$, leaving reluctance and inductance unchanged. RMS and peak quantities must not be mixed in one calculation.

What does the nonlinear curve predict at one, two, and three times rated flux?

For $a = 0.75$, $m = 9$, and a reference current of $I_{\mu N} = 0.5\,\mathrm{A}$ in a consistent amplitude convention,

$$i_\mu(1\Phi_N) = 0.5\,\mathrm{A}, \tag{E.15}$$

$$i_\mu(2\Phi_N) = 64.75\,\mathrm{A}, \tag{E.16}$$

$$i_\mu(3\Phi_N) = 2\,461.5\,\mathrm{A}. \tag{E.17}$$

For $l_m = 10\,\mathrm{m}$ and $B_N = 1.8\,\mathrm{T}$, the corresponding approximate magnetic quantities are

Φ/Φ_N	B	H	μ_r	L_{1h}
1	1.8 T	50 A/m	2.86×10^4	1 470 H
2	3.6 T	6 475 A/m	442	22.7 H
3	5.4 T	$2.46 \cdot 10^5$ A/m	17.5	0.90 H

These are point estimates based on the assumed algebraic curve. In a dynamic simulation, winding voltage drop, leakage flux, and waveform distortion reduce the effective flux and RMS current. Even at three times rated flux, the estimated main inductance remains above the example leakage inductance $L_\sigma \approx 0.169\,7\,\mathrm{H}$.

How can tank reluctance be estimated?

The tank path is strongly geometry- and frequency-dependent. A placeholder inductance of a few henries, or roughly 10 to 100 times the winding leakage inductance, may be useful for sensitivity studies, but it is not a validated transformer parameter. Measurements or field calculations are required for an accurate value.

How can transformer voltage and phase be regulated?

The rudimentary tap parameter `l_` changes the voltage in steps. For example, `l_=3` and `stproz=1.5` correspond to a nominal 4.5 % change, subject to the model sign convention.

The models `trafo_tap_l` and `trafo_tap_s` provide voltage and phase-angle regulation. Versions with fixed parameters and versions with control pins are available; the latter can be driven by a PWL source during a transient simulation.

E.3 Selecting and Analysing Models

How are alternative models selected?

Right-click the element and open the **SpiceModel** drop-down list. It contains compatible models from the associated `.lib` file. The model pin count and pin order must match the symbol. Examples include `l`, `l_int`, and `l_flux`.

How is a transient model converted to AC analysis?

Replace transient sources by sources with AC magnitude and phase definitions, and replace the `.tran` directive by an `.ac` directive. AC analysis linearizes the circuit about its operating point; it is generally not suitable for studying switching events or strongly nonlinear large-signal behaviour.

How can reactive-power transfer be recognized?

Reactive-power transfer produces a phase displacement between the fundamental voltage and current. For sinusoidal RMS phasors,

$$Q = \sqrt{3}\, U_{LL} I_L \sin\varphi. \tag{E.18}$$

For distorted waveforms, calculate average active power from $u(t)i(t)$ and use a harmonic-aware definition of reactive power.

Why do many elements have control pins?

Control pins allow parameters such as voltage level, tap position, phase shift, switch state, or model coefficient to change during the simulation. A PWL source is commonly used to drive a discrete or continuous control input.

E.4 Inductive Coupling and Flux

How is leakage between two coupled inductors calculated?

For self-inductances L_1, L_2 and mutual inductance M,

$$k = \frac{M}{\sqrt{L_1 L_2}}, \tag{E.19}$$

$$\sigma = 1 - \frac{M^2}{L_1 L_2} = 1 - k^2. \tag{E.20}$$

If the individual leakage factors are σ_1 and σ_2,

$$\sigma = \sigma_1 + \sigma_2 - \sigma_1\sigma_2. \tag{E.21}$$

For a symmetric transformer, $\sigma_1 = \sigma_2$, hence

$$\sigma = 2\sigma_1 - \sigma_1^2, \qquad k = 1 - \sigma_1, \qquad L_{1\sigma} = (1-k)L_1. \tag{E.22}$$

With $L_{1\sigma} = 0.169\,7\,\mathrm{H}$ and $L_1 = 1\,470\,\mathrm{H}$,

$$\sigma_1 = 1.1544 \times 10^{-4}, \tag{E.23}$$

$$\sigma = 2.3087 \times 10^{-4}, \tag{E.24}$$

$$k = \boxed{0.9998846}. \tag{E.25}$$

The voltage ratio and inductances satisfy

$$\frac{N_1}{N_2} = \sqrt{\frac{L_1}{L_2}} \tag{E.26}$$

when both inductances use the same magnetic path and reference convention.

What is the rated peak flux for 1000 turns?

For sinusoidal phase voltage $U_{N,\mathrm{ph}}$,

$$\hat{\Psi}_N = \frac{\sqrt{2}\,U_{N,\mathrm{ph}}}{\omega}, \qquad \hat{\Phi}_N = \frac{\hat{\Psi}_N}{N}. \tag{E.27}$$

At 50 Hz and $N = 1000$:

Voltage applied to winding	Connection interpretation	$\hat{\Phi}_N$
$10\,\mathrm{kV}/\sqrt{3}$	10 kV star phase	0.026 0 Wb
10 kV	Winding voltage	0.045 0 Wb
$20\,\mathrm{kV}/\sqrt{3}$	20 kV star phase	0.052 0 Wb
20 kV	Winding voltage	0.090 0 Wb
$110\,\mathrm{kV}/\sqrt{3}$	110 kV star phase	0.286 Wb
110 kV	Winding voltage	0.495 Wb
$220\,\mathrm{kV}/\sqrt{3}$	220 kV star phase	0.572 Wb
220 kV	Winding voltage	0.990 Wb
$400\,\mathrm{kV}/\sqrt{3}$	400 kV star phase	1.039 6 Wb

What are the time constants of RL and RC circuits?

$$\tau_{RL} = \frac{L}{R}, \qquad \tau_{RC} = RC. \tag{E.28}$$

A first-order response reaches approximately 99.3 % of its final change after 5τ and 99.995 % after 10τ.

What do `tpsid` and `ron` mean?

The parameter `tpsid` specifies how long the initialization source is connected to the 1 F flux-integrator capacitor. The parameter `ron` is the switch on-resistance. Suitable example values are

```
tpsid=0.1m  ron=1e-5
```

giving $\tau = R_{\mathrm{on}}C = 0.01\,\mathrm{ms}$ and an initialization interval of 10τ.

How can the magnetization curve be plotted?

Open the transformer secondary. In the waveform viewer, change the horizontal axis from time to primary current and plot internal flux or flux linkage versus that current. A nonzero iron-loss parameter `p0` produces a loop-like separation because the loss current is superimposed; it should not be confused with a fully physical hysteresis model unless the model explicitly contains hysteresis state equations.

E.5 Lines, Cables, and Compensation

How is cable charging power calculated?

For line-to-line RMS voltage U_N, phase-to-earth capacitance C_E, and phase-to-phase capacitance C_L per phase set,

$$Q_0 = Q_{CE} + Q_{CL} = U_N^2 \omega C_E + 3U_N^2 \omega C_L, \tag{E.29}$$

$$I_0 = \frac{Q_0}{\sqrt{3} U_N}. \tag{E.30}$$

For a shielded single-core cable, C_L is commonly negligible and $Q_0 \approx Q_{CE}$. The capacitive earth-fault current is then approximately

$$I_E = 3I_0 = \frac{3Q_{CE}}{\sqrt{3} U_N}. \tag{E.31}$$

How are Petersen coils and shunt reactors sized?

For a Petersen coil connected from the transformer neutral to earth,

$$X_L = \frac{U_N/\sqrt{3}}{I_N}, \qquad L = \frac{X_L}{\omega}. \tag{E.32}$$

For an example with $U_N = 110\,\mathrm{kV}$ and $I_N = 250\,\mathrm{A}$, the coil loss may initially be estimated with $R \approx X_L/100$ and later replaced by measured data.

For a three-phase reactor with total rating Q_N:

$$L_\Delta = \frac{3U_N^2}{\omega Q_N}, \tag{E.33}$$

$$L_Y = \frac{U_N^2}{\omega Q_N}. \tag{E.34}$$

Each delta branch sees U_N, whereas each star branch sees $U_N/\sqrt{3}$.

If $C_E = 1\,000\,\text{nF}$ at 110 kV,

$$Q_0 = U_N^2 \omega C_E \approx 3.80\,\text{Mvar}, \tag{E.35}$$

$$I_0 = \frac{Q_0}{\sqrt{3} U_N} \approx 19.95\,\text{A}, \tag{E.36}$$

$$I_E = 3 I_0 \approx 59.85\,\text{A}. \tag{E.37}$$

A Petersen coil rated approximately 110 kV and 60 A therefore provides the required compensation.

How does coupling influence parallel lines?

Lines on the same route have mutual inductive and capacitive coupling. Depending on current direction and phase arrangement, mutual induction can reduce or increase the effective series inductance. In the idealized same-direction case, $k \to 1$ makes the differential leakage inductance approach zero.

The toolbox contains coupled-line models for one, two, and four circuits. Their coupling matrices were calculated with ATP/ATPDraw and converted to LTspice format with the supplied conversion tools.

How is source impedance obtained from short-circuit power?

For a balanced three-phase network,

$$Z_N = \frac{U_N^2}{S_k}. \tag{E.38}$$

Examples are

$$U_N = 400\,\text{kV}, \quad S_k = 40\,\text{GV A} \Rightarrow Z_N = 4.0\,\Omega, \tag{E.39}$$

$$U_N = 110\,\text{kV}, \quad S_k = 7.5\,\text{GV A} \Rightarrow Z_N = 1.613\,\Omega. \tag{E.40}$$

If no resistance information is available, the first approximation is an inductive source impedance jX_N with $X_N \approx Z_N$.

How is the total capacitance of an overhead line calculated?

With phase-to-earth capacitance C_E and phase-to-phase capacitance C_L, the effective positive-sequence charging capacitance is

$$C_{\text{sum}} = C_E + 3C_L. \tag{E.41}$$

The capacitive reactive power is

$$Q_C = U_N^2 \omega C_{\text{sum}} = \frac{U_N^2}{X_C}. \tag{E.42}$$

How is the reactive power of a shunt reactor calculated?

For a balanced three-phase reactor connected in star,

$$Q_L = \frac{U_N^2}{X_L} = \frac{U_N^2}{\omega L}, \tag{E.43}$$

where L is the inductance of each phase branch.

What is the difference between star and delta loads with identical branch impedances?

If the same branch resistance or impedance is used in both connections at the same line-to-line voltage, each delta branch receives $\sqrt{3}$ times the star branch voltage. Therefore,

$$P_\Delta = 3P_Y, \qquad Q_\Delta = 3Q_Y. \tag{E.44}$$

E.6 Plot Files and Verification

How is a saved plot configuration displayed?

Many examples include saved plot-setting files. In the waveform viewer, use Plot Settings → Open Plot Settings and select the required file. If several configurations are supplied, open them one at a time to reproduce the intended trace groups.

How should a transient simulation be verified?

Always repeat the simulation with a smaller maximum time step. A credible result should change only slightly as the time step is reduced. Also check:

- energy and power balance;
- expected steady-state RMS values and phase sequence;
- sensitivity to tolerances and parasitic resistances;
- agreement between transient, AC, and hand calculations where applicable;
- correct current reference directions and voltage bases;
- plausibility of resonant frequencies and decay time constants.

Appendix Summary

Reliable transient simulation begins with consistent global parameters, correct units, suitable time steps, and physically meaningful damping. Source ramps and coordinated residual flux reduce initialization transients, while the alternate solver and carefully selected tolerances improve numerical robustness. Transformer calculations must use consistent RMS or peak conventions, and coupled-line and compensation studies require correct three-phase voltage

bases. Every important result should be repeated with a smaller time step and compared with analytical estimates.

F LTspice Schematic Editor: Menu Bar, Toolbar, and Basic Operation

> Version note: This overview refers to the Windows schematic editor used in LTspice 24.0.12. The illustration shows the default LTspice 24 toolbar. Users who prefer the older, smaller icon set can select the legacy toolbar with Tools → Settings → Operation → Toolbar Style → Legacy.

F.1 Main Areas of the Schematic Editor

1. Menu bar

The menu bar provides access to all drawing, simulation, display, and program configuration commands. Toolbar icons are available for the most commonly used commands.

2. Toolbar

The toolbar gives direct access to frequently used commands such as opening a schematic, starting a simulation, inserting components, drawing wires, and editing circuit elements. Moving the mouse pointer over an icon displays its function as a tooltip.

3. Drawing area

Components, wires, node labels, comments, and SPICE directives are placed in the drawing area. The electrical reference node must be the ground node named 0. A schematic without node 0 normally cannot be simulated.

F.2 Menu Bar

Menu	Purpose
File	Creates, opens, closes, saves, and prints schematic files. It also provides access to recently used files and program exit.
Edit	Contains undo and editing operations such as delete, duplicate, move, stretch, rotate, mirror, search, and selection commands.
Hierarchy	Creates and manages hierarchical blocks, symbols, ports, and multi-level schematics.
View	Controls zoom, grid, anchors, unconnected-pin markers, netlist display, SPICE error log, and other schematic or waveform views.
Simulate	Starts, pauses, or stops a simulation and opens the dialog for defining transient, AC, DC, noise, and operating-point analyses.
Tools	Opens program settings and provides component, symbol, and library management functions.
Window	Selects and arranges open schematic, waveform, and text windows.
Help	Opens the LTspice documentation, keyboard-shortcut information, and version information.

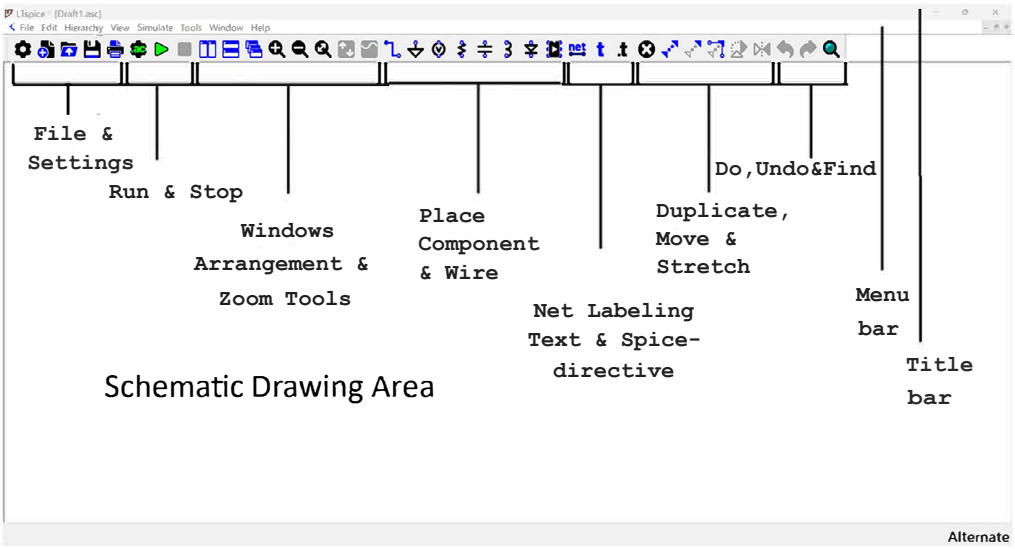


Figure F.1: Annotated LTspice 24 schematic-editor window showing the title and menu bars, functional toolbar groups, schematic drawing area, grid, and status bar. The appearance of individual icons depends on the selected toolbar style and LTspice version.

F.3 Toolbar Symbols

Command	Shortcut	Function
New Schematic	Ctrl+N	Creates a new empty schematic.
Open	Ctrl+O	Opens an existing schematic or supported LTspice file.
Save	Ctrl+S	Saves the active schematic.
Control Panel / Settings	---	Opens the general LTspice configuration.
Run	---	Starts the selected simulation.
Stop / Halt	---	Stops the active simulation.
Zoom Area	Ctrl+Z	Enlarges a rectangular part of the schematic.
Zoom Back	Ctrl+B	Returns to the preceding zoom level.
Zoom to Fit	Space	Fits the complete schematic into the window.
Wire	F3	Draws an electrical connection between nodes.
Ground	G	Places the mandatory reference node 0.
Label Net	F4	Assigns a name to a node or electrical network.
Component	F2	Opens the component and symbol selection dialog.
Resistor	R	Places a resistor.
Capacitor	C	Places a capacitor.
Inductor	L	Places an inductor.
Diode	D	Places a diode.
SPICE Directive	S	Places commands such as .tran, .param, .options, or .measure.
Text Comment	T	Places explanatory text without electrical effect.
Delete	F5	Deletes selected components, wires, labels, or text.
Duplicate / Copy	F6	Creates a copy of the selected object.
Move	F7	Moves an object; connected wires are not stretched with it.
Stretch / Drag	F8	Moves an object while retaining and adapting its wires.
Rotate	Ctrl+R	Rotates the selected component by 90 degrees.
Mirror	Ctrl+E	Mirrors the selected component.
Undo	F9*	Reverses the last editing action.

* The effective shortcut can depend on the active window and the configured hotkeys. LTspice displays the currently assigned keys in its menus and tooltips.

F.4 Basic Workflow

Creating and simulating a circuit

1. Create or open a schematic.
2. Insert the required components and assign their parameters.
3. Connect the circuit with the Wire tool.
4. Insert ground node 0 and assign useful net labels.
5. Define the simulation with Simulate → Configure Analysis or place the corresponding SPICE directive directly on the schematic.

6. Press Run and select node voltages or branch currents in the waveform viewer.

Move and Stretch: Move changes the position of a component without preserving the exact wire geometry. Stretch moves the component together with its connected wires. Press Esc or the right mouse button to leave an active editing mode.

F.5 References

- Analog Devices, "LTspice Basics: Schematic Capture."
- Analog Devices EngineerZone, information about the LTspice 24 toolbar and the legacy toolbar style, "Toolbar is Incomplete."
- Interface illustration adapted from the LTspice schematic-entry tutorial by LEnsE, Institut d'Optique Graduate School, "LTspice: Schematic Entry."

Figure F.2: Most used symbols for transient toolbox

Over the past 25 years, I have developed many macromodels for LTspice. This work resulted in two toolboxes: one for nonlinear load-flow calculations and one for transient simulations of electrical power networks. Adapting a SPICE simulator for nonlinear load-flow calculations (with a mathematical trick) was an important part of this development. Both toolboxes are intended for educational use, especially by students of electrical engineering. Their source files are freely available, together with hundreds of examples. This book focuses on transient simulations of electrical power networks with a detailed description and simulation of 9 handpicked examples. The transient toolbox includes the essential components needed to model power systems: voltage sources, synchronous generators with saturation, nonlinear transformers with residual-flux initialisation and phase-angle regulation, Petersen coils for earth-fault compensation, shunt reactors, loads, overhead lines and cables, coupled lines, grounding transformers and many other components. LTspice is a simulator developed by Analog Devices and is available free of charge. Having worked for more than 35 years as a load dispatcher in Vienna's electricity distribution system, I have drawn on my practical experience in developing these toolboxes. I hope that you enjoy exploring electrical power systems through simulation as much as I do. Some simulations can be challenging, but the results offer valuable insights. Good luck with your simulations!

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About the Author

Friedrich Schmidt has been working for more than 25 years on the practical modelling and simulation of electrical power systems, with a particular focus on load-flow calculations and transient simulations.

For questions, comments, or suggestions concerning this book and the LTspice toolboxes, he can be contacted by e-mail:

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LTspice Toolboxes

The LTspice toolboxes described and used throughout this book can be downloaded free of charge from the following sources:

- **Author's website:** to download the simulator, the toolboxes and hundreds of examples

  https://www.friedrichschmidt.at
- **LTspice User Group:**

  After signing in, the toolboxes with hundreds of examples can be downloaded from:
  https://groups.io/g/LTspice/files/z_yahoo/adventures_with_loadflow
- **GitHub:**

  After signing in the toolboxes with the nine examples and the ltspice simulator can also be found on GitHub by searching for `LTspice loadflow`.

The toolboxes contain models, libraries, and numerous simulation examples for the analysis of electrical power systems with LTspice.